

SingleRAN

EPC-based NSA Basics Feature Parameter Description

Issue	Draft B
Date	2026-01-30



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....1

2 About This Document.....6

2.1 General Statements.....6

2.2 Applicable RAT.....7

2.3 Features in This Document.....7

2.4 Differences.....8

3 Overview.....13

3.1 Introduction.....13

3.2 Related Concepts.....13

3.3 Architecture.....15

3.4 EN-DC Band Combinations.....17

4 NSA Networking Based on EPC.....23

4.1 Principles.....23

4.1.1 Carrier Management in NSA DC.....23

4.1.1.1 MCG Carrier Management.....23

4.1.1.1.1 SCC Management.....23

4.1.1.1.2 Uplink CA in EN-DC.....26

4.1.1.2 SCG Carrier Management.....27

4.1.1.2.1 Measurement-based PSCell Configuration.....32

4.1.1.2.2 PSCell Management for VoLTE UEs.....36

4.1.1.2.3 PSCell SMTC on the LTE Side.....38

4.1.1.2.4 SCG Addition Based on Virtual Grid Models.....39

4.1.1.2.5 SPID-based SCG Frequency Priority Configuration.....41

4.1.1.2.6 Operator-specific SCG Frequency Priority Configuration.....42

4.1.1.2.7 Differentiated SCG Addition and Deletion Threshold Configuration for NR CA-capable UEs.....42

4.1.1.3 SPID/QCI-specific Carrier Management.....45

4.1.1.3.1 SPID-specific NSA PCC Anchoring and PSCell Management Policies.....45

4.1.1.3.2 QCI-specific NSA PCC Anchoring and PSCell Management Policies.....49

4.1.2 Data Split in NSA DC.....52

4.1.2.1 Data Split Policy Selection.....53

4.1.2.1.1 Downlink Data Split.....53

4.1.2.1.2 Uplink Data Split.....58

4.1.2.2 Uplink Fallback to LTE.....	61
4.1.2.3 Fast Retransmission for Downlink Data Split.....	62
4.1.2.4 CP/UP Separation.....	62
4.1.2.5 UL/DL Separation.....	64
4.1.2.6 NSA UE Data Split Based on CCE Usage.....	64
4.1.2.7 SPID-based NSA Uplink Primary Path Configuration.....	67
4.1.3 Uplink Power Control in NSA DC.....	68
4.1.4 Interference Avoidance in NSA DC.....	69
4.1.4.1 Interference Type Identification.....	70
4.1.4.2 Interference Avoidance Policies.....	71
4.1.4.3 Implementation of Interference Avoidance.....	71
4.1.5 NSA DC Virtual Grid Models.....	73
4.1.6 Other Functions in NSA DC.....	79
4.1.6.1 NR Data Volume Reporting.....	79
4.1.6.2 S-K _{gNB} Update.....	81
4.1.6.3 gNodeB and UE Signaling Exchange on SRB3.....	81
4.1.6.4 Uplink Single-Side Transmission (Trial).....	82
4.1.6.5 Uplink Preallocation.....	84
4.1.6.6 SCG Release and Addition Based on UE Overheating Status Reporting.....	86
4.1.6.7 Intra-Band EN-DC.....	87
4.1.6.8 EN-DC Band Combination Blacklist Function.....	90
4.1.6.9 RRC Connection Reestablishment Optimization for Defective NSA UEs.....	92
4.1.6.10 Scheduling Protection Against the Impact of NR SRS Antenna Switching on LTE Bands.....	92
4.1.6.11 NCGI Measurement Optimization in EN-DC.....	93
4.1.6.12 Penalty on Abnormal SCG Releases for NSA UEs.....	94
4.2 Network Analysis.....	95
4.2.1 Benefits.....	95
4.2.2 Impacts.....	97
4.3 Requirements.....	101
4.3.1 Licenses.....	101
4.3.2 Functions.....	102
4.3.2.1 Prerequisite Functions.....	102
4.3.2.2 Mutually Exclusive Functions.....	104
4.3.2.3 Function Impacts.....	107
4.3.3 Hardware.....	130
4.3.4 Networking.....	132
4.3.5 Cells.....	136
4.3.6 Others.....	136
4.4 Operation and Maintenance.....	137
4.4.1 When to Use.....	137
4.4.2 Data Configuration.....	137
4.4.2.1 Data Preparation.....	138

4.4.2.2 Using MML Commands..... 179

4.4.2.3 Using the MAE-Deployment..... 187

4.4.3 Activation Verification..... 188

4.4.4 Network Monitoring..... 192

5 L4S Congestion Identification..... 195

5.1 Principles..... 195

5.1.1 Uplink L4S Congestion Identification..... 195

5.1.2 Downlink L4S Congestion Identification..... 197

5.2 Network Analysis..... 198

5.2.1 Benefits..... 198

5.2.2 Impacts..... 198

5.3 Requirements..... 198

5.3.1 Licenses..... 199

5.3.2 Functions..... 199

5.3.2.1 Prerequisite Functions..... 199

5.3.2.2 Mutually Exclusive Functions..... 199

5.3.2.3 Function Impacts..... 199

5.3.3 Hardware..... 199

5.3.4 Networking..... 200

5.3.5 Others..... 200

5.4 Operation and Maintenance..... 200

5.4.1 Data Configuration..... 200

5.4.1.1 Data Preparation..... 200

5.4.1.2 Using MML Commands..... 202

5.4.1.3 Using the MAE-Deployment..... 202

5.4.2 Activation Verification..... 203

5.4.3 Network Monitoring..... 203

6 Parameters..... 204

7 Counters..... 206

8 Glossary..... 207

9 Reference Documents..... 208

1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
 - Added descriptions of events A4 and A6 and revised the description of event A3. For details, see [3.2 Related Concepts](#).
 - Added FOFD-071203 Adaptive Managed Latency to [2.3 Features in This Document](#) and added descriptions of L4S congestion identification in [2.4 Differences](#).
 - Added the impact of the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs". For details, see [4.2.2 Impacts](#).
 - Revised the RAT information of downlink non-GBR packet bundling, delay-based differentiated scheduling for extended QCI, and enhanced downlink packet length awareness scheduling in [4.3.2.3 Function Impacts](#).
 - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 06 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the function of requesting LTE-side large-packet status of ULFB UEs. For details, see: • 4.1.2.2 Uplink Fallback to LTE • 4.2.2 Impacts • 4.4.2.1 Data Preparation • 4.4.2.2 Using MML Commands • 4.4.3 Activation Verification	Modified parameter on the NR side: Added the ULFB_LTE_UL_LRG_PKT_REQ_SW option to the gNodeBParam.NsaDcOptSwitch parameter.	LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Excluded the BTS5929R from the base station models that are compatible with NSA networking based on EPC. For details, see 4.3.3 Hardware.	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added 16T16R to the description stating the mutually exclusive relationship between LTE FDD and NR Flash Dynamic Spectrum Sharing and uplink single-side transmission. For details, see 4.3.2.2 Mutually Exclusive Functions.	None	NR FDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the support for measuring the number of SCG releases for VoLTE UEs due to uplink SINR deterioration or a high uplink or downlink packet loss rate. For details, see 4.4.4 Network Monitoring .	None	LTE FDD LTE TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Added the L4S congestion identification function. For details, see 5 L4S Congestion Identification .	None	LTE FDD LTE TDD NR FDD Low-frequency NR TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Added the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs". For details, see: 4.1.1.2 SCG Carrier Management 4.2.2 Impacts 4.4.2.1 Data Preparation 4.4.2.2 Using MML Commands	Modified parameter on the LTE side: Added the NSA_REEST_FLOW_CONFLICT_OPT_SWITCH.NsaDcAlgoExtSwitch parameter.	LTE FDD LTE TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the "inter-frequency and inter-RAT measurement optimization for VoLTE UEs in NSA DC scenarios" function. For details, see: • 4.1.1.2.2 PSCell Management for VoLTE UEs • 4.2.1 Benefits • 4.2.2 Impacts • 4.3.2.3 Function Impacts • 4.4.2.1 Data Preparation • 4.4.2.2 Using MML Commands	Modified parameter on the LTE side: Added the VOLTE_NSA_DC_IF_MEAS_OPT_SW option to the NsaDcMgmtConfig.NsaDcAlgoEnhSwitch parameter.	LTE FDD LTE TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationship between NSA networking based on EPC and LNR CRS muting. For details, see 4.3.2.3 Function Impacts.	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added the configuration requirement of uplink fallback to LTE for the NRDUCell.SlotAssignment parameter. For details, see 4.3.5 Cells.	None	High-frequency NR TDD	5900 series base stations (macro base stations)
Added the probability prediction model function. For details, see: • 4.1.1.2.4 SCG Addition Based on Virtual Grid Models • 4.4.2.1 Data Preparation • 4.4.2.2 Using MML Commands	Modified parameter on the LTE side: Added the PROBABILITY_MODEL_SW option to the LnrCarrierSelection.LnrCarrierSelectionSw parameter.	LTE FDD LTE TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between interference avoidance in NSA DC and accurate uplink RB scheduling. For details, see 4.3.2.3 Function Impacts .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Optimized the activation verification of virtual grid model building by introducing new counters that indicate the numbers of good-RSRP and poor-RSRP virtual grid models. For details, see 4.4.3 Activation Verification .	None	LTE FDD LTE TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
 - Modified the flowchart of MeNB-triggered SCG addition and added the formula for calculating the number of NR frequencies that can be delivered during SCG addition. For details, see [4.1.1.2 SCG Carrier Management](#) and [4.1.1.2.1 Measurement-based PSCell Configuration](#).
 - Added an impact relationship between UL and DL Decoupling and secondary intermodulation interference avoidance. For details, see [4.3.2.3 Function Impacts](#).
 - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-131 122	NSA Networking based on EPC (LTE FDD)	• 4 NSA Networking Based on EPC • 4.1.1 Carrier Management in NSA DC • 4.1.2 Data Split in NSA DC • 4.1.3 Uplink Power Control in NSA DC • 4.1.4 Interference Avoidance in NSA DC • 4.1.5 NSA DC Virtual Grid Models • 4.1.6 Other Functions in NSA DC
LTE TDD	MRFD-131 132	NSA Networking based on EPC (LTE TDD)	

RAT	Feature ID	Feature Name	Chapter/Section
NR	MRFD-131 162	NSA Networking based on EPC (NR)	• 4 NSA Networking Based on EPC • 4.1.2 Data Split in NSA DC • 4.1.4 Interference Avoidance in NSA DC • 4.1.6 Other Functions in NSA DC
NR	FOFD-0712 03	Adaptive Managed Latency	5 L4S Congestion Identification

2.4 Differences

Differences Between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
Carrier management in NSA DC	• Uplink CA in EN-DC: supported only in LTE FDD • Other functions: no difference	4.1.1 Carrier Management in NSA DC
Data split in NSA DC	• NSA UE data split based on CCE usage: supported only in LTE FDD • Other functions: no difference	4.1.2 Data Split in NSA DC
Uplink power control in NSA DC	None	4.1.3 Uplink Power Control in NSA DC
Interference avoidance in NSA DC	• Secondary intermodulation interference avoidance: supported only in LTE FDD • Other functions: no difference	4.1.4 Interference Avoidance in NSA DC
NSA DC virtual grid model building	None	4.1.5 NSA DC Virtual Grid Models

Function Name	Difference	Chapter/Section
Other functions in NSA DC	- Uplink single-side transmission (trial): supported only in LTE FDD - Other functions: no difference	4.1.6 Other Functions in NSA DC
L4S congestion identification	None	5 L4S Congestion Identification

Differences Between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
Carrier management in NSA DC	None	4.1.1 Carrier Management in NSA DC
Data split in NSA DC	None	4.1.2 Data Split in NSA DC
Uplink power control in NSA DC	None	4.1.3 Uplink Power Control in NSA DC
Interference avoidance in NSA DC	- Secondary intermodulation interference avoidance: supported only in NR TDD - Other functions: no difference	4.1.4 Interference Avoidance in NSA DC
NSA DC virtual grid model building	None	4.1.5 NSA DC Virtual Grid Models
Other functions in NSA DC	- Uplink single-side transmission (trial): supported only in NR FDD - Other functions: no difference	4.1.6 Other Functions in NSA DC
L4S congestion identification	None	5 L4S Congestion Identification

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Carrier management in NSA DC	Supported only in NSA networking	4.1.1 Carrier Management in NSA DC
Data split in NSA DC	Supported only in NSA networking	4.1.2 Data Split in NSA DC
Uplink power control in NSA DC	Supported only in NSA networking	4.1.3 Uplink Power Control in NSA DC
Interference avoidance in NSA DC	Supported only in NSA networking	4.1.4 Interference Avoidance in NSA DC
NSA DC virtual grid model building	Supported only in NSA networking	4.1.5 NSA DC Virtual Grid Models
Other functions in NSA DC	Supported only in NSA networking	4.1.6 Other Functions in NSA DC
L4S congestion identification	This function is supported in both SA and NSA networking, with the following differences: • The function described in this document is supported only in NSA networking. For details about this function in SA networking, see <i>Adaptive Managed Latency</i> in 5G RAN feature documentation. • For NSA UEs, uplink and downlink congestion identification are supported only in Option 3x networking.	5 L4S Congestion Identification

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Carrier management in NSA DC	• SCG addition based on virtual grid models: supported only in low frequency bands of NR • NR FR2 frequency measurement in polling mode: supported only in high frequency bands of NR • Other functions: no difference	4.1.1 Carrier Management in NSA DC
Data split in NSA DC	• Downlink data split: – Basic mechanism of downlink dynamic data split: no difference – Other downlink data split functions: supported only in low frequency bands of NR • Fast retransmission for downlink data split: supported only in low frequency bands of NR • Other functions: no difference	4.1.2 Data Split in NSA DC
Uplink power control in NSA DC	Supported only in low frequency bands of NR	4.1.3 Uplink Power Control in NSA DC
Interference avoidance in NSA DC	Supported only in low frequency bands of NR	4.1.4 Interference Avoidance in NSA DC
NSA DC virtual grid model building	Supported only in low frequency bands of NR	4.1.5 NSA DC Virtual Grid Models

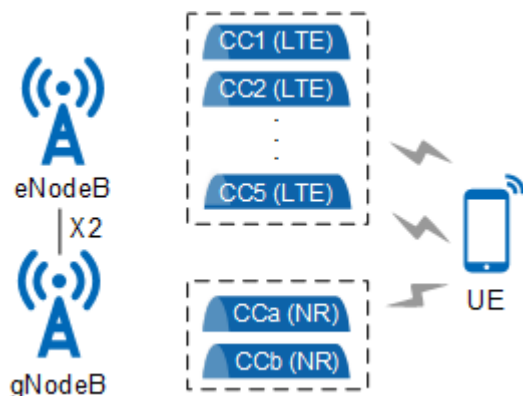
Function Name	Difference	Chapter/Section
Other functions in NSA DC	The following functions are supported only in low frequency bands of NR: • Uplink single-side transmission (trial) • Uplink preallocation • Intra-Band EN-DC • Scheduling protection against the impact of NR SRS antenna switching on LTE bands • NCGI measurement optimization in EN-DC Other functions: no difference	4.1.6 Other Functions in NSA DC
L4S congestion identification	Supported only in low frequency bands of NR	5 L4S Congestion Identification

3 Overview

3.1 Introduction

In non-standalone (NSA) networking based on evolved packet core (EPC), UEs that support NSA dual connectivity (DC) can connect to both an LTE eNodeB and a New Radio (NR) gNodeB and use radio resources provided by these base stations for data transmission. Data can be split between the two base stations. DC can be implemented after component carriers (CCs) are aggregated separately on the eNodeB and gNodeB sides, as shown in [Figure 3-1](#).

Figure 3-1 NSA networking based on EPC



3.2 Related Concepts

[Figure 3-2](#) is used as an example to illustrate concepts involved in NSA networking and [Table 3-1](#) describes these concepts. In the figure, cells 1 and 3 are co-coverage cells under the eNodeB, and cells 2 and 4 are co-coverage cells under the gNodeB. Each cell corresponds to a carrier. DC is provided for the NSA UE after two carriers are aggregated on the eNodeB and two carriers are aggregated on the gNodeB.

Figure 3-2 NSA networking

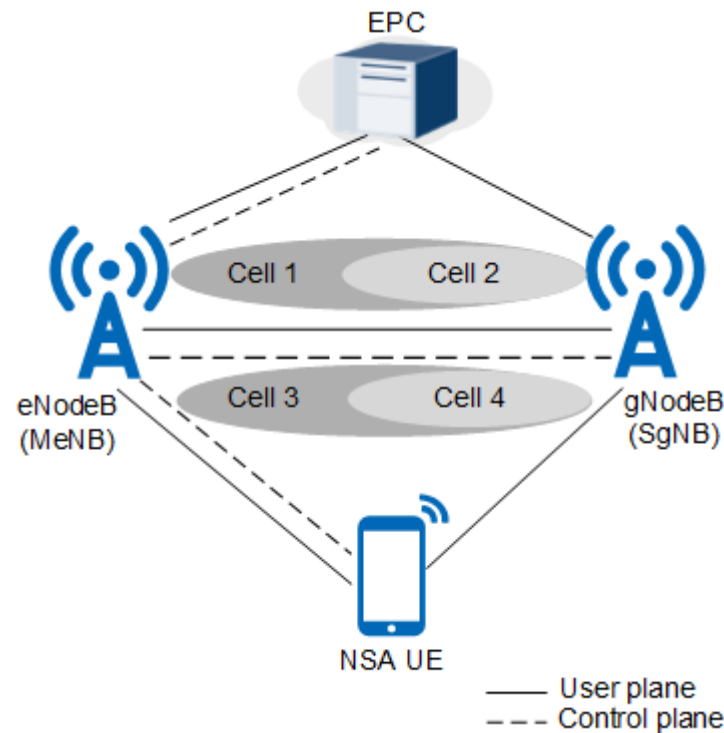


Table 3-1 Related concepts

Category	Concept	Description
Base station	MeNB	The master eNodeB (MeNB) of an NSA UE in the DC state is an LTE base station that serves the cell on which the UE is currently camping. Only an eNodeB can be configured as a master base station.
	SgNB	The secondary gNodeB (SgNB) of the NSA UE is an NR base station configured for the UE through an RRC message sent by the MeNB. Only a gNodeB can be configured as a secondary base station.
Cell group	MCG	The master cell group (MCG) of the NSA UE is an LTE cell group configured on the LTE side. In Figure 3-2 , cells 1 and 3 form the MCG.
	SCG	The secondary cell group (SCG) of the NSA UE is an NR cell group configured on the NR side. In Figure 3-2 , cells 2 and 4 form the SCG.
Cell	PCell	The primary cell (PCell) of the NSA UE is a cell that is served by the MeNB and that the UE is camping on. In Figure 3-2 , cell 1 is the PCell.

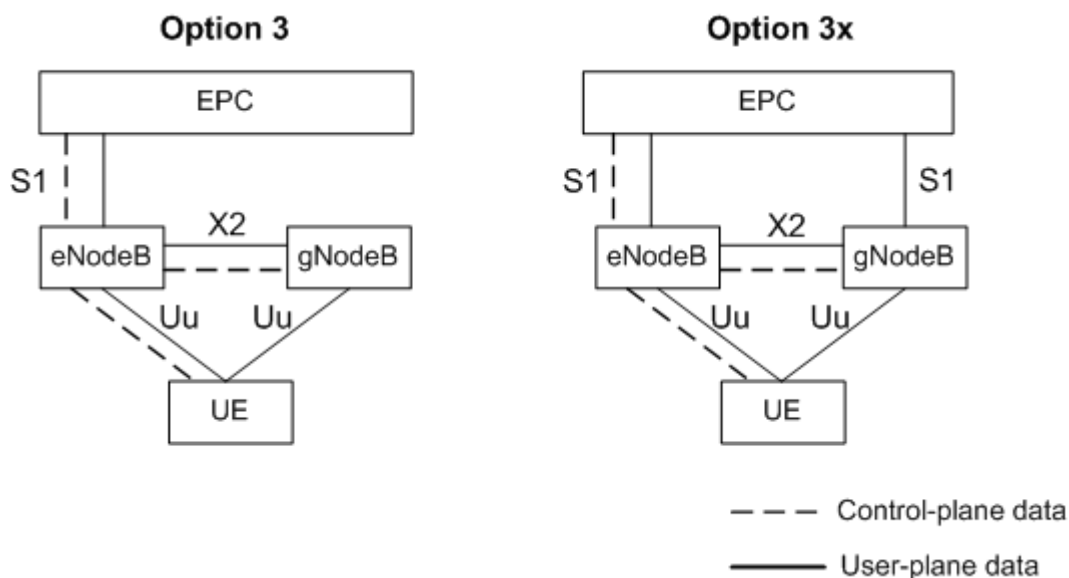
Cate gory	Conce pt	Description
	PSCell	The primary SCG cell (PSCell) of the NSA UE is a primary cell that is served by the SgNB and configured for the UE through an RRC message sent by the MeNB. The PSCell stays active once it is configured successfully. In Figure 3-2 , cell 2 is the PSCell.
	SCell	A secondary cell (SCell) of the NSA UE is a cell served by the MeNB or SgNB and configured for the UE through an RRC message sent by the MeNB. This cell can provide the UE with additional radio resources. In Figure 3-2 , cells 3 and 4 are SCells. The PUCCH is available in each PCell and PSCell but not available in any SCell.
Carri er	CC	CCs are the carriers that are aggregated for the UE.
	PCC	The primary component carrier (PCC) is the carrier of the PCell. In Figure 3-2 , the carrier of cell 1 is the PCC.
	PSCC	The primary secondary component carrier (PSCC) is the carrier of the PSCell. In Figure 3-2 , the carrier of cell 2 is the PSCC.
	SCC	A secondary component carrier (SCC) is the carrier of an SCell. In Figure 3-2 , the carriers of cells 3 and 4 are SCCs.
Event	Event A1	The signal quality of the serving cell becomes higher than a threshold.
	Event A2	The signal quality of the serving cell becomes lower than a threshold.
	Event A3	The signal quality of a neighboring cell is offset better than the signal quality of the serving cell.
	Event A4	The signal quality of a neighboring cell becomes higher than a threshold.
	Event A5	The signal quality of the serving cell becomes lower than threshold 1, while the signal quality of a neighboring cell becomes higher than threshold 2.
	Event A6	The signal quality of a neighboring cell is offset better than the signal quality of a secondary cell.
	Event B1	The signal quality of an inter-RAT neighboring cell becomes higher than a threshold.

3.3 Architecture

NSA networking based on EPC can be deployed in areas where LTE coverage and NR coverage overlap. The current version supports two network

architectures: Option 3 and Option 3x. The specific network architecture depends on how the eNodeB, gNodeB, and EPC are interconnected.

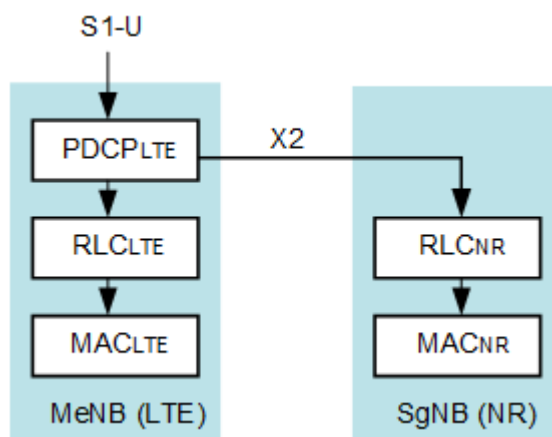
Figure 3-3 Network architectures defined by 3GPP specifications



The eNodeB is an MeNB, the gNodeB is an SgNB, user-plane data can be transmitted in either Option 3 or Option 3x, as shown in [Figure 3-3](#).

- In Option 3, data split is performed on the MeNB. The supported bearers include MCG bearer, MCG_Bearer_EUTRA_PDCP, and MCG split bearer.
 - MCG bearer and MCG_Bearer_EUTRA_PDCP: indicating that user-plane data is carried by the MeNB.
 - MCG split bearer: indicating that user-plane data is split at the PDCP layer of the MeNB, distributed to the RLC layers of the SgNB and MeNB, as shown in [Figure 3-4](#), and then aggregated at the PDCP layer of the UE.

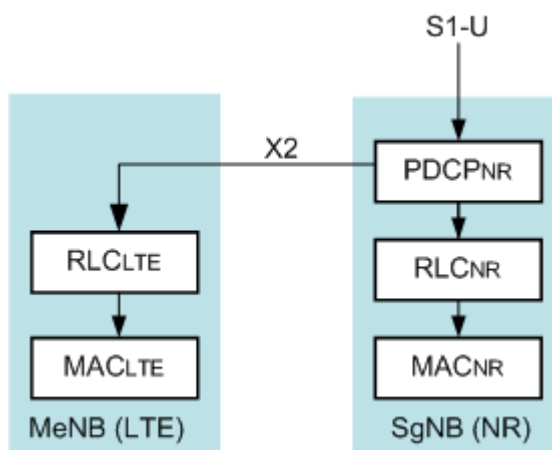
Figure 3-4 MCG split bearer



- In Option 3x, data split is performed on the SgNB. The supported bearers include SCG split bearer and SN-terminated MCG bearer.

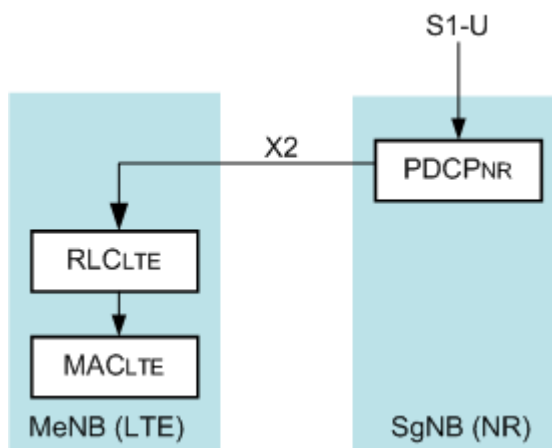
- SCG split bearer: indicating that user-plane data is split at the PDCP layer of the SgNB, distributed to the RLC layers of the MeNB and SgNB, as shown in [Figure 3-5](#), and then aggregated at the PDCP layer of the UE.

Figure 3-5 SCG split bearer



- SN-terminated MCG bearer: indicating that all user-plane data is distributed at the PDCP layer of the SgNB to the RLC layer of the MeNB, as shown in [Figure 3-6](#). For details about the SN-terminated MCG bearer, see *EPC-based NSA Performance Enhancement*.

Figure 3-6 SN-terminated MCG bearer



3.4 EN-DC Band Combinations

Supported Band Combinations

In NSA networking, the E-UTRA-NR dual connectivity (EN-DC) combinations of NR frequency bands and the corresponding LTE frequency bands are specified in 3GPP specifications. The DC_13A-n66A combination is used as an example. [Table 3-2](#) lists the LTE and NR frequency bands included in this combination. For details about EN-DC band combinations defined in 3GPP specifications, see section 5.5B "Configuration for DC" in 3GPP TS 38.101-3 V16.6.0.

Table 3-2 EN-DC band combination example (DC_13A-n66A combination)

EN-DC Band Combination	LTE or NR Band	Uplink Frequency (MHz)	Downlink Frequency (MHz)	RAT
DC_13A_n66A	B13	777 to 787	746 to 756	LTE FDD
	n66	1710 to 1780	2110 to 2200	NR FDD

Specified Band Configuration

When querying the EN-DC band combination capability of an NSA UE, the eNodeB will send the *requestedFreqBandsNR-MRDC-r15* IE to the NSA UE so that the NSA UE reports only the EN-DC band combinations consisting of the frequency bands specified by this IE. This IE contains the *bandEUTRA* and *bandNR* fields, which specify the LTE and NR bands in the EN-DC band combinations, respectively. As described in Table 3-3, the values of *bandEUTRA* and *bandNR* are determined by the frequencies configured on the eNodeB or independently configured bands. An independently configured band is specified using the **OperatorBand** MO. The **OperatorBand.BandId**, **OperatorBand.BandType**, **OperatorBand.CnOperatorList**, and **OperatorBand.BandAttribute** parameters in this MO specify the ID, type (RAT), operator list, and attribute of a band, respectively. The **OperatorBand.BandAttribute** parameter is valid only when the **OperatorBand.BandType** parameter is set to **LTE**.

If no **OperatorBand** MO is configured, the values of *bandEUTRA* and *bandNR* are determined by the frequencies configured on the eNodeB. When the frequencies configured on neighboring eNodeBs are different, the bands specified by the neighboring eNodeBs are different.

If **OperatorBand** MOs are configured, the values of *bandEUTRA* and *bandNR* are determined by whether bands of the corresponding RATs are configured in the **OperatorBand** MOs and whether the **INDEP_UE_BAND_ENQUIRY_SW** option of the **GlobalProcSwitch.ProtocolMsgOptExtSwitch** parameter is selected.

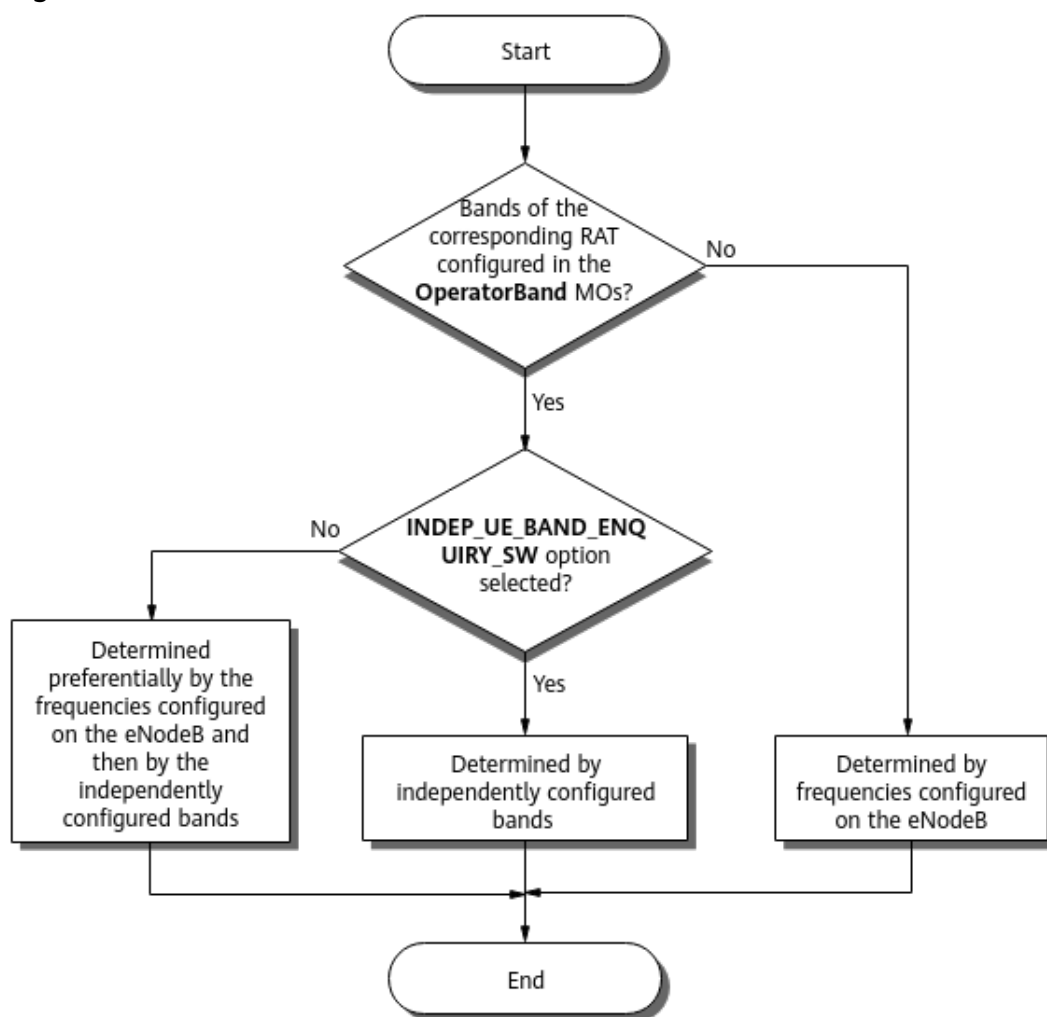
Table 3-3 Sources and descriptions of the *bandEUTRA* and *bandNR* values

Field Value	When Sourced from Frequencies Configured on the eNodeB	When Sourced from Independently Configured Bands
Value of <i>bandEUTRA</i>	Determined by the LTE bands to which the following frequencies belong: • Frequencies identified by PccFreqCfg.PccDLEarfcn corresponding to NrScgFreqConfig.ScgDLArfcn • Frequencies identified by SccFreqCfg.SccDLEarfcn corresponding to the previous PCC downlink EARFCNs	Determined by the independently configured bands identified by the IDs in the OperatorBand MOs with Band Type set to LTE. The bands are further filtered as follows: • The bands with Band Attribute set to CA are filtered out. • The bands that do not belong to the operator of the UE (indicated by CN Operator List) are filtered out. If all LTE bands are filtered out, the value of this field is empty. In this case, the UE will not report any EN-DC band combination, and the NSA DC function will not take effect.

Field Value	When Sourced from Frequencies Configured on the eNodeB	When Sourced from Independently Configured Bands
Value of <i>bandNR</i>	Determined by the NR bands to which the frequencies identified by NrScgFreqConfig.ScgDlArfcn belong.	Determined by the independently configured bands identified by the IDs in the OperatorBand MOs with Band Type set to NR . For the preceding bands that are configured with an operator list, the eNodeB will filter out the bands that do not belong to the operator of a UE. If all NR bands are filtered out, the value of this field is empty. In this case, the UE will not report any EN-DC band combination, and the NSA DC function will not take effect.

If the **OperatorBand** MOs have been configured, the final sources of the *bandEUTRA* and *bandNR* values are illustrated in [Figure 3-7](#).

Figure 3-7 Final sources of the *bandEUTRA* and *bandNR* values



The procedure is described as follows:

1. For *bandEUTRA* and *bandNR*, the eNodeB checks whether the bands of the corresponding RATs are configured in the **OperatorBand** MOs.
 - If so, the procedure goes to 2.
 - If not, the values of *bandEUTRA* or *bandNR* are determined by the frequencies configured on the eNodeB. The procedure ends.
2. The eNodeB checks whether the **INDEP_UE_BAND_ENQUIRY_SW** option is selected.
 - If this option is selected, the values of *bandEUTRA* or *bandNR* are determined by the independently configured bands. The procedure ends. In this scenario, the bands specified by neighboring eNodeBs can be the same even if the frequencies configured on the neighboring eNodeBs are different.
 - If this option is deselected, the values of *bandEUTRA* or *bandNR* are determined preferentially by the frequencies configured on the eNodeB and then by the independently configured bands. The procedure ends.

If the **QUERY_SPEC_BAND_UE_CAP_SW** option of the **UeCompat.WhitelistCtrlExtSwitch1** parameter is selected, **OperatorBand** MOs

can be used to specify bands only for whitelisted UEs. If this option is deselected, **OperatorBand** MOs can be used to specify bands for all UEs. If both this option and the **CaBandCombQueryOptSwitch** option of the **GlobalProcSwitch.ProtocolMsgOptSwitch** parameter are selected, the eNodeB cannot identify whether a UE is whitelisted when the UE performs initial access and the core network does not deliver E-UTRAN capability information. Therefore, the eNodeB considers that the UE is not whitelisted, and does not specify bands for it using the **OperatorBand** MOs.

 NOTE

If the **OperatorBand.BandType** parameter is set, network-requested CA band combination capability signaling for CA UEs will be affected. For details, see Carrier Aggregation in eRAN feature documentation.

If an LTE or NR band in an EN-DC band combination is not configured using the **OperatorBand.BandId** parameter, NSA DC cannot take effect for a UE when the UE moves to an area covered only by this band.

If the **INDEP_UE_BAND_ENQUIRY_SW** option of the **GlobalProcSwitch.ProtocolMsgOptExtSwitch** parameter is deselected, the *bandEUTRA* value is preferentially sourced from the frequencies configured on the eNodeB and then from the independently configured bands. As such, the value of *bandEUTRA* may include the bands with **OperatorBand.BandAttribute** set to **CA**.

4 NSA Networking Based on EPC

NSA networking based on EPC can implement basic functions such as carrier management, data split, uplink power control, and interference avoidance in NSA DC.

NSA networking based on EPC is controlled by the **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter on the eNodeB side and by the **NRCellAlgoSwitch.NsaDcSwitch** parameter on the gNodeB side. It takes effect when both of the switches are turned on.

4.1 Principles

4.1.1 Carrier Management in NSA DC

Carrier management in NSA DC includes carrier addition, change, removal, and frequency configuration. It is classified into MCG carrier management and SCG carrier management. It can be performed based on subscriber profile ID (SPID) or QoS class identifier (QCI). This document describes the basic functions of carrier management in NSA DC. For details about NSA PCC anchoring and blind PSCell addition, see *EPC-based NSA Performance Enhancement*.

4.1.1.1 MCG Carrier Management

This section describes how to add, activate, remove, and deactivate SCCs in the MCG configured for an NSA UE. For details about PCC anchoring in the MCG, see section "EN-DC Optimal Carrier Selection" in *EPC-based NSA Performance Enhancement*.

4.1.1.1.1 SCC Management

SCC Frequency Selection

SCC frequency selection for an MCG is the same as that in LTE carrier management. For details, see *Carrier Aggregation* in eRAN feature documentation.

SCell Activation

The eNodeB triggers LTE SCell activation based on events or at intervals when an NSA UE is in a specific state and certain conditions are met.

- If the NSA UE is not in the NSA DC state, all the following conditions must be met:
 - Data volume buffered at the RLC layer > max (Uu data rate at the RLC layer x **CaMgtCfg.ActiveBufferDelayThd**, **CaMgtCfg.ActiveBufferLenThd**)
 - First packet delay at the RLC layer > **CaMgtCfg.ActiveBufferDelayThd**

In this scenario, the interval for triggering periodic LTE SCell activation is specified by the **CaMgtCfg.SccCfgInterval** parameter.

- If the NSA UE is in the NSA DC state, all the following conditions must be met:
 - Data volume buffered at the RLC layer > max (Uu data rate at the RLC layer x **NsaDcMgmtConfig.NsaDcLteScellActBfrDelThld**, **NsaDcMgmtConfig.NsaDcLteScellActBfrLenThld**)
 - First packet delay at the RLC layer > **NsaDcMgmtConfig.NsaDcLteScellActBfrDelThld**

In this scenario, the interval for triggering periodic LTE SCell activation is specified by the **NsaDcMgmtConfig.NsaDcLteScellConfigIntvl** parameter. The duration of a single round of LTE SCell measurement is specified by the **NsaDcMgmtConfig.NsaDcLteScellMeasDuration** parameter.

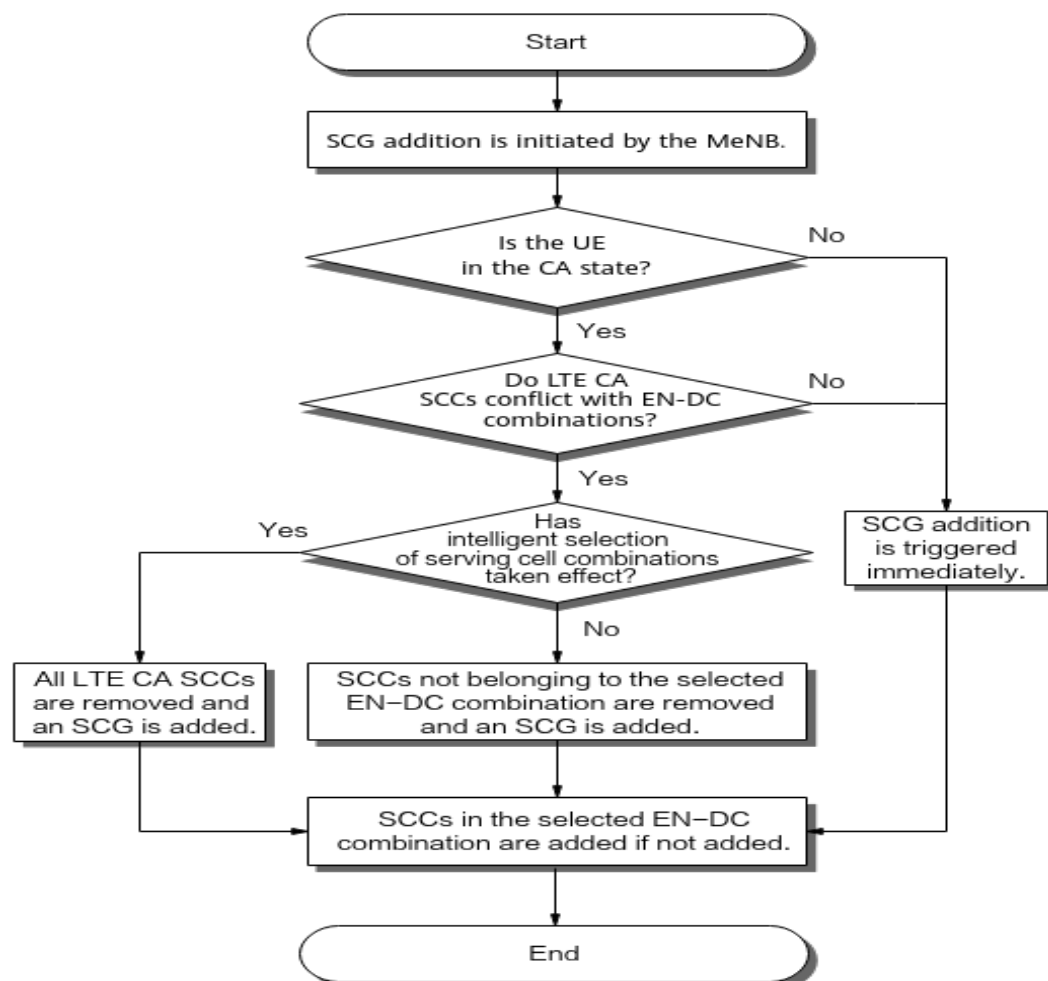
SCC Removal

It is possible that SCCs cannot be added together with an SCG for an NSA UE with compatibility issues. For such an NSA UE, the **NSA_CA_FORBID_SW** option of the **UeCompat.BlacklistControlExtSwitch2** parameter can be used to specify whether to enable blacklist control for SCC management in NSA networking. If this option is selected and the NSA UE is in the corresponding blacklist, all configured SCCs must be removed when an SCG is added for this UE, and no SCCs can be added after an SCG is added and before the SCG is released.

If this option is deselected or the NSA UE is not in the corresponding blacklist, the following are true: (1) if the combination of NR bands and configured LTE CA bands does not support NSA DC, LTE CA SCCs need to be removed; (2) if such a combination supports NSA DC, SCG addition is triggered directly. During SCG addition for the NSA UE, the SCC removal policy used depends on the setting of the **SCC_RMV_AFT_SCG_ADD_OPT_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter.

- If the **SCC_RMV_AFT_SCG_ADD_OPT_SW** option is deselected, the base station checks whether intelligent selection of serving cell combinations has taken effect. If this function has taken effect, all LTE CA SCCs are removed and an SCG is added; after the addition, SCCs in the selected EN-DC combination are added (if not added yet). For details about intelligent selection of serving cell combinations, see *Carrier Aggregation* in eRAN feature documentation. If intelligent selection of serving cell combinations has not taken effect, SCCs not belonging to the selected EN-DC combination are removed and an SCG is added; after the addition, SCCs in the selected EN-DC combination are added (if not added yet). **Figure 4-1** shows the procedure.

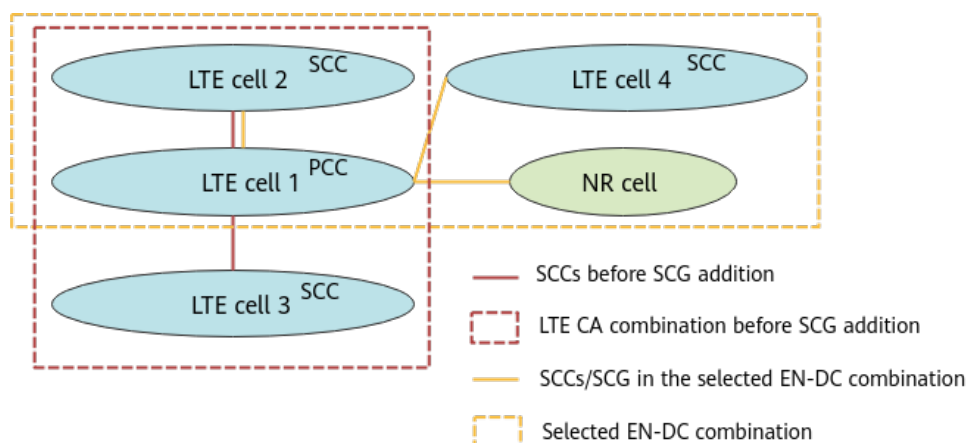
Figure 4-1 Cooperation between intelligent selection of serving cell combinations and SCG addition in NSA networking



- If the **SCC_RMV_AFT_SCG_ADD_OPT_SW** option is selected, the base station selects the EN-DC combination that is supported and has the best LTE CA performance based on a number of factors and then removes the SCCs that do not belong to the selected EN-DC combination. The factors that need to be considered include the number of carriers, SCC priority, and bandwidth. After the SCG addition, SCCs in the selected EN-DC combination are added (if not added yet).

Figure 4-2 shows the LTE CA combination before SCG addition and the EN-DC band combination after SCG addition if SCCs need to be removed before SCG addition.

Figure 4-2 LTE CA combination before SCG addition and EN-DC combination after SCG addition



NOTE

The total bandwidth of the selected EN-DC combination may be less than that of the LTE CA combination before SCG addition. In this case, NSA UE experience will be affected.

4.1.1.1.2 Uplink CA in EN-DC

Uplink CA in EN-DC means that an NR SCG and uplink LTE CA can be both configured for an NSA UE. The implementation of uplink CA in EN-DC is similar to that of uplink CA in LTE. This document describes only the differences between them. For details about uplink CA in LTE, see *Carrier Aggregation* in eRAN feature documentation.

This function allows for only uplink FDD+FDD CA. Based on the NSA UE's state, event-based or periodic LTE SCC configuration or activation will be triggered once specific conditions are met.

If the NSA UE is not in the EN-DC state, uplink 2CC aggregation is performed as described in *Carrier Aggregation* in eRAN feature documentation.

If the NSA UE is in the EN-DC state, the following functions are supported:

- Uplink SCC addition
 - Triggered based on events, including SCG addition, change, and removal
 - Triggered at intervals and based on data volume
 - If the NSA UE in the EN-DC state has no data split bearer, all the following conditions must be met:
 - Data volume buffered at the RLC layer > max (Uu data rate at the RLC layer x **CaMgtCfg.ActiveBufferDelayThd**, **CaMgtCfg.ActiveBufferLenThd**)
 - First packet delay at the RLC layer > **CaMgtCfg.ActiveBufferDelayThd**

The interval is specified by the **CaMgtCfg.SccCfgInterval** parameter.
 - If the NSA UE in the EN-DC state has a data split bearer, all the following conditions must be met:

- Data volume buffered at the RLC layer > max (Uu data rate at the RLC layer x **NsaDcMgmtConfig.NsaDcLteScellActBfrDelThld**, **NsaDcMgmtConfig.NsaDcLteScellActBfrLenThld**)
- First packet delay at the RLC layer > **NsaDcMgmtConfig.NsaDcLteScellActBfrDelThld**

The interval is specified by the **NsaDcMgmtConfig.NsaDcLteScellConfigIntvl** parameter. The duration of a single round of LTE SCell measurement is specified by the **NsaDcMgmtConfig.NsaDcLteScellMeasDuration** parameter.

- Uplink SCC removal
 - This function is the same as uplink 2CC aggregation in LTE. For details, see *Carrier Aggregation* in eRAN feature documentation.
 - If the EN-DC band combinations supported by the UE cannot be used for uplink CA and the configuration of uplink CA in LTE has been completed for the UE, the uplink SCC will be released after the SCG is added.
- Uplink SCC activation, which requires that all of the following conditions be met:
 - Data volume buffered at the RLC layer > max (Uu data rate at the RLC layer x **NsaDcMgmtConfig.NsaDcLteScellActBfrDelThld**, **NsaDcMgmtConfig.NsaDcLteScellActBfrLenThld**)
 - First packet delay at the RLC layer > **NsaDcMgmtConfig.NsaDcLteScellActBfrDelThld**
- Uplink SCC deactivation, which requires that all of the following conditions be met:
 - Buffered data volume in the BSR reported by the NSA UE ≤ **NsaDcMgmtConfig.NsaLteScellDeactBfrLenThld**
 - Uplink Uu data rate on the NSA UE at the RLC layer ≤ **NsaDcMgmtConfig.NsaLteScellDeactThptThld**

Uplink CA in EN-DC is controlled by the **NSA_LTE_UL_2CC_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoExtSwitch** parameter.

The **NSA_LTE_UL_CA_SW** option of the **UeCompat.BlacklistControlExtSwitch1** parameter can be used to disable this function for defective UEs. The **NSA_LTE_UL_CA_DIFF_DPS_SW** option of the **UeCompat.BlacklistControlExtSwitch2** parameter can be used to prohibit UEs from occupying all power on their LTE side if the UEs have no power reserved on their NR side.

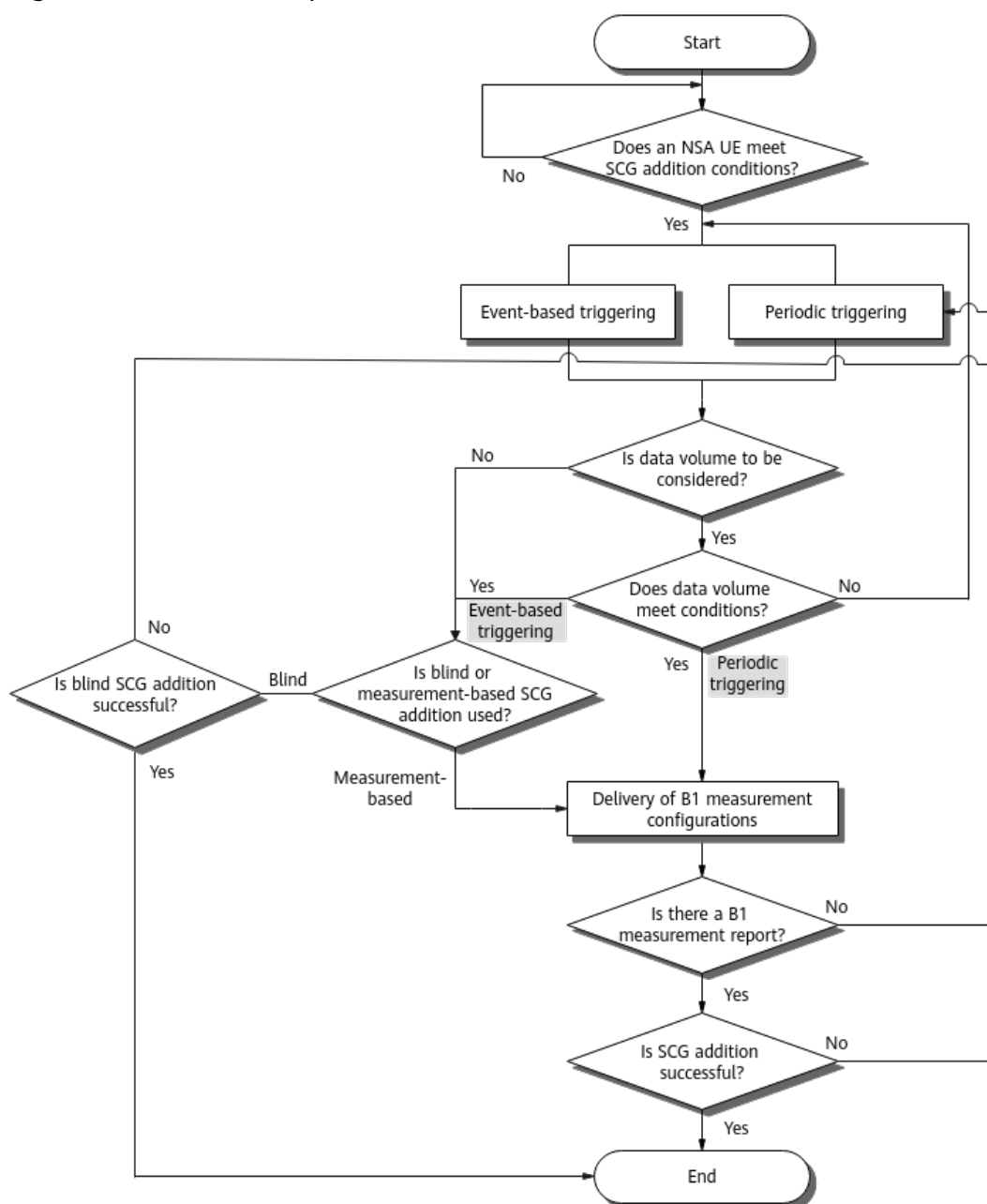
4.1.1.2 SCG Carrier Management

SCG addition is triggered if an NSA UE in the LTE-only state requests NSA DC services. SCG carrier management mainly includes PSCell configuration. For details about the SCG addition procedure, see [Figure 4-3](#).

 NOTE

- If a UE supports both NSA networking and SA networking, the UE preferentially returns to an NR cell through an LTE-NR interoperability procedure in SA networking. For details about LTE-NR interoperability, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.
- A UE cannot receive and send data at the same time in the EN-DC band combination of B41 and N41 according to 3GPP TS 38.101-3. Therefore, if the **NrScgFreqConfig.PccDLEarfcn** parameter indicates a band in B41, then the **NrScgFreqConfig.ScgDLArfcn** parameter can be set to a frequency in N41 only when the uplink subframes and downlink subframes of B41 and N41 are aligned.

Figure 4-3 SCG addition procedure



The procedure is described as follows:

1. Checks whether an NSA UE meets SCG addition conditions.

SCG addition is triggered when the NSA UE in the LTE-only state meets all the following conditions:

- There are QCI that are supported by the MCG split bearer or SCG split bearer. For the relationships between QCIs and bearers, see [4.1.2 Data Split in NSA DC](#).
- The UE is not running any emergency call.
- The VoLTE services can coexist with the SCG in the case that the UE is performing VoLTE services. For details, see [4.1.1.2.2 PSCell Management for VoLTE UEs](#).

2. Determines whether to trigger SCG addition based on events or at intervals.

- Event-based triggering

SCG addition is triggered when an NSA UE performs initial access, incoming handover, or incoming RRC connection reestablishment. The **NSA_DC_VOLUME_BASED_SCG_ADD_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter specifies whether the MeNB triggers SCG addition based on data volume. The data volume is evaluated in the same way as that for periodic SCG addition.

 NOTE

If the **NsaDcMgmtConfig.ScgAddPenaltyPeriodCount** parameter is set to a non-zero value and an incoming RRC connection reestablishment is triggered for an NSA UE during an SCG addition procedure, this UE can be barred from entering the SCG addition procedure again within a period. This period equals **NsaDcMgmtConfig.ScgAddPenaltyPeriodCount** × **NsaDcMgmtConfig.ScgAdditionInterval**.

- Periodic triggering

An SCG may fail to be added initially, or cannot be added due to no trigger source after SCG release. To trigger SCG addition for an NSA UE in the LTE-only state, the MeNB can start an SCG addition timer and trigger the addition at intervals specified by the **NsaDcMgmtConfig.ScgAdditionInterval** parameter.

Whether to check data volume for periodic triggering of SCG addition is specified by the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter. If this option is selected, SCG addition is triggered based on data volume. If this option is deselected, SCG addition is not triggered based on data volume.

Whether data volume meets requirements is determined by the SCG addition buffer length threshold (specified by **NsaDcMgmtConfig.ScgAdditionBufferLenThld**) and the SCG addition buffer delay threshold (specified by **NsaDcMgmtConfig.ScgAdditionBufferDelayThld**).

- If the SCG addition buffer length threshold is set to **0**, SCG addition is triggered when the UE is transmitting data.
- If the SCG addition buffer length threshold is set to a value other than **0**, SCG addition is triggered when the following conditions are met:

- Data volume buffered at the RLC layer > max (Air interface data rate at the RLC layer x SCG addition buffer delay threshold, SCG addition buffer length threshold)
- First packet delay at the RLC layer > SCG addition buffer delay threshold

If the UE inactivity timer (specified by **NRDUCellQciBearer.UeInactivityTimer**) on the NR side expires, the SCG will be released. When there is no data transmission on the LTE and NR sides, the following two issues arise:

- SCG addition is triggered periodically with data volume not considered.
There is a possibility that the MeNB repeatedly adds an SCG and releases it. To prevent this issue, it is recommended that the UE inactivity timer on the NR side be set to 0.
- SCG addition is triggered periodically with data volume considered.
The MeNB sends PDCP status report packets to the UE. When the data volume of such packets buffered at the RLC layer of the MeNB exceeds the buffer length threshold for SCG addition, the MeNB triggers SCG addition based on data volume. As a result, there is a possibility that the MeNB repeatedly adds an SCG and releases it. To prevent this issue, it is recommended that at least one of the following operations be performed:
 - Set the SCG addition interval to a value greater than the value of the UE inactivity timer on the NR side.
 - Set the SCG addition buffer length threshold to a value greater than the data volume of PDCP status report packets buffered at the RLC layer of the MeNB.
 - Select the **SCG_ADD_PCC_ANCHOR_VOL_OPT_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoExtSwitch** parameter so that the MeNB deducts the data volume of PDCP status report packets during data volume evaluation.

NOTE

- The **INSTANT_JUDGEMENT_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoExtSwitch** parameter specifies whether instantaneous millisecond-level values or filtered second-level values are used for estimating the data volume buffered at the RLC layer during data-volume-based SCG addition. If both this option and the **VOLUME_BASED_SCG_ADD_OPT_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter are selected, the eNodeB automatically determines whether to use the SCG addition buffer delay threshold when checking whether an NSA UE meets the conditions for data-volume-based SCG addition.
 - The **A2_HO_SCG_ADD_PCC_ANCHOR_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoExtSwitch** parameter specifies whether to allow SCG addition triggering after the reception of LTE A2 measurement reports from NSA UEs. If this option is selected, SCG addition triggering is allowed. If this option is deselected, SCG addition triggering is not allowed.
3. Performs the SCG addition procedure.
Measurement-based PSCell configuration (described in [4.1.1.2.1 Measurement-based PSCell Configuration](#)) or blind PSCell addition

(described in *EPC-based NSA Performance Enhancement*) can be performed, depending on the setting of the **NSA_BLIND_SCG_ADDITION_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter.

- If this option is deselected, measurement-based PSCell configuration is performed and an SCG is added.
- If this option is selected and the **NrNRRelationship.BlindConfigIndicator** parameter is set to **TRUE**, B1 measurement configurations are not delivered; instead, an SCG addition request is directly sent for blind PSCell addition and SCG addition in the following scenarios:
 - A UE initially accesses the network.
 - In the preparation for a handover from the source MeNB to the target MeNB, the target MeNB meets the conditions for blind SCG addition.
 - A UE has just been handed over from an LTE-only cell to an NSA anchor cell.

It is recommended that the **NrNRRelationship.BlindConfigIndicator** parameter be set to **TRUE** when the LTE cell and the neighboring NR cell are co-sited and cover the same area or when the coverage area of the neighboring NR cell is larger than that of the LTE cell. Otherwise, it is not recommended that this parameter be set to **TRUE** to prevent NR access failure due to no NR coverage.

NOTE

- When the MeNB triggers SCG addition, it performs flow control based on its CPU usage. That is, the MeNB limits the probability of SCG addition to prevent the CPU from being overloaded. The NSA flow control policy is specified by the **eNodeBFlowCtrlPara.NsaFlowCtrlStrategy** parameter. When the CPU usage reaches or exceeds 80%, flow control is performed.
 - When the CPU usage reaches 80%, the MeNB starts to limit the probability of SCG addition. When the CPU usage is 80%, 81%, 82%, 83%, or 84%, the probability of SCG addition is 80%, 60%, 40%, 20%, or 10%, respectively.
 - When the CPU usage reaches or exceeds 85%, the MeNB does not add any SCG, except for VIP users.
 - When the CPU usage falls below 80% but reaches or exceeds 70%, the probability of SCG addition is 95%.
 - When the CPU usage falls below 70%, flow control is not performed.
- The **NSA_NR_SINGLE_BC_PREFER_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter specifies whether to enable the preferential selection of a band combination with only one NR carrier for NSA UEs function. If this option is selected, a band combination that involves only one NR carrier is preferentially selected for SCG addition. This function can take effect only for whitelisted UEs if the **NSA_NR_SINGLE_BC_PREFER_SW** option of the **UeCompat.WhitelistCtrlExtSwitch1** parameter is selected.

When delivering NR frequency measurement configurations or adding NR cells in a blind manner for a UE, the MeNB preferentially considers NR frequencies or NR cells that belong to the same PLMN as the UE. If LTE and NR belong to different PLMNs, the **INTER_PLMN_ADD_SCG_SW** or **EQUIVALENT_PLMN_ADD_SCG_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoSwitch** parameter can be selected. Under this setting, when delivering B1 measurement configurations or adding NR cells in a blind manner, the MeNB can select NR frequencies or NR cells

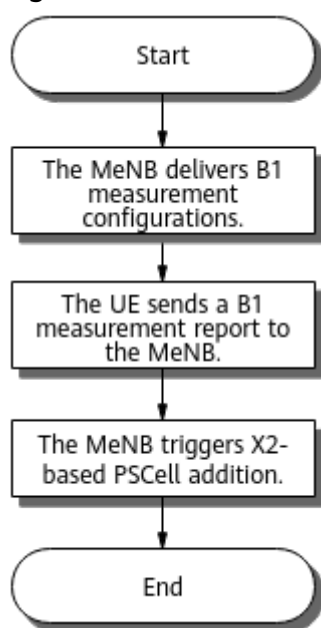
in a PLMN that is in the equivalent PLMN list or inter-PLMN list (configured using the **InterPlmnHoList** MO). To allow the SgNB to perform admission control when LTE and NR belong to different PLMNs, the **SCG_ADD_PLMN_SELECT_POLICY_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter also needs to be selected. Under this setting, the SgNB performs admission decision based on the *Selected PLMN* IE in the SgNB Addition Request message sent by the MeNB to the SgNB.

The PSCell modification success rate and PSCell change success rate may decrease due to conflicts between the LTE RRC connection reestablishment procedure and the gNodeB-initiated SCG modification or SCG change procedure, following successful SCG additions and the initiation of reestablishment on the LTE side. To reduce such procedure conflicts, enabling the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs" is recommended. This function is controlled by the **NSA_REEST_FLOW_CONFLICT_OPT_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch** parameter.

4.1.1.2.1 Measurement-based PSCell Configuration

Figure 4-4 shows the measurement-based PSCell configuration procedure.

Figure 4-4 Measurement-based PSCell configuration procedure



Measurement Configuration Delivery

Measurement configuration delivery involves frequency selection and frequency-priority-based sorting.

The eNodeB complies with the following principles when selecting frequencies to be delivered in the measurement configurations:

- The eNodeB filters out frequencies not supported based on the UE capability.
- When selecting NR frequencies for measurement configuration delivery, the eNodeB filters out the NR frequencies for which the

NR_DL_SCC_FREQ_ONLY_IND option of the **NrNFreq.AggregationAttribute** parameter is selected.

- An NR frequency can be included in measurement configurations delivered to UEs only if the **FREQ_MEAS_FLAG** option of the **NrNFreq.AggregationAttribute** parameter is selected.
- In NSA mmWave networking, if there are multiple co-sector co-coverage NR FR2 SCG downlink frequencies (specified by **NrScgFreqConfig.ScgDLArfcn**), the **NR_FR2_MEAS_POLLING_FLAG** option of the **NrScgFreqConfig.AggregationAttribute** parameter on the eNodeB can be selected for these frequencies. Under this setting, the eNodeB can randomly select an SCG downlink frequency for measurement from these frequencies for the same downlink EARFCN of the PCC (specified by **NrScgFreqConfig.PccDLEarfcn**).
- In NSA and SA hybrid networking, the MeNB filters out the SCG frequencies serving the neighboring NR cells only in SA networking mode based on the setting of the **NrExternalCell.NrNetworkingOption** or **NrExternalCellPlmn.NrNetworkingOption** parameter. In this scenario, it is recommended that the **UPT_NR_EXT_CELL_NW_OPT_CFG_SW** option of the **GlobalProcSwitch.X2BasedUptNcellCfgSwitch** parameter be selected so that the value of **NrExternalCell.NrNetworkingOption** can be automatically updated based on received X2 messages, without the need for manual configuration.

The eNodeB performs frequency-priority-based sorting as follows for the frequencies to be delivered in the measurement configurations:

After frequency selection is complete, the MeNB can configure multiple frequencies for an SCG. The priority of each frequency is specified by the **NrScgFreqConfig.ScgDLArfcn** and **NrScgFreqConfig.ScgDLArfcnPriority** parameters. The validation rules are as follows:

- If different priorities are set for the frequencies, the highest-priority frequency is set as the first measurement frequency, the second-highest-priority frequency is set as the second measurement frequency, and so on.
- If the same priority or no priority is set for the frequencies, the frequency sequence in measurement configurations is random.
- If priorities are set for some frequencies but not for others, the frequencies with no priorities configured have the lowest priority by default.

NOTE

- Assume that the frequency indicated by the **NrScgFreqConfig.ScgDLArfcn** parameter belongs to multiple NR frequency bands and no frequency band is configured in the **NrMfbiFreq** MO for this frequency. Then, the frequency band specified in NR measurement configurations is the NR frequency band in the first EN-DC-capable band combination indicated in the *supportedBandCombinationList* IE reported by the UE.
- In the current version, the MeNB supports only RSRP-based B1 measurement for SCG addition. Therefore, the **NrCellMobilityConfig.MeasTrigQuantity** parameter must be set to **RSRP** in NSA networking. Otherwise, if the UE immediately reports NR RSRQ-based event A2 after an SCG addition, the SgNB will immediately trigger a coverage-based inter-frequency handover.
- If the **NR_B1_NO_GAP_SW** option of the **EnodeBAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter is selected and a UE reports that it supports non-gap-assisted measurement of an NR frequency, the eNodeB does not include gap configurations when it delivers B1 measurement configurations for this frequency to the UE.

Maximum number of NR frequencies that can be delivered on the LTE side = $\min(\text{CellUeMeasControlCfg.MaxNrMeasFreqNum}, \text{CellUeMeasControlCfg.NsaMaxLteInterFMeasObjNum})$ – Number of non-NR frequencies)

On the LTE side, the **NrScgFreqConfig.ForbiddenSpidGrpId** parameter can be used to specify an SPID group and cells on the specified NR frequency cannot be added as SCG cells for UEs with SPIDs in this SPID group. The **NrScgFreqConfig.ForbiddenSpidGrpId** parameter setting takes effect only during the process of adding SCGs for UEs. If SCGs have been added, the setting does not take effect. As the same NR frequency can be configured in the **NrScgFreqConfig** MOs of different PCCs, the same **NrScgFreqConfig.ForbiddenSpidGrpId** parameter setting is recommended for this NR frequency.

Measurement Reporting and Processing

The **NsaDcQciParamGroup.NsaDcB1RsrpThldOffset** and **SpidCfg.NsaDcSpidB1RsrpThldOffset** parameters can be set on the MeNB to configure the QCI- and SPID-specific RSRP threshold offsets, respectively, for NSA DC event B1. Then, these offsets are included in the measurement configurations and delivered to a UE through the SgNB. The UE performs measurement according to the SgNB measurement configurations. If the measured RSRP value is greater than the RSRP threshold for event B1 (which is equal to **NrScgFreqConfig.NsaDcB1ThldRsrp** plus the preceding two offsets) throughout a period that exceeds the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, the UE reports the B1 measurement result. If an NR cell is shared by multiple operators, the offset specified by **CellOp.NsaDcB1RsrpThldOffset** can be added to the preceding RSRP threshold for event B1 so that different RSRP thresholds for event B1 can be set for different operators.

A UE supporting both NSA and SA is likely to report NSA DC B1 measurement results first as the **NrScgFreqConfig.NrB1TimeToTrigger** parameter of NSA DC B1 has a smaller default value than the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter of NR B1. To prioritize NR SA network measurement, select the **NrB1TimeToTrigAdjSw** option of the **CellHoParaCfg.CellHoAlgoSwitch** parameter. Under this setting, when the delivery of both NR B1 and NSA DC B1 measurement configurations is supported and the time to trigger for NR B1 is greater than that for NSA DC B1, the time to trigger for NSA DC B1 is adjusted to be consistent with that for NR B1. If the time to trigger for NR B1 is shorter than that for NSA DC B1, adjustment to the same value is not required.

If multiple QCIs exist, the MeNB selects the B1 threshold configured for the QCI with the highest priority (specified by **CellQciPara.QciPriorityForHo**) corresponding to the data split bearer when delivering B1 measurement configurations for SCG addition to the UE.

After the MeNB delivers neighboring NR frequency measurement configurations, it starts a waiting timer (specified by **NrScgFreqConfig.NrB1ReportWaitingTimer**) for measurement reporting. If the **NsaDcMgmtConfig.NsaDcScgAdditionPolicy** parameter is set to **HIGHEST_PRIORITY_FREQ**, the base station waits for the measurement report of only the highest-priority NR frequency. If this parameter is set to **ALL_FREQ**, the base station waits for the measurement results of all delivered NR frequencies.

- If the MeNB receives the measurement results of all the neighboring NR frequencies before the timer expires or receives the measurement results of only some neighboring NR frequencies before the timer expires, it processes the received results as follows:
 - a. It filters out some neighboring NR cells as follows:
 - First, it filters out neighboring NR cells in SA networking mode based on the setting of **NrExternalCell.NrNetworkingOption** or **NrExternalCellPlmn.NrNetworkingOption**.
 - Then, it filters out neighboring NR cells whose frequency bands and additional frequency bands are not supported by the UE, based on the UE MR-DC capability and external NR cell configurations.
 - After that, it filters out neighboring NR cells for which the **OVER_DISTANCE_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter is selected.
 - It filters out NR cells that do not support the uplink mode when the **NR_SCG_UL_NO_CHECK_SW** option of the **NsaDcAlgoParam.NsaDcAlgoExtSwitch** parameter is deselected.
 - In Option 3x, it filters out neighboring NR cells under any SgNB whose S1-U interface is unavailable.
When both the **S1U_STATUS_BASED_SCG_ADD_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoSwitch** parameter of the MeNB and the **S1U_STATUS_SEND_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter of an SgNB are selected, the SgNB periodically updates S1-U status information and sends it to the MeNB. After receiving the SI-U status information, the MeNB filters out the neighboring NR cells under the SgNB if the S1-U interface of the SgNB is unavailable.
 - b. Among the filtered neighboring NR cells, the MeNB identifies the cell with the strongest signal quality and highest frequency priority. It selects the SgNB that serves this cell, and fills the measurement results for the frequencies of the filtered neighboring NR cells served by this SgNB in the *candidateCellInfoListMN* field of the *MeNB to SgNB Container* IE in the SgNB Addition Request message for reporting.
- If the MeNB does not receive any neighboring NR frequency measurement report until the timer expires, the MeNB does not initiate an SgNB addition request.

PSCell Addition

1. The MeNB sends an SgNB Addition Request message to the target SgNB.
For the first UE that initiates NSA DC, the NR neighbor relationship needs to be automatically added or the EN-DC X2 interface needs to be automatically set up, which requires a longer time than manual configuration. For this UE, PSCell addition can be completed only after multiple B1 measurements are performed.
 - If the NR neighbor relationship is not added or the EN-DC X2 interface is not set up, two B1 measurements need to be performed. The first

measurement is used to add the NR neighbor relationship or set up the X2 interface, and the second is used to add the PSCell.

- If neither the NR neighbor relationship is added nor the EN-DC X2 interface is set up, three B1 measurements need to be performed. The first measurement is used to add the NR neighbor relationship, the second is used to set up the X2 interface, and the third is used to add the PSCell.

In other scenarios where the NR neighbor relationship has been added and the EN-DC X2 interface has been automatically set up, only one B1 measurement needs to be triggered for an NSA UE to add the PSCell. For details about automatic neighboring cell addition, see *ANR Management* in eRAN feature documentation. For details about EN-DC X2 self-setup, see *X2 and S1 Self-Management in NSA Networking*.

2. The gNodeB tries to select an NR cell that meets the SCG addition conditions as the PSCell from the NR cells according to the sequence in the *MeasResult2NR* IE of *candidateCellInfoListMN* in the SgNB Addition Request message. If an NR cell is selected, the gNodeB sends an SgNB Addition Ack message to notify the MeNB of the SgNB addition success. If the *MeasResult2NR* IE in the SgNB Addition Request message does not contain any NR cell that meets the SCG addition conditions, the gNodeB sends an SgNB Addition Reject message to notify the MeNB of an SgNB addition failure.

4.1.1.2.2 PSCell Management for VoLTE UEs

Three VoLTE-UE-specific PSCell management modes are supported. The PSCell management mode to be used is specified by the **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** parameter.

- If the **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** parameter is set to **VOLTE_PREFERRED** and an SgNB has been configured, the eNodeB immediately releases the SgNB when a VoLTE service is initiated. If no SgNB is configured, the eNodeB will not trigger an SgNB addition before the VoLTE service is released. If the **NSA_DC_VOLUME_BASED_SCG_ADD_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter is deselected, the eNodeB will trigger an SgNB addition immediately after the VoLTE service is released.
- If the **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** parameter is set to **ADAPTIVE** and a VoLTE service is initiated, then:
 - Assume that the SINR of the VoLTE UE is less than or equal to the value of **NsaDcMgmtConfig.UIQualityThldForScgRelease** or the uplink or downlink packet loss rate of services with QCI 1 is greater than or equal to the value of **NsaDcMgmtConfig.VoltePlrThldForScgRelease**. Then, if an SgNB exists, the eNodeB immediately releases it; if no SgNB exists, no SgNB addition is triggered before the VoLTE service is released. If the **NSA_DC_VOLUME_BASED_SCG_ADD_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter is deselected, the eNodeB will trigger an SgNB addition immediately after the VoLTE service is released.
 - Assume that the SINR of the VoLTE UE is greater than the value of **NsaDcMgmtConfig.UIQualityThldForScgRelease** and the uplink and downlink packet loss rates of services with QCI 1 are less than the value

of **NsaDcMgmtConfig.VoltePlrThldForScgRelease**. Then, if an SgNB exists, the eNodeB does not release it; if no SgNB exists, an SgNB addition can be triggered. The **VOLTE_NSA_DC_IF_NR_MEAS_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter can be used to control whether to deliver NR B1 measurement configurations to the UE when SgNB addition is triggered. If this option is deselected, no NR B1 measurement configurations are delivered to the UE for SgNB addition, preventing the impact on VoLTE services. If this option is selected, NR B1 measurement configurations are delivered for SgNB addition.

 NOTE

After the UE is handed over to a cell, the eNodeB determines whether to add an SgNB or delete the SgNB if it has been added, based on the SINR measurement result or the uplink or downlink packet loss rate of services with QCI 1 in this cell.

- If the **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** parameter is set to **VOLTE_SCG_COEXISTENCE**, the eNodeB does not process as it does in the preceding two modes for VoLTE services.

For VoLTE services, when **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** is set to **VOLTE_SCG_COEXISTENCE**, the **VOLTE_NSA_DC_IF_NR_MEAS_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter can be used to control whether to deliver NR B1 measurement configurations to the UE. If this option is deselected, no NR B1 measurement configurations are delivered to the UE, and an SgNB is directly added in the case that blind SCG addition is in effect, preventing the impact on VoLTE services. If this option is selected, NR B1 measurement configurations are delivered for SgNB addition.

The QCI-specific VoLTE and SCG coexistence function can be enabled for VoLTE UEs with bearers of a specific QCI if the VoLTE services and SCGs need to coexist for these UEs. If the **QCI_VOLTE_SCG_COEXIST_SW** option of the **CellQciPara.NsaDcOptSwitch** parameter is selected for any QCI of the bearers set up for a UE, this function takes effect, even if the **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** parameter is set to **VOLTE_PREFERRED** or **ADAPTIVE**. After this function takes effect: if an SgNB has been configured for the UE, the eNodeB does not release the SgNB when the VoLTE service is started; if an SgNB has not been configured for the UE, an SgNB can be added for the UE. During the SgNB addition, whether NR B1 measurement configurations are delivered to the UE is determined by the **VOLTE_NSA_DC_IF_NR_MEAS_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter. If this option is deselected, no NR B1 measurement configurations are delivered to the UE, and an SgNB is directly added in the case that blind SCG addition is in effect, preventing the impact on VoLTE services. If this option is selected, NR B1 measurement configurations are delivered for SgNB addition.

If the **QCI_VOLTE_SCG_COEXIST_SW** option of the **CellQciPara.NsaDcOptSwitch** is deselected for all QCIs of bearers set up for VoLTE UEs, QCI-specific VoLTE and SCG coexistence does not take effect. In this situation, if **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** is set to **VOLTE_PREFERRED** or **ADAPTIVE**, the eNodeB will still deliver A5 measurement configurations for NSA PCC anchoring and B1 measurement configurations for SCG addition, potentially leading to voice quality degradation. For the preceding scenario, inter-frequency and inter-RAT measurement optimization for VoLTE UEs in NSA DC scenarios can be enabled. Once this function is enabled, the eNodeB will no longer deliver A5

measurement configurations for NSA PCC anchoring and B1 measurement configurations for SCG addition to VoLTE UEs. This optimization is controlled by the **VOLTE_NSA_DC_IF_MEAS_OPT_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoEnhSwitch** parameter.

4.1.1.2.3 PSCell SMTC on the LTE Side

SMTC stands for SSB-based measurement timing configuration. The **NrNFreq** MO on the eNodeB side is used to configure SSB information for an NR frequency. The information is contained in B1 measurement configurations to instruct UEs to correctly scan SSBs. It is also used to calculate B1 measurement gap positions. SSB information includes **SMTC Period** and **SSB Offset**.

NOTE

If only two or fewer SSB positions are contained in one gap period, SSB positions may conflict with the positions of other resources such as PUCCH, SRS, and DRX resources. If SSB positions are unavailable in a gap period, a UE fails to start gap-assisted measurements on NR frequencies and the MeNB sends an SgNB Modification Refuse message to the SgNB. To address this issue, the **GAP_OFS_ASSIGNMENT_OPT_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter can be selected to reduce the conflicts between SSB positions and other resources' positions.

SMTC Period

The **NrNFreq.SsbPeriod** parameter must be set to the same value as the **NRDUCell.SsbPeriod** parameter. If multiple neighboring NR cells work on the same NR frequency and have different SSB periods, it is recommended that the LTE side use the maximum SSB period as the SMTC period for NR frequency measurement.

If the **NrNFreq.SsbPeriod** parameter is set to a value greater than 20 ms, some NSA UEs have compatibility issues and fail to find NR networks. In this case, if the **SMTC_CONFIG_SW** option of the **EnodebAlgoExtSwitch.NsaProtocolCompatSw** parameter is selected, the eNodeB sends an RRC Connection Reconfiguration message to the UEs to assist them in searching for NR networks.

SSB Offset

An SSB offset is specified by the LTE parameter **NrNFreq.SsbOffset**. This parameter can be set in two scenarios: LTE time synchronization and LTE frequency synchronization. The **LST CLKSYNCMODE** command can be executed to query the clock synchronization mode of a base station.

- LTE time synchronization

On the NG-RAN side, the frame offset is specified by the **gNodeBParam.FrameOffset** or **gNBFreqBandConfig.FrameOffset** parameter. The value ranges of the two parameters are the same. The former applies to a gNodeB, and the latter applies to a frequency band. If both parameters are configured, the latter takes effect. The following uses the **gNBFreqBandConfig.FrameOffset** parameter as an example.

- a. The time difference between the start time of the subframe for the NR SSB and the global navigation satellite system (GNSS) standard time, also known as the offset of SSB to the clock source, is calculated based on the NG-RAN frame offset.

- If the value of the **gNBFreqBandConfig.FrameOffset** parameter is less than or equal to 261120, the offset of SSB to the clock source is calculated as follows: $(\text{gNBFreqBandConfig.FrameOffset}/30720) \text{ MOD } \text{NrNFreq.SsbPeriod}$.
- If the value of the **gNBFreqBandConfig.FrameOffset** parameter is greater than or equal to 275943, the offset of SSB to the clock source is calculated as follows: $(\text{gNBFreqBandConfig.FrameOffset}/30720 - 10 + \text{NrNFreq.SsbPeriod}) \text{ MOD } \text{NrNFreq.SsbPeriod}$.

If the NR cells operate on different frequencies, the **gNBFreqBandConfig.FrameOffset** parameter must be set to an identical value across the frequencies.

- b. The SSB offset is corrected based on the eNodeB frame offset.

The offset of SSB to the clock source calculated in the previous step is used as the SSB offset before correction. Perform the following steps to correct the SSB offset:

- i. The NG-RAN frame offset remainder and E-UTRAN frame offset remainder are calculated as follows:
 - NG-RAN frame offset remainder = $(\text{gNBFreqBandConfig.FrameOffset} \text{ MOD } 30720)/30720$
 - E-UTRAN frame offset remainder = $(\text{CellFrameOffset.FrameOffset} \text{ MOD } 30720)/30720$
 - ii. If the E-UTRAN frame offset remainder is greater than the NG-RAN frame offset remainder and is less than 0.5 ms, the corrected SSB offset is calculated as follows: $(\text{SSB offset before correction} - 0.5 + \text{NrNFreq.SsbPeriod}) \text{ MOD } \text{NrNFreq.SsbPeriod}$.
 - iii. If the preceding condition is not met, the corrected SSB offset is equal to the SSB offset before correction.
 - c. The corrected SSB offset is rounded down to its nearest integer, and the integer is used as the value of the **NrNFreq.SsbOffset** parameter.
- LTE frequency synchronization
No configuration is required.

4.1.1.2.4 SCG Addition Based on Virtual Grid Models

For an NSA UE capable of EN-DC but with no SCG added, the eNodeB delivers A3 measurement configurations about the serving frequency with a measurement period of 2048 ms to the UE. After receiving measurement reports, the eNodeB obtains the coverage information of candidate SCG frequencies based on LNR virtual grid models. The time to trigger for event A3 is specified by the **LnrCarrierSelection.IntraFreqMeasA3TimeToTrig** parameter. SCG addition based on virtual grid models is controlled by the **NSA_CARRIER_SEL_WITH_VG_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter.

The accuracy requirement on virtual grid models can be configured using the **LnrCarrierSelection.VgMdlPredNrScgMeasRsrpThld** parameter. If an SCG needs to be added for an NSA UE in the LTE-only state, LNR virtual grid RSRP prediction models are available, and the accuracy of the models reaches the level required by this function, the eNodeB performs as follows:

- If **LnrCarrierSelection.VgMdlPredNrScgMeasRsrpThld** is set to a value other than **ACCURACY100_ENH**:

For a target neighboring NR frequency, the eNodeB triggers one SCG addition procedure based on event B1 measurement. During the period specified by the **NrScgFreqConfig.NrB1ReportWaitingTimer** parameter, the eNodeB does not add an SCG through virtual grid model query based on the event A3 measurement report. If no SCG is added based on event B1 measurement before the timer expires, the eNodeB does not deliver event B1 measurement configurations in subsequent procedures and directly uses the RSRP value predicted by the virtual grid model for the neighboring NR frequency based on the event A3 measurement report.

The eNodeB instructs the NSA UE to add an SCG blindly when the virtual-grid-model-based prediction result for the preceding frequency meets the entering condition of event B1. The threshold of event B1 for SCG addition is specified by the **NrScgFreqConfig.NsaDcb1ThldRsrp** parameter. When the prediction result meets the leaving condition of event B1, the eNodeB does not deliver B1 measurement configurations for this frequency.

If the preceding blind SCG addition based on virtual grid models fails due to UE incompatibility, network performance will be affected. To mitigate the impacts, the measurement and penalty mechanisms are introduced to the SCG addition based on virtual grid models function. These mechanisms are controlled by the **VG_SCG_ADD_MEAS_AND_PUNISH_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter.

- If this option is selected, after the eNodeB instructs the NSA UE to add an SCG blindly, it performs the following operations upon reception of a UE-reported SCGFailureInformationNR message based on which the eNodeB determines that the SCG addition using the virtual-grid-model-based prediction result fails:
 - The eNodeB immediately triggers an SCG addition procedure based on event B1 measurement.
 - The eNodeB imposes a penalty on the NSA UE. With this penalty, when the virtual-grid-model-based prediction result meets the entering condition of event B1 in subsequent procedures of SCG addition based on virtual grid models, SCG addition based on event B1 measurement is triggered for this UE.
 - The eNodeB stops the penalty on the NSA UE only after the UE performs a re-access or inter-eNodeB handover.
- If this option is deselected, the eNodeB does not trigger an SCG addition procedure based on event B1 measurement or impose a penalty on the NSA UE.

When the **PROBABILITY_MODEL_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected, the probability prediction model function is enabled. If the probability prediction model evaluates that the probability of a UE's RSRP being good meets the requirements of the LNR virtual grid models for RSRP prediction, the virtual grid will be used to predict the UE's RSRP, thereby improving the prediction accuracy of the LNR virtual grid models for RSRP prediction. After this option is selected, the LNR virtual grid models for RSRP prediction take effect for fewer UEs, and the prediction accuracy improves.

- If **LnrCarrierSelection.VgMdlPredNrScgMeasRsrpThld** is set to **ACCURACY100_ENH**:

For a target neighboring NR frequency, the eNodeB does not deliver event B1 measurement configurations for this frequency only when the virtual-grid-based prediction result about this frequency meets the leaving condition of event B1. When the prediction result meets the entering condition of event B1, the eNodeB delivers event B1 measurement configurations for this NR frequency. After the NSA UE reports event B1 measurement results, the eNodeB instructs the UE to add an SCG. For details about MeNB-initiated SgNB addition, see *NSA Mobility Management*.

SCG addition based on virtual grid models is not recommended in certain scenarios.

- In high-speed mobility scenarios such as high-speed railway dedicated networks, it is recommended that this function be disabled. This is to prevent network performance deterioration because of inaccurate NR coverage evaluation due to fast UE movement.
- During key event assurance, it is recommended that this function be disabled. This is to prevent network performance deterioration as the function increases the CPU usage and the number of SRBs.
- During long-term network reconstruction, it is recommended that this function be disabled or the accuracy required for SCG addition based on virtual grid models be set to 100% (that is, the **LnrCarrierSelection.VgMdlPredNrScgMeasRsrpThld** parameter be set to **ACCURACY100_OFFSET0**). This is to prevent network performance deterioration caused by low accuracy.

4.1.1.2.5 SPID-based SCG Frequency Priority Configuration

The eNodeB can configure different SCG frequency priorities for different types of UEs. As such, different types of UEs preferentially select different SCG carriers, improving user experience.

On the LTE side, **SPIDCFG.NrScgFreqPriorityGroupId** and **NrScgFreqPriGrp.NrScgFreqPriorityGroupId** are associated to configure SCG frequency priorities for UEs with different SPIDs.

- If **SPIDCFG.NrScgFreqPriorityGroupId** is set to a value in the range of 0 to 255 for a specified SPID, the priorities of SCG frequencies corresponding to **NrScgFreqPriGrp.PccDlEarfcn** and **NrScgFreqPriGrp.NrScgDlArfcn** are determined by **NrScgFreqPriGrp.NrScgDlArfcnPriority**. If a UE is not assigned an SPID on the core network or its SPID is not configured on the eNodeB, the priorities of SCG frequencies in the UE's PCC and SCG frequency combination are determined by **NrScgFreqConfig.ScgDlArfcnPriority**.
- If **SPIDCFG.NrScgFreqPriorityGroupId** is set to **65535**, the SPID-based SCG frequency priority configuration function does not take effect and the priorities of all SCG frequencies are determined by **NrScgFreqConfig.ScgDlArfcnPriority**.

If both SPID-based SCG frequency priority configuration and user-experience-based NSA carrier combination selection are enabled, the latter does not take effect for UEs with a specified SPID. In addition, no handover for user-experience-based NSA carrier combination selection will be initiated towards a base station

with the SPID-based SCG frequency priority configuration function enabled. If user-experience-based NSA carrier combination selection is required, SPID-based SCG frequency priority configuration needs to be disabled.

4.1.1.2.6 Operator-specific SCG Frequency Priority Configuration

The eNodeB supports operator-specific SCG frequency priority configuration so that UEs of different operators can preferentially select different SCG carriers, improving user experience.

On the LTE side, **CnOperator.NrScgFreqPriorityGroupId** and **NrScgFreqPriGrp.NrScgFreqPriorityGroupId** are associated to configure SCG frequency priorities for different operators.

- If **CnOperator.NrScgFreqPriorityGroupId** is set to a value in the range of 0 to 255 for an operator, the priority of the SCG frequency indicated by **NrScgFreqPriGrp.PccDLEarfcn** and **NrScgFreqPriGrp.NrScgDLArfcn** is determined by **NrScgFreqPriGrp.NrScgDLArfcnPriority**.
- If **CnOperator.NrScgFreqPriorityGroupId** is set to **65535**, the operator-specific SCG frequency priority configuration function does not take effect and the priorities of all SCG frequencies are determined by **NrScgFreqConfig.ScgDLArfcnPriority**.

4.1.1.2.7 Differentiated SCG Addition and Deletion Threshold Configuration for NR CA-capable UEs

In NSA networking where NR FDD+TDD is used, when the NR FDD cell has a small bandwidth (for example, 10 MHz), a high event B1 threshold is set for the NR FDD frequency to ensure the downlink user-perceived rate of NSA UEs. However, in areas with poor NR FDD coverage, a low event B1 threshold can also ensure the downlink user-perceived rate of NR FDD+TDD CA-capable UEs because NR TDD SCCs can be configured for them. Therefore, differentiated configuration of NR FDD SCG addition and deletion thresholds can increase the downlink user-perceived rate of CA-capable UEs in areas with poor NR FDD coverage.

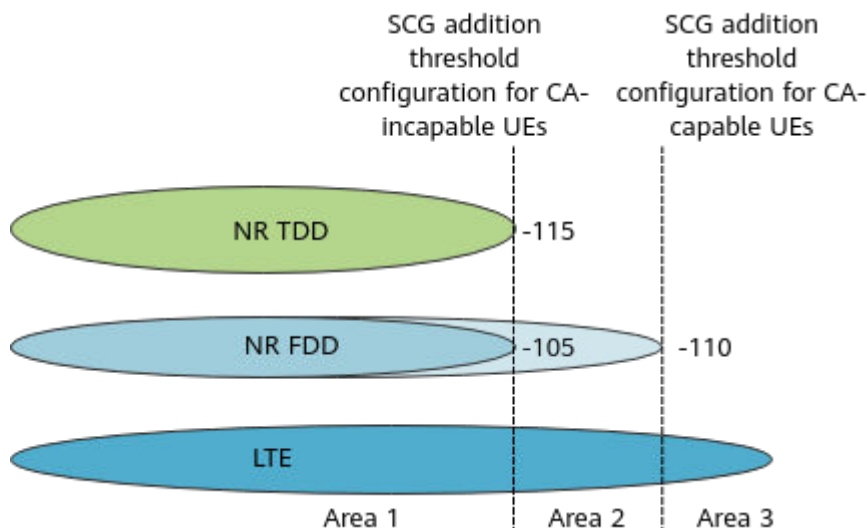
For NR FDD SCG addition, the "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function can evaluate the coverage of TDD SCCs to further improve the downlink user-perceived rate of NR FDD+TDD CA-capable UEs.

Differentiated SCG Addition Threshold Configuration for NR CA-capable UEs

"Differentiated SCG addition threshold configuration for NR CA-capable UEs" takes effect when the **SCG_ADD_B1_THLD_DIFF_SW** option of the **NsaDcAlgoParam.NsaDcAlgoExtSwitch** parameter is selected on the eNodeB side. With this function, the MeNB uses the sum of the original effective RSRP threshold for event B1 and the offset specified by the **NrScgFreqConfig.CaUeScgAddRsrpThldOffset** parameter as the FDD threshold in event B1 measurement configurations delivered to NSA UEs that support NR FDD+TDD CA during measurement-based PSCell configuration. This achieves differentiated NR FDD SCG addition threshold configuration. The original effective RSRP threshold for event B1 is specified by the **NrScgFreqConfig.NsaDcB1ThldRsrp** parameter for a non-SUL cell and by the **NrScgFreqConfig.NsaDcSulB1ThldRsrp** parameter for an SUL cell.

Figure 4-5 shows how this function works.

Figure 4-5 Differentiated SCG addition threshold configuration for NR CA-capable UEs



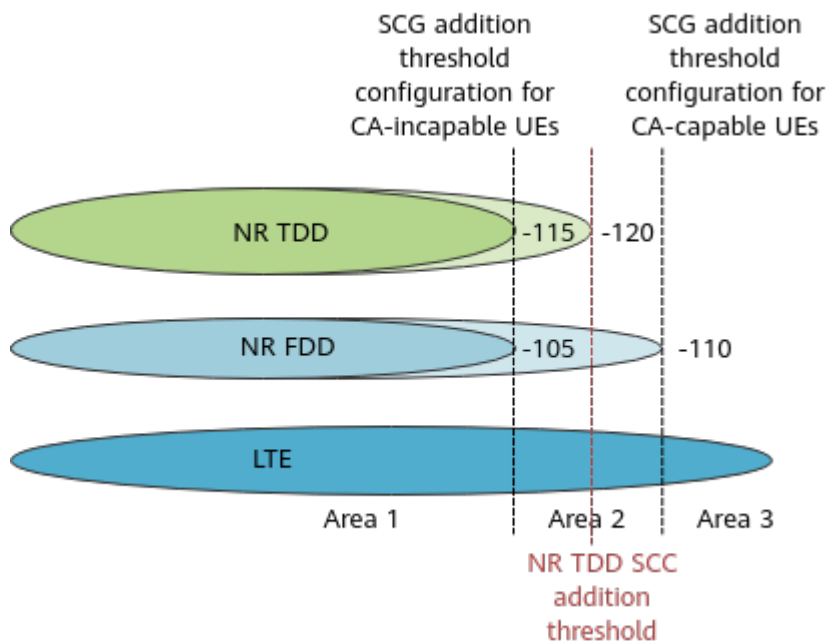
When the **SCG_ADD_B1_THLD_DIFF_SW** option of the **UeCompat.WhitelistCtrlExtSwitch2** parameter on the eNodeB side is selected, the "whitelist control on differentiated B1 threshold for SCG addition" function takes effect. In this case, the "differentiated SCG addition threshold configuration for NR CA-capable UEs" function takes effect only for whitelisted NSA UEs that support NR FDD+TDD CA, regardless of whether the **SCG_ADD_B1_THLD_DIFF_SW** option of the **NsaDcAlgoParam.NsaDcAlgoExtSwitch** parameter is selected.

FDD SCG Addition for NR CA-Capable UEs Based on the RSRP of Candidate NR TDD SCCs

When the **NsaDcAlgoParam.NrTddSccRsrpThld** parameter on the eNodeB side is not set to **255**, the "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function takes effect. This function is implemented based on "differentiated SCG addition threshold configuration for NR CA-capable UEs". After this function takes effect, the eNodeB evaluates the RSRP values of TDD SCCs when adding an FDD SCG for an NR FDD+TDD CA-capable UE to further improve the downlink user-perceived rate of such UEs.

Figure 4-6 shows how this function works.

Figure 4-6 FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs



This function works for NR FDD+TDD CA-capable UEs as follows:

1. During measurement-based PSCell configuration, the MeNB configures different event B1 RSRP thresholds for FDD and TDD frequencies when delivering event B1 measurement configurations based on frequency priorities. For FDD frequencies, an extended RSRP threshold for event B1 is delivered, which equals the original effective RSRP threshold for event B1 plus the offset specified by the **NrScgFreqConfig.CaUeScgAddRsrpThldOffset** parameter. The original effective RSRP threshold for event B1 is specified by the **NrScgFreqConfig.NsaDcB1ThldRsrp** parameter for a non-SUL cell and by the **NrScgFreqConfig.NsaDcSulB1ThldRsrp** parameter for an SUL cell. For TDD frequencies, the delivered threshold is the NR TDD SCC addition threshold, which equals the smaller one between the original effective RSRP threshold for event B1 and the NR TDD SCC RSRP threshold specified by the **NsaDcAlgoParam.NrTddScsRsrpThld** parameter.
2. After event B1 measurement configurations are delivered, the MeNB implements an SCG addition policy corresponding to the **ALL_FREQ** setting of the **NsaDcMgmtConfig.NsaDcScgAdditionPolicy** parameter. That is, it will wait for event B1 measurement reports for all delivered NR frequencies within the length of the NR B1 measurement report waiting timer (specified by the **NrScgFreqConfig.NrB1ReportWaitingTimer** parameter).
3. If the MeNB has received event B1 measurement reports for all NR frequencies or the NR B1 measurement report waiting timer has expired, the MeNB evaluates FDD frequencies based on both the existing filtering rules and the following conditions. It allows the addition of an FDD SCG corresponding to the frequency meeting any of the following conditions:
 - The RSRP in the measurement report meets the original effective RSRP threshold for event B1.

- The RSRP in the measurement report meets the extended RSRP threshold for event B1, and the MeNB receives measurement reports for TDD frequencies that meet all of the following conditions:
 - The UE supports NR CA between the current FDD frequency and the TDD frequencies.
 - The RSRP in the measurement reports for the TDD frequencies meets the NR TDD SCC addition threshold.

For details about measurement-based PSCell configuration, see [4.1.1.2.1 Measurement-based PSCell Configuration](#).

When the "whitelist control on differentiated B1 threshold for SCG addition" function is enabled, the "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function takes effect only for whitelisted NSA UEs that support NR FDD+TDD CA.

Differentiated SCG Deletion Threshold Configuration for NR CA-capable UEs

"Differentiated SCG deletion threshold configuration for NR CA-capable UEs" takes effect when the **NRCellNsaDcConfig.CaCapUePscellA2RsrpThldOfs** parameter is set to a non-zero value on the gNodeB side. With this function, the SgNB uses the sum of the original effective RSRP threshold for event A2 (specified by the **NRCellNsaDcConfig.PscellA2RsrpThld** parameter) and the offset specified by the **NRCellNsaDcConfig.CaCapUePscellA2RsrpThldOfs** parameter as the threshold in event A2 measurement configurations delivered to NSA UEs that support NR FDD+TDD CA during RSRP-based SgNB release. This achieves differentiated NR FDD SCG deletion threshold configuration.

For details about SgNB-initiated RSRP-based SgNB release, see *NSA Mobility Management*.

4.1.1.3 SPID/QCI-specific Carrier Management

When adaptive CA (adaptive configuration mode for CA) is enabled, the eNodeB can perform carrier management for different types of UEs based on SPIDs or for different service types based on QCIs, making carrier management more targeted in NSA DC. For details about adaptive configuration mode, see *Carrier Aggregation* in eRAN feature documentation.

4.1.1.3.1 SPID-specific NSA PCC Anchoring and PSCell Management Policies

Multi-Carrier Configuration Group

The eNodeB MO **CellMultiCarrCfgGrp** can be used to configure a multi-carrier configuration group and set the measurement configurations related to a downlink frequency in the group. [Table 4-1](#) lists the parameters that jointly define a downlink frequency in the multi-carrier configuration group.

Table 4-1 Parameters that define a frequency in the multi-carrier configuration group

Parameter ID	Description
CellMultiCarrCfgGrp.DlArfcn	Downlink frequency number
CellMultiCarrCfgGrp.RatType	RAT type of the downlink frequency
CellMultiCarrCfgGrp.CnOperatorId	ID of an operator for the downlink frequency
CellMultiCarrCfgGrp.LocalCellId	ID of a local cell for the downlink frequency

A multi-carrier configuration group is identified by the **CellMultiCarrCfgGrp.MultiCarrCfgGroupId** parameter. After an **SpidCfg.MultiCarrCfgGroupId** parameter is set to the ID of a specific multi-carrier configuration group, the involved SPID is associated with this group to implement different SPID-specific NSA PCC anchoring and PSCell management policies.

SPID-specific NSA PCC Anchoring Policy

For a downlink frequency in the multi-carrier configuration group associated with a specific SPID, when the RAT type of this frequency is **EUTRAN** and this frequency is already configured using the **PccFreqCfg.PccDlEarfcn** parameter, then the NSA PCC anchoring policy for the SPID-specific UEs on this frequency can be configured using the following parameters and options:

- **CellMultiCarrCfgGrp.NsaPccAnchoringPriority**: It specifies the NSA PCC anchoring priority of the frequency. If it is set to a value ranging from **0** to **31**, the **PccFreqCfg.NsaPccAnchoringPriority** parameter takes this value. If it is set to **255**, the actual value of **PccFreqCfg.NsaPccAnchoringPriority** is not affected.
- **FORBID_MEAS_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter: It specifies whether PCC anchoring measurement on the frequency is prohibited. If this option is selected, PCC anchoring measurement on the frequency is prohibited. If this option is deselected, PCC anchoring measurement on the frequency is allowed.
- **HIGH_LOAD_CELL_NSA_PCC_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter: It specifies whether the eNodeB performs special processing on the frequency during PCC anchoring for RRC_CONNECTED UEs or RRC_IDLE UEs in NSA networking.
 - During PCC anchoring for RRC_CONNECTED UEs in NSA networking, if this option is selected, the eNodeB selects this frequency as a candidate NSA PCC anchor for the SPID-specific UEs only when the average downlink PRB usage of the cell where the UEs camp exceeds **NsaDcMgmtConfig.DlPrbThldForNsaPccAnchor** and the UEs are running large-packet services. If this option is deselected, the eNodeB always selects this frequency as a candidate NSA PCC anchor for the SPID-specific UEs.

- During NSA PCC anchoring for RRC_IDLE UEs in NSA networking, if this option is selected, the eNodeB does not adjust the priority of the frequency as a candidate NSA PCC anchor for the SPID-specific UEs. If this option is deselected, the eNodeB adjusts the priority of the frequency as a candidate NSA PCC anchor for the SPID-specific UEs.

NOTE

If both SPID-specific and QCI-specific NSA PCC anchoring policies are configured for a downlink frequency, the SPID-specific NSA PCC anchoring policy preferentially takes effect.

SPID-specific PSCell Management Policy

For a downlink frequency in the multi-carrier configuration group associated with a specific SPID, when the RAT type of this frequency is **NR** and this frequency is already configured using the **PccFreqCfg.PccDLEarfcn** or **NrScgFreqConfig.ScgDLArfcn** parameter, then the PSCell management policy for the SPID-specific UEs camping on the current serving cell on this frequency can be configured using the following parameters and options:

- **FORBID_MEAS_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter: It specifies whether the frequency can be treated as an SCG frequency for measurement. If this option is selected, the frequency cannot be treated as an SCG frequency for measurement. If this option is deselected, the frequency can be treated as an SCG frequency for measurement.
- **FORBID_BLIND_CFG_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter: It specifies whether the frequency can be blindly configured as an SCG frequency. If this option is selected, the frequency cannot be blindly configured as an SCG frequency. If this option is deselected, the frequency can be blindly configured as an SCG frequency.

During measurement-based PSCell configuration, the eNodeB can configure multiple frequencies and corresponding priorities for the SCG after filtering frequencies based on UE capabilities. If the downlink frequency in the multi-carrier configuration group for SPID-specific UEs is already configured using the **NrScgFreqConfig.ScgDLArfcn** parameter, the priority of this frequency is equal to **NrScgFreqConfig.ScgDLArfcnPriority** plus **CellMultiCarrCfgGrp.NrScgPriorityOffset**.

After the SCG frequencies are configured, the eNodeB delivers measurement configurations of these frequencies to the NSA UE. For each of these frequencies, if the RSRP value measured by the NSA UE is greater than the B1 RSRP threshold, the UE will report the B1 measurement result. The B1 RSRP threshold is calculated as follows:

$$\text{B1 RSRP threshold} = A + B + C + D - E$$

- A: **NrScgFreqConfig.NsaDcB1ThldRsrp**
- B: **CellOp.NsaDcB1RsrpThldOffset**
- C: **NsaDcQciParamGroup.NsaDcB1RsrpThldOffset**
- D: **SpidCfg.NsaDcSpidB1RsrpThldOffset**
- E: **NrNFreq.FreqSpecificOffset**

If the downlink frequency in the multi-carrier configuration group for SPID-specific UEs is already configured using the **NrScgFreqConfig.ScgDlArfcn** parameter, the RSRP threshold for event B1 is equal to the preceding sum plus **CellMultiCarrCfgGrp.NsaDcB1RsrpThldOffset**.

 NOTE

If both SPID-specific and QCI-specific PSCell management policies are configured for a downlink frequency, the SPID-specific PSCell management policy preferentially takes effect.

Configuration Requirements

For different cells (specified by **CellMultiCarrCfgGrp.LocalCellId**) in a multi-carrier configuration group (specified by **CellMultiCarrCfgGrp.MultiCarrCfgGroupId**), if their downlink frequency numbers (specified by **CellMultiCarrCfgGrp.DlArfcn**), operator IDs (specified by **CellMultiCarrCfgGrp.CnOperatorId**), and RAT types (specified by **CellMultiCarrCfgGrp.RatType**) are identical, then the following parameters must also have the same settings for these cells. Otherwise, ping-pong handovers occur.

- **CellMultiCarrCfgGrp.NsaPccAnchoringPriority**
- **CellMultiCarrCfgGrp.NrScgPriorityOffset**
- **CellMultiCarrCfgGrp.NsaDcB1RsrpThldOffset**

SPID-specific SCG Addition at Short Intervals Without Considering Traffic Volume

With the function of SPID-specific SCG addition at short intervals without considering traffic volume, the eNodeB can add a PSCell for an NSA UE working in the LTE-only state. This implements more flexible and differentiated SCG addition policies. It is recommended that this function be enabled when the value of **L.Traffic.User.NsaDc.PCell.Avg** is greater than or equal to 10 and the value of **L.NsaDc.Capable.5gUser.RRC.Avg** is greater than or equal to 20. Otherwise, network data fluctuation may affect function observation.

The **SPID_SHT_PRD_NO_TFC_SCG_ADD_SW** option of the **SpidCfg.SpIdAlgoSwitch** parameter controls whether to enable this function for UEs with a specified SPID. If this option is selected, the eNodeB triggers SCG addition at short intervals without evaluating the traffic volume. In this scenario, the short interval is specified by the **NsaDcMgmtConfig.ShortPrdNoTfcScgAddIntvl** parameter. If this option is deselected, this function is disabled. In this case, see [4.1.1.2 SCG Carrier Management](#) to learn configurations related to periodic triggering of SCG addition.

 NOTE

When the **PERIODIC_SCG_ADD_OPT_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoExtSwitch** parameter is selected, periodic SCG addition and periodic NSA PCC anchoring for NSA DC UEs are decoupled. As such, the **SPID_SHT_PRD_NO_TFC_SCG_ADD_SW** option of the **SpidCfg.SpIdAlgoSwitch** parameter controls only SCG addition at short intervals without considering traffic volume.

When the **PERIODIC_SCG_ADD_OPT_SW** option is deselected, the **SPID_SHT_PRD_NO_TFC_SCG_ADD_SW** option controls both SCG addition at short intervals without considering traffic volume and periodic NSA PCC anchoring.

4.1.1.3.2 QCI-specific NSA PCC Anchoring and PSCell Management Policies

Multi-Carrier Configuration Group

The eNodeB MO **CellMultiCarrCfgGrp** can be used to configure a multi-carrier configuration group and set the measurement configurations related to a downlink frequency in the group. **Table 4-2** lists the parameters that jointly define a downlink frequency in the multi-carrier configuration group.

Table 4-2 Parameters that define a frequency in the multi-carrier configuration group

Parameter ID	Description
CellMultiCarrCfgGrp.DlArfcn	Downlink frequency number
CellMultiCarrCfgGrp.RatType	RAT type of the downlink frequency
CellMultiCarrCfgGrp.CnOperatorId	ID of an operator for the downlink frequency
CellMultiCarrCfgGrp.LocalCellId	ID of a local cell for the downlink frequency

A multi-carrier configuration group is identified by the **CellMultiCarrCfgGrp.MultiCarrCfgGroupId** parameter. After a **QciPara.MultiCarrCfgGroupId** parameter is set to the ID of a specific multi-carrier configuration group, the involved QCI is associated with this group to implement different QCI-specific NSA PCC anchoring and PSCell management policies.

If a UE runs services corresponding to multiple QCIs, the processing is as follows. The priority of each QCI is specified by the **CellQciPara.QciPriorityForHo** parameter.

- If only one QCI has the highest priority, the NSA PCC anchoring and PSCell management policies corresponding to this QCI take effect.
- If multiple QCIs have the highest priority, the eNodeB sorts these QCIs based on the QCI priorities recommended in 3GPP specifications and selects a QCI with the highest 3GPP-recommended priority. The NSA PCC anchoring and PSCell management policies corresponding to this selected QCI take effect. For details about the QCI priorities recommended in 3GPP specifications, see section 6.1.7 "Standardized QoS characteristics" in 3GPP TS 23.203 V15.0.0.

QCI-specific NSA PCC Anchoring Policy

For a downlink frequency in the multi-carrier configuration group associated with a specific QCI, when the RAT type of this frequency is **EUTRAN** and this frequency is already configured using the **PccFreqCfg.PccDlEarfcn** parameter, then the NSA PCC anchoring policy for the QCI-specific UEs on this frequency can be configured using the following parameters and options:

- **CellMultiCarrCfgGrp.NsaPccAnchoringPriority**: It specifies the NSA PCC anchoring priority of the frequency. If it is set to a value ranging from **0** to **31**, the **PccFreqCfg.NsaPccAnchoringPriority** parameter takes this value. If it is set to **255**, the actual value of **PccFreqCfg.NsaPccAnchoringPriority** is not affected.
- **FORBID_MEAS_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter: It specifies whether PCC anchoring measurement on the frequency is prohibited. If this option is selected, PCC anchoring measurement on the frequency is prohibited. If this option is deselected, PCC anchoring measurement on the frequency is allowed.
- **HIGH_LOAD_CELL_NSA_PCC_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter: It specifies whether the eNodeB performs special processing on the frequency during PCC anchoring for RRC_CONNECTED UEs or RRC_IDLE UEs in NSA networking.
 - During PCC anchoring for RRC_CONNECTED UEs in NSA networking, if this option is selected, the eNodeB selects this frequency as a candidate NSA PCC anchor for the QCI-specific UEs only when the average downlink PRB usage of the cell where the UEs camp exceeds **NsaDcMgmtConfig.DlPrbThldForNsaPccAnchor** and the UEs are running large-packet services. If this option is deselected, the eNodeB always selects this frequency as a candidate NSA PCC anchor for the QCI-specific UEs.
 - During NSA PCC anchoring for RRC_IDLE UEs in NSA networking, if this option is selected, the eNodeB does not adjust the priority of the frequency as a candidate NSA PCC anchor for the QCI-specific UEs. If this option is deselected, the eNodeB adjusts the priority of the frequency as a candidate NSA PCC anchor for the QCI-specific UEs.

 NOTE

If both SPID-specific and QCI-specific NSA PCC anchoring policies are configured for a downlink frequency, the SPID-specific NSA PCC anchoring policy preferentially takes effect.

QCI-specific PSCell Management Policy

For a downlink frequency in the multi-carrier configuration group associated with a specific QCI, when the RAT type of this frequency is **NR** and this frequency is already configured using the **PccFreqCfg.PccDlEarfcn** or **NrScgFreqConfig.ScgDlArfcn** parameter, then the PSCell management policy for the QCI-specific UEs camping on the current serving cell on this frequency can be configured using the following parameters and options:

- **FORBID_MEAS_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter: It specifies whether the frequency can be treated as an SCG frequency for measurement. If this option is selected, the frequency cannot be treated as an SCG frequency for measurement. If this option is deselected, the frequency can be treated as an SCG frequency for measurement.
- **FORBID_BLIND_CFG_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter: It specifies whether the frequency can be blindly configured as an SCG frequency. If this option is selected, the frequency cannot be blindly configured as an SCG frequency. If

this option is deselected, the frequency can be blindly configured as an SCG frequency.

During measurement-based PSCell configuration, the eNodeB can configure multiple frequencies and corresponding priorities for the SCG after filtering frequencies based on UE capabilities. If the downlink frequency in the multi-carrier configuration group for QCI-specific UEs is already configured using the **NrScgFreqConfig.ScgDLArfcn** parameter, the priority of this frequency is equal to **NrScgFreqConfig.ScgDLArfcnPriority** plus **CellMultiCarrCfgGrp.NrScgPriorityOffset**.

After the SCG frequencies are configured, the eNodeB delivers measurement configurations of these frequencies to the NSA UE. For each of these frequencies, if the RSRP value measured by the NSA UE is greater than the B1 RSRP threshold, the UE will report the B1 measurement result. The B1 RSRP threshold is calculated as follows:

$$\text{B1 RSRP threshold} = A + B + C + D - E$$

- A: **NrScgFreqConfig.NsaDcB1ThldRsrp**
- B: **CellOp.NsaDcB1RsrpThldOffset**
- C: **NsaDcQciParamGroup.NsaDcB1RsrpThldOffset**
- D: **SpidCfg.NsaDcSpidB1RsrpThldOffset**
- E: **NrNFreq.FreqSpecificOffset**

If the downlink frequency in the multi-carrier configuration group for QCI-specific UEs is already configured using the **NrScgFreqConfig.ScgDLArfcn** parameter, the RSRP threshold for event B1 is equal to the preceding sum plus **CellMultiCarrCfgGrp.NsaDcB1RsrpThldOffset**.

NOTE

If both SPID-specific and QCI-specific PSCell management policies are configured for a downlink frequency, the SPID-specific PSCell management policy preferentially takes effect.

Configuration Requirements

For different cells (specified by **CellMultiCarrCfgGrp.LocalCellId**) in a multi-carrier configuration group (specified by **CellMultiCarrCfgGrp.MultiCarrCfgGroupId**), if their downlink frequency numbers (specified by **CellMultiCarrCfgGrp.DLArfcn**), operator IDs (specified by **CellMultiCarrCfgGrp.CnOperatorId**), and RAT types (specified by **CellMultiCarrCfgGrp.RatType**) are identical, then the following parameters must also have the same settings for these cells. Otherwise, ping-pong handovers occur.

- **CellMultiCarrCfgGrp.NsaPccAnchoringPriority**
- **CellMultiCarrCfgGrp.NrScgPriorityOffset**
- **CellMultiCarrCfgGrp.NsaDcB1RsrpThldOffset**

QCI-specific SCG Addition at Short Intervals Without Considering Traffic Volume

With the function of QCI-specific SCG addition at short intervals without considering traffic volume, the eNodeB can add a PSCell for an NSA UE working in the LTE-only state. This implements more flexible and differentiated SCG addition

policies. It is recommended that this function be enabled when the value of **L.Traffic.User.NsaDc.PCell.Avg** is greater than or equal to 10 and the value of **L.NsaDc.Capable.5gUser.RRC.Avg** is greater than or equal to 20. Otherwise, network data fluctuation may affect function observation.

The **QCI_SHT_PRD_NO_TFC_SCG_ADD_SW** option of the **CellQciPara.QciAlgoSwitch** parameter controls whether to enable this function for UEs with a specified QCI. If this option is selected, the eNodeB triggers SCG addition at short intervals without evaluating the traffic volume. In this scenario, the short interval is specified by the **NsaDcMgmtConfig.ShortPrdNoTfcScgAddIntvl** parameter. If this option is deselected, this function is disabled. In this case, see [4.1.1.2 SCG Carrier Management](#) to learn configurations related to periodic triggering of SCG addition.

NOTE

When the **PERIODIC_SCG_ADD_OPT_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoExtSwitch** parameter is selected, periodic SCG addition and periodic NSA PCC anchoring for NSA DC UEs are decoupled. As such, the **QCI_SHT_PRD_NO_TFC_SCG_ADD_SW** option of the **CellQciPara.QciAlgoSwitch** parameter controls only SCG addition at short intervals without considering traffic volume.

When the **PERIODIC_SCG_ADD_OPT_SW** option is deselected, the **QCI_SHT_PRD_NO_TFC_SCG_ADD_SW** option controls both SCG addition at short intervals without considering traffic volume and periodic NSA PCC anchoring.

4.1.2 Data Split in NSA DC

Data split can be started after the MeNB receives an SgNB Addition Request Acknowledge message (indicating that an X2 tunnel has been set up) from the SgNB. The data split architecture can be configured based on the service type by specifying a bearer type, and then the uplink and downlink data split modes can be configured.

In this version, all types of services, except the following, are supported by the MCG bearer, MCG_Bearer_EUTRA_PDCP, MCG split bearer, and SCG split bearer:

- Voice services and TCP/IP data services that use the ROHC function: supported only by the MCG_Bearer_EUTRA_PDCP.
- E-RABs with QCIs of 1 or 5: supported only by the MCG bearer and MCG_Bearer_EUTRA_PDCP.
- E-RABs with QCIs of 2–4, 65–67, 75, or 82–85: supported only by the MCG bearer, MCG_Bearer_EUTRA_PDCP, and SCG split bearer. In the case of SCG split bearer, the uplink and downlink data is distributed only to the SCG. That is, uplink or downlink data of GBR services can be transmitted only on the SCG side in Option 3x. For details about the relationship between QCIs and GBRs, see *QoS Management* in 5G RAN feature documentation.
- For details about the processing of extended QCIs, see *QoS Management* in 5G RAN feature documentation.

 NOTE

- In the Option 3x architecture, if data distribution only to MCG is configured in the downlink or uplink, it is recommended that the **NRDUCellQciBearer.UelInactivityTimer** parameter be set to **0** on the gNodeB side. This prevents the SCG release triggered by the expiry of the UE inactivity timer when there is no traffic at the RLC layer on the NR side.
- The PDCP layer will discard timeout data packets and the data split performance will be affected in the following situations:
 - In Option 3, the **RlcPdcPParaGroup.DiscardTimer** parameter is set to a value other than **DiscardTimer_Infinity**.
 - In Option 3x, the **gNBpdcPParamGroup.DlPdcPDiscardTimer** parameter is set to a value other than **INFINITY**.
- In NSA DC, data transmission delay is not guaranteed.
- In NSA DC, the transport network cannot ensure fairness among UEs. If data split is performed for multiple UEs at the same time, the data split rates for the UEs may be different.
- In NSA DC, assume that uplink transmission is switched towards the MCG due to uplink fallback to LTE or uplink data transmission path selection. Then, uplink transmission is also switched towards the MCG for NSA UEs whose E-RABs with QCIs in the range of 2–4, 65–67, 75, and 82–85 are set up on SCG split bearers. In this case, one-way X2 delay is introduced.

4.1.2.1 Data Split Policy Selection

The **CellQciPara.NsaDcDefaultBearerMode** parameter specifies the bearer mode:

- When this parameter is set to **MCG_SPLIT_BEARER**, the data split architecture is Option 3.
- When this parameter is set to **SCG_SPLIT_BEARER**, the data split architecture is Option 3x.
- When this parameter is set to **MCG_BEARER** or **MCG_BEARER_EUTRA_PDCP**, data is transmitted only on the LTE side.

For QCI 1 or QCI 5, it is recommended that the **CellQciPara.NsaDcDefaultBearerMode** parameter be set to **MCG_BEARER_EUTRA_PDCP**.

 NOTE

- To ensure the consistency of user experience with services that have the same QCI between eNodeBs, it is recommended that the same bearer mode be set on these eNodeBs for the same QCI.
- When the value of the **CellQciPara.NsaDcDefaultBearerMode** parameter changes, the data volume allocated to the LTE or NR side may change significantly due to the change of the bearer mode.

4.1.2.1.1 Downlink Data Split

Downlink Data Split Mode Selection in Option 3

In Option 3, the downlink data split mode is specified by the QCI-level parameter **NsaDcQciParamGroup.DlDataPdcPSplitMode** on the LTE side.

- Downlink data is transmitted only on the MCG side.

When the **NsaDcQciParamGroup.DlDataPdcpsplitMode** parameter is set to **MCG_ONLY**, downlink PDCP PDUs are all distributed to the MCG side.

- Downlink data is transmitted only on the SCG side.

When the **NsaDcQciParamGroup.DlDataPdcpsplitMode** parameter is set to **SCG_ONLY**, downlink PDCP PDUs are all distributed to the SCG side.

- Downlink data is distributed dynamically.
 - If the **NsaDcQciParamGroup.DlDataPdcpsplitMode** parameter is set to **SCG_AND_MCG**, the primary path is on the PDCP data split anchor side. Data is preferentially distributed to the primary path and then distributed between the MeNB and the SgNB.
 - If the **NsaDcQciParamGroup.DlDataPdcpsplitMode** parameter is set to **SCG_PREFERRED**, the primary path is on the SgNB side. Data is preferentially distributed to the primary path and then distributed between the MeNB and the SgNB.
 - If the **NsaDcQciParamGroup.DlDataPdcpsplitMode** parameter is set to **ADAPT_PRIMARY_PATH**, the primary path is adaptively determined based on air interface capabilities during PDCP data split. Data is preferentially distributed to the primary path and then distributed between the MeNB and the SgNB.

Dynamic PDCP data split is based on the LTE and NR air interface status (such as the RLC transmission rate, air interface transmission delay, and RLC buffer size) and X2 interface status (such as the transmission delay and congestion status). The UE PDCP layer re-orders downlink PDCP PDUs and then transmits them to the upper layer.

On the LTE side, the **CellOpQciPara.NsaDcQciParamGroupId** parameter can be associated with the **NsaDcQciParamGroup.NsaDcQciParamGroupId** parameter to configure operator-specific downlink data split policies.

- If the **CellOpQciPara.NsaDcQciParamGroupId** parameter is set to a value in the range of 0 to 9, the downlink data split policy for the operator is determined by the associated **NsaDcQciParamGroup.DlDataPdcpsplitMode** parameter.
- If the **CellOpQciPara** MO is not configured, the downlink data split policies for all operators are determined by the **CellQciPara.NsaDcQciParamGroupId** parameter.

NOTE

The **NSA_TNLLOAD_BASED_TCSPLIT_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter specifies whether to enable the function of stopping data distribution to the LTE side based on transport load. When the transport load of the main control board on the LTE side is high, the MeNB does not trigger SCG addition. This prevents the transport load from increasing due to RLC status reporting for data split.

Downlink Data Split Mode Selection in Option 3x

In Option 3x, the downlink data split mode is specified by the **gNBpdcpsplitMode** parameter on the NR side.

- Downlink data is transmitted only on the MCG side.

When the **gNBpdcpsplitMode** parameter is set to **MCG_ONLY**, downlink PDCP PDUs are all distributed to the MCG.

- Downlink data is transmitted only on the SCG side.
When the **gNBPDcpParamGroup.DlDataPdcPsplitMode** parameter is set to **SCG_ONLY**, downlink PDCP PDUs are all distributed to the SCG.
- Downlink data is distributed dynamically.
 - If the **gNBPDcpParamGroup.DlDataPdcPsplitMode** parameter is set to **SCG_AND_MCG**, the primary path is on the PDCP data split anchor side. Data is preferentially distributed to the primary path and then distributed between the MeNB and the SgNB.
 - If the **gNBPDcpParamGroup.DlDataPdcPsplitMode** parameter is set to **MCG_PREFERRED**, the primary path is on the MeNB side. Data is preferentially distributed to the primary path and then distributed between the MeNB and the SgNB.
 - If the **gNBPDcpParamGroup.DlDataPdcPsplitMode** parameter is set to **ADAPT_PRIMARY_PATH**, the primary path is adaptively determined based on air interface capabilities during PDCP data split. Data is preferentially distributed to the primary path and then distributed between the MeNB and the SgNB.

 NOTE

- A QCI-specific service type is associated with a PDCP parameter group by adding an **NRCELLQCI BEARER** MO.
- The **NSA_TNLLOAD_BASED_TCSPPLIT_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter specifies whether to enable the function of stopping data distribution to the LTE side based on transport load. Assume that the transport load of the main control board on the LTE side is high. If the data split mode is not set to **MCG_ONLY**, data split on the LTE side will be stopped, preventing load increasing due to RLC status reporting for data split. If the data split mode is set to **MCG_ONLY**, data split will not be stopped, preventing data transmission interruption.

When NSA DC carries TCP services, the downlink PDCP PDU reordering time on the UE side increases if all the following conditions are met: the throughput on the LTE and NR sides differs by more than three times, the throughput on the side with high throughput is stable, and the throughput on the side with low throughput fluctuates (due to factors such as interference or load). This affects the TCP traffic and may cause negative impacts.

In NR low-frequency scenarios, the **PDCP_SN_OFFSET_SW** option of the **NRCeIlNsaDcConfig.NsaDcAlgoSwitch** parameter can be used to specify whether to add an offset to the SNs of packets sent to the side with low throughput. If this option is selected, the PDCP SN offset is used and data is preferentially distributed to the side with high throughput between LTE and NR. In this case, downlink traffic volume of this side increases and that of the other side decreases. If this option is deselected, the PDCP SN offset is not used.

In NR high-frequency scenarios where the LTE bandwidth is less than 20% of the NR bandwidth, downlink dynamic data split is not recommended on the NR side. If the LTE bandwidth is 20% or more of the NR bandwidth, it is recommended that downlink dynamic data split be configured on the NR side when all of the following conditions are met:

- The LTE side is lightly loaded.
- The X2 delay between LTE and NR is less than 3 ms.

- The uplink SINR of both LTE and NR is greater than 5 dB.

In addition, it is recommended that the function of stopping data split when data congestion occurs due to air interface exceptions on the LTE or NR side be enabled and the **NRCellNsaDcConfig.HighDelaySplitPenaltyPrd** parameter be set to **600**. These settings aim to reduce the impact of significant air interface performance differences between NR and LTE on the downlink UE throughput after data split.

On the NR side, the **NRCellOpQciBearer.AmPdcParamGroupId** or **NRCellOpQciBearer.UmPdcParamGroupId** parameter can be associated with the **gNBpdcParamGroup.PdcParamGroupId** parameter to configure operator-specific downlink data split policies.

- If the **NRCellOpQciBearer.AmPdcParamGroupId** or **NRCellOpQciBearer.UmPdcParamGroupId** parameter is set to a value in the range of 0 to 9, the downlink data split policy for the operator is determined by the **gNBpdcParamGroup.DlDataPdcSplitMode** parameter corresponding to the associated **gNBpdcParamGroup.PdcParamGroupId** parameter.
- If the **NRCellOpQciBearer** MO is not configured, the downlink data split policies for all operators are determined by the **gNBpdcParamGroup.PdcParamGroupId** parameter.

Basic Mechanism of Downlink Dynamic Data Split

If downlink dynamic data split is used (that is, the **gNBpdcParamGroup.DlDataPdcSplitMode** parameter is set to **SCG_AND_MCG**), the LTE and NR RLC layers send transmission capability information such as the RLC transmission rate and buffered data volume to the PDCP layer. Then, the PDCP layer performs dynamic data split based on the LTE and NR air interface status and X2 interface status (such as the transmission delay and congestion status). For details about X2 interface status, see *Transmission Resource Management* in 5G RAN feature documentation.

The basic principle of data split is that data is preferentially distributed to the primary path and then distributed between the MeNB and the SgNB. The main steps are as follows:

1. Data is preferentially distributed to the RLC layer of the primary path based on the RLC transmission rate and other factors.
2. Data is then distributed to the RLC layer of the other side based on the desired buffer size reported by the RLC layer to align the delays on the two sides. This way, the distributed data can reach the UE side at the same time. For details about the desired buffer size, see section 5.5.3.5 "Desired buffer size for the data radio bearer" in 3GPP TS 38.425 V15.6.0.
3. After the delays are aligned, data is distributed by the PDCP layer to the eNodeB and gNodeB sides in turn based on their RLC transmission rates and other factors.

Enhanced Mechanism of Downlink Dynamic Data Split

The enhanced mechanism of downlink dynamic data split includes the following functions:

- Downlink real-time data split
- Downlink dynamic data split stop and resumption
- Downlink data split optimization in DRX scenarios

In Option 3x (not in Option 3) networking, downlink real-time data split can take effect to fully utilize the data transmission capabilities of LTE and NR air interfaces and improve user experience when all the following options are selected:

- On the LTE side: **NSA_SPLIT_RATE_OPT_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoExtSwitch** parameter
- On the NR side: **NSA_SPLIT_RATE_OPT_SW** option of the **NRCellNsaDcConfig.NsaDcAlgoSwitch** parameter, and **NSA_PDCP_SPLIT_OPT_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter

The real-time data split function shortens the reporting period of RLC transmission capability information by exchanging messages between the RLC and PDCP layers in real time. In this way, the PDCP layer can deliver data in a timely manner, reducing air interface resource waste. This function estimates the transmission rate based on the MAC scheduling capability in the event of service interruption, improving the data split accuracy.

Dynamic data split is stopped at the PDCP layer in certain scenarios and this penalty lasts for a period of time, as listed in [Table 4-3](#).

Table 4-3 Data split stop scenarios and penalty period parameters

Data Split Stop Scenario	Penalty Period Parameter	Architecture
Data rates differ considerably between LTE and NR sides.	NRCellNsaDcConfig.DlDataSplitPenaltyPeriod	Option 3x
	NsaDcAlgoParam.DlDataSplitPenaltyPeriod	Option 3
Packet loss occurs over the X2 interface, or RLC status reports are not updated on the NR side in a timely manner.	gNodeBParam.AbnDlDataSplitPenaltyPrd	Option 3x
	NsaDcAlgoParam.AbnDlDataSplitPenaltyPrd	Option 3
Data congestion occurs because the air interface on the LTE or NR side is abnormal.	NRCellNsaDcConfig.HighDelaySplitPenaltyPrd	Option 3x

If downlink data split is stopped when data rates differ considerably between LTE and NR sides or data congestion occurs because the air interface on one side is abnormal, downlink data split can be resumed when the PDCP detects that the air interface condition becomes better. This function takes effect only if it is enabled as follows:

- In Option 3x, the **NSA_SPLIT_ADAPT_RECOVERY_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter is selected.
- In Option 3, the **NSA_SPLIT_ADAPT_RECOVERY_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter is selected.

 NOTE

- In NSA DC, an IP performance monitoring (IP PM) session is automatically created over the X2 interface. The X2 interface status information required by PDCP data split is obtained through the IP PM session.
- If the downlink data split mode needs to be set to **SCG_AND_MCG**, it is recommended that IP PM be configured in advance for X2 delay observation.
- If the number of IP PM sessions to be set up exceeds the limit, IP PM sessions fail to be set up and the PDCP layer cannot obtain X2 delay information. In this case, data split performance deteriorates.

When data split is stopped due to data congestion caused by an air interface exception on one side, the PDCP migrates data from the congested side to the other side for delay-based retransmission if the **NSA_RETRANS_BASED_ON_DELAY_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter is selected.

NSA downlink data split optimization in DRX scenarios is controlled by the **DRX_NSA_SPLIT_OPT_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter on the LTE side and the **DRX_NSA_SPLIT_OPT_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter on the NR side.

- If the primary path for uplink data split is on the SCG side, the PDCP migrates data packets that fall into the LTE sleep time to the NR side for retransmission, stops data split on the LTE side, and resumes data split in the active time.
- If the primary path for uplink data split is on the MCG side, the PDCP migrates data packets that fall into the NR sleep time to the LTE side for retransmission, stops data split on the NR side, and resumes data split in the active time.

For details about the sleep time and active time, see *DRX and Signaling Control* in eRAN feature documentation and *DRX* in 5G RAN feature documentation.

 NOTE

In DRX scenarios, it is recommended that NSA downlink data split optimization be enabled together with NSA uplink preallocation so that LTE and NR are in active time if possible when there are services.

4.1.2.1.2 Uplink Data Split

Uplink Data Split Mode Selection in Option 3

In Option 3, the uplink data split mode is specified by the QCI-level parameters **NsaDcQciParamGroup.UIDataSplitPrimaryPath** and **NsaDcQciParamGroup.UIDataSplitThreshold** on the LTE side:

- Uplink data is transmitted only on the MCG side.
The **NsaDcQciParamGroup.UIDataSplitThreshold** parameter is set to **INFINITY**, and the **NsaDcQciParamGroup.UIDataSplitPrimaryPath** parameter is set to **MCG**.

- Uplink data is transmitted only on the SCG side.
The **NsaDcQciParamGroup.UIDataSplitThreshold** parameter is set to **INFINITY**, and the **NsaDcQciParamGroup.UIDataSplitPrimaryPath** parameter is set to **SCG**.
- Uplink data is distributed dynamically.
If the **NsaDcQciParamGroup.UIDataSplitThreshold** parameter is set to a value other than **INFINITY** and the UE PDCP buffer data size is greater than or equal to this threshold, uplink data is distributed dynamically.

On the LTE side, the **CellOpQciPara.NsaDcQciParamGroupId** parameter can be associated with the **NsaDcQciParamGroup.NsaDcQciParamGroupId** parameter to configure operator-specific uplink data split policies.

- If the **CellOpQciPara.NsaDcQciParamGroupId** parameter is set to a value in the range of 0 to 9, the uplink data split policy for the operator is determined by the **NsaDcQciParamGroup.UIDataSplitPrimaryPath** and **NsaDcQciParamGroup.UIDataSplitThreshold** parameters corresponding to the associated **NsaDcQciParamGroup.NsaDcQciParamGroupId** parameter.
- If the **CellOpQciPara** MO is not configured, the uplink data split policies for all operators are determined by the **CellQciPara.NsaDcQciParamGroupId** parameter.

NOTE

- If the **NsaDcQciParamGroup.UIDataSplitThreshold** parameter is set to a value other than **INFINITY** and the data volume buffered at the UE PDCP layer is less than this threshold, the data transmission mode is determined by the **NsaDcQciParamGroup.UIDataSplitPrimaryPath** parameter.
- If the NSA UE does not support dynamic uplink data split, the data transmission mode that takes effect for the UE is determined by the **NsaDcQciParamGroup.UIDataSplitPrimaryPath** parameter as in scenarios where the **NsaDcQciParamGroup.UIDataSplitThreshold** parameter is set to **INFINITY**.
- If the **NsaDcQciParamGroup.UIDataSplitThreshold** parameter is set to **INFINITY** and the **NsaDcQciParamGroup.UIDataSplitPrimaryPath** parameter is set to **SCG**, there is still uplink data transmission on the LTE side even when the uplink data split primary path is switched to the MCG or when the SCG is released.

Uplink Data Split Mode Selection in Option 3x

In Option 3x, the uplink data split mode is specified by the parameters **gNBPDcpParamGroup.UIDataSplitPrimaryPath** and **gNBPDcpParamGroup.UIDataSplitThreshold** on the NR side:

- Uplink data is transmitted only on the MCG side.
The **gNBPDcpParamGroup.UIDataSplitThreshold** parameter is set to **INFINITY**, and the **gNBPDcpParamGroup.UIDataSplitPrimaryPath** parameter is set to **MCG**.
- Uplink data is transmitted only on the SCG side.
The **gNBPDcpParamGroup.UIDataSplitThreshold** parameter is set to **INFINITY**, and the **gNBPDcpParamGroup.UIDataSplitPrimaryPath** parameter is set to **SCG**.
- Uplink data is distributed dynamically.

If the **gNBPDcpParamGroup.UIDataSplitThreshold** parameter is set to a value other than **INFINITY** and the UE PDCP buffer data size is greater than or equal to this threshold, uplink data is distributed dynamically.

On the NR side, the **NRCellOpQciBearer.AmPdcPParamGroupId** or **NRCellOpQciBearer.UmPdcPParamGroupId** parameter can be associated with the **gNBPDcpParamGroup.PdcPParamGroupId** parameter to configure operator-specific uplink data split policies.

- If the **NRCellOpQciBearer.AmPdcPParamGroupId** or **NRCellOpQciBearer.UmPdcPParamGroupId** parameter is set to a value in the range of **0** to **9**, the uplink data split policy for the operator is determined by the **gNBPDcpParamGroup.UIDataSplitPrimaryPath** and **gNBPDcpParamGroup.UIDataSplitThreshold** parameters corresponding to the associated **gNBPDcpParamGroup.PdcPParamGroupId** parameter.
- If the **NRCellOpQciBearer** MO is not configured, the uplink data split policies for all operators are determined by the **gNBPDcpParamGroup.PdcPParamGroupId** parameter.

NOTE

- If the **gNBPDcpParamGroup.UIDataSplitThreshold** parameter is set to a value other than **INFINITY** and the data volume buffered at the UE PDCP layer is less than this threshold, the data transmission mode is determined by the **gNBPDcpParamGroup.UIDataSplitPrimaryPath** parameter.
- If the NSA UE does not support dynamic uplink data split, the data transmission mode that takes effect for the UE is determined by the **gNBPDcpParamGroup.UIDataSplitPrimaryPath** parameter as in scenarios where the **gNBPDcpParamGroup.UIDataSplitThreshold** parameter is set to **INFINITY**.
- If the **gNBPDcpParamGroup.UIDataSplitThreshold** parameter is set to **INFINITY** and the **gNBPDcpParamGroup.UIDataSplitPrimaryPath** parameter is set to **SCG**, there is still uplink data transmission on the LTE side even when the uplink data split primary path is switched to the MCG or when the SCG is released.
- If NSA is selected for an NSA UE during NSA/SA selection based on uplink coverage, the uplink primary path of the UE is switched to the MCG side (that is, uplink data is transmitted only on the MCG side). For details about NSA/SA selection based on uplink coverage, see *NSA-SA Selection Based on User Experience*.

Uplink AMBR Control for Data Split

During initial SCG configuration of a UE, the aggregate maximum bit rate (AMBR) of the MeNB and SgNB is configured based on the UE capability.

- If the UE supports uplink data split, the AMBR is configured for the MeNB and SgNB based on the proportion specified by the **NsaDcMgmtConfig.NsaDcUeMcgULAmbrRatio** parameter.
 - If uplink data is distributed dynamically or transmitted only to the SCG, the **NsaDcMgmtConfig.NsaDcUeMcgULAmbrRatio** parameter cannot be set to **100**.
 - If uplink data is distributed dynamically or transmitted only to the MCG, the **NsaDcMgmtConfig.NsaDcUeMcgULAmbrRatio** parameter cannot be set to **0**.
 - If uplink data is distributed only to the SCG or MCG, the **NsaDcMgmtConfig.NsaDcUeMcgULAmbrRatio** parameter can be set to **255**. In this case, the AMBRs of both the MeNB and the SgNB are configured based on the AMBRs delivered by the core network.

- If the UE does not support uplink data split, the AMBR is configured for the MeNB and SgNB based on the AMBRs delivered by the core network.

The MAC layers of the MeNB and SgNB schedule data based on the configured AMBRs.

 NOTE

When the uplink data split mode is switched to MCG only, the AMBR of the MeNB is configured based on the AMBR delivered by the core network.

4.1.2.2 Uplink Fallback to LTE

In NSA DC scenarios, when the **CellQciPara.NsaDcDefaultBearerMode** parameter is set to **MCG_SPLIT_BEARER** or **SCG_SPLIT_BEARER**, the network side controls the UE to dynamically send uplink data to the gNodeB or eNodeB based on the uplink SINR of NR. In this way, the uplink coverage of LTE can be used to compensate for the insufficient uplink coverage of NR. It is recommended that this function be enabled when LTE and NR cells are co-sited and uplink data is distributed only to the SCG. After this function takes effect, the uplink data transmission of cell edge users (CEUs) in the NR cell is switched to the LTE network to improve their experience.

This function is controlled by the **UL_FALLBACK_TO_LTE_SWITCH** option of the **NRCellNsaDcConfig.NsaDcAlgoSwitch** parameter.

If this option is selected in Option 3:

- When the uplink NR SINR is less than **NRDUCellSrsMeas.NsaUlFackToLteSinrThld** minus **NRDUCellSrsMeas.NsaUlFackToLteSinrHyst** five consecutive times at an interval specified by **NRDUCellSrsMeas.NsaUlToLteSinrTimeToTrig**, the eNodeB instructs the UE to perform data transmission on the LTE side by sending an RRC connection reconfiguration message to the UE.
- When the uplink NR SINR is greater than **NRDUCellSrsMeas.NsaUlFackToLteSinrThld** plus **NRDUCellSrsMeas.NsaUlFackToLteSinrHyst** five consecutive times at an interval specified by **NRDUCellSrsMeas.NsaUlToLteSinrTimeToTrig**, the eNodeB instructs the UE to perform data transmission in the original mode by sending an RRC connection reconfiguration message to the UE.

If this option is selected in Option 3x:

- When the uplink NR SINR is less than **NRDUCellSrsMeas.NsaUlFackToLteSinrThld** minus **NRDUCellSrsMeas.NsaUlFackToLteSinrHyst** five consecutive times at an interval specified by **NRDUCellSrsMeas.NsaUlToLteSinrTimeToTrig**, the gNodeB triggers a change in the uplink data split mode by sending an SgNB Modification Required message to the eNodeB. The eNodeB instructs the UE to transmit data on the LTE side by sending an RRC connection reconfiguration message to the UE.
- When the uplink NR SINR is greater than **NRDUCellSrsMeas.NsaUlFackToLteSinrThld** plus **NRDUCellSrsMeas.NsaUlFackToLteSinrHyst** five consecutive times at an interval specified by **NRDUCellSrsMeas.NsaUlToLteSinrTimeToTrig**, the gNodeB triggers a change in the uplink data split mode by sending an SgNB

Modification Required message to the eNodeB. The eNodeB instructs the UE to use the original uplink data split mode by sending an RRC connection reconfiguration message to the UE.

 NOTE

If the **NRDUCellSrsMeas.NsaUlFackToLteSinrThld** parameter is set to a value too large, uplink fallback to LTE will be triggered too early. If the LTE uplink throughput is lower than the pre-fallback NR uplink throughput, the uplink throughput of NSA DC will decrease. When uplink data of NSA UEs is transmitted only on the LTE side, the uplink load of LTE cells will increase.

If the gNodeB is required to obtain the packet status (large or small) of UEs that have undergone uplink fallback to LTE (ULFB UEs), the function of requesting LTE-side large-packet status of ULFB UEs needs to be enabled. Once this function is enabled, the gNodeB reports the function switch status to the eNodeB. After learning that the switch is on, and both the **UL_LARGE_PKT_SERV_DECT_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter and the **ULFB_UL_LRG_PKT_FLAG_TO_NR_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoSwitch** parameter are selected, the eNodeB provides the uplink large- and small-packet status of ULFB UEs to the gNodeB. The function of requesting LTE-side large-packet status of ULFB UEs is controlled by the **ULFB_LTE_UL_LRG_PKT_REQ_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter.

4.1.2.3 Fast Retransmission for Downlink Data Split

In NSA DC downlink data split scenarios, data may be overstocked on the LTE or NR side if the air interface on this side is interfered. In this case, fast retransmission for downlink data split can be used to quickly migrate the overstocked data to the other side for fast retransmission. This reduces the impact of data overstock on the TCP sliding window and increases UE throughput.

This function is controlled by the **NSA_DC_FAST_RETRANS_SWITCH** option of the **gNodeBParam.NsaDcOptSwitch** parameter on the NR side. When this option is selected and the PDCP layer detects that the difference between the time taken to empty the RLC buffer on the LTE side and that on the NR side is greater than a certain threshold throughout a period, data is migrated from the rate drop side to the other side for fast retransmission.

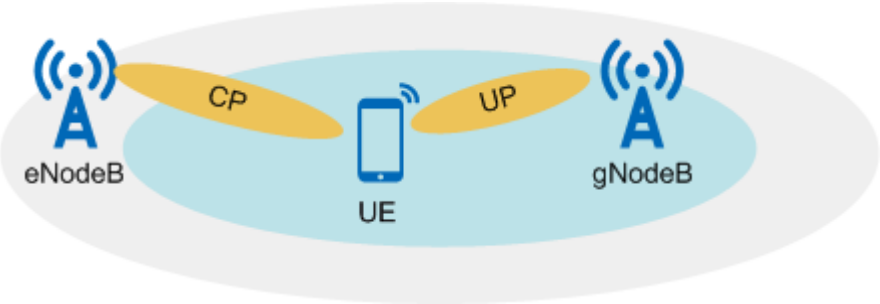
 NOTE

- Data migration is not triggered in the following situations: the RLC transmission mode is UM or TM, the X2 one-way delay is greater than or equal to 10 ms, the CPU usage is greater than 70%, the PDCP SN length is not 18 bits, the downlink data split mode is MCG only or SCG only, and the downlink data split bearer type is SRB.
- When packet loss occurs on the X2 link, if data split is performed on the LTE side, data is not migrated from the LTE side to the NR side; if data split is performed on the NR side, data is not migrated from the NR side to the LTE side.

4.1.2.4 CP/UP Separation

In the initial phase of NR deployment, 3.5 GHz or 28 GHz frequency bands are generally used. Their coverage ranges are far less than those of LTE frequencies. However, NR can provide higher-rate services. Therefore, LTE can be used to carry control-plane (CP) data to ensure mobility, while NR carries user-plane (UP) data, as shown in [Figure 4-7](#), rendering higher throughput and larger capacity.

Figure 4-7 CP/UP separation



- Parameter configurations for downlink CP/UP separation

Data Split Architecture	CP	UP	Parameter Setting
Option 3	LTE	NR	On the LTE side: - Set the CellQciPara.NsaDcDefaultBearer-Mode parameter to MCG_SPLIT_BEARER. - Set the NsaDcQciParamGroup.DlDataPdcpsplit-Mode parameter to SCG_ONLY.
Option 3x	LTE	NR	On the LTE side: - Set the CellQciPara.NsaDcDefaultBearer-Mode parameter to SCG_SPLIT_BEARER. On the NR side: - Set the gNBpdcpsplit-Mode parameter to SCG_ONLY.

- Parameter configurations for uplink CP/UP separation

Data Split Architecture	CP	UP	Parameter Setting
Option 3	LTE	NR	On the LTE side: - Set the NsaDcQciParamGroup.UlDataSplitThreshhold parameter to INFINITY. - Set the NsaDcQciParamGroup.UlDataSplitPrimaryPath parameter to SCG.

Data Split Architecture	CP	UP	Parameter Setting
Option 3x	LTE	NR	On the NR side: - Set the gNBPDcpParamGroup.UIDataSplitThreshold parameter to INFINITY. - Set the gNBPDcpParamGroup.UIDataSplitPrimaryPath parameter to SCG.

4.1.2.5 UL/DL Separation

In NSA DC, when NR uplink coverage is limited but LTE uplink coverage is not limited, uplink and downlink user-plane services are carried respectively on the LTE side and NR side. LTE uplink coverage is used to compensate for the insufficient NR uplink coverage.

UL/DL separation can be implemented by setting the parameters listed in the following table.

Data Split Architecture	Uplink MCG Transmission	Downlink SCG Transmission
Option 3	On the LTE side: - Set the NsaDcQciParamGroup.UIDataSplitThreshold parameter to INFINITY. - Set the NsaDcQciParamGroup.UIDataSplitPrimaryPath parameter to MCG.	On the LTE side: - Set the CellQciPara.NsaDcDefaultBearerMode parameter to MCG_SPLIT_BEARER. - Set the NsaDcQciParamGroup.DIDataPdcpsplitMode parameter to SCG_ONLY.
Option 3x	On the NR side: - Set the gNBPDcpParamGroup.UIDataSplitThreshold parameter to INFINITY. - Set the gNBPDcpParamGroup.UIDataSplitPrimaryPath parameter to MCG.	On the LTE side: - Set the CellQciPara.NsaDcDefaultBearerMode parameter to SCG_SPLIT_BEARER. On the NR side: - Set the gNBPDcpParamGroup.DIDataPdcpsplitMode parameter to SCG_ONLY.

4.1.2.6 NSA UE Data Split Based on CCE Usage

A low-frequency LTE cell has a large coverage area and can easily admit a large number of UEs at the cell edge. If NSA UEs can only camp on this cell for NSA DC,

the cell is likely to be congested. In this case, the user experience of non-NSA non-CA UEs in the cell deteriorates.

This function controls downlink data split of NSA DC and CA based on the CCE usage of the PCell. It alleviates cell congestion by degrading the user experience of NSA DC and CA UEs and improves the user experience of non-NSA non-CA UEs in the cell. It is recommended that this function be enabled when the downlink data split mode is set to **SCG_ONLY** or **SCG_AND_MCG**.

This function takes effect only in LTE FDD cells with a bandwidth less than or equal to 5 MHz. It is controlled by the **CellDlschAlgo.DataSplitCceUsageHighThld** parameter. [Table 4-4](#) lists the configured values and effective values of CCE usage threshold parameters for data split.

Table 4-4 Configured values and effective values of CCE usage threshold parameters for data split

Parameter	Configured Value	Effective Value
CellDlschAlgo.DataSplitCceUsageHighThld (an upper threshold of CCE usage for data split)	0–100	Equal to the configured value
CellDlschAlgo.DataSplitCceUsageLowThld (a lower threshold of CCE usage for data split)	0–100	• The effective value of this parameter is 0 if the configured value of this parameter is greater than or equal to the effective value of CellDlschAlgo.DataSplitCceUsageHighThld. • The effective value of this parameter is equal to the configured value of this parameter if the configured value of this parameter is less than the effective value of CellDlschAlgo.DataSplitCceUsageHighThld.

The eNodeB determines whether the PCell of NSA DC is congested based on the measured and filtered CCE usage of the PCell.

The CCE usage of the PCell of an NSA UE is calculated as follows:

$$\text{CCE usage} = \max((\text{L.ChMeas.CCE.ULUsed.Equivalent} + \text{L.ChMeas.CCE.DLUsed.Equivalent}) / \text{L.ChMeas.CCE.AvailPower.Equivalent} \times 100\%, (\text{L.ChMeas.CCE.CommUsed} + \text{L.ChMeas.CCE.ULUsed} + \text{L.ChMeas.CCE.DLUsed}) / \text{L.ChMeas.CCE.Avail} \times 100\%)$$

The calculated CCE usage is then filtered and compared with the values of **CellDlschAlgo.DataSplitCceUsageHighThld** and **CellDlschAlgo.DataSplitCceUsageLowThld** to determine whether the PCell is congested.

- If the CCE usage of the PCell of NSA DC is greater than the effective value of **CellDlschAlgo.DataSplitCceUsageHighThld**, the PCell is congested.
- If the CCE usage of the PCell of NSA DC is less than or equal to the effective value of **CellDlschAlgo.DataSplitCceUsageLowThld**, the PCell is not congested.
- If the CCE usage of the PCell of NSA DC is less than or equal to the effective value of **CellDlschAlgo.DataSplitCceUsageHighThld** and greater than the effective value of **CellDlschAlgo.DataSplitCceUsageLowThld**, the congestion status of the PCell is the same as that in the last second.

Table 4-5 describes the handling mechanisms based on the congestion status of the PCell.

Table 4-5 Handling mechanisms based on the congestion status of the PCell

Scenario	PCell Congested	PCell Not Congested
A UE has only one carrier on the LTE side.	• When the CCE_USAGE_BASED_SPLIT_OPTION option of the CellAlgoExtSwitch.DlschEnhSwitch parameter is selected: – In Option 3, downlink data of the UE is distributed only to the NR side through the PDCP layer of the eNodeB. – In Option 3x, downlink data of the UE is distributed only to the NR side through the PDCP layer of the gNodeB. • When the CCE_USAGE_BASED_SPLIT_OPTION option of the CellAlgoExtSwitch.DlschEnhSwitch parameter is deselected, downlink data is dynamically distributed for the UE.	Downlink data of the UE is distributed as usual. In Option 3x, however, if the data split mode is not set to MCG_ONLY and backpressure of NSA data split based on transport load is triggered, downlink data of the UE is transmitted only on the NR side.

Scenario	PCell Congested	PCell Not Congested
A UE has multiple carriers on the LTE side.	Downlink data of the UE on the LTE side is transmitted only on LTE SCCs.	Downlink data of the UE on the LTE side is transmitted as usual. In Option 3x, however, if the data split mode is not set to MCG_ONLY and backpressure of NSA data split based on transport load is triggered, downlink data of the UE is transmitted only on the NR side.

If the downlink data of an NSA UE cannot be transmitted in the PCell:

- The user experience of this NSA UE deteriorates. If the number of NSA UEs in the cell is very large, the **Cell Downlink Average Throughput** decreases.
- If the LTE SCC or NR SCG coverage or the air interface quality deteriorates, the user experience of this NSA UE deteriorates significantly.
- The downlink retransmission rate and downlink scheduling delay increase for the NSA UE but decrease for non-NSA UEs. Therefore, the downlink retransmission rate and downlink packet delay of the cell change.
- More resources in the PCell are available to non-NSA non-CA UEs. If the proportion of small-packet services is high, the number of scheduling times in the PCell increases and accordingly the CCE usage in the PCell increases.

NOTE

- After NSA UE data split based on CCE usage takes effect, the **L.CA.Traffic.bits.DL.PCell** counter value of the PCell may decrease if some NSA UEs in the PCell have multiple carriers on the LTE side.
- Assume that the **CCE_USAGE_BASED_SPLIT_OPT_SW** option of the **CellAlgoExtSwitch.DLSchEnhSwitch** parameter is selected, and an NSA UE has only one carrier on the LTE side and its downlink data is distributed only to the NR side. Then, data-volume-based LTE SCC addition and NSA PCC anchoring cannot be triggered as the data volume is insufficient.

4.1.2.7 SPID-based NSA Uplink Primary Path Configuration

This function allows operators to configure uplink primary path selection policies and uplink data split thresholds for different types of NSA UEs based on SPIDs, thereby providing differentiated experience for these UEs. This function takes effect only for non-GBR bearers. It does not affect the configuration of uplink primary paths and uplink data split thresholds for GBR bearers.

When an NSA UE with a specific SPID accesses or is handed over to the network, the uplink primary path and uplink data split threshold for the NSA UE are specified by the parameters listed in [Table 4-6](#).

Table 4-6 SPID-based NSA uplink primary path configuration

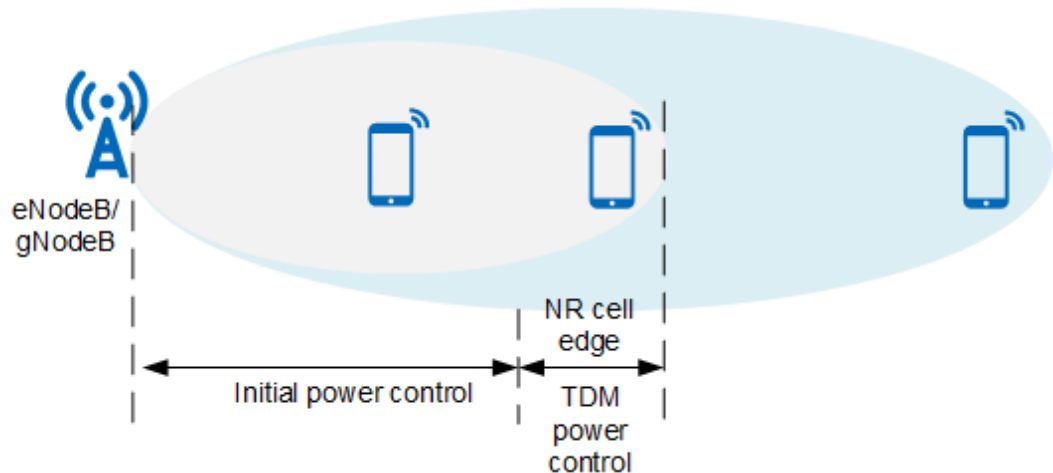
Parameter Name	Parameter Setting
Uplink Primary Path Selection Mode (gNB RfspConfig. <i>UlPrimaryPathSelMode</i>)	• MCG: Data is preferentially distributed to the MCG. • SCG: Data is preferentially distributed to the SCG. • DEFAULT: The SPID-based NSA uplink primary path configuration function does not take effect. The QCI-specific configuration takes effect by default.
Uplink Split Threshold (gNB RfspConfig. <i>UlDataSplitThreshold</i>)	• This parameter does not take effect when the gNBRfspConfig.<i>UlPrimaryPathSelMode</i> parameter is set to DEFAULT. • When this parameter is set to INFINITY: – If the gNBRfspConfig.<i>UlPrimaryPathSelMode</i> parameter is set to MCG, uplink data of the NSA UE is transmitted only on the MCG side. – If the gNBRfspConfig.<i>UlPrimaryPathSelMode</i> parameter is set to SCG, uplink data of the NSA UE is transmitted only on the SCG side. • When this parameter is set to a value other than INFINITY and the UE PDCP buffer data size is greater than or equal to the threshold specified by this parameter, uplink data is distributed dynamically.

4.1.3 Uplink Power Control in NSA DC

The eNodeB and the gNodeB can control the uplink transmit power of NSA UEs in the NSA DC state through parameter configuration.

Initial power control is performed for NSA UEs that initially access the network, and time division multiplexing (TDM) power control is performed to improve uplink coverage when trigger conditions are met, as shown in [Figure 4-8](#). TDM power control is an enhancement to uplink power control in NSA DC. For details, see *EPC-based NSA Performance Enhancement*.

Figure 4-8 Uplink power control in NSA DC



The maximum transmit power for initial power control is configured as follows:

- If the maximum uplink transmit power supported by an NSA UE that initially accesses the network is 23 dBm, the following parameters need to be set on the LTE side to specify the maximum transmit power for the LTE and NR sides of the UE in NSA DC:
 - **NsaDcMgmtConfig.NsaDcUeMcgUlMaximumPower**, indicating the maximum uplink transmit power on the MCG side
 - **NsaDcMgmtConfig.NsaDcUeScgUlMaximumPower**, indicating the maximum uplink transmit power on the SCG side
- It is recommended that the sum of the maximum uplink transmit power on the MCG side and that on the SCG side do not exceed 23 dBm.
- If the maximum uplink transmit power supported by an NSA UE that initially accesses the network is 26 dBm, the maximum transmit power values configured for the LTE and NR sides of the UE are both 23 dBm.

NOTE

After an NSA UE accesses the network, a maximum transmit power is configured for the UE on the NR side. The total power of uplink channels at the same time cannot exceed the maximum transmit power. For details about power control on each channel, see *Power Control* in 5G RAN feature documentation.

4.1.4 Interference Avoidance in NSA DC

In NSA DC scenarios, if secondary intermodulation signals generated during uplink concurrence of LTE and NR on the UE side fall in the downlink frequency band of LTE, the UE will experience interference when receiving downlink signals of LTE and the UE receiver sensitivity will decrease by a maximum of 28.6 dB. If secondary harmonic signals generated during uplink transmission of LTE on the UE side fall in the downlink frequency band of NR, the UE will experience interference when receiving downlink signals of NR and the UE receiver sensitivity will decrease by a maximum of 23.9 dB. **Interference avoidance in NSA DC is implemented based on interference type identification. It uses certain interference avoidance policies for secondary intermodulation interference and secondary harmonic interference to reduce the impact of interference.**

4.1.4.1 Interference Type Identification

Table 4-7 describes the parameters required for determining the interference type.

Table 4-7 Parameters required for determining the interference type

Parameter	Description	Calculation Method
LTE_UL_low	Minimum value of the LTE uplink frequency range	LTE uplink center frequency – LTE bandwidth/2
LTE_UL_high	Maximum value of the LTE uplink frequency range	LTE uplink center frequency + LTE bandwidth/2
LTE_DL_low	Minimum value of the LTE downlink frequency range	LTE downlink center frequency – LTE bandwidth/2
LTE_DL_high	Maximum value of the LTE downlink frequency range	LTE downlink center frequency + LTE bandwidth/2
NR_UL_low	Minimum value of the NR uplink frequency range	NR uplink center frequency – NR bandwidth/2
NR_UL_high	Maximum value of the NR uplink frequency range	NR uplink center frequency + NR bandwidth/2
NR_DL_low	Minimum value of the NR downlink frequency range	NR downlink center frequency – NR bandwidth/2
NR_DL_high	Maximum value of the NR downlink frequency range	NR downlink center frequency + NR bandwidth/2

Method for determining whether secondary intermodulation interference exists

- Condition 1: $\text{LTE_DL_low} > \text{NR_UL_high} - \text{LTE_UL_low}$
- Condition 2: $\text{LTE_DL_high} < \text{NR_UL_low} - \text{LTE_UL_high}$

If neither condition is met, secondary intermodulation interference exists.

Method for determining whether secondary harmonic interference exists

- Condition 1: $\text{NR_DL_low} > \text{LTE_UL_high} \times 2$
- Condition 2: $\text{NR_DL_high} < \text{LTE_UL_low} \times 2$

If neither condition is met, secondary harmonic interference exists.

4.1.4.2 Interference Avoidance Policies

Interference avoidance policies in this version are implemented by using time division scheduling on the base station side in the time and frequency domains.

Secondary Intermodulation Interference Avoidance

- Time-domain avoidance

In secondary intermodulation interference avoidance, the LTE side avoids NR uplink control channels and data channels based on TDM.

As shown in [Figure 4-9](#), the LTE FDD+NR TDD scenario is used as an example. Assume that the slot configuration of NR TDD is DDDSU and the scheduling unit is 0.5 ms. When LTE uplink subframes conflict with NR uplink subframes, the LTE side does not perform uplink transmission. The time-domain avoidance solution applies to scenarios with both LTE FDD and NR TDD.

Figure 4-9 TDM for intermodulation interference avoidance

	Scheduling Unit	Subframe Number													
		0	1	2	3	4	5	6	7	8	9				
LTE 1.8G UL	1 ms	U	U		U		U	U		U					
NR 3.5G UL	0.5 ms	D	D	D	S	U	D	D	S	U	D	D	D	S	U

NOTE

Intermodulation interference avoidance requires time synchronization between the LTE and NR cells in NSA networking. The TDM function triggered by secondary intermodulation interference uses the same TDM pattern selection policy as TDM power control.

- Frequency-domain avoidance
If only some RBs in the LTE downlink frequency band experience secondary intermodulation interference, the eNodeB preferentially allocates the RBs that are not interfered by secondary intermodulation signals to UEs in the cell, reducing the interference impact.

Secondary Harmonic Interference Avoidance

The eNodeB preferentially schedules the uplink RBs that do not cause interference to NR downlink data channels and does not schedule the uplink RBs that cause interference to NR downlink common channels (PBCH/common PDCCH).

4.1.4.3 Implementation of Interference Avoidance

In NSA DC, the NR side determines the interference type based on the frequency ranges of the PCell and PSCell. Based on the interference type, the NR side sends an LTE-NR joint scheduling pattern to the LTE side for interference avoidance.

Secondary Intermodulation Interference Avoidance

- Time-domain avoidance
 - On the NR side, if the **CROSS_MDLT_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch.NsaDcAlgoSwitch** parameter is selected, the

gNodeB determines the interference type as described in [4.1.4.1 Interference Type Identification](#). If the interference type is secondary intermodulation interference, the NR side sends an LTE-NR joint scheduling time sequence to the eNodeB through a pattern. This mode requires that LTE FDD and NR TDD meet the time synchronization requirements and relative frame offset constraints. When the **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter is selected, the LTE side can send its frame offset to the NR side, and the NR side calculates the relative frame offset.

- On the LTE side, if the **TDM_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter is selected, the eNodeB performs uplink and downlink scheduling based on the pattern received from the gNodeB. When the **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter is selected, the relative frame offset between LTE and NR can be automatically obtained.

When secondary intermodulation interference avoidance in the time domain is enabled, it is recommended that the **TDM_CHG_WITH_INTRA_CELL_HO_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter be selected. Under this setting, an intra-MeNB-cell handover is triggered when there is a TDM status change, ensuring that the TDM status of the MeNB is consistent with that of the NSA UE. TDM status changes include TDM activation, TDM deactivation, and TDM pattern change.

- Frequency-domain avoidance

If the **INTERFERENCE_AVOID_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter is selected and the **NsaDcMgmtConfig.lmdIntrfFreqAvoidanceType** parameter is set to **PRIOR_ALLOC_NO_INTRF_RB**, the eNodeB preferentially schedules the downlink RBs that are not interfered by secondary intermodulation signals.

NOTE

- In scenarios with small-packet services or a small number of UEs, the eNodeB preferentially schedules the downlink RBs that are not interfered by secondary intermodulation signals. Therefore, interference with the downlink of LTE can be avoided.
- In scenarios with large-packet services or a large number of UEs, it is likely that UEs occupy all RBs. Therefore, the interference avoidance effect is not obvious.
- The gain of interference avoidance depends on the anti-interference capability of the terminal. The better the anti-interference capability of the terminal, the lower the gain.
- When nonstandard-bandwidth-related features (such as compact bandwidth, flexible bandwidth, manual RB blocking, Horizontal SuperBAND, and spectrum sharing) are enabled on the LTE side, the interference avoidance effect is not obvious in multi-user scenarios.

Secondary Harmonic Interference Avoidance

- Avoiding NR common channels
 - On the NR side, if the **HARMONIC_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch.NsaDcAlgoSwitch** parameter is selected and the **NRDUCellNsaDcConfig.HarmonicIntrfAvoidRange** parameter is set to

NR_DL_COMMON_CHANNEL, the interference type is determined as described in [4.1.4.1 Interference Type Identification](#). If the interference type is secondary harmonic interference, the gNodeB sends the time-frequency positions of NR downlink common channels to the eNodeB through a pattern.

- On the LTE side, if the **INTERFERENCE_AVOID_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter is selected and the **NsaDcMgmtConfig.HarmonicIntrfAvoidRange** parameter is set to **NR_DL_COMMON_CHANNEL**, the eNodeB does not schedule the RBs indicated by the pattern received from the gNodeB.
- Avoiding both NR common channels and data channels
 - On the NR side, if the **HARMONIC_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch.NsaDcAlgoSwitch** parameter is selected and the **NRDUCellNsaDcConfig.HarmonicIntrfAvoidRange** parameter is set to **NR_DL_COMMON_AND_DATA_CHANNEL**, the gNodeB sends the time-frequency positions of NR downlink common channels to the eNodeB through a pattern and preferentially schedules the downlink RBs that are not interfered by LTE secondary harmonic signals.
 - On the LTE side, if the **INTERFERENCE_AVOID_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter is selected and the **NsaDcMgmtConfig.HarmonicIntrfAvoidRange** parameter is set to **NR_DL_COMMON_AND_DATA_CHANNEL**, the eNodeB does not schedule the RBs indicated by the pattern received from the gNodeB and preferentially schedules the uplink RBs that do not cause secondary harmonic interference to the downlink of NR.

NOTE

- In scenarios with small-packet services or a small number of UEs, the gNodeB preferentially schedules the downlink RBs that are not interfered by secondary harmonic signals, and the eNodeB preferentially schedules the uplink RBs that do not cause secondary harmonic interference to the downlink of NR. Therefore, interference with the downlink of NR can be avoided.
- In scenarios with large-packet services or a large number of UEs, it is likely that UEs occupy all RBs. Therefore, the interference avoidance effect is not obvious.

4.1.5 NSA DC Virtual Grid Models

Basic Principles

UEs with the same radio signal characteristics can be classified into a category based on multi-dimensional measurements. The eNodeB considers the UEs with the same RSRP measurement result on a frequency to be in the same virtual grid. For example, if UE 1's measurement result on a frequency is [(Cell 1, RSRP 1), (Cell 2, RSRP 2), ...] and UE 2's measurement result is the same, the eNodeB considers the two UEs to be in the same virtual grid. The eNodeB uses machine learning technology with virtual grids as features to construct a signal characteristics mapping from all virtual grids in a cell to a frequency. This mapping is contained in an entity called virtual grid model.

The eNodeB supports the building of LNR virtual grid models (only RSRP prediction models) in NSA DC. After obtaining the virtual grid information of a UE in an LTE cell, the eNodeB can query the RSRP prediction model for a

specific neighboring NR frequency to quickly predict the RSRP of the UE on this frequency.

Functions related to virtual grid model building and application are controlled by the following options of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter:

- **LNR_VIRTUAL_GRID_SW**: If this option is selected, LNR virtual grid model building is enabled.
- **NSA_ANCHOR_VIRTUAL_GRID_SW**: If this option is selected, NSA LTE anchor virtual grid model building is enabled.

For details about LTE virtual grid models, see *Multi-carrier Unified Scheduling* in eRAN feature documentation.

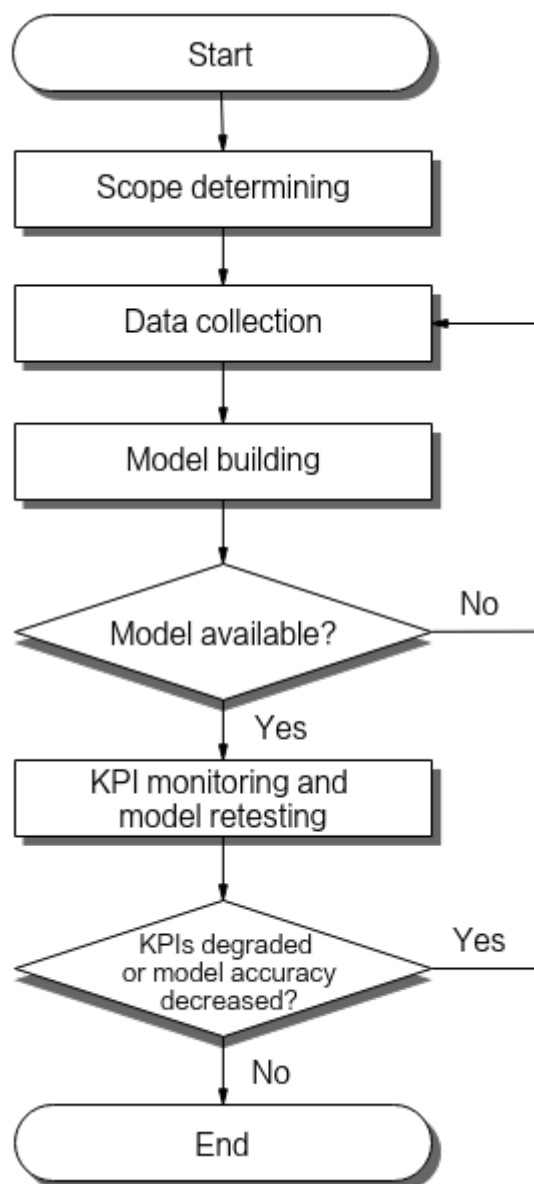
Functions related to virtual grid model building require a large number of main control board resources. Different switches need to be turned on for different main control boards:

- For the UMPTe/UMPTga/UMPTga, it is recommended that the **AI_ENHANCEMENT_OPT_SWITCH** option of the **eNodeBResModeAlgo.ServiceMode** parameter be selected.
- For the UMPTg, it is recommended that the **UMPTG_ENHANCEMENT_SWITCH** option of the **eNodeBResModeAlgo.ServiceMode** parameter be selected.

Virtual Grid Model Building and Update

Figure 4-10 illustrates how virtual grid models are built and updated in NSA scenarios.

Figure 4-10 Process for building and updating virtual grid models



The building process is as follows:

1. The eNodeB determines the scope of LNR RSRP prediction models and uses the format of *[LTE cell, Neighboring NR frequency]* to indicate the scope.
The following cells will not be selected for virtual grid model building:
 - Cells with multiple physical cells each with a different ID, for example, a single frequency network (SFN) cell or a cell with the **Cell.MultiRruCellFlag** parameter set to **BOOLEAN_TRUE**
 - High-speed-mobility cells, for example, cells with high speed mobility enabled and cells with ultra high speed mobility enabled
 - Cells in which UEs are stationary and therefore handovers are not involved, such as eMTC-only cells
 - Cells that can only serve as SCells, where UE admission is not supported and therefore samples cannot be collected, for example, cells with the

Cell.WorkMode parameter set to **DL_ONLY** or **LAA** and cells with the **ForbidCellSiSwitch** option of the **CellSiMap.SiSwitch** parameter selected

- Cells with a cell radius greater than 100 km
- FDD NB-IoT cells
- Cells in a group with the **PATTERN_ADJ_SWITCH** option of the **SectorSplitGroup.SectorSplitSwitch** parameter selected
- Cells having the same frequency as cells in a group with the **BEAM_NUM_ADJ_SWITCH** option of the **SectorSplitGroup.SectorSplitSwitch** parameter selected

The following neighboring NR frequencies will not be selected for virtual grid model building:

- Frequencies for which the **VG_MODEL_FORBID_BUILD_FLAG** option of the **NrNFreq.AggregationAttribute** parameter is selected
- Frequencies for which the **FREQ_MEAS_FLAG** option of the **NrNFreq.AggregationAttribute** parameter is deselected

If the **VG_MDL_PRED_NCELL_ENH_SW** option of the **MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw** parameter is selected and virtual grid model building is triggered, the following neighboring NR cells reported by UEs during data collection are identified as unknown neighboring cells and included in model training:

- Neighboring NR cells with physical cell identifier (PCI) conflicts
- Neighboring NR cells not in the table of neighbor relationships with NR

If the number of virtual grid models to be built exceeds the specifications, these models are prioritized based on the following rules, and then higher-priority models are selected for building as long as the specifications allow.

- A larger cell bandwidth indicates a higher model building priority.
- A higher NSA anchoring priority indicates a higher model building priority.
- A higher SCG frequency priority indicates a higher model building priority.

NSA LTE anchor virtual grid model building can also be enabled for cells with LNR virtual grid model building enabled if the

NSA_ANCHOR_VIRTUAL_GRID_SW option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected. The building process is the same as that for LTE virtual grid models. For details, see *Multi-carrier Unified Scheduling* in eRAN feature documentation.

NOTE

LTE and LNR virtual grid models share the same model specifications and are built based on the feature activation sequence.

The virtual grid model specifications are as follows:

If an eNodeB serves S cells and each cell has N neighboring NR frequencies, a total of $S \times N$ LNR RSRP prediction models are required.

- The eNodeB equipped with a UMPTga or UMPTHa supports 90 LNR RSRP prediction models if the **AI_ENHANCEMENT_SWITCH** or **AI_ENHANCEMENT_OPT_SWITCH** option of the **eNodeBResModeAlgo.ServiceMode** parameter is selected or 36 such models if the options are deselected.

- When equipped with a UMPTe:
 - The eNodeB supports 90 LNR RSRP prediction models if the **AI_ENHANCEMENT_SWITCH** option of the **eNodeBResModeAlgo.ServiceMode** parameter is selected.
 - The eNodeB supports 18 LNR RSRP prediction models if the **AI_ENHANCEMENT_OPT_SWITCH** option of the **eNodeBResModeAlgo.ServiceMode** parameter is selected in the case of LN or GULN multimode scenarios.
 - If both of the preceding options are deselected, the eNodeB supports 36 LNR RSRP prediction models.
 - The eNodeB equipped with a UMPTg supports 168 LNR RSRP prediction models if the **UMPTG_ENHANCEMENT_SWITCH** option of the **eNodeBResModeAlgo.ServiceMode** parameter is selected or 90 such models if the option is deselected.
2. The base station randomly selects 15 UEs (supporting EN-DC or SA) every 40s in each cell for which models are to be built, and periodically collects their 24-hour measurement reports about each frequency as the sample data of RSRP prediction models.

LTE measurement configurations are generally delivered by the eNodeB to a UE through an RRC Connection Reconfiguration message. The UE performs measurement based on the measurement configurations and reports measurement results to the eNodeB through a measurement report. Measurements are triggered periodically or based on events. By default, the eNodeB periodically triggers measurements. If the **VG_MDL_NCELL_MR_OPT_SW** option of the **MultiCarrUnifiedSch.MultiCarrierUnifiedSchEnSw** parameter is selected, inter-RAT neighboring cell measurements are triggered by event B1 during model training. Event B1 indicates that the signal quality of an inter-RAT neighboring cell becomes higher than a specific threshold, which is fixed at – 156 dBm.

When event B1 is used for measurements, a UE not covered by a neighboring frequency does not report inter-frequency measurement results. If the **SAM_EXPAND_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected, samples without such measurement results are included in virtual grid model training. If this option is deselected, these samples are excluded from the training.

Samples without intra-frequency neighboring cell measurement results impair virtual grid model training. If the **NO_INTRA_FREQ_SAM_DEL_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected, samples without such measurement results are excluded from virtual grid model training. If this option is deselected, these samples are included in the training.

3. After completing data collection, the base station starts model training and calculates the accuracy of each model.

If the **SAM_BALANCING_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected, the learning weight is adjusted for the samples with good RSRP prediction results and those with poor results. Specifically, samples in the majority are given a lower weight while those in the minority a higher weight. This improves the

accuracy of RSRP prediction models. If this option is deselected, the sample weights are not adjusted during model training.

4. The base station evaluates whether each model can be used based on their model accuracy. If the accuracy of an RSRP prediction model is greater than or equal to 80%, the model has been built successfully and can be put into service. Otherwise, the model has failed to be built and the building process goes back to 2.

Model building stops when either of the conditions listed below is met. Data can be collected again for model building only after a penalty period of 28 days.

- Model building fails seven consecutive times.
 - The time since the initial data collection exceeds the maximum allowed duration for building an LNR virtual grid model specified by the **LnrCarrierSelection.VgMdlMaxBuildDuration** parameter if this parameter is set to a value other than 0. (Under such parameter setting, penalty optimization for LNR virtual grid model building takes effect.)
5. After completing the model building based on 24-hour data, the base station randomly selects a number of UEs every 40s in each cell for which models are to be built, collects 7x24-hour data, and builds new models. If a new model is more accurate than the old one, the new one is put into service. The number of UEs to be selected is specified by the **LnrCarrierSelection.SampleUeNum** parameter.

When the **LNR_VIRTUAL_GRID_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is deselected, the eNodeB stops the building and use of virtual grid models, and stores the models that have been successfully built. These models can be stored for a maximum of seven days if the eNodeB is not reset. They can be used again if this option is selected within the seven days. They are no longer stored if the eNodeB is reset within the seven days.

The process for monitoring the performance of virtual grid models and updating models is as follows:

1. After a model is put into service, the eNodeB starts the model monitoring and updating process. It performs 2 and 4 in parallel.
2. The eNodeB monitors the success rate of PSCell additions based on virtual grid models in real time and goes to 3.
3. If the success rate of PSCell additions based on a virtual grid model is less than the value of **LnrCarrierSelection.VgMdlKpiMonitorThld** throughout two hours, the eNodeB suspends the use of this model, goes to 6 to rebuild a model, and uses the new one after the rebuilding succeeds.
4. The eNodeB calculates the model accuracy every two days and goes to 5.
5. If the accuracy of a model is lower than expected, the eNodeB considers that this model is aged. It suspends the use of this model, goes to 6 to rebuild a model, and uses the new one after the rebuilding succeeds.
6. The eNodeB randomly selects a number of UEs every 40s in each cell for which a model is to be built, collects 7x24-hour data, and builds a new model. The number of UEs to be selected is specified by the **LnrCarrierSelection.SampleUeNum** parameter.

Running the **RST APP** command will delete the successfully built models if one of the following options of the **eNodeBResModeAlgo.ServiceMode** parameter is deselected: (1) **AI_ENHANCEMENT_SWITCH** for the UMPTe/UMPTga/UMPTga; (2) **AI_ENHANCEMENT_OPT_SWITCH** for the UMPTe/UMPTga/UMPTga; (3) **UMPTG_ENHANCEMENT_SWITCH** for the UMPTg.

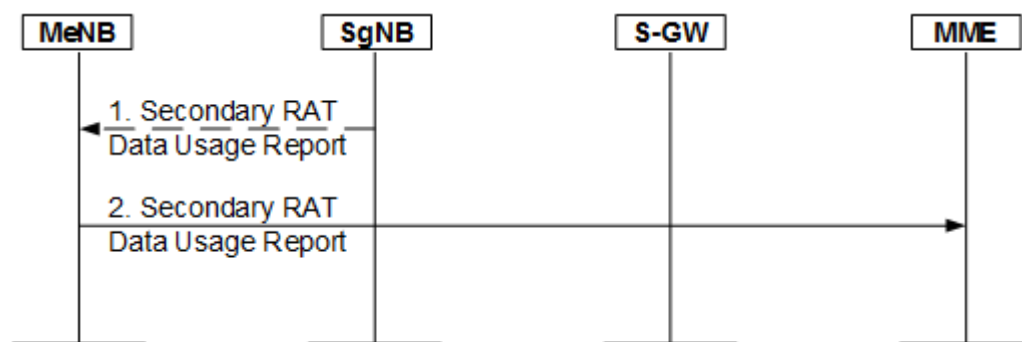
If the eNodeB is reset after the LNR virtual grid model is successfully built, the eNodeB will use the pre-reset model and then rebuilds the model. If the grid model fails to be built due to insufficient grid samples collected after the reset, the eNodeB will no longer use the pre-reset model when the **LnrCarrierSelection.VgMdlMaxBuildDuration** parameter is not set to **0** and the eNodeB fails to build the LNR grid model two consecutive times after the reset. Otherwise, the eNodeB will continue to use the pre-reset model.

4.1.6 Other Functions in NSA DC

4.1.6.1 NR Data Volume Reporting

According to 3GPP TS 37.340, the RAN reports NR data volumes to the core network for independent charging of 5G RAN data volumes in NSA DC scenarios, as shown in [Figure 4-11](#).

Figure 4-11 NR data volume reporting from the RAN side



Uplink and downlink NR data volumes are reported on a per bearer basis. The base station measures the number of bytes in SDU packets to calculate the air-interface data volumes distributed to the NR side. It then reports the volumes through the Secondary RAT Usage Report List IE in a Secondary RAT Data Usage Report message or another message, as shown in [Figure 4-11](#).

- In Option 3, the data split anchor is on the MeNB. Therefore, the MeNB directly measures the data volumes distributed to the NR side and does not require data volume reporting by the SgNB over the X2 interface. Then, the MeNB directly reports the volumes to the MME on the core network.
- In Option 3x, the SgNB measures the air interface data volume distributed to the NR side and notifies the MeNB through an X2 interface message. If the LTE parameter **NsaDcMgmtConfig.NrDataVolumeRptCfg** is not set to 0, the LTE side will report the NR data volumes notified by the NR side to the core network.

Both the SgNB and MeNB support event-triggered NR data volume reporting and periodic NR data volume reporting.

 NOTE

The internal timer of the eNodeB or gNodeB has a deviation of one period, and therefore the time for the SgNB or MeNB to report the NR data volume possibly has a deviation.

- In Option 3, the **NsaDcMgmtConfig.NrDataVolumeRptCfg** parameter on the LTE side specifies whether NR data volumes are reported to the core network based on events or at intervals.
 - Event-triggered NR data volume reporting
If this parameter is set to **65535**, the eNodeB reports NR data volumes only based on events. After MCG split bearer setup, the eNodeB starts timing and measures NR data volumes. Upon a bearer change or release, the eNodeB records the end time, calculates the total data volumes distributed to the NR side, and reports the uplink and downlink NR data volumes to the core network through the Secondary RAT Usage Report List IE in a Secondary RAT Data Usage Report, Context Release Request, Context Release Complete, E-RAB Modification Indication, E-RAB Release Indication, or E-RAB Release Response message.
 - Periodic NR data volume reporting
If this parameter is set to a value within the range of 60 to 36000, the eNodeB reports NR data volumes not only based on events but also based on the period specified by this parameter. That is, the eNodeB periodically reports NR data volumes to the core network through the Secondary RAT Data Usage Report message after the timing starts.
- In Option 3x, the **NRCellNsaDcConfig.NrDataVolumeRptCfg** parameter on the NR side specifies whether NR data volumes are reported to the eNodeB based on events or at intervals.
 - Event-triggered NR data volume reporting
If this parameter is set to **65535**, the gNodeB reports NR data volumes based on events. After SCG split bearer setup, the gNodeB starts timing and measures NR data volumes. Upon a bearer change (due to inter-gNodeB handover) or SCG release (including SCG release triggered by inter-MeNB handover), the gNodeB records the end time, calculates the total data volumes distributed to the NR side, and reports the data volumes to the eNodeB through the Secondary RAT Data Usage Report message over the X2 interface. The eNodeB reports the uplink and downlink NR data volumes to the core network through the Secondary RAT Usage Report List IE in a Secondary RAT Data Usage Report, Context Release Request, Context Release Complete, E-RAB Modification Indication, E-RAB Release Indication, or E-RAB Release Response message.
 - Periodic NR data volume reporting
If this parameter is set to a value within the range of 60 to 36000, the gNodeB reports NR data volumes not only based on events but also based on the period specified by this parameter. That is, the gNodeB periodically reports NR data volumes to the eNodeB through the Secondary RAT Data Usage Report message after the timing starts. After receiving the message, the eNodeB forwards it to the MME.

In Option 3x, if the eNodeB is required to report NR data volumes to the MME, the **NsaDcMgmtConfig.NrDataVolumeRptCfg** parameter also needs to be set to a value other than 0 on the LTE side. This parameter specifies

whether NR data volumes are reported to the MME, but does not specify whether NR data volumes are reported based on events or at intervals.

 NOTE

In Option 3x, the NR data volume reporting functions on the LTE and NR sides must be both disabled or enabled. The following situations must be avoided:

- This function is enabled on the NR side but disabled on the LTE side. In this case, the gNodeB sends data volume information to the eNodeB, but the eNodeB cannot process the data volume information.
- This function is disabled on the NR side but enabled on the LTE side. In this case, when an inter-gNodeB handover is performed, the eNodeB starts a timer to wait for NR data volume reporting but the gNodeB does not report the information. However, in the 0.5s waiting period, the bearer path is not switched to the target SgNB, and therefore the UE has no data volume on the target SgNB side. After the timer expires, a bearer path change is triggered. NSA PCC anchoring cannot be triggered when an RRC connection is reestablished on the LTE side.

If the UE has multiple E-RABs that use different data split bearers (MCG split bearer or SCG split bearer), NR data volumes are reported on a per bearer basis according to the methods specified in Option 3 and Option 3x, respectively.

4.1.6.2 S-K_{gNB} Update

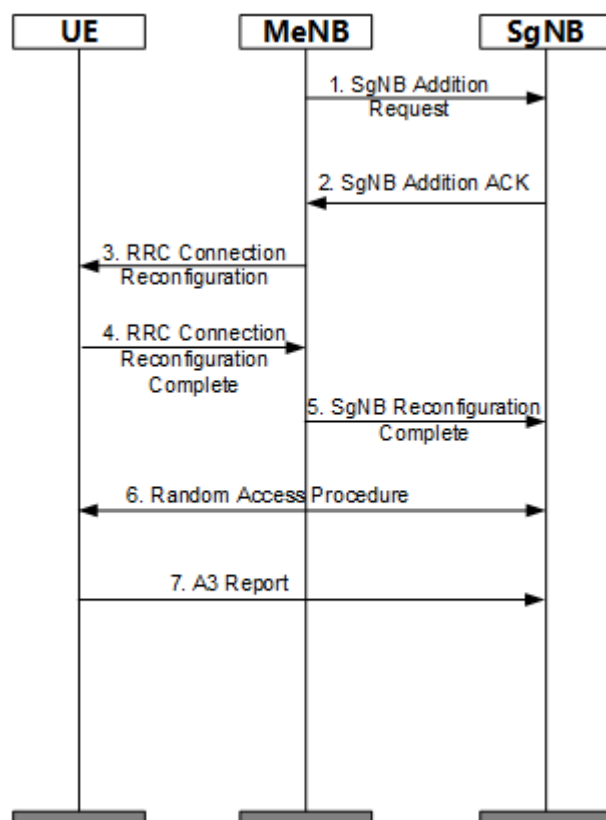
To improve security in data transmission in NSA DC, the security key S-K_{gNB} at the PDCP layer cannot remain unchanged. When detecting that the number of data packets transmitted by NSA DC reaches a specified threshold, the SgNB sends an S-K_{gNB} update request to the MeNB over the X2 interface. Considering that there may be UE compatibility issues, the SCG needs to be deleted first and an SCG is added after a period of time. The time does not exceed the value of **NsaDcMgmtConfig.ScgAdditionInterval**.

4.1.6.3 gNodeB and UE Signaling Exchange on SRB3

According to section 7.5 "SRB3" in 3GPP TS 37.340 V15.7.0, the gNodeB can directly exchange eNodeB-unrelated signaling (such as intra-NR measurement configuration or intra-NR resource configuration) with a UE on SRB3.

When the **NSA_DC_SRB3_SWITCH** option of the **gNodeBParam.NsaDcOptSwitch** parameter is selected, NSA DC SRB3 is supported. During an SgNB addition, the gNodeB determines whether the UE supports SRB3 (that is, whether the *UE-MRDC-Capability* IE reported by the UE contains SRB3 information). If the UE supports SRB3, the gNodeB sends SRB3 configurations to the UE. After the configurations are applied, the UE can send measurement reports directly to the gNodeB on SRB3, as shown in [Figure 4-12](#). The gNodeB can also directly send intra-NR resource configurations to the UE through an RRC Reconfiguration message. If the UE does not support SRB3, signaling cannot be directly exchanged between the gNodeB and the UE; instead, signaling needs to be forwarded by the eNodeB.

Figure 4-12 Signaling exchange on SRB3



During inter-SgNB PSCell change or SgNB release, the gNodeB directly deletes the SRB3 configurations.

NOTE

If some UEs on the live network are incompatible with SRB3, the probability of UE-triggered abnormal SgNB releases increases after NSA DC SRB3 is enabled. The NSA DC SRB3 blacklist function can be enabled for these UEs so that NSA DC SRB3 does not take effect for them. If the blacklist function is disabled, no differentiated handling is performed on these UEs. The blacklist function is controlled by the **NSA_DC_SRB3_SWITCH_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter.

4.1.6.4 Uplink Single-Side Transmission (Trial)

This is a trial feature and applies only to combinations formed by LTE FDD and NR FDD frequency bands. If this feature is enabled in other scenarios, SCGs cannot be added and the number of SCG addition attempts and the SCG addition success rate decrease in NSA networking.

NSA UEs can send data only on the LTE or NR side in some NSA DC frequency band combinations, which requires adaptation on the network side. This feature allocates uplink channel resources at the TTI level on the LTE and NR sides in TDM mode, and implements uplink single-side transmission on the LTE or NR side through TDM. This feature is similar to TDM power control described in *EPC-based NSA Performance Enhancement*. The difference is the scenarios where TDM patterns take effect. The eNodeB delivers a TDM pattern to the UE in the following scenarios:

- SgNB addition: The frequency band of the PSCell of the SgNB and the frequency band of the PCell of the MeNB must be an uplink single-side transmission frequency band combination.
- SgNB or MeNB change: The frequency band combination of the PCell and PSCell after the change must be an uplink single-side transmission frequency band combination.

This feature is controlled by the **TDM_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter on the LTE side and the **SINGLE_UL_NSA_SW** option of the **NRCellNsaDcConfig.NsaDcAlgoSwitch** parameter on the NR side. In addition, uplink single-side transmission frequency band combinations need to be configured on the two sides.

- On the LTE side, use the **NrScgFreqConfig.PccDLearfcn** and **NrScgFreqConfig.ScgDLArfcn** parameters to specify EN-DC frequency combinations that support only uplink single-side transmission (by converting bands to frequencies), and select the **SINGLE_UL_NSA_FLAG** option of the **NrScgFreqConfig.AggregationAttribute** parameter.
- On the NR side, use the **gNBDedNsaDcBandComb.LteBandId** and **gNBDedNsaDcBandComb.NrBandId** parameters to specify EN-DC band combinations that support only uplink single-side transmission, and select the **SINGLE_UL_NSA_FLAG** option of the **gNBDedNsaDcBandComb.AggregationAttribute** parameter.

When uplink single-side transmission is enabled, it is recommended that the **TDM_CHG_WITH_INTRA_CELL_HO_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter be selected. Under this setting, an intra-MeNB-cell handover is triggered when there is a TDM status change, ensuring that the TDM status of the MeNB is consistent with that of the NSA UE. TDM status changes include TDM activation, TDM deactivation, and TDM pattern change.

This feature requires that the radio frames and subframes of LTE and NR be completely aligned. The alignment can be achieved by setting frame offsets in the case of time synchronization.

- On the LTE side, use the **CellFrameOffset.FrameOffset** parameter to set a frame offset.
- On the NR side, use the **gNBFreqBandConfig.FrameOffset** parameter to set a frame offset.

NOTE

This feature requires time synchronization between the eNodeB and the gNodeB, with a relative frame offset of 0 ms.

This feature requires that the NR cell parameter **NRDUCellPrach.PrachConfigurationIndex** be set only to 1, 2, 5, 6, 9, 10, 13, 14, 17, 18, or 21.

With this feature, the network side selects a TDM pattern from [Figure 4-13](#) and [Figure 4-14](#), and sends it to the UE. Based on the pattern, the UE performs uplink single-side transmission in the time domain where LTE and NR subframes are staggered in each TTI.

Figure 4-13 TDM pattern 1

Harqoffset=7		Subframe number									
		0	1	2	3	4	5	6	7	8	9
TDM-Pattern:sa3	1ms	U	U	D	D	D	D	D	D	S	U
LTE FDD	1ms	U	U	⊗	⊗	⊗	⊗	⊗	⊗	⊗	U
NR FDD	1ms	⊗	⊗	U	U	U	U	U	U	U	⊗

Figure 4-14 TDM pattern 2

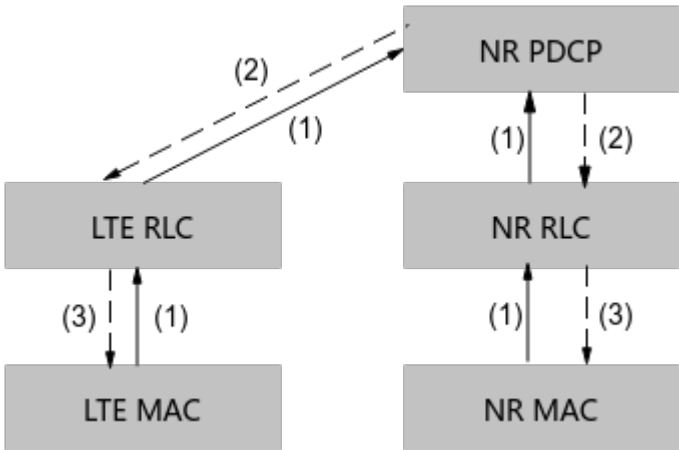
Harqoffset=9		Subframe number									
		0	1	2	3	4	5	6	7	8	9
TDM-Pattern:sa4	1ms	S	U	U	D	D	D	D	D	D	D
LTE FDD	1ms	⊗	U	U	⊗	⊗	⊗	⊗	⊗	⊗	⊗
NR FDD	1ms	U	⊗	⊗	U	U	U	U	U	U	U

4.1.6.5 Uplink Preallocation

In a single mode (LTE or NR), uplink preallocation is triggered by downlink packets. In NSA DC scenarios, if uplink and downlink data split modes are inconsistent, preallocation on the LTE or NR side cannot be triggered, increasing the end-to-end delay.

In NSA DC scenarios, this function increases the number of times that the base station proactively schedules UEs to reduce the duration of uplink data packet buffering on UEs, speed up the response to UE services, and improve user experience. The following uses Option 3x as an example to describe the NSA uplink preallocation procedure.

Figure 4-15 Uplink preallocation procedure in Option 3x



- The NSA uplink preallocation shown in [Figure 4-15](#) is described as follows:
- (1) The NR PCP determines whether to send uplink preallocation indications to the LTE and NR sides based on the uplink and downlink packets transmitted on the LTE and NR sides.
 - (2) The NR PCP instructs the LTE and NR sides to enable uplink preallocation.

(3) The LTE MAC and NR MAC start uplink preallocation after receiving the uplink preallocation instruction from the NR PDCP.

The related configuration is as follows:

- In Option 3, this function requires the selection of the **NSA_DC_PREALLOCATION_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter. This function also requires the enabling of uplink smart preallocation on the LTE side and uplink preallocation on the NR side.
- In Option 3x, this function requires the selection of both the **NSA_DC_PREALLOCATION_SW** option of the **NRCellNsaDcConfig.NsaDcAlgoSwitch** parameter and the **NSA_DC_PREALLOCATION_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter. This function also requires the enabling of uplink smart preallocation on the LTE side and uplink preallocation on the NR side.

 NOTE

- For sparse packets, the NSA uplink preallocation switch does not need to be turned on if the following conditions are met: (1) uplink and downlink data is transmitted only on the MCG side in Option 3 or only on the SCG side in Option 3x; (2) fallback to LTE and uplink data transmission path selection are not enabled.
- For common packets, the NSA uplink preallocation switch does not need to be turned on if the following conditions are met: (1) uplink and downlink data is transmitted only on the MCG or SCG side, or downlink data is dynamically distributed and uplink data is transmitted on the data split anchor side; (2) fallback to LTE and uplink data transmission path selection are not enabled.

For details about uplink smart preallocation on the LTE side, see *Uplink Scheduling* in eRAN feature documentation. For details about uplink preallocation on the NR side, see *Uplink Scheduling* in 5G RAN feature documentation.

Table 4-8 lists the settings of the UE count threshold for NSA preallocation on the LTE side and the NSA uplink preallocation switch and their results.

Table 4-8 Settings of the UE count threshold for NSA preallocation and the NSA uplink preallocation switch and their results

UE Count Threshold for NSA Preallocation (<i>NsaDcMgmtConfig.NsaDcUePreUserCntThld</i>)	NSA Uplink Preallocation Switch (NSA_DC_PREALLOCATION_SW option of <i>NsaDcMgmtConfig.NsaDcAlgoSwitch</i>)	Result
Not 0	On	NSA uplink preallocation takes effect. The number of UEs for which the function takes effect is limited by the <i>NsaDcMgmtConfig.NsaDcUePreUserCntThld</i> parameter setting.
0	On	NSA uplink preallocation takes effect.
Not 0	Off	NSA uplink preallocation does not take effect, but preallocation for NSA UEs on the LTE side takes effect. The number of NSA UEs for which preallocation takes effect is limited by the <i>NsaDcMgmtConfig.NsaDcUePreUserCntThld</i> parameter. For details, see <i>Uplink Scheduling</i> in eRAN feature documentation.
0	Off	Neither NSA uplink preallocation nor preallocation for NSA UEs on the LTE side takes effect.

4.1.6.6 SCG Release and Addition Based on UE Overheating Status Reporting

This function supports SCG release and addition based on overheating messages and overheating clearance messages reported by UEs to improve user experience in NSA scenarios.

This function is controlled by the **OVERHEATING_RPOTECTION_SW** option of the ***NsaDcAlgoParam.NsaDcAlgoSwitch*** parameter on the LTE side.

- After a UE detects an overheating state, it notifies the eNodeB by sending a UEAssistanceInformation message with the *OverheatingAssistance* IE containing the *reducedMaxCCs* field, which indicates the maximum number of

SCCs supported by the UE. Then, the eNodeB triggers an SCG release procedure.

- After the UE is no longer in the overheating state, it notifies the eNodeB by sending a UEAssistanceInformation message with the *OverheatingAssistance* IE not containing the *reducedMaxCCs* field. After receiving the overheating clearance indication, the eNodeB checks the interval between the overheating clearance reporting time and the overheating reporting time.
 - If the interval is greater than or equal to the **NsaDcMgmtConfig.ScgAdditionInterval** parameter value, an SCG addition procedure is triggered.
 - If the interval is less than the **NsaDcMgmtConfig.ScgAdditionInterval** parameter value, an SCG addition procedure is triggered only after the interval reaches the parameter value.

After reporting an overheating message, the UE can report another overheating message only after the penalty timer specified by the **RrcConnStateTimer.OverheatingIndProhibitTmr** parameter expires.

4.1.6.7 Intra-Band EN-DC

Whether an EN-DC band combination supports intra-band EN-DC can be configured, according to section 4.2.7.9 "MRDC-Parameters" in 3GPP TS 38.306 V16.1.0, as listed in [Table 4-9](#).

Table 4-9 Intra-band EN-DC capabilities

UE Capability Reporting ^a	Description
The UE reports an <i>intraBandENDC-Support</i> IE with a value of "non-contiguous".	Supports intra-band EN-DC with only non-contiguous spectrum.
The UE does not report the <i>intraBandENDC-Support</i> IE.	Supports intra-band EN-DC with only contiguous spectrum.
The UE reports the <i>intraBandENDC-Support</i> IE with a value of "both".	Supports intra-band EN-DC with both contiguous and non-contiguous spectrum.
NOTE a: The preceding UE capabilities are reported at EN-DC band combination level.	

When intra-band EN-DC is required, the eNodeB and the gNodeB determine whether intra-band LTE and NR frequencies conflict with each other based on NSA UE capabilities and base station configurations. The **LnrFreqRelationship** MO is used to specify whether an LTE frequency and an NR frequency are intra-band contiguous or intra-band non-contiguous. In addition, the **NSA_FREQ_CONFLICT_AVOID_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter on the NR side needs to be selected. If intra-band LTE and NR frequencies conflict with each other, the eNodeB and the gNodeB perform the following processing:

- eNodeB
 - The eNodeB does not deliver measurement configurations for an NR frequency that cannot coexist with the NSA anchor and does not add any NR cell on this frequency as a PSCell.
 - If the eNodeB detects that an LTE SCell cannot coexist with the added NR SCG, it will remove the LTE SCell and will not deliver measurement configurations for the frequency of this LTE SCell.
 - If the target LTE cell for an inter-frequency handover conflicts with the PSCell in the NR SCG, the eNodeB performs a handover with an SgNB change.
- gNodeB
 - The gNodeB does not add an NR cell that cannot coexist with the NSA anchor as an SCell.
 - The gNodeB does not select an NR cell that cannot coexist with the NSA anchor as the target cell for an inter-frequency handover.
 - If an NR SCell cannot coexist with the target cell for an LTE inter-frequency handover, the gNodeB removes this SCell.

 NOTE

Whether a UE supports inter-band EN-DC with contiguous spectrum is specified in an *interBandContiguousMRDC* IE, according to section 4.2.7.9 "MRDC-Parameters" in 3GPP TS 38.306 V16.1.0. The **LnrFreqRelationship** MO can be used to configure whether an LTE frequency and an NR frequency are inter-band contiguous. Based on the parameter settings and UE capabilities, the eNodeB and the gNodeB determine whether the LTE and NR frequencies can coexist. If not, the base stations perform the processing above.

If an LTE frequency and an NR frequency of a specified NR bandwidth overlap in terms of frequency band, the LTE frequency, NR frequency and NR bandwidth can be configured using the **LnrFreqRelationship.DlEarfcn**, **LnrFreqRelationship.NrDlEarfcn** and **LnrFreqRelationship.NrDlBandwidth** parameters, respectively, and the **OVERLAPPED** option of the **LnrFreqRelationship.FreqRelationship** parameter must be selected. Under these settings, the base stations will determine that the two frequencies conflict with each other for NSA DC and perform the previous processing.

 NOTE

- If the frequency configured for an LTE cell is a multi-band frequency, this LTE frequency is considered to conflict with an NR frequency if the **OVERLAPPED** option is selected for the relationship between the primary or secondary frequency and the NR frequency. In this case, the previous processing method is performed.
- It is recommended that the **NR_SCG_BW_CHECK_OPT_SW** option of the **NsaDcAlgoParam.NsaDcAlgoSwitch** parameter be selected. If the UE capability information does not include the *channelBWs-UL* or *channelBWs-DL* IE that specifies the NR bandwidths supported by a UE, the eNodeB considers that the UE supports bandwidths supported by each NR frequency band defined in section 5.3.5 in 3GPP TS 38.101-1 V15.7.0 and 3GPP TS 38.101-2 V15.7.0 by default when evaluating the NR cell bandwidth during an NR SCG addition. In addition, if the NR cell bandwidth is 30 MHz or 70 MHz, an NR SCG addition can be performed only when the UE supports the 30 MHz or 70 MHz bandwidth, respectively.

If an LTE frequency and an NR frequency with specified bandwidth 1 are intra-band contiguous, the LTE frequency, NR frequency and NR bandwidth 1 can be configured using the **LnrFreqRelationship.DlEarfcn**,

LnrFreqRelationship.NrDLArfcn and **LnrFreqRelationship.NrDLBandwidth** parameters, respectively, and the **LnrFreqRelationship.FreqRelationship** parameter must be set to **INTRA_BAND_CONTIGUOUS**. If at the same time this LTE frequency and the same NR frequency with specified bandwidth 2 are intra-band non-contiguous, the LTE frequency, NR frequency and NR bandwidth 2 can be configured using the **LnrFreqRelationship.DLEarfcn**, **LnrFreqRelationship.NrDLArfcn** and **LnrFreqRelationship.NrDLBandwidth** parameters, respectively, and the **LnrFreqRelationship.FreqRelationship** parameter must be set to **INTRA_BAND_NON_CONTIGUOUS**. If a UE does not support intra-band contiguous EN-DC using the configured LTE frequency and the NR frequency with specified NR bandwidth 1 or the UE does not support intra-band non-contiguous EN-DC using the configured LTE frequency and the NR frequency with specified NR bandwidth 2, set the **UeCompat.LnrFreqRelpWhitelist** parameter to configure a function whitelist for the UE and then associate this whitelist with the **LnrFreqRelationship** MO. That is, set **FREQ_RELATIONSHIP_ID_N** of the **UeCompat.LnrFreqRelpWhitelist** parameter to **ON**, where *N* is equal to the value of **LnrFreqRelationship.RelationshipId**. In this way, the relationship between LTE and NR frequencies is configured to take effect only for a certain type of NSA UE. The gNodeB determines whether the UE supports each LTE-NR frequency relationship configured on the eNodeB. The processing method is the same as above.

Intra-band EN-DC requires that LTE and NR cells be deployed on the same site, according to 3GPP TS38.201 V16.1.0.

On the LTE side, co-site intra-band combinations such as the NSA DC combination of B3 and N3 can be configured when the **CO_DEPLOYMENT_NSA_FLAG** option of the **NrScgFreqConfig.AggregationAttribute** parameter is selected. Co-site cell pairs can be configured when the **CO_DEPLOYMENT_NSA_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter is selected.

If the LTE anchor cell frequency belongs to a band (for example, B3) in a co-site intra-band combination, then:

- During an SCG addition by the MeNB, co-site cell filtering is performed based on the measurement result and blind SCG addition procedure.
- During an intra-MeNB handover, an SgNB Modification Request message is sent. Co-site cells are selected from the neighboring NR cells reported along with LTE cell measurement results or from the blind-configurable neighboring NR cells. (When the **NsaDcLteMeasCtrlwithNbrMeasSw** option of the **GlobalProcSwitch.ProtocolSupportSwitch** parameter is selected, the MeNB includes a *reportAddNeighMeas* IE in delivered LTE measurement configurations and the NSA UE reports neighboring NR cell measurement results along with LTE cell measurement results.) If there are qualified neighboring NR cells, the SgNB side is notified and determines whether to trigger an intra-SgNB cell change. If there are no qualified neighboring NR cells and the source SgNB cell and the target MeNB cell are in the same co-site intra-band combination but not in the same cell pair, the SCG is released.
- During an inter-MeNB handover, a Handover Request message is sent. If the target MeNB cell and the source SgNB cell are not in the same cell pair, a handover with an SgNB change is performed and the target MeNB determines whether to add an SCG.

On the NR side, co-site cells need to be selected when the following conditions are met: (1) an intra-frequency or inter-frequency cell change is required on the NR side; (2) the frequency band (for example, band 3) of the LTE anchor cell belongs to a co-site intra-band combination; (3) the SgNB receives a measurement report from the UE. Co-site cells are selected from measurement cells based on the co-site intra-band combinations and co-site cell pairs configured on the NR side.

On the NR side, co-site intra-band combinations can be configured when the **CO_DEPLOYMENT_NSA_FLAG** option of the **gNBdedNsaDcBandComb.AggregationAttribute** parameter is selected. Co-site cell pairs can be configured when the **CO_DEPLOYMENT_NSA_FLAG** option of the **gNBnrLteCellComb.AggregationAttribute** parameter is selected.

A reported cell is filtered out if its frequency is in the NR band (for example, N3) of a co-site intra-band combination but it is not in a co-site cell pair.

NOTE

If LTE+NR co-site intra-band combinations are configured on the NR side, it is recommended that the band information of co-site external NR cells be specified by the **NRExternalNCell.FrequencyBand** parameter so that the gNodeB can select target cells based on the band information.

4.1.6.8 EN-DC Band Combination Blacklist Function

When some NSA UEs access certain EN-DC band combinations, their performance deteriorates if UE compatibility issues occur. In this case, the EN-DC band combination blacklist function can be used to prohibit the use of specific LTE or NR bands in certain EN-DC band combinations, improving user experience of NSA UEs.

A blacklist can be configured by setting **NSA_BAND_COMB_ID_N** of the **UeCompat.NsaBandCombBlacklist1** or **UeCompat.NsaBandCombBlacklist2** parameter. Specifically, select **NSA_BAND_COMB_ID_N**, where *N* is equal to the value of **PrivateCaBandComb.PrivateCaCombId** in a **PrivateCaBandComb** MO. If this **PrivateCaBandComb** MO is associated with a **PrivateBand** MO where **Bandwidth_0M** of the **PrivateBand.CombBandBw** parameter is selected, the specified band cannot be added to the EN-DC band combination associated with the **PrivateCaBandComb** MO.

For example, an NSA UE reports an EN-DC band combination B3+B20+n41. If the base station is not expected to configure this combination for the UE, perform the following configurations. Set the **UeCompat** MO to blacklist this UE, select **NSA_BAND_COMB_ID_0** of the **UeCompat.NsaBandCombBlacklist1** parameter, and set the parameters in the **PrivateBand** MO associated with the band combination whose **PrivateCaBandComb.PrivateCaCombId** parameter is set to **0**, by referring to the configurations below.

- Configure band 3 in the EN-DC band combination.

Parameter ID	Parameter Value
PrivateBand.PrivateCaCombId	0
PrivateBand.CombBandIndex	1

Parameter ID	Parameter Value
PrivateBand.CombBandId	3
PrivateBand.BandType	LTE
PrivateBand.CombBandBw	Bandwidth_0M is not set to On.

- Configure band 20 in the EN-DC band combination.

Parameter ID	Parameter Value
PrivateBand.PrivateCaCombId	0
PrivateBand.CombBandIndex	2
PrivateBand.CombBandId	20
PrivateBand.BandType	LTE
PrivateBand.CombBandBw	Bandwidth_0M is not set to On.

- Configure N41 in the EN-DC band combination.

Parameter ID	Parameter Value
PrivateBand.PrivateCaCombId	0
PrivateBand.CombBandIndex	3
PrivateBand.CombBandId	41
PrivateBand.BandType	NR
PrivateBand.CombBandBw	Bandwidth_0M is set to On and other options are set to Off, indicating that the band is not allowed in this combination.

If an LTE frequency in EN-DC band combinations with a specified NR bandwidth that include a certain NR frequency is not expected to be configured as an LTE SCC, the **LnrFreqRelationship** MO can be set to prohibit this LTE frequency from serving as an LTE SCC. To be specific, the LTE frequency, NR frequency, and NR bandwidth can be configured using the **LnrFreqRelationship.DlEarfcn**, **LnrFreqRelationship.NrDlArfcn**, and **LnrFreqRelationship.NrDlBandwidth** parameters, respectively, and **LnrFreqRelationship.FreqRelationship** must be set to **SCC_AVOIDANCE**. Under these settings, in any available EN-DC band combination, if this NR frequency exists, the configured LTE frequency will not be added as an LTE SCC. The relationship between LTE and NR frequencies can be configured to take effect only for a certain type of NSA UE. First, set the **UeCompat.LnrFreqRelpWhitelist** parameter to configure a function whitelist for the UE. Then, associate this whitelist with the **LnrFreqRelationship** MO, that is, select **FREQ_RELATIONSHIP_ID_N** of the **UeCompat.LnrFreqRelpWhitelist** parameter, where *N* is equal to the value of **LnrFreqRelationship.RelationshipId**.

The **NSA_FREQ_CONFLICT_AVOID_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter needs to be selected on the gNodeB side. Under this setting, after the eNodeB receives information about the NR PSCell and SCell frequencies in EN-DC band combinations from the gNodeB, the eNodeB will determine whether to prohibit the addition of an NR cell as the PSCell or an SCell if the NR cell conflicts with the LTE PCell.

4.1.6.9 RRC Connection Reestablishment Optimization for Defective NSA UEs

When the BWP actually used on an SCG frequency is modified (intra-base-station scenarios), changed (inter-base-station scenarios), or added for a defective UE on the network, the BWP bandwidth is strictly checked. If the BWP bandwidth is not a 3GPP-defined standard bandwidth, the defective UE will repeatedly trigger RRC connection reestablishment. To address this issue, Huawei eNodeB supports RRC connection reestablishment optimization for defective NSA UEs. When the **NSA_NR_CARRIER_FORBID_SW** option of the **UeCompat.BlacklistControlExtSwitch2** parameter is selected and the defective UEs are blacklisted, the eNodeB checks the causes of RRC connection reestablishments for the blacklisted UEs. If an RRC connection reestablishment for a blacklisted UE is caused by a modification, change, or addition of the BWP actually used on an SCG frequency and the reestablishment cause is **reconfigurationFailure**, a frequency that actually uses a non-protocol-defined BWP cannot be added as an SCG frequency for the blacklisted UE.

When there are multiple configurable SCG frequencies on the network, the **NON_STANDARD_BWP_FLAG** option of the **NrScgFreqConfig.AggregationAttribute** parameter can be used to specify whether the BWP actually used on an SCG frequency is a non-protocol-defined BWP. When an RRC connection reestablishment is triggered for a blacklisted UE because the BWP actually used on the SCG frequency is modified, changed, or added, the eNodeB:

- Prohibits addition of a frequency with a non-protocol-defined BWP (the frequency for which the **NON_STANDARD_BWP_FLAG** option is selected) as an SCG frequency for the blacklisted UE.
- Allows addition of a frequency with a protocol-defined BWP (the frequency for which the **NON_STANDARD_BWP_FLAG** option is deselected) as an SCG frequency for the blacklisted UE.

The preceding differentiated processing makes it possible to configure SCG frequencies for blacklisted UEs.

For more details about blacklist control, see *Terminal Awareness Differentiation* in eRAN feature documentation.

4.1.6.10 Scheduling Protection Against the Impact of NR SRS Antenna Switching on LTE Bands

If a UE is configured with an EN-DC combination (for example, a combination of LTE 1800 MHz and NR 2600 MHz) and SRS antenna switching on the NR side affects uplink transmission and downlink reception on the LTE side, then the **txSwitchImpactToRx** and **txSwitchWithAnotherBand** IEs need to be included in the UE capability information according to 3GPP TS 38.306.

Assume that the **SRS_RES_CONFIG_TO_LTE_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter is selected on the NR side, the **NR_UL_ANT_SEL_OPT_SW** option of the **NsaDcAlgoParam.NsaDcAlgoExtSwitch** parameter is selected on the LTE side, and the gNodeB detects that SRS antenna switching in the NR cell imposes an impact on the LTE band in the case of EN-DC. Then, the gNodeB notifies the eNodeB of time-domain positions of SRS antenna switching. After receiving the message, the eNodeB stops uplink and downlink scheduling in the affected subframes. This reduces the bit error rate and increases the uplink and downlink throughput on the LTE side.

 NOTE

- The preceding function can be enabled only when time synchronization is achieved on both LTE and NR sides.
- Uplink scheduling avoidance is not performed in the LTE cell if the SRS antenna switching period in the NR cell exceeds 10 ms. Downlink scheduling avoidance is not performed in the LTE cell if the SRS antenna switching period in the NR cell exceeds 40 ms.

4.1.6.11 NCGI Measurement Optimization in EN-DC

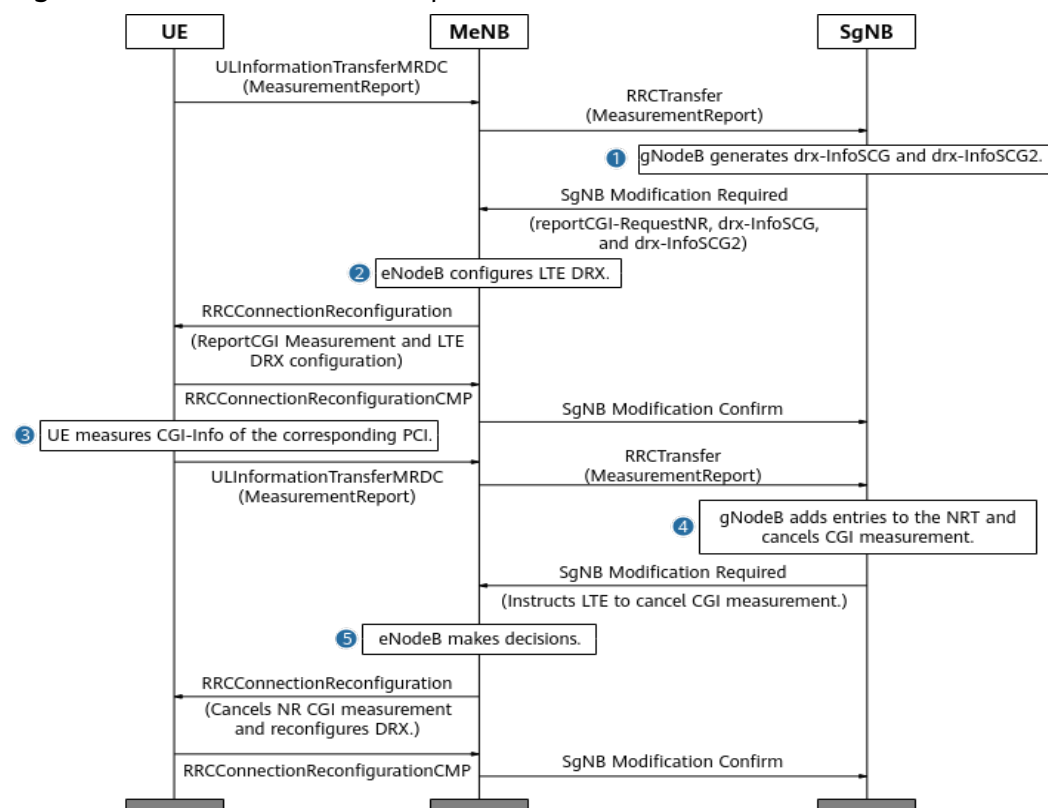
A UE uses the DRX mechanism by default when measuring the CGI of a neighboring cell. Specifically, the base station configures a set of DRX parameters for the UE, and the UE reads the CGI of the neighboring cell after entering the DRX state. When an NSA UE measures CGIs on the NR side, the UE expects a long sleep time on both the NR and LTE sides. Currently, DRX parameters are configured separately on the NR and LTE sides for the NSA UE and take effect independently. In this case, if the NSA UE does not have enough common sleep time for measuring CGIs on the NR side, CGI measurement will fail on the NR side.

This function enables the NR and LTE sides to cooperate with each other when an NSA UE performs CGI measurement on the NR side, maximizing the duration in which the UE is in the common sleep time and increasing the probability of CGI measurement by the UE. This function applies to time synchronization scenarios. For details about time synchronization, see *Synchronization*.

This function requires the following options to be selected:

- On the LTE side: **NSA_NR_NCGI_DRX_OPT_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter
- On the NR side: **NSA_NCGI_MEAS_DRX_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter

Figure 4-16 NCGI measurement procedure for NSA UEs



When the function is enabled on both sides and an NSA UE triggers a CGI measurement on the NR side, the related procedure is as follows:

1. The gNodeB sends an SgNB Modification Required message to the eNodeB over the X2 interface. This message contains the measurement configuration with the measurement report type of "reportCGI", and the DRX parameters of the UE for CGI measurement on the NR side.
2. If the UE is not performing VoIP services (QCI = 1/65/66) on the LTE side, the eNodeB uses the DRX parameters sent from the gNodeB to trigger LTE DRX in ANR scenarios.
3. The UE reads SIB1 of the corresponding cells based on the measurement configuration and sends a measurement report containing the CGI-Info to the gNodeB.
4. After receiving the measurement report containing the CGI information, the gNodeB deletes the CGI measurement configuration and instructs the eNodeB to cancel the CGI measurement.
5. After receiving the CGI measurement cancellation notification, the eNodeB stops LTE DRX in ANR scenarios.

4.1.6.12 Penalty on Abnormal SCG Releases for NSA UEs

The "penalty on abnormal SCG releases for NSA UEs" function is introduced to NSA networking to reduce the abnormal SgNB release rate (also referred to as the SCG service drop rate). If an NSA UE reports an SCGFailureInformationNR message to the MeNB due to an exception such as an access failure on the NR side, the MeNB immediately releases the SCG, causing an abnormal SCG release.

This function imposes an SCG addition penalty on a UE that experiences frequent abnormal SCG releases due to SCGFailureInformationNR. SCG addition is prohibited for the UE until the penalty ends, reducing the abnormal SgNB release rate.

The MeNB starts an abnormal SCG release statistic timer when a UE experiences the first abnormal SCG release caused by SCGFailureInformationNR. If the number of consecutive abnormal SCG releases due to SCGFailureInformationNR is greater than or equal to the abnormal SCG release number threshold until this timer expires, the eNodeB imposes an SCG addition penalty on this UE. With the penalty, SCG addition is prohibited for this UE until the abnormal SCG release penalty timer expires. After an SCG addition penalty is imposed on the UE, the abnormal SCG release statistic timer will be reset and the number of abnormal SCG releases will be changed to zero.

Related parameters are described as follows:

- The "penalty on abnormal SCG releases for NSA UEs" function is controlled by the **NSA_SCG_ABN_REL_PUNISH_SW** option of the **NsaDcMgmtConfig.NsaDcAlgoEnhSwitch** parameter.
- The length of the abnormal SCG release statistic timer is specified by the **NsaDcMgmtConfig.ScgAbnRelStatDuration** parameter.
- The abnormal SCG release number threshold is specified by the **NsaDcMgmtConfig.ScgAbnRelNumThld** parameter.
- The abnormal SCG release penalty timer is specified by the **NsaDcMgmtConfig.ScgAbnRelPunishTimer** parameter.

4.2 Network Analysis

4.2.1 Benefits

- Overall throughput of the entire network
NSA DC does not directly affect network capacity. However, if the resources of the network have not been used up, NSA DC can increase the resource usage and throughput of the network.
- NSA UE throughput
If the resources of the network have not been used up, NSA DC can increase the throughput of UEs.
- Overall PRB usage for the network
Most UE services on commercial networks are burst services, and there is a very low probability that PRBs in all the CCs of a UE are used up simultaneously. When NSA DC is enabled, loads can be balanced rapidly using data split policies and scheduling, effectively utilizing idle resources on the network and increasing the overall PRB usage of the network.
- Downlink real-time data split
In NR FDD scenarios, this function can fully utilize the air interface transmission capabilities of LTE and NR to improve user experience of NSA UEs.
- Uplink preallocation

This function increases the number of times the base station proactively schedules UEs in NSA scenarios, shortens the duration of uplink data packet buffering on the NSA UE, accelerates the response to UE services, and improves user experience.

- Uplink fallback to LTE

After this function takes effect, the uplink data transmission of cell edge users (CEUs) in the NR cell is switched to the LTE network, increasing the uplink user-perceived rate of these UEs.

- Fast retransmission for downlink data split

This function reduces the impact of data overstock due to single-side rate drop on the TCP sliding window and increases UE throughput.

- SCG addition based on virtual grid models

For UEs in an area without NR coverage, this function reduces invalid gap-assisted measurements for periodic SCG addition based on measurement. For UEs moving from an area without NR coverage to an area with NR coverage, this function can quickly detect NR coverage and add SCGs, increasing the 5G online duration of these UEs.

- Uplink CA in EN-DC

This function allows an NSA UE to be configured with both an NR SCG and LTE uplink CA, improving uplink experience of the UE at the cell center or at a medium distance from the cell center.

- Preferential selection of a band combination with only one NR carrier for NSA UEs

This function increases the throughput of NSA UEs when NR CA cannot be performed.

- Differentiated SCG addition and deletion threshold configuration for NR CA-capable UEs

This function allows differentiated configuration of NR FDD SCG addition and deletion thresholds for NSA UEs that support NR FDD+TDD CA, increasing the downlink user-perceived rate of these UEs.

- Inter-frequency and inter-RAT measurement optimization for VoLTE UEs in NSA DC scenarios

This function reduces the delivery of inter-frequency and inter-RAT measurement configurations to VoLTE UEs and the associated reporting, possibly improving voice quality.

- QCI-specific VoLTE and SCG coexistence

This function ensures the user-perceived rate of data services for UEs with bearers of the specific QCI when they are performing VoLTE services.

- Penalty on abnormal SCG releases for NSA UEs

This function decreases the SCG service drop rate in NSA networking. The SCG service drop rate in NSA networking is calculated as follows:

- On the LTE side: $(L.NsaDc.SgNB.Rmv.Att - L.NsaDc.MenbTrig.SgNB.NormRel - L.NsaDc.SgnbTrig.SgNB.NormRel) / L.NsaDc.SgNB.Rmv.Att \times 100\%$
- On the NR side: $(N.NsaDc.SgNB.Rel - N.NsaDc.SgNB.Rel.SgNBTrigger - N.NsaDc.SgNB.Rel.MeNBTrigger.NormalRel + N.NsaDc.SgNB.AbnormRel) / N.NsaDc.SgNB.Rel \times 100\%$

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- If a UE is configured with NSA DC, the UE has one RRC connection with the MeNB and another RRC connection with the SgNB.
- NSA DC performance factor = NSA DC rate / (LTE_ONLY rate + NR_ONLY rate). The greater the NSA DC performance factor, the better the performance of data split in NSA DC. When the delay on the X2 interface increases, the NSA DC performance factor decreases. In the preceding formula, LTE_ONLY indicates that services are carried only on the LTE network; NR_ONLY indicates that services are carried only on the NR network; NSA DC indicates that services are carried on both the LTE and NR networks.
- In scenarios with a smaller LTE bandwidth and a higher load or with a larger NR bandwidth and a lower load, it is more likely that the NSA DC rate is less than the NR_ONLY rate.
- In Option 3x, when the value of the **gNBPDcpParamGroup.DlDataPdcpSplitMode** parameter is changed from SCG_ONLY to SCG_AND_MCG, the RLC-layer processing delay of the gNodeB (which equals $(N.QoS.DL.PktDelaygNBDU.Time / N.QoS.DL.PktDelaygNBDU.Num) / 1000$) may decrease. In addition, if the proportion of CEUs in the NR cell is high, the average downlink UE throughput on the NR side (which equals $(N.ThpVol.DL - N.ThpVol.DL.LastSlot) / N.ThpTime.DL.RmvLastSlot$) may increase.
- If frequency synchronization is configured on the LTE side and gap-assisted measurement positions are not aligned between LTE and NR, some downlink data on the NR side will be discarded, decreasing the downlink throughput on the NR side.
- Assume that the UE power saving function is triggered after an NSA UE reports an SCGFailureInformationNR message in which the *failureType-r15* IE is set to t310-Expiry and the first *ARFCN-ValueNR-r15* IE and first *pci-r15* IE in the *measResultFreqListNR* IE are both set to 0. Then, if the SCG_REL_MEAS_OPT_SW option of the **ENodebCounterParaGrp.ENodebCounterAlgoSwitch** parameter is selected and the eNodeB releases the SCG after receiving that message, the release is counted as a normal release. The **L.NsaDc.MenbTrig.SgNB.NormRel** counter value will increase and the KPIs calculated based on this counter will also change.
- After the function of SCG release and addition based on UE overheating status reporting takes effect, the values of SgNB-release-related counters on the LTE side will increase. Such counters include **L.NsaDc.MenbTrig.SgNB.Rel.UeException**, **L.NsaDc.SgNB.Rmv.Att**, and **L.NsaDc.MenbTrig.SgNB.NormRel**. The KPIs calculated based on the preceding counters will also change.
- After downlink real-time data split takes effect in Option 3x, the data volume on the LTE and NR sides changes, and the values of user-plane performance indicators, such as the average MCS index, BLER, load, and average user-perceived rate, also change.

After downlink real-time data split takes effect in NR FDD scenarios where UEs are evenly distributed, the data volume distributed from the NR PDCP to

the LTE side increases, that is, the value of the **N.PDCP.Vol.DL.X2U.TrfPDU.Tx** counter increases.

- On the LTE side, there are increases in the downlink cell data volume, cell throughput, PRB usage, and average MCS index.
- On the NR side, there are decreases in the downlink cell data volume, cell throughput, PRB usage, and average MCS index.
- After downlink real-time data split, downlink dynamic data split stop, or fast retransmission for downlink data split takes effect, there is a possibility that no new packets or a decreased number of new packets are received on the NR side. As a result, a timer update is not triggered and the timer expires. A retransmission is performed upon timer expiry. If the number of retransmissions reaches 32, service drops occur. Therefore, after the preceding functions take effect, the service drop rate may increase.
- After uplink dynamic data split takes effect in Option 3x, a UE transmits data on both the MCG and SCG sides. Due to factors such as the uplink data split algorithm on the UE side and air interface signal quality on the MCG and SCG sides, packet disorder or loss may occur when data packets transmitted through the MCG and SCG reach the NR PDCP layer. As a result, the value of the **N.PDCP.UL.TrfSDU.RxPacket.Loss** counter increases.
- After the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs" is enabled, the PSCell modification success rate and PSCell change success rate may increase.
- Enabling uplink fallback to LTE has the following impacts on the network:
 - The proportion of downlink traffic at the cell edge is likely to increase as the uplink data transmission performance at the cell edge improves. This may decrease the downlink data rates of NR UEs and increase the proportion of user data distributed to the LTE side.
 - The uplink data volume on the LTE side is likely to increase as NR UEs running uplink services at the cell edge fall back to LTE. In addition, the increase in the number of LTE CEUs may decrease the LTE uplink rate and increase the packet loss rate.
 - The number of SgNB Modification Required messages sent by the NR side increases, and the probability of procedure conflicts increases. As a result, the SgNB modification success rate decreases.
 - The service drop rate on the LTE side is likely to increase as the number of RRC connection reconfiguration messages sent on the LTE side increases.
 - The rate of abnormal SgNB releases triggered by the SgNB increases, but the total rate of abnormal SgNB releases triggered by the MeNB and SgNB decreases.
- Enabling the function of requesting LTE-side large-packet status of ULFB UEs increases the X2 signaling overhead.
- Virtual grid model building and update require the selection of a certain number of UEs for gap-assisted inter-frequency measurement. Therefore, the uplink and downlink throughputs of these UEs decrease. In addition, the number of SgNB Modification Request messages sent by the eNodeB to the gNodeB increases, and the values of counters related to network interference slightly change. As the number of selected UEs is small, the impact on the average uplink and downlink throughputs of UEs is negligible when there are more than 15 online UEs in a cell.

- Measurement reports need to be collected during virtual grid model building. This may have the following network impacts:
 - The number of signaling messages over the air interface increases.
 - The number of SRBs increases.
 - The average MCS index decreases because small-index MCSs are used for SRBs.
 - The service drop rate increases because RRC connection reconfiguration messages may fail to be sent.
 - The traffic volume fluctuates because gap-assisted measurement causes fluctuations in channel quality measurement results.
- The CPU usage of the main control board increases when the eNodeB builds virtual grid models and frequently queries virtual grid models.
- The building and query of virtual grid models consume more main control board resources. When the number of UEs served by a base station is too large, congestion occurs earlier. As a result, the RRC connection setup success rate decreases and the service drop rate increases.
- After penalty optimization for LNR virtual grid model building takes effect, a longer maximum allowed duration for building an LNR virtual grid model results in a lower probability of virtual grid model building failures, a higher probability of long-term UE measurement data collection, and a higher probability of average user-perceived rate decrease; a shorter maximum allowed duration results in a higher probability of virtual grid model building failures and a lower probability of average user-perceived rate decrease.
- If the required accuracy of LNR virtual grid models for SCG addition is set to a value other than 100%, the model-based predictions can be used for blind SCG addition. Indicators related to SCG addition are likely to deteriorate in the following scenarios:
 - The blind SCG addition performance of certain UEs is poor.
 - The eNodeB monitors and updates virtual grid models when the network structure changes due to RF optimization or other reasons.
 - The predicted RSRP values in some grids are inaccurate.
 - Some UEs in power saving mode report the SCGFailureInformationNR message to the MeNB.
- In DRX scenarios, the downlink packet transmission delay decreases when NSA downlink data split optimization takes effect. If the uplink non-primary path side enters the DRX sleep state, the data volume decreases on this side, whereas the data volume (for example, **L.Thrp.bits.DL.NsaDc**, **N.PDCP.Vol.DL.X2U.TrfPDU.Tx**, or **N.ThpVol.DL**) increases and the number of active UEs (**L.Traffic.ActiveUser.DL.Avg**) decreases on the uplink primary path side. After the data volume changes, the values of user-plane indicators (such as the PRB usage, MCS index, or CQI) also change. For example, the PRB usage and average MCS index decrease on the uplink non-primary path side but increase on the uplink primary path side.
- In DRX scenarios, UEs are more likely to enter the DRX state as SCG addition based on virtual grid models will reduce inter-frequency measurement configurations delivered by the base station to UEs. As a result, the values of the **L.Traffic.User.Cdrx.Avg** and **L.Cdrx.Active.TtiNum** counters will increase.

- If SCG addition based on virtual grid models has been enabled on the network, enabling measurement and penalty for SCG addition based on virtual grid models has the following network impacts:
 - The eNodeB will deliver more event B1 measurement configurations. If there are UEs that are incompatible with blind SCG addition on the network, the NR data volume will increase.
 - If blind SCG addition would fail but SCG addition based on event B1 measurement would succeed for some UEs on the network, the number of SCG failures will decrease.
 - If both blind SCG addition and SCG addition based on event B1 measurement would fail for some UEs on the network, the number of SCG addition failures will increase for such UEs during their first SCG addition procedures.
- After NCGI measurement optimization in EN-DC takes effect, the throughput of NSA UEs on the LTE side may decrease.
- Enabling SPID- or QCI-specific NSA PCC anchoring and PSCell management policies has the following impacts on the network:
 - The number of NSA DC PCC-anchoring-based handover preparation attempts (indicated by **L.NsaDC.PCCAnchor.HHO.PrepareAttOut**) decreases.
 - If the **FORBID_MEAS_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter is selected, the number and total duration of inter-frequency measurements for NSA UEs decrease. As a result, PCC anchoring to the frequency specified by **CellMultiCarrCfgGrp.DLArfcn** cannot be triggered or the frequency specified by **CellMultiCarrCfgGrp.DLArfcn** cannot be added as an SCG frequency, and the UE throughput on the LTE and NR sides may decrease. If this option is deselected, network performance is not affected.
 - If the **FORBID_BLIND_CFG_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter is selected, the number and total duration of inter-frequency measurements for NSA UEs increase. If this option is deselected, network performance is not affected.
 - If the **HIGH_LOAD_CELL_NSA_PCC_FLAG** option of the **CellMultiCarrCfgGrp.AggregationAttribute** parameter is selected, NSA PCC anchoring to the frequency specified by the **CellMultiCarrCfgGrp.DLArfcn** parameter is allowed only for UEs in high-load cells. As a result, the load of the cell operating on this frequency decreases, and the number of NSA UEs also decreases. If this option is deselected, network performance is not affected.
- After SPID- or QCI-specific SCG addition at short intervals without considering traffic volume is enabled, the duration in which UEs with the specified SPID or QCI stay in the NSA DC state and the number of inter-RAT measurements increase. Inter-RAT measurements decrease the user-perceived rate and increase UE power consumption.
- After "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" takes effect, a smaller value of the **NsaDcAlgoParam.NrTddSccRsrpThld** parameter results in a higher probability of adding FDD SCGs for NR CA-capable UEs, an increase in the number of NR SCG additions, an increase in the number of NR CA-capable UEs, and a decrease in the downlink user-perceived rate of 5G UEs; a larger

value of the **NsaDcAlgoParam.NrTddSccRsrpThld** parameter results in the opposite effects.

- When NSA DC SRB3 is enabled on the live network and some UEs are incompatible with SRB3, enabling the NSA DC SRB3 blacklist function for these UEs reduces the probability of UE-triggered abnormal SgNB releases. This decreases the total number of abnormal SgNB releases triggered by the MeNB due to a UE-reported SCG failure in LTE-NR NSA DC scenarios (indicated by the **N.NsaDc.SgNB.Rel.MeNBTrigger.AbnormRel.ScgFail** counter).
- When QCI-specific VoLTE and SCG coexistence takes effect and VoLTE services and the SgNB coexist, the transmit power on the LTE side may decrease for UEs with bearers of the specific QCI. As a result, VoLTE service experience deteriorates for the UEs.
- After "penalty on abnormal SCG releases for NSA UEs" takes effect, SCG addition penalty will be imposed on UEs that experience frequent SCG releases due to SCGFailureInformationNR. As a result, the 5G NSA camping proportion decreases. The 5G NSA camping proportion equals **L.Traffic.User.NsaDc.PCell.Avg** divided by **L.NsaDc.Capable.5gUser.RRC.Avg**.
- Once the "MR/MDT and multi-carrier unified scheduling measurement" function is activated, inter-frequency and inter-RAT measurements are started for NSA UEs during NSA virtual grid model building. These measurements will reduce the downlink user-perceived rate of NSA UEs. The "MR/MDT and multi-carrier unified scheduling measurement" function is controlled by the **MR_MDT_MC_UNIFIED_SCH_MEAS_SW** option of the **EnodebAlgoExtSwitch.NsaProtocolCompatSw** parameter.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

If a UE is configured with NSA DC, the UE has one RRC connection with the MeNB and another RRC connection with the SgNB. Therefore, both LTE and NR sides require an RRC Connected User license unit. The licensing principle for the number of RRC_CONNECTED UEs on the LTE side in NSA networking is the same as that in LTE-only scenarios. For details, see *License Control Item Lists (FDD)*, *License Control Item Lists (CioT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation. For the licensing principle for the number of RRC_CONNECTED UEs on the NR side, see *License Control Item Lists* in 5G RAN feature documentation.

In Option 3, the eNodeB receives more data because of EN-DC data split. Therefore, you need to apply for a license for eNodeB traffic on the LTE side. For details, see *License Control Item Lists (FDD)*, *License Control Item Lists (CioT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Uplink preallocation	- LTE: SmartPreAllocationSwitch option of the CellAlgoSwitch.UlSwitch parameter - NR: UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i> in eRAN feature documentation <i>Uplink Scheduling</i> in 5G RAN feature documentation	In Option 3 or Option 3x, NSA uplink preallocation requires that both LTE uplink smart preallocation and NR uplink preallocation be enabled.
NR FDD Low-frequency NR TDD	Fast retransmission for downlink data split	NSA_DC_FAST_RETRANS_SWITCH option of the gNodeBParam.NsaDcOptSwitch parameter	<i>EPC-based NSA Basics</i>	If delay-based retransmission is enabled, fast retransmission for downlink data split also needs to be enabled.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast retransmission for downlink data split	NSA_DC_FAST_RETRANS_SWITCH option of the gNodeBParameter.NsaDcOptSwitch parameter	<i>EPC-based NSA Basics</i>	In DRX scenarios, if NSA downlink data split optimization is enabled, fast retransmission for downlink data split also needs to be enabled.
LTE FDD LTE TDD NR FDD Low-frequency NR TDD	NSA data split rate optimization	• LTE: NSA_SPLIT_RATE_OPT_SW option of the NsaDcMgmtConfig.NsaDcAlgorithmExtSwitch parameter • NR: NSA_SPLIT_RATE_OPT_SW option of the NRCellNsaDcConfig.NsaDcAlgorithmSwitch parameter	<i>EPC-based NSA Basics</i>	If preferred path adaption in downlink data split is enabled, NSA data split rate optimization also needs to be enabled. The RLC performs rate estimation based on the MAC scheduling capability and reports the RLC status to the PDCP more frequently.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Uplink 2CC aggregation	CaUl2CCSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i> in eRAN feature documentation	- In intra-eNodeB uplink CA scenarios, uplink CA in EN-DC requires uplink 2CC aggregation to be enabled. - In inter-eNodeB uplink CA scenarios, uplink CA in EN-DC requires inter-eNodeB CA based on relaxed backhaul or inter-eNodeB CA based on eNodeB coordination on both eNodeBs and uplink 2CC aggregation to be enabled.
LTE FDD	Inter-eNodeB CA based on relaxed backhaul	RelaxedBackhaulCaSwitch option of the ENodeBAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i> in eRAN feature documentation	
LTE FDD	Inter-eNodeB CA based on eNodeB coordination	FreqCfgCaOverBBUsSwitch option of the ENodeBAlgoSwitch.OverBBUsSwitch parameter	<i>Carrier Aggregation</i> in eRAN feature documentation	

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Blind PSCell addition for EPS fallback UEs	EPS_FALLBACK_BLIND_SCG_ADD_SW option of the NsaDcMgmtConfig.NsaDcAlgoExtSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	The NsaDcMgmtConfig.VolteUeScgMgmtStrategy parameter cannot be set to VOLTE_PREFERRED if blind PSCell addition for EPS fallback UEs is enabled.
LTE FDD LTE TDD	High speed mobility	Cell.HighSpeedFlag	<i>High Speed Mobility</i> in eRAN feature documentation	Virtual grids are not built for high-speed cells.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Collaboration between ROHC and NR PDCP	ROHC_NR_PDCP_COLLAB_SW option of the NsaDcMgmtConfig.NsaDcAlgoExtSwitch parameter	None	If the ROHC_NR_PDCP_COLLAB_SW option of the NsaDcMgmtConfig.NsaDcAlgoExtSwitch parameter is selected, the NSA_DC_CAPABILITY_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter must be deselected.
LTE FDD	Ultra-low-latency scheduling for inter-eNodeB CA based on relaxed backhaul	INTER_ENB_ULTRA_LOW_LATENCY_SW option of the MultiCarrUnifiedSch.MultiCarrierUnifiedSchEnSw parameter	<i>Multi-carrier Unified Scheduling</i> in eRAN feature documentation	If ultra-low-latency scheduling for inter-eNodeB CA based on relaxed backhaul is enabled, the TDM_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter must be deselected for the uplink single-side transmission and secondary intermodulation interference avoidance functions.
Low-frequency NR TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i> in 5G RAN feature documentation	SRS carrier switching is not compatible with secondary intermodulation interference avoidance.
NR FDD	Extended Cell Range	NRDUCell.CellRadius	<i>Extended Cell Range</i> in 5G RAN feature documentation	Extended Cell Range is not compatible with uplink single-side transmission.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Uplink data transmission path selection	NSA_DC_UL_PATH_SELECT ION_SW option of the NRCellNsaDc Config.NsaDc AlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	Uplink data transmission path selection is not compatible with uplink fallback to LTE.
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.Spec trumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	LTE FDD and NR Flash Dynamic Spectrum Sharing is not compatible with uplink single-side transmission in 16T16R or 32T32R cells.
NR FDD	Hybrid DSS Based on Asymmetric Bandwidth	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.Spec trumCloudSwitch parameter and LTE_NR_FDD_SPCT_SHR_AS YM_SW option of the NRDUCellAlgoSwitch.Spec trumCloudEnhSwitch parameter	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	Hybrid DSS Based on Asymmetric Bandwidth is not compatible with uplink single-side transmission in 16T16R or 32T32R cells.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Traffic model-based performance optimization	DLPacketLenAwareSchSwitch option of the CellAlgoSwitch . DISchSwitch parameter	<i>Downlink Scheduling</i> in eRAN feature documentation	The throughput of LTE UEs decreases, and the throughput of NSA UEs increases.
LTE FDD LTE TDD	Downlink non-GBR packet bundling	NonGbrBundlingSwitch option of the CellAlgoSwitch . DISchSwitch parameter	<i>Downlink Scheduling</i> in eRAN feature documentation	The throughput of LTE UEs increases, and the throughput of NSA UEs decreases.
LTE FDD LTE TDD	Delay-based differentiated scheduling for extended QCI	QciPara.DLPdbForExtendedQci	<i>Downlink Scheduling</i> in eRAN feature documentation	The throughput of LTE UEs increases, and the throughput of NSA UEs decreases.
LTE TDD	Downlink scheduling priority enhancement	DL_SCH_PRIORITY_ENHANCE_SWITCH option of the CellDLSchExtParam . DISchExtSwitch parameter	<i>LTE AHR (TDD)</i> in eRAN feature documentation	Downlink scheduling priority enhancement decreases the throughput of non-NSA UEs and increases the throughput of NSA UEs.
LTE FDD	Enhanced downlink packet length awareness scheduling	DL_PKT_LENGTH_AWARE_SCH_ENHANCE_SWITCH option of the CellAlgoExtSwitch . DISchEnhSwitch parameter	<i>Downlink Scheduling</i> in eRAN feature documentation	Enhanced downlink packet length awareness scheduling decreases the throughput of non-NSA UEs and increases the throughput of NSA UEs.
LTE FDD LTE TDD	Emergency call	None	<i>Emergency Call</i> in eRAN feature documentation	When a UE is identified as an emergency call UE, it cannot perform NSA DC.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Breathing Pilot	BreathingPilotSwitch option of the CellDlschAlgo.BreathingPilotAlgoSwitch parameter	<i>Breathing Pilot</i> in eRAN feature documentation	After this feature is enabled, the data volume of an NSA UE may be lower than the downlink data volume threshold for large-packet services and consequently the NSA UE is identified as a non-large-packet UE. As a result, the gains brought by this feature are affected.
LTE FDD LTE TDD	Uplink FDD +TDD CA	InterFddTddCaSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i> in eRAN feature documentation	- When NSA DC services are in progress, uplink FDD+TDD CA does not take effect. - When uplink FDD +TDD CA is in progress, the uplink secondary carrier needs to be removed if NSA DC is required.
LTE FDD LTE TDD	ROHC	CellAlgoSwitch.RohcSwitch	<i>ROHC</i> in eRAN feature documentation	This function is not compatible with the MCG bearer, MCG split bearer, or SCG split bearer. If such a bearer is set up for an NSA UE, the PDCP of the bearer will be set to NR PDCP (according to 3GPP TS 38.323). However, if ROHC-based services (voice services or TCP/IP data services) are used, only the bearer named MCG_Bearer_EUTRA_PDCP can be set up and the PDCP of the bearer must be set to E-UTRA PDCP (according to 3GPP TS 36.323).

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Turbo start video	PdcpRohcPara.Profiles set to Profile0x0006	<i>Video Experience Optimization</i> in eRAN feature documentation	This function is not compatible with the MCG bearer, MCG split bearer, or SCG split bearer. If such a bearer is set up for an NSA UE, the PDCP of the bearer will be set to NR PDCP (according to 3GPP TS 38.323). However, if ROHC-based services (voice services or TCP/IP data services) are used, only the bearer named MCG_Bearer_EUTRA_PDCP can be set up and the PDCP of the bearer must be set to E-UTRA PDCP (according to 3GPP TS 36.323).
LTE FDD	Super combined cell	SuperCombCellSwitch option of the CellAlgoSwitch.SfnAlgoSwitch parameter	<i>Super Combined Cell (FDD)</i> in eRAN feature documentation	If an LTE cell has a larger coverage than an NR cell, a UE that is in the LTE cell but not in the NR cell cannot measure the NR cell.
LTE FDD LTE TDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC</i> in eRAN feature documentation	eMTC UEs do not support NR.
LTE FDD LTE TDD	In-band relay	InBandRelayDeNbSwitch and InBandRelayReNbSwitch options of the CellAlgoSwitch.RelaySwitch parameter	<i>Relay</i> in eRAN feature documentation	In-band relay requires the support of Huawei customized UEs. Huawei customized UEs supporting in-band relay do not support NSA DC.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	VoLTE	ENodeBAlgoSwitch.EutranVoipSupportSwitch	<i>VoLTE</i> in eRAN feature documentation	If an SCG is added for a UE running VoLTE services, the UE occupies power on the NR side, which decreases the available power on the LTE side and affects the VoLTE uplink coverage.
LTE FDD LTE TDD	System information broadcast	Sib24Switch option of the CellSiMap.SiS parameter	<i>Idle Mode Management</i> in eRAN feature documentation	If system information broadcast is enabled together with EPC-based NSA DC or NSA PCC anchoring, then the CellSiMap.SibxPeriod parameter ($x = 2, 3, \dots$) must be greater than or equal to 16 radio frames, the CellSiMap.SibTransCtrlSwitch parameter must be set to OFF , and the CellSiMap.SiSchResRatio parameter must be set to 0 . Under other configurations, NSA DC or NSA PCC anchoring may fail.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Energy conservation and emission reduction	None	<i>Energy Conservation and Emission Reduction</i> in eRAN feature documentation	The accuracy of inter-frequency signal quality prediction based on virtual grid models decreases in cells with energy conservation and emission reduction enabled. If a cell of a base station is blocked, data collection about this cell will be stopped in all virtual grids of its intra-base-station intra-frequency cells and data collection about the frequency of this cell will also be stopped in the virtual grids of the other cells.
LTE FDD LTE TDD	Multi-carrier coordinated energy saving	CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN , ON_MULTICARRIER_HIERMANUAL_SHUTDOWN , or ON_CARRIER_PRECISE_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i> in eRAN feature documentation	Enabling multi-carrier coordinated energy saving affects the progress of virtual grid model building.
LTE FDD LTE TDD	Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SW option of the MultiCarrierUnifiedSch.MultiCarrierUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling</i> in eRAN feature documentation	If LTE and LNR virtual grid models are built at the same time in a cell, a UE chosen by both models for sampling randomly selects one of the target frequencies for measurement. This affects the building duration.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Smart carrier selection based on virtual grids	SMART_CARRIER_SELECTION_SW option of the MultiCarrierUnifiedSch.MultiCarrierUnifiedSchSw parameter	<i>Multi-carrier Unified Scheduling</i> in eRAN feature documentation	- The LTE_CELL_MEAS_OPT_SW option of the EnodebAlgoExtSwitch.NsaDcAlgoSwitch parameter needs to be selected when smart carrier selection based on virtual grids is enabled. Then the duration of a single round of LTE SCell measurements for a UE in the NSA DC state can be specified by the NsaDcMgmtConfig.NsaDcLteScellMeasDuration parameter. - After smart carrier selection based on virtual grids takes effect, the CA capability of individual NSA UEs for which an SCG has been added is restricted to the CA capability corresponding to the currently selected EN-DC band combination. As a result, an optimal CA band combination may not be configured for the NSA UEs on the LTE side.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	AQM	CellAlgoSwitch.AqmAlgoSwitch	<i>AQM</i> in eRAN feature documentation	This function does not take effect within 1s after an SCG is released if real-time data split is enabled in downlink dynamic data split mode in Option 3x. This reduces the probability of inappropriate packet discard.
LTE FDD LTE TDD	Management of neighboring NG-RAN frequency measurement flags	NR_NFREQ_MEAS_MGMT_SW option of the CellAlgoExtSwitch.AnrOptSwitch parameter	<i>ANR Management</i> in eRAN feature documentation	If measurement of a neighboring frequency is disabled by setting the corresponding frequency measurement flag, the MeNB does not select this frequency as a target frequency for SCG addition.
LTE FDD LTE TDD	Uplink full-antenna reception	UL_COVERAGE_BOOST_SW option of the SectorSplitGroup.SectorSplitSwitch parameter	<i>Massive MIMO Enhancements (FDD)</i> in eRAN feature documentation	After uplink full-antenna reception takes effect, the number of SgNB releases triggered by LTE uplink coverage may change because uplink coverage on the LTE side improves. This affects PSCell management for VoLTE UEs.
LTE FDD LTE TDD	QCI-specific VoLTE and SCG coexistence	QCI_VOLTE_SCG_COEXIST_SW option of the CellQciPara.NsaDcOptSwitch parameter for any QCI	<i>EPC-based NSA Basics</i>	If QCI-specific VoLTE and SCG coexistence is in effect, inter-frequency and inter-RAT measurement optimization for VoLTE UEs in NSA DC scenarios will not take effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	PSCell management for VoLTE UEs	NsaDcMgmtConfig.VolteUeScgMgmtStrategy	<i>EPC-based NSA Basics</i>	If the NsaDcMgmtConfig.VolteUeScgMgmtStrategy parameter is set to VOLTE_SCG_COEXISTENCE , inter-frequency and inter-RAT measurement optimization for VoLTE UEs in NSA DC scenarios does not take effect.
LTE FDD	LNR CRS muting	LNR_CRS_MUTE_SW option of the CellAlgoExtSwitch.LnrZeroBufferzoneSw parameter	<i>LTE and NR Zero Bufferzone</i> in eRAN feature documentation	After LNR CRS muting is enabled, the data volume of an NSA UE may be lower than the downlink data volume threshold for large-packet services and consequently this UE is considered as a non-large-packet UE. As a result, the gains brought by LNR CRS muting are affected.
LTE FDD	Accurate uplink RB scheduling	UL_RB_PRECISE_SCH_SW option of the CellAlgoExtSwitch.UlSchedulingSwitch parameter	<i>Uplink Scheduling</i> in eRAN feature documentation	Interference avoidance in NSA DC may provide reduced benefits when accurate uplink RB scheduling takes effect.
LTE FDD LTE TDD	Prohibition of IMNCI delivery for MLB upon UE release	UE_REL_FOR_BID_IMNCI_FOR_MLB_SW option of the CellQciPara.QciAlgoSwitch parameter	None	After prohibition of IMNCI delivery for MLB upon UE release is enabled, the RRC Connection Release message does not carry the <i>IMNCI</i> IE when the UE enters idle mode. As a result, the PCC anchoring procedure for NSA UEs in idle mode does not take effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	TDM power control	• LTE: NSA_DC_E NH_UL_POWER_CONTROL_SW and TDM_SWITCH options of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter • NR: NSA_DC_E NH_UL_POWER_CONTROL_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If both uplink CA in EN-DC and TDM power control are enabled, TDM power control preferentially takes effect.
LTE FDD	Uplink single-side transmission	• LTE: TDM_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter • NR: SINGLE_UL_NSA_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	If both uplink CA in EN-DC and uplink single-side transmission are enabled, uplink single-side transmission preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Uplink CA in EN-DC	NSA_LTE_UL_2CC_SW option of the NsaDcMgmt Config.NsaDcAlgoExtSwitch parameter	<i>EPC-based NSA Basics</i>	If both uplink CA in EN-DC and secondary intermodulation interference avoidance are enabled, secondary intermodulation interference avoidance preferentially takes effect.
LTE FDD	Uplink CA in EN-DC	NSA_LTE_UL_2CC_SW option of the NsaDcMgmt Config.NsaDcAlgoExtSwitch parameter	<i>EPC-based NSA Basics</i>	If both uplink CA in EN-DC and secondary harmonic interference avoidance are enabled, secondary harmonic interference avoidance preferentially takes effect.
LTE FDD LTE TDD	NSA PCC anchoring based on NR coverage	NSA_PCC_ANCHORING_SWITCH option of the NsaDcMgmt Config.NsaDcAlgoSwitch parameter and NsaDcMgmt Config.NsaDcPccAnchoring Policy set to BASED_ON_NR_COVERAGE	<i>EPC-based NSA Performance Enhancement</i>	Differentiated SCG addition threshold configuration for NR CA-capable UEs has the following impacts on NSA PCC anchoring based on NR coverage for NSA UEs that support NR FDD+TDD CA: The sum of the original effective RSRP threshold and the offset specified by the NrScgFreqConfig.CaUeScgAddRsrpThldOffset parameter is used as the actual RSRP threshold for event B1 to determine whether an SCG can be added with the candidate target LTE cell during PCC anchoring.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Differentiated SCG addition threshold configuration for NR CA-capable UEs	SCG_ADD_B1_THLD_DIFF_SW option of the NsaDcAlgoParam.NsaDcAlgoExtSwitch parameter	<i>EPC-based NSA Basics</i>	When both the "differentiated SCG addition threshold configuration for NR CA-capable UEs" function and the "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function are both enabled, only the latter function takes effect.
LTE FDD LTE TDD	NSA DC SCG Addition Policy	NsaDcMgmtConfig.NsaDcScgAdditionPolicy	<i>EPC-based NSA Basics</i>	When the "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function takes effect, the MeNB implements an SCG addition policy corresponding to the ALL_FREQ setting of the NsaDcMgmtConfig.NsaDcScgAdditionPolicy parameter for NR FDD+TDD CA-capable UEs.
LTE FDD LTE TDD	SCG addition based on virtual grid models	NSA_CARRIER_SEL_WITH_VG_SW option of the LnrCarrierSelection.LnrCarrierSelectionSw parameter	<i>EPC-based NSA Basics</i>	The "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function does not take effect when SCG addition based on virtual grid models takes effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA PCC anchoring based on NR coverage	NSA_PCC_ANCHORING_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter and NsaDcMgmtConfig.NsaDcPccAnchoringPolicy set to BASED_ON_NR_COVERAGE	<i>EPC-based NSA Performance Enhancement</i>	The "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function does not take effect when NSA PCC anchoring based on NR coverage takes effect.
LTE FDD LTE TDD	NSA carrier combination selection based on user experience	NSA_LTE_FDD_SEL_OPT_SW or NSA_LTE_TDD_SEL_OPT_SW option of the EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw parameter	<i>EPC-based NSA Experience Improvement</i>	The "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function does not take effect when NSA carrier combination selection based on user experience takes effect.
LTE FDD LTE TDD	NSA/SA selection based on user experience	LTE_FDD_NSA_SA_DL_SEL_OPT_SW , LTE_TDD_NSA_SA_DL_SEL_OPT_SW , or LTE_FDD_NSA_SA_UL_SEL_OPT_SW option of the EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>	The "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function does not take effect when NSA/SA selection based on user experience takes effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA-related measurement report check cancelation	NSA_MEAS_RPT_CHECK_CANCEL_SW option of the NsaDcAlgoParam.NsaDcAlgorithmSwitch parameter	None	The "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function does not take effect when NSA-related measurement report check cancelation is enabled.
NR FDD Low-frequency NR TDD High-frequency NR TDD	RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode	<i>Multi-Operator Sharing</i> in 5G RAN feature documentation	When both NSA networking based on EPC and RAN sharing with common carrier are enabled, data split is supported between an E-UTRAN cell and an NR cell only if they belong to the same operator.
NR FDD Low-frequency NR TDD High-frequency NR TDD	RAN sharing with dedicated carrier	gNBSharingMode.gNBMultiOpSharingMode	<i>Multi-Operator Sharing</i> in 5G RAN feature documentation	When both NSA networking based on EPC and RAN sharing with dedicated carrier are enabled, data split is supported between an E-UTRAN cell and an NR cell only if they belong to the same operator.
NR FDD Low-frequency NR TDD	Configuration of low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgorithmSwitch.HighReliabilitySwitch parameter	<i>URLLC</i> in 5G RAN feature documentation	Secondary harmonic interference avoidance does not take effect if it is enabled together with configuration of low latency and high reliability.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Downlink interference randomization-based scheduling	DL_INTRF_RANDOM_SWITCH option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Downlink Scheduling</i> in 5G RAN feature documentation	If both the NR frequency-domain avoidance function for secondary harmonic interference avoidance and the downlink interference randomization-based scheduling function are enabled, the former does not take effect.
NR FDD Low-frequency NR TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i> in 5G RAN feature documentation	In NSA networking, if uplink fallback to LTE is enabled in a combined cell, the number of fallbacks to the LTE network may decrease, and the NR cell may serve more UEs in weak coverage areas. The increase in the number of UEs in weak coverage areas causes the uplink and downlink cell throughput to decrease and the service drop rate to increase. To mitigate this issue, it is recommended that the NRDUCellSrsMeas.NsaUlFackToLteSinrThld parameter be set to a larger value.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i> in 5G RAN feature documentation	In NSA networking, if uplink fallback to LTE is enabled in a hyper cell, the number of fallbacks to the LTE network may decrease, and the NR cell may serve more UEs in weak coverage areas. The increase in the number of UEs in weak coverage areas causes the uplink and downlink cell throughput to decrease and the service drop rate to increase. To mitigate this issue, it is recommended that the NRDUCellSrsMeas.NsaUlfackToLteSnrThreshold parameter be set to a larger value.
NR FDD	CCE-based data split stop for NSA DC UEs	CCE_BASED_SPLIT_STOP_SW option of the NRDUCellAlgoSwitch.NsaDcAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i> in 5G RAN feature documentation	• LTE side: CCE-based data split stop for NSA DC UEs takes effect only when the CellQciPara.NsaDcDefaultBearerMode parameter is set to SCG_SPLIT_BEARER. • NR side: CCE-based data split stop for NSA DC UEs does not take effect for UEs for which the gNBPDcpParamGroup.DlDataPdcplitMode parameter is set to MCG_ONLY or SCG_ONLY.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	CCE-based data split stop for NSA DC UEs	CCE_BASED_SPLIT_STOP_SWITCH.NsaDcAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i> in 5G RAN feature documentation	It is recommended that NSA uplink preallocation also be enabled when CCE-based data split stop for NSA DC UEs and uplink smart preallocation are both enabled. When downlink scheduling stops in the NR cell, NSA uplink preallocation increases the number of times UE data packets are preallocated and accelerates the gNodeB's response to UE services.
NR FDD	Uplink smart preallocation	UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPush.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i> in 5G RAN feature documentation	
Low-frequency NR TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UlDlDecouplingSwitch	<i>UL and DL Decoupling</i> in 5G RAN feature documentation	If UL and DL Decoupling is enabled in an NR cell, uplink fallback to LTE does not take effect.
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	IP performance monitoring	None	<i>IP Performance Monitor</i>	If the IP PM function is abnormal, the base station cannot obtain the X2 interface delay. As a result, the downlink data split performance of NSA DC is poor or downlink data cannot be split.

RAT	Function Name	Function Switch	Reference	Description
NR FDD	LTE key event assurance	LTE_BIG_EVENT_ASSURANCE_SW option of the LteNrSpctShrCellGrp.LteNrSpctShrSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	If LTE key event assurance is enabled, uplink fallback to LTE does not take effect.
LTE FDD	UL coordinated channel shutdown	CellRfShutdown.MultiRatJointChnShutdownSw	<i>Multi-RAT Coordinated Energy Saving</i>	If UL coordinated channel shutdown is enabled, sample collection for virtual grid model building is affected.
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Uplink preallocation	- LTE: PreAllocationSwitch option of the CellAlgoSwitch.UlSwitch parameter - NR: UL_BASIC_PREALLOCATION_SWITCH and UL_SMART_PREALLOCATION_SWITCH options of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i> in eRAN feature documentation <i>Uplink Scheduling</i> in 5G RAN feature documentation	- If the NsaDcMgmtConfig.NsaDcUeMcgUlambrRatio parameter is set to 0, preallocation does not take effect on the LTE side. - If the NsaDcMgmtConfig.NsaDcUeMcgUlambrRatio parameter is set to 100, preallocation does not take effect on the NR side.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA carrier combination selection based on user experience	- LTE: NSA_LTE_FDD_SEL_OPT_SW or NSA_LTE_TDD_SEL_OPT_SW option of the ENodeBAlgorithm.MultiNetworkingOptionOptSw parameter - NR: NSA_LTE_SEL_OPT_SW option of the gNodeBParam.NetworkingOptionOptSw parameter	<i>EPC-based NSA Experience Improvement</i>	After SPID-based SCG frequency priority configuration is enabled, NSA carrier combination selection based on user experience does not take effect for UEs with the specified SPIDs.
LTE FDD	DSS and NR Flexible Refarming	- LTE: DSS_FLEX_REFARM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter - NR: DSS_FLEX_REFARM_SW option of the NRDUCellAlgorithm.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	After refarming from spectrum sharing to NR, the number of LTE frequencies decreases by one and therefore the number of NSA UEs may decrease.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD	PC2 UE uplink NR power configuration maximization	- LTE: PC2_UE_NR_UL_PWR_CFG_MAX_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter - NR: PC2_UE_NR_UL_PWR_CFG_MAX_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	When PC2 UE uplink NR power configuration maximization takes effect for PC2 UEs, initial power control in NSA DC does not take effect.
NR FDD Low-frequency NR TDD	PDCP SN offset	PDCP_SN_OFFSET_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	When the PDCP SN offset function is enabled together with fast retransmission for downlink data split, the SN offset function preferentially takes effect.
Low-frequency NR TDD	TDM power control	NSA_DC_ENH_UL_POWER_CONTROL_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If TDM is enabled, uplink fallback to LTE does not take effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Cloud X services	CLOUD_VR_SERVICE_SWITCH option of the NRCellAlgoSwitch.CloudVrSwitch parameter	<i>Ultimate Cloud X Experience</i> in 5G RAN feature documentation	When uplink fallback to LTE takes effect on the uplink path of the Cloud X bearer, uplink data transmission is redirected to the MCG.
LTE FDD Low-frequency NR TDD	Network-coordinated dynamic UE power sharing	- LTE: NSA_DC_COOPERATION_DPS_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter - NR: NSA_DC_COOPERATION_DPS_SWITCH option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If secondary intermodulation interference avoidance has taken effect, network-coordinated dynamic UE power sharing does not take effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD Low-frequency NR TDD	Uplink data transmission path selection	• LTE: NSA_DC_UL_PATH_SELECTION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter • NR: NSA_DC_UL_PATH_SELECTION_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If secondary intermodulation interference avoidance has taken effect, uplink data transmission path selection does not take effect.
LTE FDD NR FDD	Network-coordinated dynamic UE power sharing	• LTE: NSA_DC_COOPERATION_DPS_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter • NR: NSA_DC_COOPERATION_DPS_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If uplink single-side transmission has taken effect, network-coordinated dynamic UE power sharing does not take effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD NR FDD	Uplink data transmission path selection	• LTE: NSA_DC_UL_PATH_SELECTION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter • NR: NSA_DC_UL_PATH_SELECTION_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If uplink single-side transmission has taken effect, uplink data transmission path selection does not take effect.
LTE FDD LTE TDD NR FDD Low-frequency NR TDD	Uplink data transmission path selection	• LTE: NSA_DC_UL_PATH_SELECTION_SW option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter • NR: NSA_DC_UL_PATH_SELECTION_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	If secondary harmonic interference avoidance has taken effect, uplink data transmission path selection does not take effect.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Downlink intra-FR inter-band CA	INTRA_FR_IN TER_BAND_C A_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i> in 5G RAN feature documentation	TDM power control and secondary intermodulation interference avoidance allow for the LTE-NR relative frame offset of 3 ms. If the frame offset of an NR cell is changed and the downlink intra-FR inter-band CA feature is enabled for the NR cell, the frame offset of the PSCell must also be changed because this feature requires that the absolute value of the frame offset difference between NR carriers be less than or equal to 625 Ts. For details about the requirements of downlink intra-FR inter-band CA on the frame offset difference between the PCell and SCells, see "Downlink Intra-FR Inter-Band CA" in <i>Carrier Aggregation</i> in 5G RAN feature documentation.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	UL and DL Decoupling	NRDUCellAlgoSwitch.UIDLDecouplingSwitch	<i>UL and DL Decoupling</i> in 5G RAN feature documentation	If UL and DL Decoupling in SUL independent deployment mode has taken effect, secondary intermodulation interference avoidance does not take effect. The TDM pattern corresponding to UL and DL Decoupling in SUL independent deployment mode is used.
Low-frequency NR TDD	Proactive avoidance of remote interference	SRS_RIM_INT RF_AVOID_SW and SRS_INTRF_COORD_S_SLOTT_SW options of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i> in 5G RAN feature documentation	Proactive avoidance of remote interference reduces the interference present to SRS, which lowers the uplink SINR that is considered in the determination of whether to use uplink fallback to LTE. This in turn increases the number of UEs that fall back from NR to LTE for uplink data transmissions. Accordingly, the value of N.NsaDc.ULChangetoMeNB increases.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with a BBU model other than the BBU5900C.

- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

The compatible boards are listed below.

RAT	Board Type	Board Name	Option 3	Option 3x
LTE	Main control board	UMPTb	Supported	Supported
LTE	Main control board	UMPTe	Supported	Supported
LTE	Main control board	UMPTg/ UMPTga	Supported	Supported
LTE	Main control board	UMPTHa	Supported	Supported
LTE	Baseband processing unit	All UBBP boards	Supported	Supported
LTE	Integrated board	UMDU	Supported	Supported
NR	Main control board	UMPTe	Supported	Supported
NR	Main control board	UMPTg/ UMPTga	Supported	Supported
NR	Main control board	UMPTHa	Supported	Supported
NR	Baseband processing unit	UBBPg	Supported	Supported
NR	Baseband processing unit	UBBPfw1	Supported	Supported
NR	Baseband processing unit	UBBPh	Supported	Supported

 NOTE

LNR virtual grid model building requires that the main control board of the eNodeB be UMPTe/UMPTga/UMPTg/UMPTHa series.

RF Modules

This function does not depend on RF modules.

4.3.4 Networking

The following describes the networking scenarios and interconnection modes supported by NSA DC.

- In co-site co-BBU or separate-BBU scenarios, the LTE and NR base stations in NSA networking support CI interconnection, backplane interconnection, and IP transmission interconnection. For LTE-NR control-plane data, different transmission paths can be selected in different networking scenarios, including CI interconnection, backplane interconnection (requiring tunnels to be configured), and IP transmission interconnection. For LTE-NR user-plane data, different transmission paths can be selected in different networking scenarios, including IP transmission interconnection, CI interconnection, backplane interconnection (not requiring tunnels to be configured).

 NOTE

IP transmission is recommended in co-site inter-BBU scenarios. If CI interconnection is used, it is recommended that the same type of main control board (for example, UMPTg) be configured for two BBUs to ensure optimal performance.

The transmission path of LTE-NR user-plane data is specified by the **gNodeBParam.X2uTransmissionType** parameter:

- If this parameter is set to **ROUTING_TRANS**, LTE-NR user-plane data is transmitted through IP.
- In co-site separate-BBU CI interconnection scenarios, if this parameter is set to **INTRA_TRANS**, LTE-NR user-plane data is transmitted through CI interconnection.
- In co-site co-BBU scenarios, if this parameter is set to **INTRA_TRANS**, LTE-NR user-plane data is transmitted through the backplane.

 NOTE

In LTE-NR co-MPT and co-BBU separate-MPT using co-transmission through backplane interconnection scenarios, the **gNodeBParam.X2uTransmissionType** parameter must be set to **INTRA_TRANS**. Otherwise, SCG additions will fail in NSA DC. This restriction does not apply to other scenarios.

- In inter-site scenarios, LTE- and NR-only base stations in NSA networking support only IP transmission interconnection.

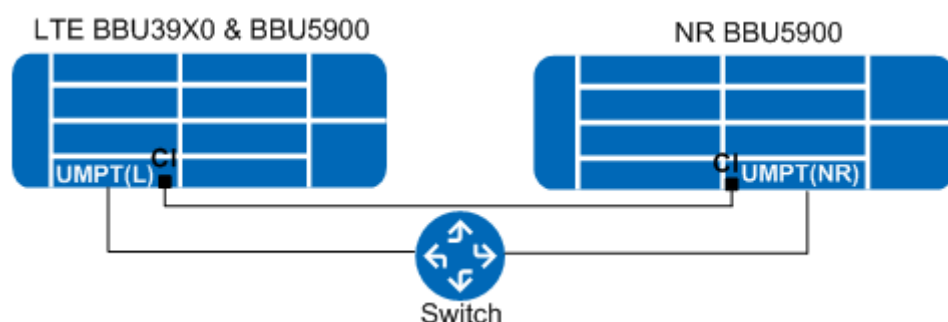
LTE BBU and NR BBU

LTE BBU and NR BBU networking scenarios are classified into **LTE-NR separate-BBU separate-MPT**, **LTE-NR co-BBU separate-MPT** and **LTE-NR co-BBU co-MPT**, and **IP transmission interconnection between LTE- and NR-only base stations**.

- LTE-NR separate-BBU separate-MPT

Figure 4-17 shows an NR base station using a BBU5900 or BBU5910 and an LTE base station using another BBU. The NR and LTE base stations use different main control boards. In this scenario, use a CI interconnection cable to connect the main control boards, and configure a **CTRLINK** MO for interconnected UMPT boards to implement infrastructure interconnection between separate-MPT base stations. For details, see *BBU Interconnection in 3900 & 5900 Series Base Station Product Documentation*. X2 control-plane data between LTE and NR is transmitted through IP transmission interconnection, and X2 user-plane data is transmitted through CI interconnection or IP transmission interconnection, depending on the setting of **gNodeBParam.X2uTransmissionType**.

Figure 4-17 LTE-NR separate-BBU networking

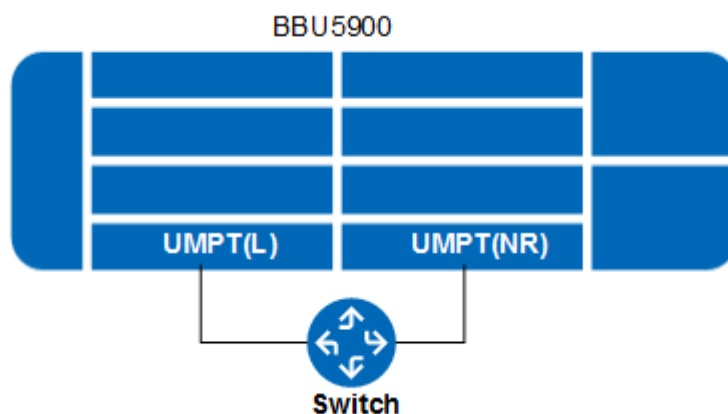


- LTE-NR co-BBU separate-MPT and LTE-NR co-BBU co-MPT

As shown in **Figure 4-18**, **Figure 4-19**, **Figure 4-20**, and **Figure 4-21**, the LTE and NR base stations share the same BBU5900 or BBU5910. They can use different main control boards or share the same main control board.

- As shown in **Figure 4-18**, X2 control-plane data between LTE and NR is transmitted through IP transmission interconnection, and X2 user-plane data is transmitted through the backplane or IP transmission interconnection.

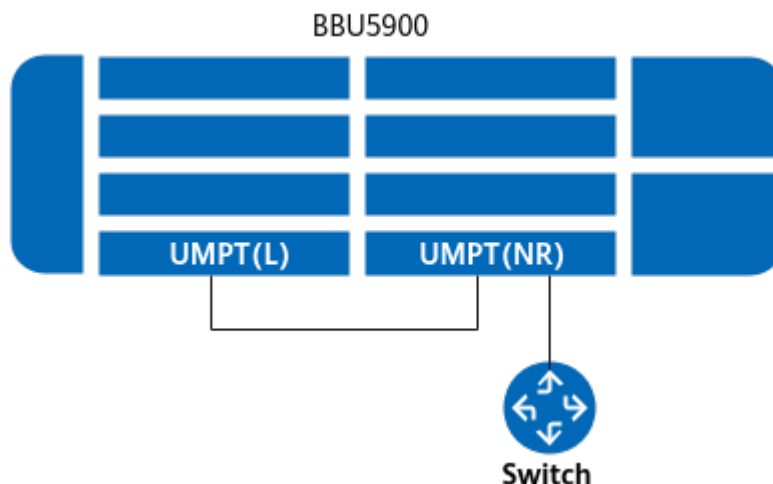
Figure 4-18 LTE-NR co-BBU separate-MPT networking



- As shown in **Figure 4-19**, X2 control-plane data between LTE and NR is transmitted in IP mode through the panel interconnection cable between LTE and NR, and X2 user-plane data between LTE and NR is transmitted in IP mode through the panel interconnection cable or the backplane. For

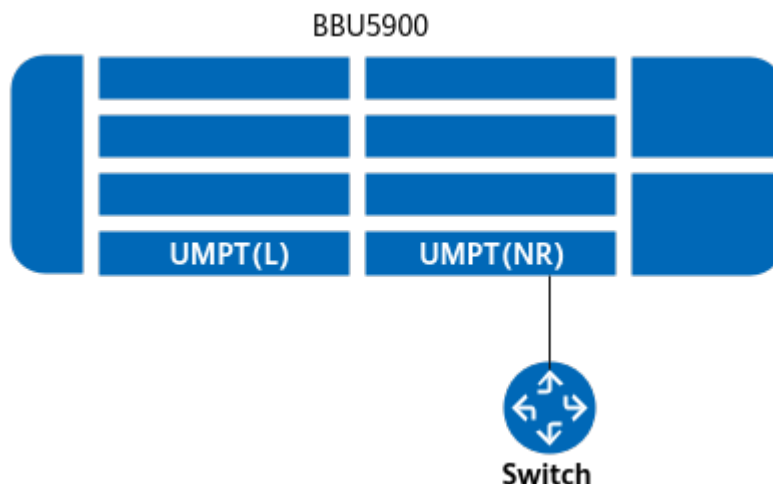
details about main-control-board-based co-transmission through panel interconnection, see *Common Transmission*.

Figure 4-19 Main-control-board-based LTE-NR co-transmission through panel interconnection



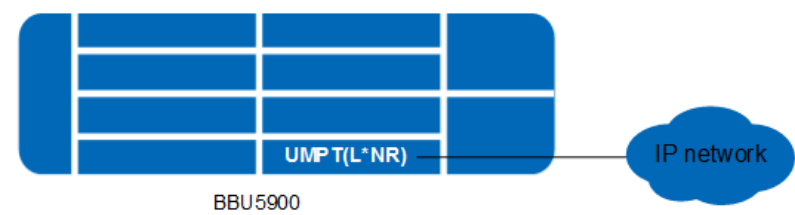
- As shown in [Figure 4-20](#), X2 control-plane data between LTE and NR is transmitted through the backplane between LTE and NR (requiring tunnels to be configured), and X2 user-plane data between LTE and NR is transmitted through the backplane. For details about tunnels and main-control-board-based co-transmission through backplane interconnection, see *Common Transmission*.

Figure 4-20 Main-control-board-based LTE-NR co-transmission through backplane interconnection



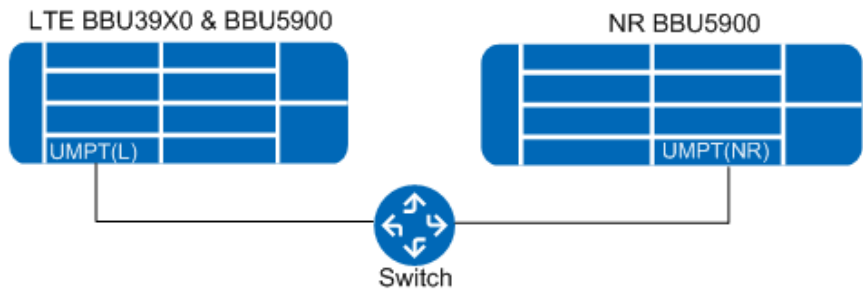
- As shown in [Figure 4-21](#), X2 control-plane and user-plane data between LTE and NR is exchanged within the UMPT.

Figure 4-21 LTE-NR co-BBU co-MPT networking



- IP transmission interconnection between LTE- and NR-only base stations
As shown in Figure 4-22, the NR base station uses a BBU5900 or BBU5910, LTE and NR use different BBUs, and both NR and LTE base stations are single-mode ones.
X2 control-plane and user-plane data between LTE and NR is transmitted through IP transmission interconnection.

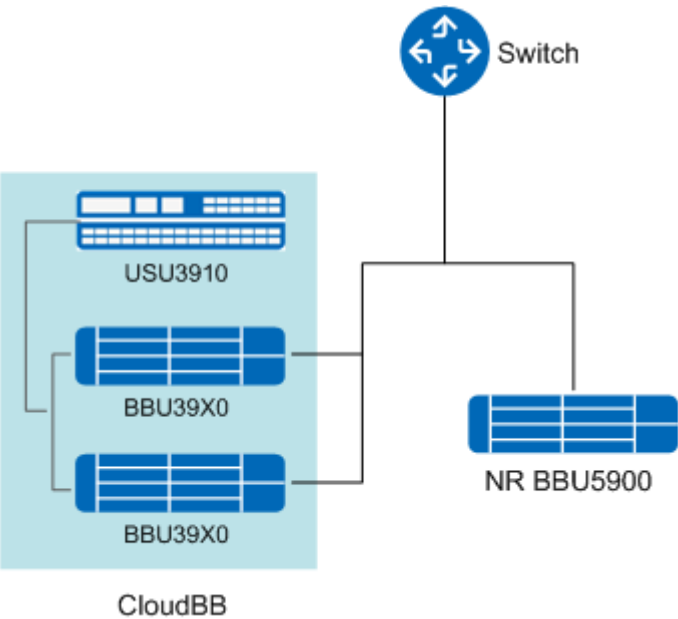
Figure 4-22 Interconnection between LTE and NR



LTE Cloud BB and NR BBU

Figure 4-23 shows a newly deployed NR network using a BBU5900 (not a BBU5910) and the existing LTE network using Cloud BB. In this scenario, only transmission interconnection is supported.

Figure 4-23 LTE Cloud BB and NR BBU networking



4.3.5 Cells

On the LTE side:

For uplink single-side transmission and secondary intermodulation interference avoidance, the **TDM_SWITCH** option of the **NsaDcMgmtConfig.NsaDcAlgoSwitch** parameter can be selected only when either of the following conditions is met: (1) **RACHCfg.PrachConfigIndexCfglnd** is set to **CFG** and **RACHCfg.PrachConfigIndex** is set to **0, 3, 6, 9, 13, 14, 16, 19, 22, 25, 29, 32, 35, 38, 41, 45, 48, 51, 54, or 57**; (2) **RACHCfg.PrachConfigIndexCfglnd** is set to **NOT_CFG**, **ENodeBAlgoSwitch.PrachTimeStagSwitch** is set to **OFF**, and the **RachAdjSwitch** option of **CellAlgoSwitch.RachAlgoSwitch** is deselected.

On the NR side:

Uplink fallback to LTE is not supported when the **NRDUCell.FrequencyBand** parameter is set to **N257, N258, N260, or N261**, and the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU**.

Secondary intermodulation interference avoidance

- The **NRDUCellPrach.PrachConfigurationIndex** parameter must be set to **0, 4, 8, 12, 16, 19, or 65535** when secondary intermodulation interference avoidance is enabled and all of the following conditions are met:
 - The **NRDUCell.DuplexMode** parameter is set to **CELL_FDD**.
 - The **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter is selected.
 - The **NRDUCellAlgoSwitch.NsaTdmPatternTypeStrategy** parameter is set to **LTE_UL_PREFERRED**.
- The **NRDUCellPrach.PrachConfigurationIndex** parameter must be set to **1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15, 17, 18, or 20** when secondary intermodulation interference avoidance is enabled and all of the following conditions are met:
 - The **NRDUCell.DuplexMode** parameter is set to **CELL_FDD**.
 - The **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter is selected.
 - The **NRDUCellAlgoSwitch.NsaTdmPatternTypeStrategy** parameter is set to **NR_UL_PREFERRED**.

Uplink single-side transmission: When this function is enabled, the **NRDUCellPrach.PrachConfigurationIndex** parameter must be set to **1, 2, 5, 6, 9, 10, 13, 14, 17, 18, or 21**.

4.3.6 Others

- UE requirements
 - The UE must support NSA DC specified in 3GPP Release 15.
 - The UE must have subscribed to LTE and NR services.
 - The UE must match the gNodeB and eNodeB versions.
 - TDM-based intermodulation interference avoidance, uplink power control, and uplink single-side transmission require the UE to support TDM.

- Uplink dynamic data split requires the support from the UE.
- Uplink CA in EN-DC requires the support from the UE. You can determine whether an NSA UE supports this function by checking *ca-BandwidthClassUL-EUTRA* under *NRBandCombination* of the *NRUE-MRDC-Capability* IE. For details about *ca-BandwidthClassUL-EUTRA*, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V16.19.0, section 4.2.7.1 "BandCombinationList parameters" in 3GPP TS 38.306 V16.14.0, and Table 5.6A-1 in 3GPP TS 36.101 V16.10.0.
- EPC requirements
 - The EPC must be CloudEPC to support Option 3 and Option 3x.
 - The EPC must support NSA DC. If the core network is provided by Huawei, see *WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management* for details.
 - If NSA DC is enabled on an eNodeB, the connected MMEs need to support NSA DC. If a connected MME does not support NSA DC, the **MmeCapInfo.MmeNsaDcCapability** parameter for this MME must be set to **NOT_SUPPORT**.
- Clock requirements
 - TDM requires time synchronization between LTE FDD and NR.

4.4 Operation and Maintenance

4.4.1 When to Use

It is recommended that this feature be enabled when operators require dual connectivity with the eNodeB and gNodeB or service provisioning by the gNodeB. For details about RAN sharing in NSA networking, see *RAN Sharing* in eRAN feature documentation and *Multi-Operator Sharing* in 5G RAN feature documentation.

SCG addition based on virtual grid models is not recommended in the following scenarios:

- High-speed mobility scenarios such as high-speed railway dedicated networks: requiring the prevention of network performance deterioration because of inaccurate NR coverage evaluation due to fast UE movement
- Big event assurance scenarios: requiring the prevention of network performance deterioration caused by the increase in CPU usage when this function is used
- Long-term network reconstruction scenarios: requiring the prevention of network performance deterioration caused by the decrease in the accuracy of virtual grid models

4.4.2 Data Configuration

4.4.2.1 Data Preparation

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellAlgoSwitch** MO on the NR side to configure NSA DC.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	LTE: NsaDcMgmtConfig.NsaDcAlgoSwitch	Select the NSA_DC_CAPABILITY_SWITCH option.
NSA DC Switch	NR: NRCellAlgoSwitch.NsaDcSwitch	The default value is ON .

The following table describes the parameters that must be set in the **PccFreqCfg** MO on the LTE side to configure PCC frequencies (or anchor frequencies).

NOTE

This MO and the anchoring priority do not need to be configured for non-anchor frequencies.

Parameter Name	Parameter ID	Setting Notes
PCC DL EARFCN	PccFreqCfg.PccDLEarfcn	Set this parameter based on the operator's network plan.
NSA PCC Anchoring Priority	PccFreqCfg.NsaPccAnchoringPriority	Set this parameter based on the network plan if an independent camping policy needs to be configured for NSA UEs.

The following table describes the parameters that must be set in the **NrScgFreqConfig** MO on the LTE side to configure the association between the PCC and SCG frequencies.

Parameter Name	Parameter ID	Setting Notes
PCC DL EARFCN	NrScgFreqConfig.PccDLEarfcn	Set this parameter based on the operator's network plan.
SCG DL ARFCN	NrScgFreqConfig.ScgDLArfcn	Set this parameter based on the operator's network plan.

Parameter Name	Parameter ID	Setting Notes
SCG DL ARFCN Priority	NrScgFreqConfig.ScgDLArfcnPriority	Set this parameter based on the operator's network plan.
NSA DC B1 Event RSRP Threshold	NrScgFreqConfig.NsaDcB1ThresholdRsrp	It is recommended that this parameter be set to a value not less than the coverage-based PSCell release threshold or handover threshold on the NR side. This prevents a PSCell release or handover from being triggered immediately after an SgNB is added.
NR B1 Time To Trigger	NrScgFreqConfig.NrB1TimeToTrigger	A smaller value of this parameter results in earlier event B1 reporting. A larger value of this parameter results in the opposite effect.

The following table describes the parameters that must be set in the **NrExternalCell** MO on the LTE side to configure external NR cells.

 NOTE

The frequency bands and additional frequency bands configured for external NR cells on the LTE side must be the same as those configured for the corresponding NR cells on the NR side.

Parameter Name	Parameter ID	Setting Notes
Downlink ARFCN	NrExternalCell.DLArfcn	Set this parameter based on the operator's network plan.
Uplink ARFCN	NrExternalCell.UlArfcn	Set this parameter based on the operator's network plan.
Physical Cell ID	NrExternalCell.PhyCellId	Set this parameter based on the operator's network plan.
Tracking Area Code	NrExternalCell.Tac	Set this parameter based on the operator's network plan.
Frequency Band	NrExternalCell.FrequencyBand	Set this parameter based on the operator's network plan.
Additional Frequency Band	NrExternalCell.AdditionalFrequencyBand	Set this parameter based on the operator's network plan.

The following table describes the parameters that must be set in the **NrNFreq** MO on the LTE side to configure neighboring NR frequencies.

Parameter Name	Parameter ID	Setting Notes
Downlink ARFCN	NrNFreq.DlArfcn	This frequency is the NR SSB frequency. Set this parameter based on the operator's network plan.
Uplink ARFCN	NrNFreq.UlArfcn	Set this parameter based on the operator's network plan.
SSB Period	NrNFreq.SsbPeriod	See 4.1.1.2.3 PSCell SMTC on the LTE Side .
SSB Offset	NrNFreq.SsbOffset	See 4.1.1.2.3 PSCell SMTC on the LTE Side .
SSB Subcarrier Spacing	NrNFreq.SubcarrierSpacing	Set this parameter to the same value as the NRDUCell.SsbSubcarrierSpacing parameter.

As specified in section 5.4.2.3 "Channel raster entries for each operating band" in 3GPP TS 38.104 V17.9.0, a frequency may belong to multiple NR frequency bands. In such cases, the corresponding band must be configured on the eNodeB. For example, a neighboring NR frequency whose NR-ARFCN is in the range of 422000–434000 belongs to bands n1 and n66. The **NrMfbiFreq** MO on the eNodeB side must be set based on the NR network plan, with this NR frequency configured in band n1 or n66.

If a frequency does not belong to multiple NR frequency bands, the NR MFBI parameters do not need to be set.

Parameter Name	Parameter ID	Setting Notes
Downlink ARFCN	NrMfbiFreq.DlArfcn	Set this parameter to the NR SSB frequency based on the operator's network plan.
Frequency Band	NrMfbiFreq.FrequencyBand	Set this parameter based on the operator's network plan.
Additional Frequency Band	NrMfbiFreq.AdditionalFrequencyBand	Set this parameter based on the operator's network plan.

The following table describes the parameters that must be set in the **NrNRelationship** MO on the LTE side to configure neighboring NR cells.

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrNRRelationship.GnodebId	Set this parameter based on the operator's network plan.
Local cell ID	NrNRRelationship.LocalCellId	Set this parameter based on the operator's network plan.
Blind Configuration Indicator	NrNRRelationship.BlindConfigIndicator	It is recommended that this parameter be set to TRUE if the LTE cell and the neighboring NR cell are co-sited and cover the same area or if the coverage area of the neighboring NR cell is larger than that of the LTE cell. Otherwise, set this parameter to FALSE. On an NR frequency, there is only one neighboring NR cell with the same coverage as an LTE cell. It is recommended that only one neighboring NR cell be configured on an NR frequency. If multiple neighboring NR cells are configured on one NR frequency, SgNB addition may fail.

 NOTE

NR cells with the same NR-ARFCN and PCI cannot be configured as neighboring cells of the same LTE cell.

The following table describes the parameters that must be set in the **NsaDcAlgoParam** MO on the LTE side to configure the gap offset allocation optimization switch.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	NsaDcAlgoParam.NsaDcAlgoSwitch	Select the GAP_OFS_ASSIGNMENT_OPT_SW option.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side to configure periodic SCG addition based on data volume.

Parameter Name	Parameter ID	Setting Notes
SCG Addition Interval	NsaDcMgmtConfig.ScgAdditionInterval	The default value is 60 . If NSA carrier combination selection based on user experience is not enabled and SCG addition needs to be quickly triggered, set this parameter to a smaller value.
SCG Addition Buffer Length Threshold	NsaDcMgmtConfig.ScgAdditionBufferLenThld	The default value is 0 . It is recommended that this parameter be set to a value greater than 0 to avoid frequent SCG additions triggered by small-packet services.
SCG Addition Buffer Delay Threshold	NsaDcMgmtConfig.ScgAdditionBufferDelayThld	The recommended value is 5 .

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** and **UeCompat** MOs on the LTE side to configure the "preferential selection of a band combination with only one NR carrier for NSA UEs" function.

Parameter Name	Parameter ID	Setting Notes
Multi Networking Option Opt Switch	EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw	Select the NSA_NR_SINGLE_BC_PREFER_SW option of this parameter if the "preferential selection of a band combination with only one NR carrier for NSA UEs" function is required.
Whitelist Control Extension Switch 1	UeCompat.WhitelistCtrlExtSwitch1	Select the NSA_NR_SINGLE_BC_PREFER_SW option of this parameter if the "preferential selection of a band combination with only one NR carrier for NSA UEs" function needs to take effect only for whitelisted UEs.

The following table describes the parameters that must be set in the **NsaDcAlgoParam** and **NrScgFreqConfig** MOs on the LTE side to configure "differentiated SCG addition threshold configuration for NR CA-capable UEs".

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Extension Switch	NsaDcAlgoParam.NsaDcAlgoExtSwitch	Select the SCG_ADD_B1_THLD_DIFF_SW option of this parameter if "differentiated SCG addition threshold configuration for NR CA-capable UEs" is required.
SCG Addition RSRP Thld Offset for CA UEs	NrScgFreqConfig.CaUeScgAddRsrpThldOffset	Set this parameter based on the operator's network plan.
Whitelist Control Extension Switch 2	UeCompat.WhitelistCtrlExtSwitch2	Select the SCG_ADD_B1_THLD_DIFF_SW option of this parameter if "differentiated SCG addition threshold configuration for NR CA-capable UEs" only needs to take effect for whitelisted UEs.

The following table describes the parameters that must be set in the **NsaDcAlgoParam** and **NrScgFreqConfig** MOs on the LTE side to configure the "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function.

Parameter Name	Parameter ID	Setting Notes
NR TDD SCC RSRP Thld	NsaDcAlgoParam.NrTddSccRsrpThld	Set this parameter to a value other than 255 based on the operator's network plan if the "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function is required.
SCG Addition RSRP Thld Offset for CA UEs	NrScgFreqConfig.CaUeScgAddRsrpThldOffset	Set this parameter based on the operator's network plan.
Whitelist Control Extension Switch 2	UeCompat.WhitelistCtrlExtSwitch2	Select the SCG_ADD_B1_THLD_DIFF_SW option if the "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function only needs to take effect for whitelisted UEs.

The following table describes the parameters that must be set in the **NRCellNsaDcConfig** MO on the NR side to configure "differentiated SCG deletion threshold configuration for NR CA-capable UEs".

Parameter Name	Parameter ID	Setting Notes
CA-capable UE PSCell A2 RSRP Thld Offset	NRCellNsaDcConfig.CaCapUePscellA2RsrpThldOfs	Set this parameter to a non-zero value based on the operator's network plan if "differentiated SCG deletion threshold configuration for NR CA-capable UEs" is required.

The following table describes the parameters that must be set in the **CellQciPara**, **BRDRESASSIGNMENT**, and **NsaDcQciParamGroup** MOs on the LTE side and the **gNBPDcpParamGroup** MO on the NR side to configure downlink data split modes.

Parameter Name	Parameter ID	Setting Notes
NSA DC Default Bearer Mode	LTE: CellQciPara.NsaDcDefaultBearerMode	Set this parameter based on the operator's network plan.
Board Resource Assignment Mode	LTE: BRDRESASSIGNMENT.BRDASSIGNMENT	The default parameter value is MODE1 . Set this parameter to MODE3 if the CellQciPara.NsaDcDefaultBearerMode parameter is set to MCG_SPLIT_BEARER . The change in the value of this parameter may change the transmission specifications and signaling specifications of the main control board.
Downlink Data PDCP Split Mode	LTE: NsaDcQciParamGroup.DlDataPdcpsplitMode	In Option 3, use this parameter to configure the downlink data split mode at the PDCP layer on the LTE side. Set this parameter based on the operator's network plan.

Parameter Name	Parameter ID	Setting Notes
Downlink Data PDCP Split Mode	NR: gNBpdcpparamgroup.DlDataPdcpsplitmode	In Option 3x, use this parameter to configure the downlink data split mode at the PDCP layer on the NR side. Set this parameter based on the operator's network plan.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellNsaDcConfig**, **gNodeBParam**, and **gNBpdcpparamgroup** MOs on the NR side to support downlink real-time data split in Option 3x.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Extension Switch	LTE: NsaDcMgmtConfig.NsaDcAlgoExtSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.
NSA DC Algorithm Switch	NR: NRCellNsaDcConfig.NsaDcAlgoSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.
NSA DC Optimization Switch	NR: gNodeBParam.NsaDcOptSwitch	Select the NSA_PDCP_SPLIT_OPT_SW option.
Downlink Data PDCP Split Mode	NR: gNBpdcpparamgroup.DlDataPdcpsplitmode	The value SCG_AND_MCG is recommended.

The following table describes the parameters that must be set in the **NsaDcAlgoParam** and **NsaDcMgmtConfig** MOs on the LTE side and the **NRCellNsaDcConfig** MO on the NR side to enable downlink data split to be resumed adaptively in Option 3.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	LTE: NsaDcAlgoParam.NsaDcAlgoSwitch	Select the NSA_SPLIT_ADAPT_RECOVERY_SW option.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Extension Switch	LTE: NsaDcMgmtConfig.NsaDcAlgoExtSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.
NSA DC Algorithm Switch	NR: NRCellNsaDcConfig.NsaDcAlgorithmSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellNsaDcConfig** and **gNodeBParam** MOs on the NR side to enable downlink data split to be resumed adaptively in Option 3x.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Extension Switch	LTE: NsaDcMgmtConfig.NsaDcAlgoExtSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.
NSA DC Optimization Switch	NR: gNodeBParam.NsaDcOptSwitch	Select the NSA_SPLIT_ADAPT_RECOVERY_SW option.
NSA DC Algorithm Switch	NR: NRCellNsaDcConfig.NsaDcAlgorithmSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.

The following table describes the parameters that must be set in the **NsaDcQciParamGroup** MO on the LTE side and the **gNBPDcpParamGroup** MO on the NR side to configure uplink data split modes.

Parameter Name	Parameter ID	Setting Notes
Uplink Data Split Threshold	LTE: NsaDcQciParamGroup.UplinkDataSplitThreshold	The value INFINITY is recommended.
Uplink Data Split Primary Path	LTE: NsaDcQciParamGroup.UplinkDataSplitPrimaryPath	The value SCG is recommended.

Parameter Name	Parameter ID	Setting Notes
Uplink Data Split Threshold	NR: gNBpdcPParamGroup.UIDataSplitThreshold	The value INFINITY is recommended.
Uplink Data Split Primary Path	NR: gNBpdcPParamGroup.UIDataSplitPrimaryPath	The value SCG is recommended.

 NOTE

When uplink data is transmitted over the X2 link and packet loss occurs on the X2 link, the transmission rate of TCP services decreases. If the packet loss rate over the X2 link (the proportion of the number of lost packets to the total number of packets in a certain period over the X2 link) is greater than 0.0001%, it is recommended that the uplink data split mode be set to **MCG_ONLY** in the Option 3 architecture and be set to **SCG_ONLY** in the Option 3x architecture.

The following table describes the parameters that must be set in the **NRCellNsaDcConfig** and **NRDUCellSrsMeas** MOs on the NR side to configure uplink fallback to LTE.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	NRCellNsaDcConfig.NsaDcAlgorithmSwitch	Select the UL_FALLBACK_TO_LTE_SWITCH option.
NSA Uplink Fallback to LTE SINR Thld	NRDUCellSrsMeas.NsaUlfackToLteSinrThld	The value -30 is recommended.
NSA Uplink Fallback to LTE SINR Hyst	NRDUCellSrsMeas.NsaUlfackToLteSinrHyst	The value 30 is recommended.

The following table describes the parameter that must be set in the **gNodeBParam** MO on the NR side to configure the function of requesting LTE-side large-packet status of ULFB UEs.

Parameter Name	Parameter ID	Setting Notes
NSA DC Optimization Switch	gNodeBParam.NsaDcOptSwitch	Select the ULFB_LTE_UL_LRG_PKT_REQ_SW option if the gNodeB is required to obtain the LTE-side large- and small-packet status of ULFB UEs.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side to configure initial power control in NSA DC.

Parameter Name	Parameter ID	Setting Notes
NSA DC UE MCG Uplink Maximum Power	NsaDcMgmtConfig.NsaDcUeMcgUlMaximumPower	The value 20 is recommended.
NSA DC UE SCG Uplink Maximum Power	NsaDcMgmtConfig.NsaDcUeScgUlMaximumPower	The value 20 is recommended.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRDUCellAlgoSwitch** and **NRDUCellNsaDcConfig** MOs on the NR side to configure interference avoidance.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	LTE: NsaDcMgmtConfig.NsaDcAlgoSwitch	Select the INTERFERENCE_AVOID_SW and TDM_SWITCH options.
NSA DC Algorithm Switch	NR: NRDUCellAlgoSwitch.NsaDcAlgoSwitch	- If the interference type is secondary intermodulation interference, select the CROSS_MDLT_INTRF_AVOID_SW option. - If the interference type is secondary harmonic interference, select the HARMONIC_INTRF_AVOID_SW option.

Parameter Name	Parameter ID	Setting Notes
IMD Intrf Freq Domain Avoidance Type	LTE: NsaDcMgmtConfig.ImdIntrfFreqAvoidanceType	To configure secondary intermodulation interference avoidance in the frequency domain, select the PRIOR_ALLOC_NO_INTRF_RB option.
Harmonic Interference Avoidance Range	LTE: NsaDcMgmtConfig.HarmonicIntrfAvoidRange	In secondary harmonic interference avoidance, set this parameter to NR_DL_COMMON_CHANNEL if NR common channel avoidance is required and to NR_DL_COMMON_AND_DATA_CHANNEL if avoidance of both NR common channels and data channels is required.
Harmonic Interference Avoidance Range	NR: NRDUCellNsaDcConfig.HarmonicIntrfAvoidRange	In secondary harmonic interference avoidance, set this parameter to NR_DL_COMMON_CHANNEL if NR common channel avoidance is required and to NR_DL_COMMON_AND_DATA_CHANNEL if avoidance of both NR common channels and data channels is required.
NSA DC Algorithm Switch	LTE: NsaDcAlgoParam.NsaDcAlgoSwitch	Select the LNR_RELATIVE_FRM_OFFSET_APT_SW option if automatic acquisition of the LTE-NR relative frame offset is required.

The following table describes the parameters that must be set in the **NRCellNsaDcConfig** MO on the NR side to specify the NR data volume reporting policy in Option 3x.

Parameter Name	Parameter ID	Setting Notes
NR Data Volume Reporting Config	NRCellNsaDcConfig.NrDataVolumeRptCfg	- The recommended value is 65535 if NR data volume reporting needs to be triggered only by events. - The recommended value is 600 if both event-based triggering and periodic triggering of NR data volume reporting are required.

The following table describes the parameters that must be set in the **NRCellQciBearer** MO on the NR side to add the AM or UM PDCP parameter group ID for the corresponding QCI.

Parameter Name	Parameter ID	Setting Notes
AM PDCP Parameter Group ID	NRCellQciBearer.AmPdcparGroupID	Set this parameter based on the operator's network plan.
UM PDCP Parameter Group ID	NRCellQciBearer.UmPdcparGroupID	Set this parameter based on the operator's network plan.

The following table describes the parameters that must set in the **gNBpdcparParamGroup** MO on the NR side to specify the PDCP parameter group ID.

Parameter Name	Parameter ID	Setting Notes
PDCP Parameter Group ID	gNBpdcparParamGroup.PdcparGroupID	Set this parameter based on the value of NRCellQciBearer.AmPdcparGroupID or NRCellQciBearer.UmPdcparGroupID configured for the corresponding QCI.

The following table describes the parameters that must be set in the **NRCellOpQciBearer** MO on the NR side to configure operator-specific uplink and downlink data split modes in Option 3x.

Parameter Name	Parameter ID	Setting Notes
AM PDCP Parameter Group ID	NRCellOpQciBearer.AmPdcParamGroupId	Set this parameter based on the operator's network plan.
UM PDCP Parameter Group ID	NRCellOpQciBearer.UmPdcParamGroupId	Set this parameter based on the operator's network plan.

If the UE supports shortSN and the PDCP sequence numbers configured on the LTE and NR sides are different, then dedicated bearers with the same QCI will fail to be set up on the LTE and NR sides, and the default bearer will be released. It is recommended that the PDCP sequence numbers be set to the same value on the LTE and NR sides. For details about the shortSN, see section 4.2.4 "PDCP Parameters" in 3GPP TS 38.306 V15.5.0.

The NSA DC feature splits data at the PDCP layer. It is recommended that the PDCP SN size be set to 18 bits on both the LTE and NR sides to avoid data transmission interruption caused by PDCP SN size insufficiency. The configuration method is as follows:

- On the LTE side, set the **QciPara.NsaDcRlcPdcParamGroupId** parameter to the ID of an RLC/PDCP parameter group used in NSA DC scenarios; set the **RlcPdcParamGroup.AmPdcSnSize** parameter in the parameter group to **AmPdcSnSize_18bits**.
- On the NR side, set the **gNBPDcpParamGroup.DIPdcSnSize** parameter to **BITS18**.

The following table describes the parameters that must be set in the **RlcPdcParamGroup** MO on the LTE side and the **NRCellQciBearer** and **gNBPDcpParamGroup** MOs on the NR side to configure the RLC mode and PDCP SN size.

Parameter Name	Parameter ID	Setting Notes
RLC-UM or RLC-AM mode	RlcPdcParamGroup.RlcMode	Set this parameter based on the QCI.
AM Mode PDCP SN Length	RlcPdcParamGroup.AmPdcSnSize	It is recommended that this parameter be set to AmPdcSnSize_18bits when RlcPdcParamGroup.RlcMode is set to RlcMode_AM .

Parameter Name	Parameter ID	Setting Notes
UM PDCP-SN size	RlcPdcPParaGroup.PdcPsnSize	It is recommended that this parameter be set to PdcPsnSize_18bits when RlcPdcPParaGroup.RlcMode is set to RlcMode_UM .
RLC Mode	NrCellQciBearer.RlcMode	Set this parameter based on the QCI. If the RLC mode configured for bearers of the same QCI is inconsistent between LTE and NR, the RLC mode configured on the LTE side is used, and this parameter does not take effect.
Downlink PDCP SN Size	gNBpdcPParamGroup.DlPdcPsnSize	The value BITS18 is recommended.

The following table describes the parameters that must be set in the **NrExternalCell** and **NrExternalCellPlmn** MOs on the LTE side to configure the external NR cell networking mode in NSA and SA hybrid networking.

Parameter Name	Parameter ID	Setting Notes
NR Networking Option	NrExternalCell.NrNetworkingOption	Set this parameter based on the operator's network plan.
NR Networking Option	NrExternalCellPlmn.NrNetworkingOption	Set this parameter based on the operator's network plan.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** and **CellQciPara** MOs on the LTE side to configure an adaptive PSCell management policy for VoLTE UEs.

Parameter Name	Parameter ID	Setting Notes
VoLTE UE SCG Management Strategy	NsaDcMgmtConfig.VolteUeScgMgmtStrategy	Set this parameter based on the operator's network plan.

Parameter Name	Parameter ID	Setting Notes
VoLTE Packet Loss Rate Thld for SCG Release	NsaDcMgmtConfig.VoltePlrThldForScgRelease	The default value is 5%. If this parameter is set to a large value, voice quality deteriorates. Therefore, it is recommended that this parameter be set to a value less than or equal to 5%.
NSA DC Optimization Switch	CellQciPara.NsaDcOptSwitch	Select the QCI_VOLTE_SCG_COEXIST_SW option if VoLTE services and SCGs are expected to coexist for UEs with bearers of a specific QCI.
NSA DC Algorithm Enhancement Switch	NsaDcMgmtConfig.NsaDcAlgorithmEnhSwitch	It is recommended that the VOLTE_NSA_DC_IF_MEAS_OPTION_SW option be selected to enable inter-frequency and inter-RAT measurement optimization for VoLTE UEs in NSA DC scenarios if the eNodeB is required not to deliver A5 measurement configurations for NSA PCC anchoring and B1 measurement configurations for SCG addition to VoLTE UEs when QCI-specific VoLTE and SCG coexistence does not take effect for VoLTE UEs and the NsaDcMgmtConfig.VolteUeScgMgmtStrategy parameter is not set to VOLTE_SCG_COEXISTENCE .

The following table describes the parameters that must be set in the **NsaDcAlgoParam** MO on the LTE side to specify the maximum X2 transmission rate for PDCP packets in Option 3.

Parameter Name	Parameter ID	Setting Notes
X2 Transmission Max Rate	NsaDcAlgoParam.X2TransMaxRate	• The default value 1 is recommended to prevent the NR network from being affected by excessively high data rates allocated to the NR side. • When dynamic data split requires the NR network to support a larger air interface capability, it is recommended that this parameter be set to a value greater than the NR air interface capability and the final data transmission rate is subject to the air interface capability. If the configured value of this parameter is less than the NR air interface capability, the actual data transmission rate is the configured value of this parameter. • The value of this parameter must be smaller than the physical transmission bandwidth. • If this parameter is set to a value far less than the NR air interface capability, data forwarding will take a long time. If a handover occurs during data forwarding, packet loss occurs.

The following table describes the parameters that must be set in the **gNodeBParam** MO on the NR side to configure the maximum X2 transmission rate for PDCP packets when LTE-NR user-plane data is transmitted in Option 3x.

Parameter Name	Parameter ID	Setting Notes
Max X2 Transmission Rate	gNodeBParam.MaxX2TransRate	• The default value 1 is recommended to prevent the LTE network from being affected by excessively high data rates allocated to the LTE side. • When dynamic data split requires the LTE network to support a larger air interface capability, it is recommended that this parameter be set to a value greater than the LTE air interface capability and the final data transmission rate is subject to the air interface capability. If the configured value of this parameter is less than the LTE air interface capability, the actual data transmission rate is the configured value of this parameter. • The value of this parameter must be smaller than the physical transmission bandwidth. • The default value of the gNodeBParam.MaxX2TransRate parameter is 100Mbits. Adjust the parameter based on transmission conditions when enabling SN-terminated MCG bearer transmission. • If this parameter is set to a value far less than the NR air interface capability, data forwarding will take a long time. If a handover occurs during data forwarding, packet loss occurs.

The following table describes the parameters that must be set in the **gNodeBParam** MO on the NR side to configure the maximum X2 transmission

rate for PDCP packets when user-plane data is transmitted through the backplane or cascading interface (CI) in Option 3x scenarios where LTE and NR are co-sited.

Parameter Name	Parameter ID	Setting Notes
Intra-Site Max X2 Transmission Rate	gNodeBParam.IntraSiteMaxX2TransRate	• The default value of this parameter is 0, indicating that the maximum X2 transmission rate for PDCP packets is not configured based on the transmission path but determined by the configured value of the gNodeBParam.MaxX2TransRate parameter. • If this parameter is set to a non-zero value, the maximum X2 transmission rate for PDCP packets is configured based on the transmission path. – If LTE-NR user-plane data is transmitted through the backplane or CI, the maximum X2 transmission rate for PDCP packets is determined by the configured value of the gNodeBParam.IntraSiteMaxX2TransRate parameter. – If LTE-NR user-plane data is transmitted through IP, the maximum X2 transmission rate for PDCP packets is determined by the configured value of the gNodeBParam.MaxX2TransRate parameter. • When dynamic data split requires the LTE network to support a larger air interface capability, it is recommended that this parameter be set to a value greater than the LTE air interface capability. If the configured value of this parameter is less than the LTE air interface capability,

Parameter Name	Parameter ID	Setting Notes
		the actual data transmission rate is the configured value of this parameter. The value of this parameter must be smaller than the physical transmission bandwidth.

The following table describes the parameters that must be set in the **gNodeBParam** MO on the NR side to configure fast retransmission for downlink data split.

Parameter Name	Parameter ID	Setting Notes
NSA DC Optimization Switch	gNodeBParam.NsaDcOptSwitch	Select the NSA_DC_FAST_RETRANS_SWITCH option.

The following table describes the parameters that must be set in the **gNodeBParam** and **gNBUECompatSw** MOs on the NR side to configure the NSA DC SRB3 function and the NSA DC SRB3 blacklist function.

Parameter Name	Parameter ID	Setting Notes
NSA DC Optimization Switch	gNodeBParam.NsaDcOptSwitch	Select the NSA_DC_SRB3_SWITCH option when NSA DC SRB3 is required.
Blacklist Control Extension Switch	gNBUECompatSw.BlacklistCtrlExtSwitch	It is recommended that the NSA_DC_SRB3_SWITCH_OFF option be selected when NSA DC SRB3 is enabled on the live network and some UEs are incompatible with SRB3.

The following table describes the parameters that must be set in the **NrScgFreqConfig** MO on the LTE side and the **gNBDedNsaDcBandComb** and **NRCellNsaDcConfig** MOs on the NR side to configure uplink single-side transmission frequency band combinations.

Parameter Name	Parameter ID	Setting Notes
Frame Offset or FDD Frame Offset	LTE: CellFrameOffset.FrameOffset or ENodeBFrameOffset.FddFrameOffset	- If the CellFrameOffset.FrameOffset parameter is set, the frame offset is determined by this parameter. - If the CellFrameOffset.FrameOffset parameter is not set, the frame offset is determined by the ENodeBFrameOffset.FddFrameOffset parameter. - The frame offset is 0 ms in time synchronization scenarios if neither of the preceding parameters is set. - The parameter value must be the same as that configured on the NR side.
PCC DL EARFCN	LTE: NrScgFreqConfig.PccDLEarfcn	Set this parameter based on the operator's network plan.
SCG DL ARFCN	LTE: NrScgFreqConfig.ScgDLArfcn	Set this parameter based on the operator's network plan.
Aggregation Attribute	LTE: NrScgFreqConfig.AggregationAttribute	Set this parameter to SINGLE_UL_NSA_FLAG .
NSA DC Algorithm Switch	LTE: NsaDcMgmtConfig.NsaDcAlgorithmSwitch	Select the TDM_SWITCH option.
Frame Offset	NR: gNBFreqBandConfig.FrameOffset	The frame offset is 0 ms in time synchronization scenarios if this parameter is not set. The frame offset on the NR side must be the same as that on the LTE side.
LTE Band ID	NR: gNBDedNsaDcBandComb.LteBandId	Set this parameter based on the operator's network plan.
NR Band ID	NR: gNBDedNsaDcBandComb.NrBandId	Set this parameter based on the operator's network plan.
Aggregation Attribute	NR: gNBDedNsaDcBandComb.AggregationAttribute	Set this parameter to SINGLE_UL_NSA_FLAG .

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	NR: NRCellNsaDcConfig.NsaDcAlgorithmSwitch	Select the SINGLE_UL_NSA_SW option.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side to configure NSA uplink preallocation in Option 3.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	NsaDcMgmtConfig.NsaDcAlgorithmSwitch	Select the NSA_DC_PREALLOCATION_SW option.

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellNsaDcConfig** MO on the NR side to configure NSA uplink preallocation in Option 3x.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	LTE: NsaDcMgmtConfig.NsaDcAlgorithmSwitch	Select the NSA_DC_PREALLOCATION_SW option.
NSA DC Algorithm Switch	NR: NRCellNsaDcConfig.NsaDcAlgorithmSwitch	Select the NSA_DC_PREALLOCATION_SW option.

(Optional) The following table describes the parameters that must be set in the **NrScgFreqConfig** MO on the LTE side to configure an LTE+NR intra-band combination.

Parameter Name	Parameter ID	Setting Notes
PCC DL EARFCN	NrScgFreqConfig.PccDLEarfcn	Set this parameter based on the operator's network plan.
SCG DL ARFCN	NrScgFreqConfig.ScgDLArfcn	Set this parameter based on the operator's network plan.

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	NrScgFreqConfig.AggregationAttribute	If the eNodeB PCC band (or a neighboring eNodeB PCC band) and the downlink SCG band have the same band number (for example, B3 and N3), it is recommended that the CO_DEPLOYMENT_NSA_FLAG option of this parameter be selected.

(Optional) The following table describes the parameters that must be set in the **NrNRRelationship** MO on the LTE side to configure a co-site intra-band pair of LTE and NR cells.

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrNRRelationship.GnodebId	Set this parameter based on the operator's network plan.
Local cell ID	NrNRRelationship.LocalCellId	Set this parameter based on the operator's network plan.
Aggregation Attribute	NrNRRelationship.AggregationAttribute	If the LTE cell and the neighboring NR cell are deployed in the same band (for example, B3 and N3) and on the same site, it is recommended that the CO_DEPLOYMENT_NSA_FLAG option of this parameter be selected.

(Optional) The following table describes the parameters that must be set in the **gNBDedNsaDcBandComb** MO on the NR side to configure an LTE+NR intra-band combination.

Parameter Name	Parameter ID	Setting Notes
LTE Band ID	gNBDedNsaDcBandComb.LteBandId	Set this parameter based on the operator's network plan.
NR Band ID	gNBDedNsaDcBandComb.NrBandId	Set this parameter based on the operator's network plan.

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	gNBDedNsaDcBand-Comb.AggregationAttribute	If the gNodeB band (or a neighboring gNodeB band) and the LTE anchor band are the same (for example, B3 and N3), it is recommended that the CO_DEPLOYMENT_NSA_FLAG option of this parameter be selected.

(Optional) The following table describes the parameters that must be set in the **gNBNrLteCellComb** MO on the NR side to configure the CGIs of a co-site intra-band pair of LTE and NR cells.

Parameter Name	Parameter ID	Setting Notes
NR MCC	gNBNrLteCellComb.Mcc	Set this parameter based on the operator's network plan.
NR MNC	gNBNrLteCellComb.Mnc	Set this parameter based on the operator's network plan.
gNodeB ID	gNBNrLteCellComb.gNodeBId	Set this parameter based on the operator's network plan.
Cell ID	gNBNrLteCellComb.CellId	Set this parameter based on the operator's network plan.
LTE MCC	gNBNrLteCellComb.LteMcc	Set this parameter based on the operator's network plan.
LTE MNC	gNBNrLteCellComb.LteMnc	Set this parameter based on the operator's network plan.
eNodeB ID	gNBNrLteCellComb.eNodeBId	Set this parameter based on the operator's network plan.
LTE Cell ID	gNBNrLteCellComb.LteCellId	Set this parameter based on the operator's network plan.
Aggregation Attribute	gNBNrLteCellComb.AggregationAttribute	If the LTE cell and the neighboring NR cell are deployed in the same band (for example, B3 and N3) and on the same site, it is recommended that the CO_DEPLOYMENT_NSA_FLAG option of this parameter be selected.

The following table describes the parameters that must be set in the **NRCellNsaDcConfig** and **NRCellNsaDcConfigGrp** MOs on the NR side to configure SgNB release based on RSRP or SSB SINR.

Parameter Name	Parameter ID	Setting Notes
PSCell A2 RSRP Threshold	NRCellNsaDcConfig.PscellA2RsrpThld	It is recommended that this parameter be set to a value less than the NSA DC event B1 RSRP threshold on the LTE side to prevent PSCell addition from being triggered immediately after SgNB release.
PSCell A2 RSRP Thld Offset	NRCellNsaDcConfigGrp.PscellA2RsrpThldOffset	It is recommended that the sum of NRCellNsaDcConfig.PscellA2RsrpThld and NRCellNsaDcConfigGrp.PscellA2RsrpThldOffset be less than the NSA DC event B1 RSRP threshold on the LTE side to prevent PSCell addition from being triggered immediately after SgNB release.
PSCell A2 Time To Trigger	NRCellNsaDcConfigGrp.PscellA2TimeToTrig	A smaller value of this parameter results in earlier event A2 reporting. A larger value of this parameter results in the opposite effect.
NSA DC Algorithm Switch	NRCellNsaDcConfig.NsaDcAlgoSwitch	Select the SINR_BASED_SCG_RELEASE_SW option.
PSCell A2 SINR Threshold	NRCellNsaDcConfigGrp.PscellA2SinrThld	A smaller value of this parameter results in a lower probability of removing the SCG of an NSA UE. A larger value of this parameter results in the opposite effect.

NOTE

The PSCell A2 hysteresis has a fixed value of 1 dB and it is unconfigurable.

(Optional) In Option 3x, the MeNB filters out neighboring NR cells under any SgNB whose S1-U interface is unavailable. The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** and **gNodeBParam** MOs.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	EnodebAlgoExtSwitch.NsaDcAlgoSwitch	Select the S1U_STATUS_BASED_SCG_ADD_SW option.
NSA DC Optimization Switch	gNodeBParam.NsaDcOptSwitch	Select the S1U_STATUS_SEND_SW option.

(Optional) The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** and **NrExternalCell** MOs on the LTE side and the **gNodeBParam** MO on the NR side to configure SCG addition when LTE and NR belong to different PLMNs.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	EnodebAlgoExtSwitch.NsaDcAlgoSwitch	If the EQUIVALENT_PLMN_ADD_SCG_SW option is selected, the MeNB can select NR frequencies or NR cells based on the equivalent PLMN information delivered by the core network.
NSA DC Algorithm Switch	EnodebAlgoExtSwitch.NsaDcAlgoSwitch	If the core network does not deliver equivalent PLMN information, it is recommended that the INTER_PLMN_ADD_SCG_SW option be selected. Under this setting, the MeNB can select NR frequencies or NR cells based on the inter-PLMN handover list specified by the InterPlmnHoList MO.
NSA DC Optimization Switch	gNodeBParam.NsaDcOptSwitch	If the SCG_ADD_PLMN_SELECT_POLICY_SW option is selected, the SgNB performs admission decision based on the <i>Select PLMN</i> IE in the SgNB Addition Request message.

Parameter Name	Parameter ID	Setting Notes
Master PLMN Reserved Flag	NrExternalCell.MasterPlmnReservedFlag	Set this parameter to TRUE if the gNodeB serving an external NR cell works in RAN sharing with dedicated carrier and the external cell belongs to a secondary operator.

The following table describes the parameters that must be set in the **NsaDcAlgoParam** and **RrcConnStateTimer** MOs on the LTE side to configure SCG release and addition based on UE overheating status reporting.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	NsaDcAlgoParam.NsaDcAlgoSwitch	Select the OVERHEATING_RPOTECTION_SW option.
Overheating Indication Prohibit Timer	RrcConnStateTimer.OverheatingIndProhibitTmr	The value S120 is recommended.

The following table describes the parameters that must be set in the **eNodeBResModeAlgo** MO on the LTE side to set the service mode for the eNodeB resource allocation algorithms.

Parameter Name	Parameter ID	Setting Notes
Service Mode	eNodeBResModeAl- go.ServiceMode	• If the main control board is a UMPTg, select the UMPTG_ENHANCEMENT_SWITCH option. You are advised to select this option only for base stations where the number of RRC_CONNECTED UEs during busy hours and the number of busy hour call attempts (BHCAs) are less than 80% of their respective maximum numbers supported by the main control board. Otherwise, the RRC connection setup success rate and handover success rate may decrease. • If the main control board is a UMPTe or UMPTga, select the AI_ENHANCEMENT_OPT_SWITCH option. You are advised to select this option only for base stations where the number of RRC_CONNECTED UEs during busy hours and the number of BHCAs are less than 65% of their respective maximum numbers supported by the main control board. Otherwise, the RRC connection setup success rate and handover success rate may decrease. • If the main control board is a UMPTha, select the AI_ENHANCEMENT_OPT_SWITCH option. You are advised to select this option only for base stations where the number of RRC_CONNECTED UEs during busy hours and the number of BHCAs are less

Parameter Name	Parameter ID	Setting Notes
		than 80% of their respective maximum numbers supported by the main control board. Otherwise, the RRC connection setup success rate and handover success rate may decrease. For details about the maximum number of RRC_CONNECTED UEs and the maximum number of BHCAs supported by a main control board, see related descriptions of the main control board in the "LTE Technical Specifications of Main Control Boards" section in <i>3900 & 5900 Series Base Station Product Documentation</i>.

The following table describes the parameters that must be set in the **LnrCarrierSelection** MO on the LTE side to configure functions related to NSA DC virtual grid model building.

Parameter Name	Parameter ID	Setting Notes
LNR Carrier Selection Switch	LnrCarrierSelection.LnrCarrierSelectionSw	Select the LNR_VIRTUAL_GRID_SW option to enable LNR virtual grid model building. Select the NSA_ANCHOR_VIRTUAL_GRID_SW option to enable NSA LTE anchor virtual grid model building.

Parameter Name	Parameter ID	Setting Notes
LNR Carrier Selection Switch	LnrCarrierSelection.LnrCarrierSelectionSw	- To include samples without inter-frequency neighboring cell measurement results in LNR virtual grid model training, select the SAM_EXPAND_SW option. This is applicable when UEs cannot send inter-frequency measurement reports due to lack of inter-frequency coverage - To exclude samples without intra-frequency neighboring cell measurement results from LNR virtual grid model training, select the NO_INTRA_FREQ_SAM_DELETE_SW option. - To improve the accuracy of virtual grid models, select the SAM_BALANCING_SW option.
Sample UE Number	LnrCarrierSelection.SampleUeNum	It is recommended that this parameter be set to LOW, which indicates that three UEs are involved in sample collection. Set this parameter based on the network plan.
Virtual Grid Model Maximum Build Duration	LnrCarrierSelection.VgMdlMaxBuildDuration	Set this parameter to a value other than 0 if penalty optimization for LNR virtual grid model building is required. Set this parameter based on the network plan.

The following table describes the parameters that must be set in the **LnrCarrierSelection** MO on the LTE side to configure SCG addition based on virtual grid models.

Parameter Name	Parameter ID	Setting Notes
LNR Carrier Selection Switch	LnrCarrierSelection.LnrCarrierSelectionSw	To enable SCG addition based on virtual grid models, select the LNR_VIRTUAL_GRID_SW and NSA_CARRIER_SEL_WITH_VG_SW options. It is recommended that the VG_SCG_ADD_MEAS_AND_PUNISH_SW option be selected if there are UEs that are incompatible with blind SCG addition on the network. To enable the probability prediction model function, select the PROBABILITY_MODEL_SW option.
Intra-Freq Measurement A3 Time-to-Trigger	LnrCarrierSelection.IntraFreqMeasA3TimeToTrig	The value 320MS is recommended.
Virtual Grid Mdl Pred NR SCG Meas RSRP Thld	LnrCarrierSelection.VgMdlPredNrScgMeasRsrpThld	The value ACCURACY100_ENH is recommended.
Virtual Grid Mdl KPI Monitor Thld	LnrCarrierSelection.VgMdlKpiMonitorThld	The value 95 is recommended.

The following tables describe the parameters that must be set in the **LnrFreqRelationship**, **UeCompat**, **PrivateCaBandComb**, and **PrivateBand** MOs on the LTE side and the **gNodeBParam** MO on the NR side to configure the EN-DC band combination blacklist function, prohibiting an EN-DC band combination from being used or prohibiting a certain LTE frequency in any EN-DC band combination that includes a certain NR frequency from serving as an LTE SCC.

- Prohibiting the use of an EN-DC band combination

Parameter Name	Parameter ID	Setting Notes
NSA Band Combination Blacklist 1	UeCompat.NsaBandCombBlacklist1	Select NSA_BAND_COMB_ID_ <i>N</i> , where <i>N</i> is equal to the value of the PrivateCaBandComb.PrivateCaCombId parameter associated with the EN-DC band combination blacklist.
NSA Band Combination Blacklist 2	UeCompat.NsaBandCombBlacklist2	Select NSA_BAND_COMB_ID_ <i>N</i> , where <i>N</i> is equal to the value of the PrivateCaBandComb.PrivateCaCombId parameter associated with the EN-DC band combination blacklist.
Private CA Band Combination ID	PrivateCaBandComb.PrivateCaCombId	Set this parameter based on the operator's network plan.
NSA Flag	PrivateCaBandComb.NsaFlag	Set this parameter based on the operator's network plan.
Band Type	PrivateBand.BandType	Set this parameter based on the operator's network plan.
Combination Band Bandwidth	PrivateBand.CombBandBw	For an NR band, select the Bandwidth_0M option and deselect the other options.

- Prohibiting a certain LTE frequency in any EN-DC band combination that includes a certain NR frequency from serving as an LTE SCC

Parameter Name	Parameter ID	Setting Notes
LTE-NR Frequency Relationship Whitelist	UeCompat.LnrFreqRelWhitelist	Select the FREQ_RELATIONSHIP_ID_ <i>N</i> option, where <i>N</i> is equal to the value of the associated LnrFreqRelationship.RelationshipId parameter.

Parameter Name	Parameter ID	Setting Notes
Relationship ID	LnrFreqRelationship.RelationshipId	Set this parameter to the same value as <i>N</i> in FREQ_RELATIONSHIP_ID_N that is selected in the associated UeCompat.LnrFreqRelpWhitelist parameter.
Downlink EARFCN	LnrFreqRelationship.DLEarfcn	Set this parameter based on the operator's network plan. The LTE frequency specified by this parameter cannot serve as an LTE SCC if it is in an EN-DC combination that includes a specified NR frequency.
Downlink NR-ARFCN	LnrFreqRelationship.NrDLArfcn	Set this parameter based on the operator's network plan. The specified LTE frequency cannot serve as an LTE SCC if it is in an EN-DC combination that includes the NR frequency specified by this parameter.
Frequency Relationship	LnrFreqRelationship.FreqRelationship	Set this parameter to SCC_AVOIDANCE .
NSA DC Optimization Switch	gNodeBParam.NsaDcOptSwitch	Select the NSA_FREQ_CONFLICT_AVOID_ID_SW option.

The following table describes the parameters that must be set in the **UeCompat** and **NrScgFreqConfig** MOs on the LTE side to configure RRC connection reestablishment optimization for defective NSA UEs.

Parameter Name	Parameter ID	Setting Notes
Blacklist Control Extension Switch 2	UeCompat.BlacklistControlExtSwitch2	Configure the NSA_NR_CARRIER_FORBID_SW option based on the network plan.
Aggregation Attribute	NrScgFreqConfig.AggregationAttribute	Configure the NON_STANDARD_BWP_FLAG option based on the network plan.

The following table describes the parameters that must be set to configure preferred path adaption in downlink data split in Option 3.

Parameter Name	Parameter ID	Setting Notes
Downlink Data PDCP Split Mode	LTE: NsaDcQciParamGroup.DlDataPdcpsplitMode	The value ADAPT_PRIMARY_PATH is recommended.
NSA DC Algorithm Extension Switch	LTE: NsaDcMgmtConfig.NsaDcAlgoExtSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.
NSA DC Algorithm Switch	NR: NRCellNsaDcConfig.NsaDcAlgorithmSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.

The following table describes the parameters that must be set to configure preferred path adaption in downlink data split in Option 3x.

Parameter Name	Parameter ID	Setting Notes
Downlink Data PDCP Split Mode	NR: gNBpdcpsplitMode	The value ADAPT_PRIMARY_PATH is recommended.
NSA DC Algorithm Extension Switch	LTE: NsaDcMgmtConfig.NsaDcAlgoExtSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.
NSA DC Algorithm Switch	NR: NRCellNsaDcConfig.NsaDcAlgorithmSwitch	Select the NSA_SPLIT_RATE_OPT_SW option.

The following table describes the parameters that must be set to configure NSA downlink data split optimization in DRX scenarios.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	LTE: NsaDcAlgoParam.NsaDcAlgorithmSwitch	Select the DRX_NSA_SPLIT_OPT_SW option.
NSA DC Optimization Switch	NR: gNodeBParam.NsaDcOptSwitch	Select the NSA_DC_FAST_RETRANS_SWITCH option.

Parameter Name	Parameter ID	Setting Notes
NSA DC Optimization Switch	NR: gNodeBParam.NsaDcOptSwitch	Select the DRX_NSA_SPLIT_OPT_SW option.

The following table describes the parameters that must be set on the NR side for SPID-based NSA uplink primary path configuration.

Parameter Name	Parameter ID	Setting Notes
Uplink Primary Path Selection Mode	gNBRfspConfig.UlPrimaryPathSelMode	- It is recommended that the uplink primary path selection mode be set to DEFAULT. - The uplink primary path can be set to MCG or SCG as required.
Uplink Data Split Threshold	gNBRfspConfig.UlDataSplitThreshold	- If uplink data split is not enabled, it is recommended that the uplink data split threshold be set to INFINITY. - If uplink data split is enabled, it is recommended that the uplink data split threshold be set to BYTE1600 or set as required.

The following table describes the parameters that must be set for SPID-based SCG frequency priority configuration on the LTE side.

Parameter Name	Parameter ID	Setting Notes
NR SCG Frequency Priority Group ID	SPIDCFG.NrScgFreqPriorityGroupId	Set this parameter to the same value as the NrScgFreqPriGrp.NrScgFreqPriorityGroupId parameter.
NR SCG Frequency Priority Group ID	NrScgFreqPriGrp.NrScgFreqPriorityGroupId	Set this parameter based on the operator's network plan.

Parameter Name	Parameter ID	Setting Notes
PCC DL EARFCN	NrScgFreqPriGrp.PccDLEarfcn	Set this parameter based on the operator's network plan.
NR SCG DL ARFCN	NrScgFreqPriGrp.NrScgDLArfcn	Set this parameter based on the operator's network plan.
NR SCG DL ARFCN Priority	NrScgFreqPriGrp.NrScgDLArfcnPriority	Set this parameter based on the operator's network plan.

The following table describes the parameters that must be set on the LTE side to control operator-specific SCG frequency priority configuration.

Parameter Name	Parameter ID	Setting Notes
NR SCG Frequency Priority Group ID	CnOperator.NrScgFreqPriorityGroupId	Set this parameter to the same value as the NrScgFreqPriGrp.NrScgFreqPriorityGroupId parameter.
NR SCG Frequency Priority Group ID	NrScgFreqPriGrp.NrScgFreqPriorityGroupId	Set this parameter based on the operator's network plan.
PCC DL EARFCN	NrScgFreqPriGrp.PccDLEarfcn	Set this parameter based on the operator's network plan.
NR SCG DL ARFCN	NrScgFreqPriGrp.NrScgDLArfcn	Set this parameter based on the operator's network plan.
NR SCG DL ARFCN Priority	NrScgFreqPriGrp.NrScgDLArfcnPriority	Set this parameter based on the operator's network plan.

The following table describes the parameters that must be set on the LTE side to configure uplink CA in EN-DC.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Extension Switch	NsaDcMgmtConfig.NsaDcAlgorithmExtSwitch	To enable uplink CA in EN-DC, select the NSA_LTE_UL_2CC_SW option.

(Optional) The following table describes the parameters that must be set on the LTE side to configure SPID-specific NSA PCC anchoring and PSCell management policies.

Parameter Name	Parameter ID	Setting Notes
Multi-Carrier Config Group ID	CellMultiCarrCfgGrp.MultiCarrCfgGroupId	Set this parameter to a value ranging from 0 to 15 based on the network plan.
Multi-Carrier Config Group ID	SpidCfg.MultiCarrCfgGroupId	Set this parameter to a value ranging from 0 to 15 based on the network plan. The value of this parameter must be the same as that of CellMultiCarrCfgGrp.MultiCarrCfgGroupId .
CN Operator ID	CellMultiCarrCfgGrp.CnOperatorId	Set this parameter to a value ranging from 0 to 5 based on the network plan.
RAT Type	CellMultiCarrCfgGrp.RatType	Set this parameter to EUTRAN or NR as required.
Downlink ARFCN	CellMultiCarrCfgGrp.DlArfcn	Set this parameter based on the network plan.
NSA PCC Anchoring Priority	CellMultiCarrCfgGrp.NsaPccAnchoringPriority	Set this parameter to a value ranging from 0 to 31 based on the network plan.
Aggregation Attribute	CellMultiCarrCfgGrp.AggregationAttribute	Set this parameter based on the network plan.
NR SCG Priority Offset	CellMultiCarrCfgGrp.NrScgPriorityOffset	Set this parameter based on the network plan.
NSA DC B1 RSRP Threshold Offset	CellMultiCarrCfgGrp.NsaDcB1RsrpThldOffset	Set this parameter based on the network plan.
DL PRB Usage Thld for NSA PCC Anchor	NsaDcMgmtConfig.DlPrbThldForNsaPccAnchor	Set this parameter based on the network plan.
SPID Algorithm Switch	SpidCfg.SpidAlgoSwitch	Set the SPID_SHT_PRD_NO_TFC_SCG_ADD_SW option based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Short-Period No-Traffic SCG Add Interval	NsaDcMgmtConfig.ShortPrdNoTfcScgAddIntvl	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set on the LTE side to configure QCI-specific NSA PCC anchoring and PSCell management policies.

Parameter Name	Parameter ID	Setting Notes
Multi-Carrier Config Group ID	CellMultiCarrCfgGrp.MultiCarrCfgGroupId	Set this parameter to a value ranging from 0 to 15 based on the network plan.
Multi-Carrier Config Group ID	QciPara.MultiCarrCfgGroupId	Set this parameter to a value ranging from 0 to 15 based on the network plan. The value of this parameter must be the same as that of CellMultiCarrCfgGrp.MultiCarrCfgGroupId .
CN Operator ID	CellMultiCarrCfgGrp.CnOperatorId	Set this parameter to a value ranging from 0 to 5 based on the network plan.
RAT Type	CellMultiCarrCfgGrp.RatType	Set this parameter to EUTRAN or NR as required.
Downlink ARFCN	CellMultiCarrCfgGrp.DlArfcn	Set this parameter based on the network plan.
NSA PCC Anchoring Priority	CellMultiCarrCfgGrp.NsaPccAnchoringPriority	Set this parameter to a value ranging from 0 to 31 based on the network plan.
Aggregation Attribute	CellMultiCarrCfgGrp.AggregationAttribute	Set this parameter based on the network plan.
NR SCG Priority Offset	CellMultiCarrCfgGrp.NrScgPriorityOffset	Set this parameter based on the network plan.
NSA DC B1 RSRP Threshold Offset	CellMultiCarrCfgGrp.NsaDcB1RsrpThldOffset	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
DL PRB Usage Thld for NSA PCC Anchor	NsaDcMgmtConfig.DlPrbThldForNsaPccAnchor	Set this parameter based on the network plan.
QCI Algorithm Switch	CellQciPara.QciAlgoSwitch	Set the QCI_SHT_PRD_NO_TFC_SCG_ADD_SW option based on the network plan.
Short-Period No-Traffic SCG Add Interval	NsaDcMgmtConfig.ShortPrdNoTfcScgAddIntvl	Set this parameter based on the network plan.

(Optional) The following table describes the parameters that must be set on the LTE side to configure NR FR2 frequency measurement in polling mode in NSA mmWave networking.

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	NrScgFreqConfig.AggregationAttribute	Set the NR_FR2_MEAS_POLLING_FLAG option based on the network plan. It is recommended that the option settings be the same for FR2 frequencies with different NrScgFreqConfig.PccDlEarfcn values but with the same NrScgFreqConfig.ScgDlArfcn value.

(Optional) The following table describes the parameters that must be set to configure blacklist control for SCC management in NSA networking on the LTE side if there are abnormal UEs for which SCCs cannot be added together with SCGs.

Parameter Name	Parameter ID	Setting Notes
Blacklist Control Extension Switch 2	UeCompat.BlacklistControlExtSwitch2	Select the NSA_CA_FORBID_SW option if there are abnormal UEs for which SCCs cannot be added together with SCGs.

(Optional) The following table describes the parameters that must be set to enable "penalty on abnormal SCG releases for NSA UEs" on the LTE side to reduce the SCG service drop rate in NSA networking.

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Enhancement Switch	NsaDcMgmtConfig.NsaDcAlgorithmEnhSwitch	Select the NSA_SCG_ABN_REL_PUNISH_SW option.
SCG Abnormal Release Statistic Duration	NsaDcMgmtConfig.ScgAbnormalStatDuration	Set this parameter based on the operator's network plan.
SCG Abnormal Release Number Threshold	NsaDcMgmtConfig.ScgAbnormalNumThld	Set this parameter based on the operator's network plan.
SCG Abnormal Release Punish Timer	NsaDcMgmtConfig.ScgAbnormalPunishTimer	Set this parameter based on the operator's network plan.

(Optional) The following table describes the parameters that must be set on the LTE side to enable the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs".

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Extension Switch	EnodebAlgoExtSwitch.NsaDcAlgoExtSwitch	To reduce the conflicts between the LTE RRC connection reestablishment procedure and the gNodeB-initiated SCG modification or SCG change procedure, select the NSA_REEST_FLOW_CONFLICT_OPT_SW option to enable the function "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs".

4.4.2.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

NOTE

To enable or disable the main control board enhancement mode:

- If the main control board model has been determined, you are advised to select or deselect the corresponding option.
- If the main control board model is unknown, you are advised to select or deselect both the **AI_ENHANCEMENT_SWITCH** and **UMPTG_ENHANCEMENT_SWITCH** options. The base station automatically identifies the main control board model and activates or deactivates the enhancement mode of only the identified model.

On the eNodeB side

```
//Adding a neighboring NR frequency
ADD NRNFREQ: LocalCellId=21, DlArfcn=636666, UlArfcnConfigInd=NOT_CFG, SsbOffset=0,
SsbPeriod=20MS, SubcarrierSpacing=30KHZ;
//Specifying a frequency band and an additional frequency band for the neighboring NR frequency whose
NR-ARFCN falls within the range of 158200-160600, 285400-303400, 386000-398000, 402000-404000,
422000-434000, 514000-537999, 620000-653333, or 2054166-2084999.
//Setting the frequency band to which the neighboring NR frequency belongs
ADD NRMFBIFREQ: DlArfcn=636666, FrequencyBand=N77, AdditionalFrequencyBand=N78;
//Adding an external NR cell in multi-operator radio access network (MORAN) scenarios, where cells are
not shared by operators, with Mcc and Mnc set to the MCC and MNC of the primary operator
ADD NREXTERNALCELL: Mcc="262", Mnc="01", GnodebId=255, CellId=1, DlArfcn=636666,
UlArfcnConfigInd=NOT_CFG, PhyCellId=1, Tac=1, NrNetworkingOption=NSA, FrequencyBand=N77,
```

```

AdditionalFrequencyBand=N78, MasterPlmnReservedFlag=TRUE;
//Adding an external NR cell in non-MORAN scenarios, with Mcc and Mnc set to the MCC and MNC of the
primary operator
ADD NREXTERNALCELL: Mcc="262", Mnc="01", GnodebId=255, CellId=1, DlArfcn=636666,
UlArfcnConfigInd=NOT_CFG, PhyCellId=1, Tac=1, NrNetworkingOption=NSA, FrequencyBand=N77,
AdditionalFrequencyBand=N78;
//(Optional) Adding a PLMN ID for an external NR cell in sharing scenarios, with Mcc and Mnc set to the
MCC and MNC of the primary operator and SharedMcc and SharedMnc set to the MCC and MNC of the
secondary operator
ADD NREXTERNALCELLPLMN: Mcc="262", Mnc="01", GnodebId=255, CellId=1, SharedMcc="460",
SharedMnc="01", NrNetworkingOption=NSA;
//Adding a neighbor relationship with the NR cell, with Mcc and Mnc set to the MCC and MNC of the
primary operator
ADD NRNRELATIONSHIP: LocalCellId=21, Mcc="262", Mnc="01", GnodebId=255, CellId=1;
//Adding a PCC frequency
ADD PCCFREQCFG: PccDlEarfcn=1500;
//Adding an NR SCG frequency
ADD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=636666, ScgDlArfcnPriority=6,
NsaDcB1ThldRsrp=-105, NrB1TimeToTrigger=40MS;
//Modifying NSA DC management configuration
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_CAPABILITY_SWITCH-1;
MOD CELLQIPARA: LocalCellId=21, Qci=9, NsaDcDefaultBearerMode=SCG_SPLIT_BEARER;
MOD QCIPARA: Qci=9, NsaDcRlcPdcPParamGroupId=223;
MOD RLCPDCPPARAGROUP: RlcPdcPParaGroupId=223, RlcMode=RlcMode_AM,
AmPdcPsnSize=AmPdcPsnSize_18bits;
//(Optional) Enabling gap offset allocation optimization to reduce conflicts between SSB positions and the
positions of other resources such as PUCCH, SRS, and DRX resources
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=GAP_OFS_ASSIGNMENT_OPT_SW-1;
//(Optional) Modifying the period or data volume thresholds for periodic SCG addition (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, ScgAdditionInterval=60, ScgAdditionBufferLenThld=0,
ScgAdditionBufferDelayThld=5;
//(Optional) Enabling the "preferential selection of a band combination with only one NR carrier for NSA
UEs" function if such a combination needs to be preferentially selected in SCG addition
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=NSA_NR_SINGLE_BC_PREFER_SW-1;
//(Optional) Enabling whitelist control for the "preferential selection of a band combination with only one
NR carrier for NSA UEs" function if such a combination needs to be preferentially selected in SCG addition
for whitelisted UEs only
//Running the following commands if no UeCompat MO has been configured
ADD UEINFO: UeCapIndex=0, AccessStratumRelease=1, UeCategory=40, Fgi=1247894;
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhitelistCtrlExtSwitch1=NSA_NR_SINGLE_BC_PREFER_SW-1;
//Running the following command if a UeCompat MO has been configured
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhitelistCtrlExtSwitch1=NSA_NR_SINGLE_BC_PREFER_SW-1;
//(Optional) Enabling "differentiated SCG addition threshold configuration for NR CA-capable UEs" if
differentiated NR FDD SCG addition threshold configuration is required
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=SCG_ADD_B1_THLD_DIFF_SW-1;
//(Optional) Setting the NR TDD SCC RSRP threshold to enable "FDD SCG addition for NR CA-capable UEs
based on the RSRP of candidate NR TDD SCCs". It is recommended that this threshold be set when the
RSRP of candidate NR TDD SCCs needs to be used to assist in FDD SCG addition.
MOD NSADCALGOPARAM: NrTddSccRsrpThld=-105;
//(Optional) Setting the offset used to calculate the RSRP threshold for SCG addition for CA-capable UEs. It
is recommended that this offset be set when a differentiated RSRP threshold for SCG addition is required
for NR FDD+TDD CA-capable UEs.
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=636666, CaUeScgAddRsrpThldOffset=-5;
//(Optional) Enabling the "whitelist control on differentiated B1 threshold for SCG addition" function if one
of the following functions only needs to take effect for whitelisted UEs: "differentiated SCG addition
threshold configuration for NR CA-capable UEs" and "FDD SCG addition for NR CA-capable UEs based on
the RSRP of candidate NR TDD SCCs"
//Running the following commands if no UeCompat MO has been configured
ADD UEINFO: UeCapIndex=0, AccessStratumRelease=1, UeCategory=40, Fgi=1247894;
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhitelistCtrlExtSwitch2=SCG_ADD_B1_THLD_DIFF_SW-1;
//Running the following command if a UeCompat MO has been configured
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhitelistCtrlExtSwitch2=SCG_ADD_B1_THLD_DIFF_SW-1;
//(Optional) Enabling time-domain intermodulation interference avoidance (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=TDM_SWITCH-1;
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=LNR_RELATIVE_FRM_OFS_ADAPT_SW-1;

```

```

//(Optional) Enabling frequency-domain intermodulation interference avoidance (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=INTERFERENCE_AVOID_SW-1,
ImlntrFReqAvoidanceType=PRIOR_ALLOC_NO_INTRF_RB;
//(Optional) Enabling harmonic interference avoidance (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=INTERFERENCE_AVOID_SW-1,
HarmonicIntrfAvoidRange=NR_DL_COMMON_CHANNEL;
//(Optional) Configuring a band combination for uplink single-side transmission if uplink single-side
transmission in NSA DC scenarios is required. If the frame offsets in the CellFrameOffset and
EnodeBFrameOffset MOs are not set, the frame offset takes effect as 0 in time synchronization scenarios.
MOD CELLFRAMEOFFSET: LocalCellId=21, FrameOffsetMode=CustomFrameOffset, FrameOffset=0;
MOD NRSCGFREQCONFIG: PccDIEarfcn=1500, ScgDIArfcn=361000,
AggregationAttribute=SINGLE_UL_NSA_FLAG-1;
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=TDM_SWITCH-1;
//Setting the downlink PDCP-layer data split mode in the NSA DC QCI parameter group in Option 3 (The
following uses dynamic data split as an example. Set the data split mode based on the network plan.)
MOD NSADCQCIPARAMGROUP: LocalCellId=21, NsaDcQciParamGroupId=0,
DLDataPdcpsplitMode=SCG_AND_MCG;
//Setting the QCI-specific primary path and threshold for uplink data split in Option 3 (The following uses
data split only to the SCG as an example. Set the data split mode based on the network plan.)
MOD NSADCQCIPARAMGROUP: LocalCellId=21, NsaDcQciParamGroupId=0, ULDataSplitPrimaryPath=SCG,
ULDataSplitThreshold=INFINITY;
//(Optional) Modifying the downlink and uplink data split modes in Option 3 for a specific operator (Set
the data split mode based on the network plan.)
MOD CELLOPQCIPARA: CnOperatorId=1, LocalCellId=21, Qci=9, NsaDcQciParamGroupId=0;
//(Optional) Enabling downlink real-time data split if the air interface transmission capabilities of LTE and
NR need to be fully utilized in NR FDD scenarios
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NSA_SPLIT_RATE_OPT_SW-1;
//(Optional) Enabling adaptive resumption of downlink data split in Option 3 (if required)
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=NSA_SPLIT_ADAPT_RECOVERY_SW-1;
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NSA_SPLIT_RATE_OPT_SW-1;
//(Optional) Enabling adaptive resumption of downlink data split in Option 3x (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NSA_SPLIT_RATE_OPT_SW-1;
//(Optional) Configuring the maximum uplink MCG transmit power and maximum uplink SCG transmit
power for NSA DC UEs for initial power control (Set the power based on the network plan.)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcUeMcgULMaximumPower=20,
NsaDcUeScgULMaximumPower=20;
//(Optional) Enabling NSA uplink preallocation if the number of times that the base station proactively
schedules UEs needs to be increased
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_PREALLOCATION_SW-1;
//(Optional) Configuring intra-band EN-DC with Mcc and Mnc set to the MCC and MNC of the primary
operator
MOD NRSCGFREQCONFIG: PccDIEarfcn=1500, ScgDIArfcn=361000,
AggregationAttribute=CO_DEPLOYMENT_NSA_FLAG-1;
ADD NRNRELATIONSHIP: LocalCellId=21, Mcc="262", Mnc="01", GnodebId=1, CellId=7,
AggregationAttribute=CO_DEPLOYMENT_NSA_FLAG-1;
//(Optional) Configuring SCG release and addition based on UE overheating status reporting if SCG release
and addition based on overheating messages and overheating clearance messages reported by UEs are
required
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=OVERHEATING_RPOTECTION_SW-1;
MOD RRCONNSTATETIMER: OverheatingIndProhibitTmr=S120;
//(Optional) Enabling SCG addition based on S1-U status in Option 3x if neighboring NR cells under SgNBs
with unavailable S1-U interfaces need to be filtered out
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=S1U_STATUS_BASED_SCG_ADD_SW-1;
//(Optional) Turning on the EQUIVALENT_PLMN_ADD_SCG_SW or INTER_PLMN_ADD_SCG_SW switch
when LTE and NR belong to different PLMNs
MOD ENODEBALGOEXTSWITCH:
NsaDcAlgoSwitch=EQUIVALENT_PLMN_ADD_SCG_SW-1&INTER_PLMN_ADD_SCG_SW-1;
//(Optional) Turning on the UMPTg enhancement switch if the main control board is a UMPTg
MOD ENODEBRESMODEALGO: ServiceMode=UMPTG_ENHANCEMENT_SWITCH-1;
RST APP: AID=1;
//(Optional) Turning on the AI enhancement optimization switch if the main control board is a UMPTe,
UMPTga, or UMPTHa
MOD ENODEBRESMODEALGO:
ServiceMode=AI_ENHANCEMENT_SWITCH-0&AI_ENHANCEMENT_OPT_SWITCH-1;
RST APP: AID=1;
//(Optional) Enabling LNR virtual grid model building if NSA DC virtual grid models are required
MOD LNRARRIERSELECTION: LnrCarrierSelectionSw=LNR_VIRTUAL_GRID_SW-1;
//(Optional) Turning on SAM_EXPAND_SW if UEs cannot send inter-frequency measurement reports due to
lack of inter-frequency coverage and samples without inter-frequency neighboring cell measurement results

```

are expected to be included in LNR virtual grid model training

MOD LNR CARRIER SELECTION: LnrCarrierSelectionSw=SAM_EXPAND_SW-1;
 //(Optional) Turning on NO_INTRA_FREQ_SAM_DEL_SW to exclude samples without intra-frequency neighboring cell measurement results from LNR virtual grid model training

MOD LNR CARRIER SELECTION: LnrCarrierSelectionSw=NO_INTRA_FREQ_SAM_DEL_SW-1;
 //(Optional) Turning on SAM_BALANCING_SW to improve the accuracy of virtual grid models

MOD LNR CARRIER SELECTION: LnrCarrierSelectionSw=SAM_BALANCING_SW-1;
 //(Optional) Setting SampleUeNum when the number of UEs involved in sample collection during virtual grid model training needs to be adjusted

MOD LNR CARRIER SELECTION: SampleUeNum=LOW;
 //(Optional) Enabling penalty optimization for LNR virtual grid model building if grid model building fails many times (In the following example, the maximum allowed duration for building an LNR virtual grid model is set to 15 days.)

MOD LNR CARRIER SELECTION: VgMdlMaxBuildDuration=15;
 //(Optional) Enabling NSA LTE anchor virtual grid model building if NSA DC virtual grid models are required

MOD LNR CARRIER SELECTION: LnrCarrierSelectionSw=NSA_ANCHOR_VIRTUAL_GRID_SW-1;
 //(Optional) Enabling SCG addition based on virtual grid models with VgMdlPredNrScgMeasRsrpThld set to ACCURACY100_ENH (if required)

MOD LNR CARRIER SELECTION: VgMdlPredNrScgMeasRsrpThld=ACCURACY100_ENH,
 LnrCarrierSelectionSw=NSA_CARRIER_SEL_WITH_VG_SW-1, IntraFreqMeasA3TimeToTrig=320MS,
 VgMdlKpiMonitorThld=95;
 //(Optional) Enabling SCG addition based on virtual grid models with VgMdlPredNrScgMeasRsrpThld set to a value other than ACCURACY100_ENH (if required)

MOD LNR CARRIER SELECTION: VgMdlPredNrScgMeasRsrpThld=ACCURACY90_OFFSET0,
 LnrCarrierSelectionSw=NSA_CARRIER_SEL_WITH_VG_SW-1, IntraFreqMeasA3TimeToTrig=320MS,
 VgMdlKpiMonitorThld=95;
 //(Optional) Enabling measurement and penalty for SCG addition based on virtual grid models if there are UEs that are incompatible with blind SCG addition on the network

MOD LNR CARRIER SELECTION: LnrCarrierSelectionSw=VG_SCG_ADD_MEAS_AND_PUNISH_SW-1;
 //(Optional) Enabling the probability prediction model function

MOD LNR CARRIER SELECTION: LnrCarrierSelectionSw=PROBABILITY_MODEL_SW-1;
 //(Optional) Prohibiting the use of a specific EN-DC band combination for NSA UEs that have compatibility issues with this combination (for example, B3+B20+n41)

ADD UE COMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, NsaBandCombBlacklist1=NSA_BAND_COMB_ID_0-1;
 ADD PRIVATE CABAND COMB: PrivateCaCombId=0, MaxAggregatedBw=200, BwCombSetId=0, NsaFlag=TRUE;
 ADD PRIVATE BAND: PrivateCaCombId=0, CombBandIndex=1, CombBandId=3,
 CombBandBw=Bandwidth_1_4M-1&Bandwidth_3M-1&Bandwidth_5M-1&Bandwidth_10M-1&Bandwidth_15M-1&Bandwidth_20M-1&Bandwidth_0M-0;
 ADD PRIVATE BAND: PrivateCaCombId=0, CombBandIndex=2, CombBandId=20,
 CombBandBw=Bandwidth_1_4M-1&Bandwidth_3M-1&Bandwidth_5M-1&Bandwidth_10M-1&Bandwidth_15M-1&Bandwidth_20M-1&Bandwidth_0M-0;
 ADD PRIVATE BAND: PrivateCaCombId=0, CombBandIndex=3, CombBandId=41,
 CombBandBw=Bandwidth_1_4M-0&Bandwidth_3M-0&Bandwidth_5M-0&Bandwidth_10M-0&Bandwidth_15M-0&Bandwidth_20M-0&Bandwidth_0M-1, BandType=NR;
 //(Optional) Forbidding a specified LTE frequency from serving as an LTE SCC when it is in an EN-DC band combination that includes a specified NR frequency for NSA UEs that have compatibility issues with such a combination

ADD UE COMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, LnrFreqRelpWhitelist=FREQ_RELATIONSHIP_ID_32-1;
 ADD LNR FREQ RELATIONSHIP: RelationshipId=0, DlEarfcn=1500, NrDlEarfcn=636666,
 NrDlBandwidth=CELL_BW_20M-0, FreqRelationship=SCC_AVOIDANCE;
 //(Optional) Forbidding the addition of NR carriers as SCG carriers if RRC connection reestablishment optimization for defective NSA UEs is required (Set the parameters based on the network plan.)

MOD UE COMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
 BlacklistControlExtSwitch2=NSA_NR_CARRIER_FORBID_SW-1;
 //(Optional) Specifying the SCG frequency with a non-protocol-defined BWP if RRC connection reestablishment optimization for defective NSA UEs is required (Set the parameters based on the network plan.)

MOD NR SCG FREQ CONFIG: PccDlEarfcn=1500, ScgDlEarfcn=151600,
 AggregationAttribute=NON_STANDARD_BWP_FLAG-1;
 //(Optional) Enabling preferred path adaption in downlink data split (Set the parameters based on the network plan.)

MOD NSADCQCIPARAM GROUP: LocalCellId=21, NsaDcQciParamGroupId=0,
 DlDataPdcPsplitMode=ADAPT_PRIMARY_PATH;
 MOD NSADCMGMT CONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NSA_SPLIT_RATE_OPT_SW-1;
 //(Optional) Enabling NSA downlink data split optimization in DRX scenarios (if required)

MOD NSADCALGOPARAM: NsaDcAlgoSwitch=DRX_NSA_SPLIT_OPT_SW-1;
 //(Optional) Enabling SPID-based SCG frequency priority configuration (Set the parameters based on the

```

network plan.)
ADD NRSCGFREQPRIGRP: NrScgFreqPriorityGroupId=0, PccDlEarfcn=1500, NrScgDlArfcn=636666,
NrScgDlArfcnPriority=7;
ADD NRSCGFREQPRIGRP: NrScgFreqPriorityGroupId=0, PccDlEarfcn=1500, NrScgDlArfcn=361000,
NrScgDlArfcnPriority=6;
ADD SPIDCFG: Spid=1, NrScgFreqPriorityGroupId=0;
//(Optional) Setting the operator-specific SCG frequency priority configuration function (Set the parameters
based on the network plan.)
ADD NRSCGFREQPRIGRP: NrScgFreqPriorityGroupId=0, PccDlEarfcn=1500, NrScgDlArfcn=636666,
NrScgDlArfcnPriority=7;
ADD NRSCGFREQPRIGRP: NrScgFreqPriorityGroupId=1, PccDlEarfcn=1500, NrScgDlArfcn=636666,
NrScgDlArfcnPriority=6;
MOD CNOOPERATOR: CnOperatorId=0, NrScgFreqPriorityGroupId=1;
MOD CNOOPERATOR: CnOperatorId=1, NrScgFreqPriorityGroupId=0;
//(Optional) Enabling uplink CA in EN-DC (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NSA_LTE_UL_2CC_SW-1;
//(Optional) Adding an SPID configuration with a multi-carrier configuration group ID specified (Set the
parameters based on the network plan.)
ADD SPIDCFG: Spid=1, MultiCarrCfgGroupId=1;
//(Optional) Modifying a QCI configuration with a multi-carrier configuration group ID specified (Set the
parameters based on the network plan.)
MOD QCIPARA: Qci=1, MultiCarrCfgGroupId=1;
//(Optional) Configuring the NSA PCC anchoring policy (Set the parameters based on the network plan.)
ADD CELLMULTICARRCFGGRP: DlArfcn=1500, RatType=EUTRAN, CnOperatorId=1, MultiCarrCfgGroupId=1,
LocalCellId=21, AggregationAttribute=FORBID_MEAS_FLAG-1&HIGH_LOAD_CELL_NSA_PCC_FLAG-1,
NsaPccAnchoringPriority=8;
MOD NSADCMGMTCONFIG: LocalCellId=21, DlPrbThldForNsaPccAnchor=95;
//(Optional) Configuring the PSCell management policy (Set the parameters based on the network plan.)
ADD CELLMULTICARRCFGGRP: DlArfcn=636666, RatType=NR, CnOperatorId=1, MultiCarrCfgGroupId=1,
LocalCellId=21, NrScgPriorityOffset=1, NsaDcB1RsrpThldOffset=1,
AggregationAttribute=FORBID_MEAS_FLAG-1;
//(Optional) Enabling SPID-specific SCG addition at short intervals without considering traffic volume (Set
the parameters based on the network plan.)
ADD SPIDCFG: Spid=1, SpidAlgoSwitch=SPID_SHT_PRD_NO_TFC_SCG_ADD_SW-1;
//(Optional) Enabling QCI-specific SCG addition at short intervals without considering traffic volume (Set
the parameters based on the network plan.)
ADD CELLQCIPARA: LocalCellId=21, Qci=11, QciAlgoSwitch=QCI_SHT_PRD_NO_TFC_SCG_ADD_SW-1;
//(Optional) Setting Short-Period No-Traffic SCG Add Interval (Set the parameters based on the network
plan.)
MOD NSADCMGMTCONFIG: LocalCellId=21, ShortPrdNoTfcScgAddIntvl=4;
//(Optional) Enabling NR FR2 frequency measurement in polling mode in NSA mmWave networking
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=151600, ScgDlArfcnPriority=7,
AggregationAttribute=NR_FR2_MEAS_POLLING_FLAG-1;
//(Optional) Configuring blacklist control for SCC management in NSA networking on the LTE side if there
are abnormal UEs for which SCCs cannot be added together with SCGs
ADD UECOMPAT: Index=0, UeInfoType=IMEISV_TAC_SVN, ImeisvTac=12345678, ImeisvSvn=12,
BlacklistControlExtSwitch2=NSA_CA_FORBID_SW-1;
//(Optional) Configuring PSCell management for VoLTE UEs (Set the parameters based on the network
plan.)
MOD NSADCMGMTCONFIG: LocalCellId=21, VolteUeScgMgmtStrategy=VOLTE_PREFERRED;
//(Optional) Setting the VoLTE packet loss rate threshold for SCG release when
NsaDcMgmtConfig.VolteUeScgMgmtStrategy is set to ADAPTIVE
MOD NSADCMGMTCONFIG: LocalCellId=21, VoltePlrThldForScgRelease=5;
//(Optional) Enabling inter-frequency and inter-RAT measurement optimization for VoLTE UEs in NSA DC
scenarios if the eNodeB is required not to deliver A5 measurement configurations for NSA PCC anchoring
and B1 measurement configurations for SCG addition to VoLTE UEs when QCI-specific VoLTE and SCG
coexistence does not take effect for VoLTE UEs and the NsaDcMgmtConfig.VolteUeScgMgmtStrategy
parameter is not set to VOLTE_SCG_COEXISTENCE
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=VOLTE_NSA_DC_IF_MEAS_OPT_SW-1;
//(Optional) Enabling QCI-specific VoLTE and SCG coexistence for UEs with bearers of a specific QCI (QCI
159 used as an example) if VoLTE services and SCGs are expected to coexist for such UEs
//Running the ADD CELLQCIPARA command if QCI 159 has not been configured for the cell and running
the MOD CELLQCIPARA command if QCI 159 has been configured for the cell
ADD CELLQCIPARA: LocalCellId=21, Qci=159, NsaDcOptSwitch=QCI_VOLTE_SCG_COEXIST_SW-1;
MOD CELLQCIPARA: LocalCellId=21, Qci=159, NsaDcOptSwitch=QCI_VOLTE_SCG_COEXIST_SW-1;
//(Optional) Enabling "penalty on abnormal SCG releases for NSA UEs" if the SCG service drop rate needs
to be reduced in NSA networking
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=NSA_SCG_ABN_REL_PUNISH_SW-1,
ScgAbnRelStatDuration=60, ScgAbnRelNumThld=5, ScgAbnRelPunishTimer=600;

```

```

//(Optional) Enabling the function "optimized handling of procedure conflicts upon RRC connection
reestablishments of NSA UEs" to reduce the conflicts between the LTE RRC connection reestablishment
procedure and the gNodeB-initiated SCG modification or SCG change procedure
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_REEST_FLOW_CONFLICT_OPT_SW-1;

```

On the gNodeB side

```

//Setting the downlink PDCP-layer data split mode and downlink PDCP SN size in the gNodeB PDCP
parameter group in Option 3x (The following uses dynamic data split as an example. Set the data split
mode based on the network plan.)
MOD GNBPDCCPARAMGROUP: PdcParamGroupId=1, DlDataPdcSplitMode=SCG_AND_MCG,
DlPdcSnSize=BITS18;
//Setting the primary path and threshold for uplink data split in the gNodeB PDCP parameter group in
Option 3x (The following uses data split only to the SCG as an example. Set the data split mode based on
the network plan.)
MOD GNBPDCCPARAMGROUP: PdcParamGroupId=1, UlDataSplitPrimaryPath=SCG,
UlDataSplitThreshold=INFINITY;
//(Optional) Configuring downlink and uplink data split modes in Option 3x for a specific operator (Set the
parameters based on the network plan.)
MOD NRCELLOPQCIBEARER: OperatorId=1, NrCellId=7, Qci=9, RlcMode=AM, AmPdcParamGroupId=1;
//(Optional) Enabling downlink real-time data split if the air interface transmission capabilities of LTE and
NR need to be fully utilized in NR FDD scenarios
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_SPLIT_RATE_OPT_SW-1;
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_PDCP_SPLIT_OPT_SW-1;
//(Optional) Enabling adaptive resumption of downlink data split in Option 3 (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_SPLIT_RATE_OPT_SW-1;
//(Optional) Enabling adaptive resumption of downlink data split in Option 3x (if required)
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_SPLIT_ADAPT_RECOVERY_SW-1;
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_SPLIT_RATE_OPT_SW-1;
//Setting NR cell algorithm switches
MOD NRCELLALGOSWITCH: NrCellId=7, NsaDcSwitch=ON;
//Modifying the bearer configuration of an NR cell with the RLC Mode parameter set to AM and the
AmPdcParamGroupId parameter set to the same value as PdcParamGroupId
MOD NRCELLQCIBEARER: NrCellId=7, Qci=9, AmPdcParamGroupId=1, RlcMode=AM;
//(Optional) Configuring event-based triggering of NR data volume reporting in Option 3x (you can specify
whether NR data volumes are reported based on events or based on events and at intervals according to
the network plan)
MOD NRCELLNSADCCONFIG: NrCellId=7, NrDataVolumeRptCfg=65535;
//(Optional) Enabling uplink fallback to LTE and setting the uplink SINR threshold (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=UL_FALLBACK_TO_LTE_SWITCH-1;
MOD NRDUCELLSRMEAS: NrDuCellId=120, NsaUlFackToLteSinrThld=-30, NsaUlFackToLteSinrHyst=30;
//(Optional) Enabling the function of requesting LTE-side large-packet status of ULFB UEs, if the gNodeB is
required to obtain the packet status (large or small) of UEs that have undergone uplink fallback to LTE
MOD GNODEBPARAM: NsaDcOptSwitch=ULFB_LTE_UL_LRG_PKT_REQ_SW-1;
//(Optional) Enabling time-domain intermodulation interference avoidance (if required)
MOD NRDUCELLALGOSWITCH: NrDuCellId=120, NsaDcAlgoSwitch=CROSS_MDLT_INTRF_AVOID_SW-1;
//(Optional) Enabling harmonic interference avoidance (if required)
MOD NRDUCELLALGOSWITCH: NrDuCellId=120, NsaDcAlgoSwitch=HARMONIC_INTRF_AVOID_SW-1;
MOD NRDUCELLNSADCCONFIG: NrDuCellId=120, HarmonicIntrfAvoidRange=NR_DL_COMMON_CHANNEL;
//(Optional) Enabling NSA DC SRB3 (if required)
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_DC_SRB3_SWITCH-1;
//(Optional) Enabling the NSA DC SRB3 blacklist function when NSA DC SRB3 is enabled on the live
network and some UEs are incompatible with SRB3. (The following example assumes that information
about UEs with UeInfoIndex set to 1 does not exist. Set UE information based on live network conditions.)
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,
UeFeatureValueContent="ec7fee769ff7e0fff1",
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION2;
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=NSA_DC_SRB3_SWITCH_OFF-1;
//(Optional) Enabling uplink single-side transmission in NSA DC scenarios (if required). If the frame offset is
not set, the frame offset takes effect as 0 in time synchronization scenarios.
MOD GNBFREQBANDCONFIG: FrequencyBand=N3, FrameOffset=0;
ADD GNBDEDNSADCBANDCOMB: NrBandId=N3, LteBandId=3,
AggregationAttribute=SINGLE_UL_NSA_FLAG-1;
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=SINGLE_UL_NSA_SW-1;
//(Optional) Enabling fast retransmission for downlink data split (if required)
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_DC_FAST_RETRANS_SWITCH-1;
//(Optional) Enabling NSA uplink preallocation (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_DC_PREALLOCATION_SW-1;
//(Optional) Configuring intra-band EN-DC with Mcc and Mnc set to the MCC and MNC of the primary

```

```

operator (if required)
ADD GNBDEDNSADCBANDCOMB: NrBandId=N3, LteBandId=3,
AggregationAttribute=CO_DEPLOYMENT_NSA_FLAG-1;
ADD GNBRLTECELLCOMB: Mcc="262", Mnc="01", gNodeBId=329, CellId=247, LteMcc="262",
LteMnc="01", eNodeBId=324, LteCellId=240, AggregationAttribute=CO_DEPLOYMENT_NSA_FLAG-1;
//(Optional) Configuring SgNB release based on RSRP to implement differentiated NR FDD SCG deletion
threshold configuration
MOD NRCELLNSADCCONFIG: NrCellId=7, PscellA2RsrpThld=-121;
MOD NRCELLNSADCCONFIGGRP: NrCellId=7, NsaDcConfigGroupId=2, PscellA2RsrpThldOffset=0,
PscellA2TimeToTrig=640MS;
//(Optional) Configuring SgNB release based on SSB SINR (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=SINR_BASED_SCG_RELEASE_SW-1;
MOD NRCELLNSADCCONFIGGRP: NrCellId=7, NsaDcConfigGroupId=2, PscellA2SinrThld=-10;
//(Optional) Configuring the offset used to calculate the RSRP threshold for PSCell event A2 for CA-capable
UEs if "differentiated SCG deletion threshold configuration for NR CA-capable UEs" is required
MOD NRCELLNSADCCONFIG: NrCellId=7, CaCapUePscellA2RsrpThldOfs=-5;
//(Optional) Enabling S1-U status sending in Option 3x if the MeNB is required to filter out neighboring NR
cells under any SgNB whose S1-U interface is unavailable
MOD GNODEBPARAM: NsaDcOptSwitch=S1U_STATUS_SEND_SW-1;
//(Optional) Setting the PLMN selection policy switch for SCG addition when LTE and NR belong to
different PLMNs
MOD GNODEBPARAM: NsaDcOptSwitch=SCG_ADD_PLMN_SELECT_POLICY_SW-1;
//(Optional) Enabling the gNodeB to send information about the NR PSCell and SCell frequencies in EN-DC
band combinations to the eNodeB so that the eNodeB can determine whether to forbid the addition of an
LTE frequency as the frequency of an LTE SCC (if intra-band EN-DC is required)
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_FREQ_CONFLICT_AVOID_SW-1;
//(Optional) Enabling preferred path adaption in downlink data split (if required)
MOD GNBPDCCPARAMGROUP: PdcParamGroupId=1, DlDataPdcSplitMode=ADAPT_PRIMARY_PATH;
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_SPLIT_RATE_OPT_SW-1;
//(Optional) Enabling NSA downlink data split optimization in DRX scenarios (if required)
MOD GNODEBPARAM: NsaDcOptSwitch=DRX_NSA_SPLIT_OPT_SW-1;
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_DC_FAST_RETRANS_SWITCH-1;
//(Optional) Configuring an SPID-specific NSA uplink primary path with the SPID specified by
gNB RfSpConfig.RfSpIndex (Set the parameters based on the network plan.)
MOD GNB RFSPPCONFIG: OperatorId=0, RfSpIndex=0, UlPrimaryPathSelMode=SCG,
UlDataSplitThreshold=BYTE1600;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

On the eNodeB side

```

//Turning off the NSA DC switch on the LTE side
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_CAPABILITY_SWITCH-0;
//Disabling downlink real-time data split
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NSA_SPLIT_RATE_OPT_SW-0;
// Disabling NSA downlink data split optimization in DRX scenarios
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=DRX_NSA_SPLIT_OPT_SW-0;
//Disabling adaptive resumption of downlink data split in Option 3
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=NSA_SPLIT_ADAPT_RECOVERY_SW-0;
//Disabling event-triggered NR data volume reporting in Option 3 on the LTE side
MOD NSADCMGMTCONFIG: LocalCellId=21, NrDataVolumeRptCfg=0;
//Turning off the interference avoidance switch
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=INTERFERENCE_AVOID_SW-0,
ImdIntrFReqAvoidanceType=NO_AVOIDANCE;
//Disabling uplink single-side transmission in NSA DC
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=361000,
AggregationAttribute=SINGLE_UL_NSA_FLAG-0;
//Disabling the co-site intra-band pair of LTE and NR cells
MOD NRSCGFREQCONFIG: PccDlEarfcn=1500, ScgDlArfcn=361000,
AggregationAttribute=CO_DEPLOYMENT_NSA_FLAG-0;
MOD NRNRELATIONSHIP: LocalCellId=21, Mcc="262", Mnc="01", GnodeBId=1, CellId=7,
AggregationAttribute=CO_DEPLOYMENT_NSA_FLAG-0;
//Disabling SCG addition based on equivalent PLMN or inter-PLMN information
MOD ENODEBALGOEXTSWITCH:

```

```
NsaDcAlgoSwitch=EQUIVALENT_PLMN_ADD_SCG_SW-0&INTER_PLMN_ADD_SCG_SW-0;
//Turning off the LTE-NR carrier selection switches
MOD LNRCARRIERSELECTION:
LnrCarrierSelectionSw=LNR_VIRTUAL_GRID_SW-0&NSA_CARRIER_SEL_WITH_VG_SW-0&NSA_ANCHOR_VIRTUAL_GRID_SW-0;
//Disabling LNR virtual grid model building
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=LNR_VIRTUAL_GRID_SW-0;
//Disabling NSA LTE anchor virtual grid model building
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=NSA_ANCHOR_VIRTUAL_GRID_SW-0;
//Disabling SCG addition based on virtual grid models
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=NSA_CARRIER_SEL_WITH_VG_SW-0;
//Disabling measurement and penalty for SCG addition based on virtual grids
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=VG_SCG_ADD_MEAS_AND_PUNISH_SW-0;
//Disabling the probability prediction model function
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=PROBABILITY_MODEL_SW-0;
//Disabling penalty optimization for LNR virtual grid model building
MOD LNRCARRIERSELECTION: VgMdlMaxBuildDuration=0;
//Turning off the UMPTg enhancement switch when the main control board is a UMPTg (The switch setting must be the same as that before SCG addition based on virtual grid models is enabled.)
MOD ENODEBRESMODEALGO: ServiceMode=UMPTG_ENHANCEMENT_SWITCH-0;
RST APP: AID=1;
//Turning off the AI enhancement optimization switch if the main control board is a UMPTe, UMPTga, or UMPTHa (The switch setting must be the same as that before SCG addition based on virtual grid models is enabled.)
MOD ENODEBRESMODEALGO: ServiceMode=AI_ENHANCEMENT_OPT_SWITCH-0;
RST APP: AID=1;
//Disabling the preferred path adaption function
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NSA_SPLIT_RATE_OPT_SW-0;
//Disabling SPID-based SCG frequency priority configuration
ADD SPIDCFG: Spid=1, NrScgFreqPriorityGroupId=65535;
//(Optional) Disabling the "preferential selection of a band combination with only one NR carrier for NSA UEs" function
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=NSA_NR_SINGLE_BC_PREFER_SW-0;
//(Optional) Turning off the whitelist control switch for the "preferential selection of a band combination with only one NR carrier for NSA UEs" function
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhitelistCtrlExtSwitch1=NSA_NR_SINGLE_BC_PREFER_SW-0;
//(Optional) Disabling "differentiated SCG addition threshold configuration for NR CA-capable UEs"
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=SCG_ADD_B1_THLD_DIFF_SW-0;
//(Optional) Disabling "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs"
MOD NSADCALGOPARAM: NrTddScsRsrpThld=255;
//(Optional) Disabling the "whitelist control on differentiated B1 threshold for SCG addition" function
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhitelistCtrlExtSwitch2=SCG_ADD_B1_THLD_DIFF_SW-0;
//(Optional) Disabling uplink CA in EN-DC
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NSA_LTE_UL_2CC_SW-0;
//(Optional) Disabling SPID-specific SCG addition at short intervals without considering traffic volume
ADD SPIDCFG: Spid=1, SpidAlgoSwitch=SPID_SHT_PRD_NO_TFC_SCG_ADD_SW-0;
//(Optional) Disabling QCI-specific SCG addition at short intervals without considering traffic volume
ADD CELLQCIPARA: LocalCellId=21, Qci=11, QciAlgoSwitch=QCI_SHT_PRD_NO_TFC_SCG_ADD_SW-0;
//(Optional) Disabling NR FR2 frequency measurement in polling mode in NSA mmWave networking
MOD NRSCGFREQCONFIG: PccDIEarfcn=1500, ScgDIEarfcn=151600, ScgDIEarfcnPriority=7,
AggregationAttribute=NR_FR2_MEAS_POLLING_FLAG-0;
//(Optional) Disabling QCI-specific VoLTE and SCG coexistence
MOD CELLQCIPARA: LocalCellId=21, Qci=159, NsaDcOptSwitch=QCI_VOLTE_SCG_COEXIST_SW-0;
//(Optional) Disabling inter-frequency and inter-RAT measurement optimization for VoLTE UEs in NSA DC scenarios
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=VOLTE_NSA_DC_IF_MEAS_OPT_SW-0;
//(Optional) Disabling "penalty on abnormal SCG releases for NSA UEs"
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=NSA_SCG_ABN_REL_PUNISH_SW-0;
//(Optional) Disabling "optimized handling of procedure conflicts upon RRC connection reestablishments of NSA UEs"
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_REEST_FLOW_CONFLICT_OPT_SW-0;
```

On the gNodeB side

```
//Turning off the NSA DC switch on the NR side
MOD NRCELLALGOSWITCH: NrCellId=7, NsaDcSwitch=OFF;
//(Optional) Disabling downlink real-time data split
```



```
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_SPLIT_RATE_OPT_SW-0;
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_PDCP_SPLIT_OPT_SW-0;
//(Optional) Disabling adaptive resumption of downlink data split in Option 3x
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_SPLIT_ADAPT_RECOVERY_SW-0;
//(Optional) Disabling fallback to LTE
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=UL_FALLBACK_TO_LTE_SWITCH-0;
//(Optional) Disabling the function of requesting LTE-side large-packet status of ULFB UEs
MOD GNODEBPARAM: NsaDcOptSwitch=ULFB_LTE_UL_LRG_PKT_REQ_SW-0;
//(Optional) Disabling event-based triggering of NR data volume reporting in Option 3x
MOD NRCELLNSADCCONFIG: NrCellId=7, NrDataVolumeRptCfg=0;
//(Optional) Turning off the intermodulation interference avoidance switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=120, NsaDcAlgoSwitch=CROSS_MDLT_INTRF_AVOID_SW-0;
//(Optional) Turning off the harmonic interference avoidance switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=120, NsaDcAlgoSwitch=HARMONIC_INTRF_AVOID_SW-0;
//(Optional) Disabling NSA DC SRB3
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_DC_SRB3_SWITCH-0;
//(Optional) Disabling the NSA DC SRB3 blacklist function
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=NSA_DC_SRB3_SWITCH_OFF-0;
//(Optional) Disabling uplink single-side transmission in NSA DC
MOD GNBDEDNSADCBANDCOMB: NrBandId=N3, LteBandId=3,
AggregationAttribute=SINGLE_UL_NSA_FLAG-0;
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=SINGLE_UL_NSA_SW-0;
//(Optional) Disabling fast retransmission for downlink data split
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_DC_FAST_RETRANS_SWITCH-0;
//(Optional) Disabling NSA uplink preallocation
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_DC_PREALLOCATION_SW-0;
//(Optional) Disabling the co-site intra-band pair of LTE and NR cells
MOD GNBDEDNSADCBANDCOMB: NrBandId=N3, LteBandId=3, AggregationAttribute=
CO_DEPLOYMENT_NSA_FLAG-0;
//(Optional) Disabling the preferred path adaption function
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_SPLIT_RATE_OPT_SW-0;
//(Optional) Disabling NSA downlink data split optimization in DRX scenarios
MOD GNODEBPARAM: NsaDcOptSwitch=DRX_NSA_SPLIT_OPT_SW-0;
MOD GNODEBPARAM: NsaDcOptSwitch=NSA_DC_FAST_RETRANS_SWITCH-0;
//(Optional) Disabling "differentiated SCG deletion threshold configuration for NR CA-capable UEs"
MOD NRCELLNSADCCONFIG: NrCellId=7, CaCapUePscellA2RsrpThldOfs=0;
```

4.4.2.3 Using the MAE-Deployment

- Fast batch activation

This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance.**

- Single/Batch configuration

This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

Monitoring Counters

Counters related to NSA DC can be subscribed on the MAE-Access in one-click mode.

- After activating NSA DC, observe the counters listed in the following tables. On a network with NSA UEs, NSA DC has taken effect only if the following counters produce non-zero values.

Counter ID	Counter Name	NE
1526747850	L.Traffic.User.NsaDc.PCell.Avg	eNodeB
1911816745	N.User.NsaDc.PSCell.Avg	gNodeB

- Check the number of times NR measurement configurations are delivered in NSA DC scenarios.
 - The counter **L.NsaDc.MeasCtrl.NR** indicates the number of times NR measurement configurations are delivered in NSA DC scenarios.
 - The counter **L.NsaDc.MeasCtrl.NR.NoGap** indicates the number of times NR gap-free measurement configurations are delivered in NSA DC scenarios.
- Check whether data split has taken effect in NSA DC.
 - The counter **N.PDCP.Vol.DL.X2U.TrfPDU.Tx** indicates the total downlink data volume that is split for NSA UEs over the X2 interface. If the counter value is greater than 0, downlink data split has taken effect.
 - The counter **N.PDCP.Vol.UL.X2U.TrfPDU.Rx** indicates the total uplink data volume that is split for NSA UEs over the X2 interface. If the counter value is greater than 0, uplink data split has taken effect.
- Check whether uplink preallocation has taken effect.
 - This function has taken effect if there is uplink data transmission but no downlink data transmission on the LTE side and the smart preallocation function takes effect on the LTE side in NSA networking. For details about the activation verification method, see *Uplink Scheduling* in eRAN feature documentation.
 - This function has taken effect if there is uplink data transmission but no downlink data transmission on the NR side and the uplink preallocation function takes effect on the NR side in NSA networking. For details about the activation verification method, see *Uplink Scheduling* in 5G RAN feature documentation.
- Check whether virtual grid model building has taken effect.

Perform monitoring from the second week after the function is enabled as it takes one week to build virtual grid models.

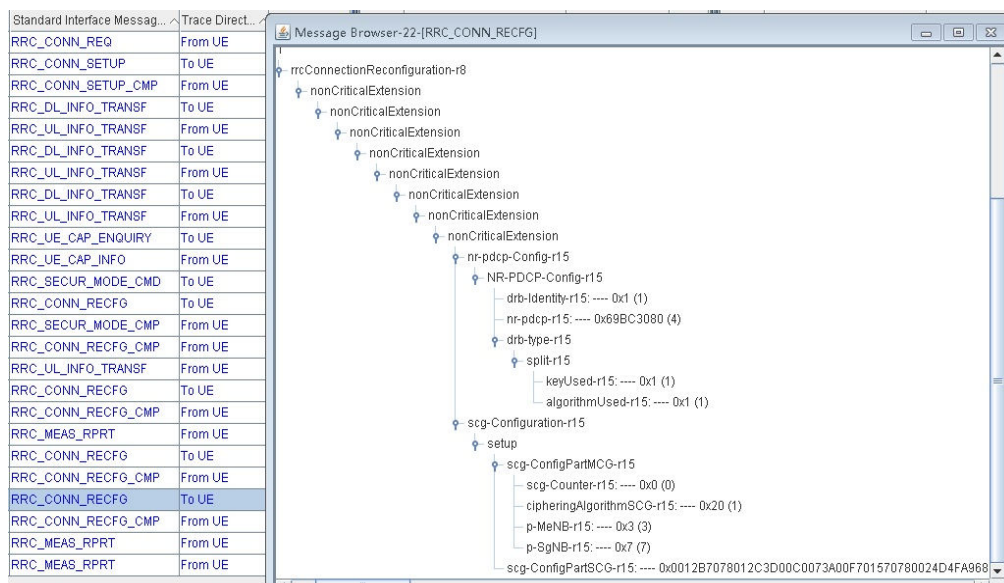
 - If the value of **L.AiGrid.LnrRsrpPred.Model.AverNum** is not 0, virtual grid model building has taken effect. You can further check the numbers of good-RSRP and poor-RSRP virtual grid models through the **L.VirtualGrid.LnrGoodRsrp.Model.Avg**,

L.VirtualGrid.LnrGoodRsrp.Model.Max,
L.VirtualGrid.LnrPoorRsrp.Model.Avg, and
L.VirtualGrid.LnrPoorRsrp.Model.Max counters.

- If the value of **L.VirtualGrid.Model.NrFreq.ScgAddAtt.Times** is not 0, SCG addition based on virtual grid models has taken effect. The number of failed SCG additions based on LNR virtual grid models is indicated by the **L.VirtualGrid.Model.NrFreq.ScgAddFail.Times** counter. The number of successful SCG additions based on LNR virtual grid models is equal to **L.VirtualGrid.Model.NrFreq.ScgAddAtt.Times** minus **L.VirtualGrid.Model.NrFreq.ScgAddFail.Times**.
 - If the value of **L.VirtualGrid.Model.NrFreq.InvalidMeasFree.Times** is not 0, invalid NR measurement configurations are not delivered because it is predicted that there is no NR coverage based on virtual grids.
 - If the **SMART_CARRIER_SELECTION_SW** option of the **MultiCarrUnifiedSch.MultiCarrierUnifiedSchSw** parameter is deselected and the values of **L.HHO.VirtualGrid.Model.PrepAttOut** and **L.NsaDC.PCCAnchor.HHO.PrepAttOut** are not 0, NSA anchoring using LTE RSRP values predicted based on virtual grids has taken effect.
 - Run the **DSP AIGRIDMOD** command to query the status of all models of the eNodeB, including Good RSRP Forecast Status, Poor RSRP Forecast Status, RSRP Model Latest Offline Time, and RSRP Model Latest Offline Reason.
- Check whether downlink real-time data split has taken effect.
Check the value of **N.PDCP.Vol.DL.X2U.TrfPDU.Tx/N.PDCP.Vol.DL.TrfPDU.Tx x 100%**. That is, check the ratio of the data volume of downlink PDCP PDUs transmitted over the X2 interface to that distributed from the PDCP to the RLC. If the value increases, downlink real-time data split has taken effect.

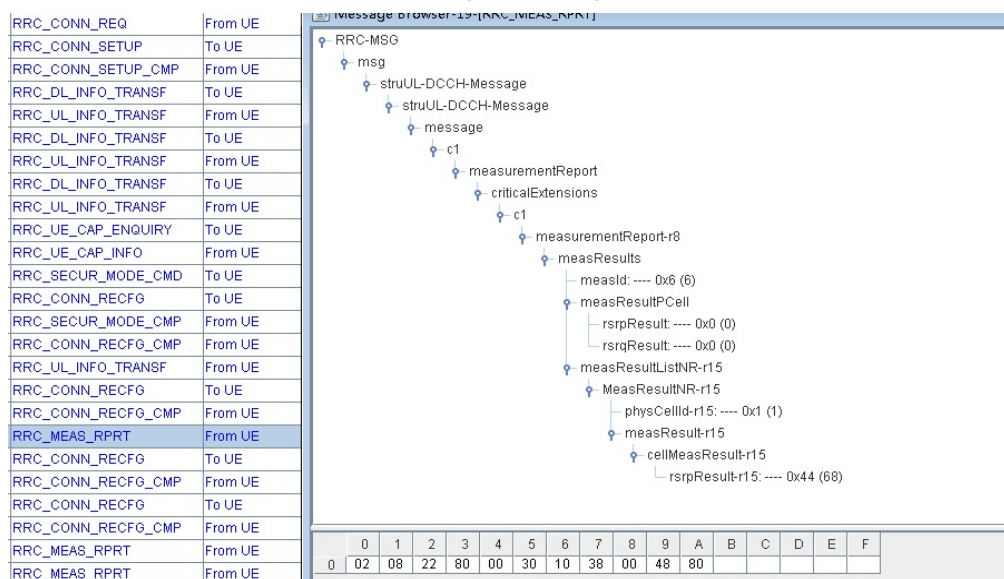
Message Tracing

1. Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** window is displayed.
2. Trace UE random access.
Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. You can observe that the UE sends an **RRC_CONN_SETUP_CMP** message to the eNodeB to initiate an LTE access procedure.
3. (Optional) The eNodeB delivers an NR measurement configuration message to the UE.
Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. You can observe that the eNodeB sends an **RRC_CONN_RECFG** message to the UE, which contains the *EventB1* IE.



4. (Optional) The UE reports measurement results.

Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. You can observe that the UE sends an RRC_MEAS_RPRT message, which contains IEs related to the measured PCIs and signal strength of NR cells.



5. The SgNB is added on the LTE side.

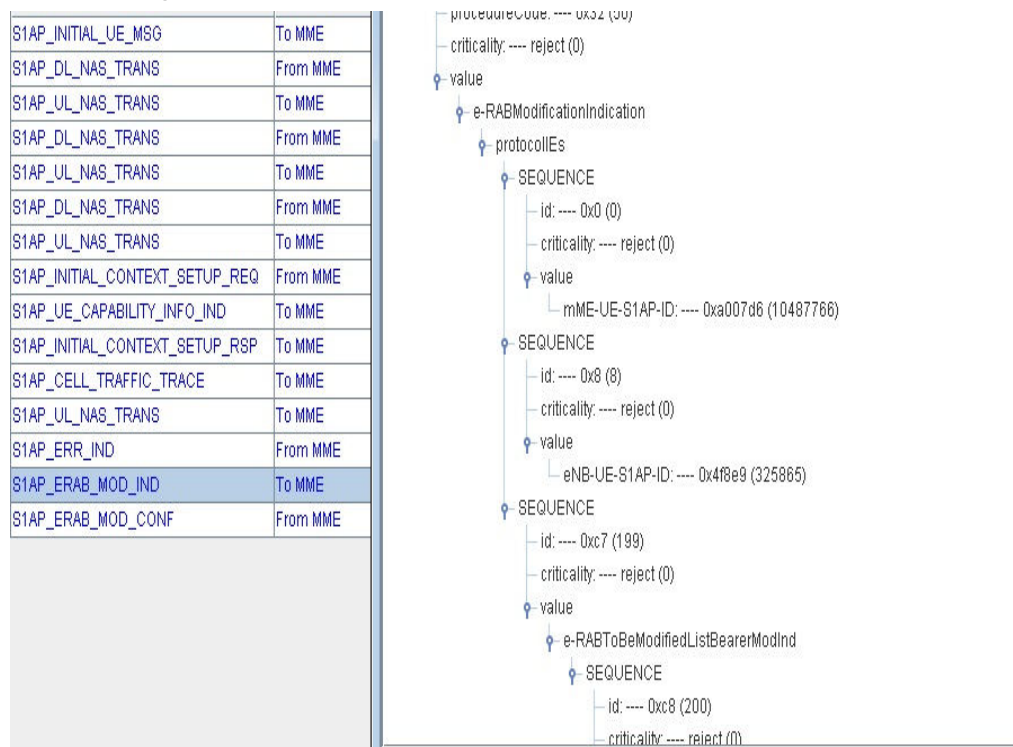
Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. You can observe that the eNodeB sends an SGNB_ADD_REQ message to the gNodeB over the X2 interface. The gNodeB responds with an SGNB_ADD_REQ_ACK message. The eNodeB sends an SGNB_RECONFIG_CMP message, indicating that the SCG addition procedure is completed.

6. The eNodeB configures the SCG for the UE. The UE returns a message indicating that the configuration is complete.

Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. You can observe that the eNodeB sends an RRC_CONN_RECFG message to the UE, which contains the *nr-config-r15* IE.

7. The eNodeB sends an ERAB MOD message to the core network over an S1 interface.

Choose **Trace Type > LTE > Application Layer > S1 Interface Trace**. You can observe that the eNodeB sends an S1AP_E_RAB_MOD_IND message to the core network, notifying the core network that all downlink data of the UE is sent to the gNodeB.



NOTE

For LTE-NR joint tracing, joint tracing must be started before the UE accesses the network.

8. (Optional) Check whether RLC rate optimization in downlink real-time data split has taken effect.
Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check whether the eNodeB sends an EN-DC Private Vendor Configuration Transfer message with an *eNB Cell Configuration Indication 1* IE whose bit 11 is set to 1, and whether the gNodeB sends the eNodeB an EN-DC Private Vendor Configuration Transfer message with a *gNB Cell Configuration Indication 1* IE whose bit 9 is set to 1. If so, the function has taken effect.
9. (Optional) Check whether uplink fallback to LTE has taken effect.
Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check whether the gNodeB sends the eNodeB an SgNB Modification Required message with IEs *SgNB to MeNB Container > CG-Config > scg-RB-Config > drb-ToAddModList > pdcp-Config* in which *primaryPath* is set to 0 (indicating the MCG) and *ul-DataSplitThreshold* is set to infinity. If so, the function has taken effect.
10. (Optional) Check whether the interference avoidance function has taken effect.
Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check whether the gNodeB sends the eNodeB an SgNB Addition Request

Acknowledge, SgNB Modification Request Acknowledge, or SgNB Modification Required message with IEs *SgNB Resource Coordination Information* > *UL Coordination Information* whose bits are not all zeros. If the bits are not all zeros, the function has taken effect. If the bits are all zeros, the function has not taken effect.

11. (Optional) Check whether uplink single-side transmission in NSA DC has taken effect.
 - Choose **Trace Type > LTE > Application Layer > Uu Interface Trace**. Check whether the eNodeB sends a UE an RRC_CONN_RECFG message with a *tdm-PatternConfig-r15* IE. If so, the function has taken effect.
 - Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check that the gNodeB sends the eNodeB an SgNB Addition Request Acknowledge, SgNB Modification Request Acknowledge, or SgNB Modification Required message with IEs *SgNB Resource Coordination Information* > *UL Coordination Information* that includes a bitmap indicating a TDM pattern.
12. (Optional) Check whether the NR data volume reporting function has taken effect.
 Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check whether the gNodeB sends the eNodeB a Secondary RAT Data Usage Report message. If so, the function has taken effect. In the message, the *Usage Count UL* IE indicates the uplink data volume, and the *Usage Count DL* IE indicates the downlink data volume.
13. (Optional) Check whether fast retransmission for downlink data split has taken effect.
 Choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check whether the gNodeB sends the eNodeB an EN-DC Private Vendor Configuration Transfer message with a *gNB Feature Configuration Indication 1* IE whose bit 0 is set to 1. If so, the function has taken effect.
14. (Optional) Check whether the function of requesting LTE-side large-packet status of ULFB UEs has taken effect.
 Once the function of requesting LTE-side large-packet status of ULFB UEs is enabled, choose **Trace Type > LTE > Application Layer > X2 Interface Trace**. Check whether the gNodeB sends the eNodeB an EN-DC Private Vendor Configuration Transfer message with a *gNB Feature Configuration Indication 1* IE whose bit 30 is set to 1. If so, the function has taken effect.

4.4.4 Network Monitoring

Monitor the counters listed below and compare them against the network plan to evaluate the feature performance. In Option 3, user-plane data is split at the PDCP layer of the MeNB. Therefore, the monitoring results are observed on the eNodeB side. In Option 3x, user-plane data is split at the PDCP layer of the SgNB. Therefore, the monitoring results are observed on the gNodeB side.

- On the eNodeB side
 - After NSA DC is enabled, observe the following counters to determine whether the handover success rate and service drop rate of NSA UEs are different from those of LTE-only UEs:

- Outgoing handover success rate of NSA UEs = $\frac{\text{L.NsaDc.HHO.ExecSuccOut}}{\text{L.NsaDc.HHO.ExecAttOut}} \times 100\%$
- Incoming handover success rate of NSA UEs = $\frac{\text{L.NsaDc.PCell.Change.Succ}}{\text{L.NsaDc.PCell.Change.Exec}} \times 100\%$
- Abnormal service drop rate of NSA UEs = $\frac{\text{L.NsaDc.E-RAB.AbnormRel}}{\text{L.NsaDc.E-RAB.NormRel}} \times 100\%$
- Calculate the total downlink data split volume of NSA UEs by adding **L.Thpt.bits.DL.McgSplit.MeNB** and **L.Thpt.bits.DL.McgSplit.SgNB**. If the volume increases continuously, this feature is running properly.
- Calculate the total uplink data split volume of NSA UEs by adding **L.Thpt.bits.UL.McgSplit.MeNB** and **L.Thpt.bits.UL.McgSplit.SgNB**. If the volume increases continuously, this feature is running properly.
- Observe the values of the **L.ChMeas.PRB.DL.NsaDc.Used.Avg** and **L.ChMeas.PRB.UL.NsaDc.Used.Avg** counters to determine the PRB usage of NSA UEs on the LTE side.
- Observe the **L.Thrp.bits.DL.NsaDc**, **L.Thrp.bits.DL.LastTTI.NsaDc**, and **L.Thrp.Time.DL.RmvLastTTI.NsaDc** counters to determine the downlink throughput of NSA UEs on the LTE side.
- Observe the **L.Thrp.bits.UL.NsaDc** and **L.Thrp.Time.UL.NsaDc** counters to determine the uplink throughput of NSA UEs on the LTE side.
- Observe the **L.NsaDc.DRB.Add.Att** and **L.NsaDc.DRB.Add.Succ** counters to obtain the number of data split bearer setup attempts and the number of successful data split bearer setups for NSA UEs, respectively.
- When the **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** parameter is set to **ADAPTIVE**, observe the **L.NsaDc.MenbTrig.SgNB.Rel.VolTE.SINR** counter to obtain the number of SCG releases for VoLTE UEs due to uplink SINR deterioration.
- When the **NsaDcMgmtConfig.VolteUeScgMgmtStrategy** parameter is set to **ADAPTIVE**, observe the **L.NsaDc.MenbTrig.SgNB.Rel.VolTE.PktLoss** counter to obtain the number of SCG releases for VoLTE UEs due to a high uplink or downlink packet loss rate.

NOTE

In NSA networking, counter reporting by PLMN is supported on the LTE side. For details, see *RAN Sharing* in eRAN feature documentation.

- On the gNodeB side
 - Observe the **N.PDCP.Vol.DL.X2U.TrfPDU.Tx** counter to obtain the total downlink data volume that is split for NSA UEs over the X2 interface. If the value of this counter increases continuously, this feature is working properly.
 - Observe the **N.PDCP.Vol.UL.X2U.TrfPDU.Rx** counter to obtain the total uplink data volume that is split for NSA UEs over the X2 interface. If the value of this counter increases continuously, this feature is working properly.
 - Observe the **N.NsaDc.InterSite.SgNB.Add.Att** counter to obtain the number of inter-site SgNB addition attempts in the LTE-NR NSA DC scenario.

- Observe the **N.NsaDc.InterSite.SgNB.Add.Succ** counter to obtain the number of successful inter-site SgNB additions in the LTE-NR NSA DC scenario.
- Observe the **N.NsaDc.SgNB.Add.Att.QCI** counter to obtain the number of SgNB addition attempts for a specific QCI in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.Add.Succ.QCI** counter to obtain the number of successful SgNB additions for a specific QCI in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.Rel.QCI** counter to obtain the total number of SgNB releases for a specific QCI in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.Rel.SgNBTrigger.QCI** counter to obtain the total number of SgNB releases triggered by SgNB for a specific QCI in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.Rel.MeNBTrigger.NormalRel.QCI** counter to obtain the total number of normal SgNB releases triggered by MeNB for a specific QCI in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.AbnormRel.QCI** counter to obtain the total number of abnormal SgNB releases triggered by SgNB for a specific QCI in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.Rel.InitialAdd.RachFail.QCI** counter to obtain the number of SCG releases due to random access failures during initial SCG addition for a specific QCI in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.UeInactive.Activity.Notification** counter to obtain the total number of SgNB Activity Notification messages sent due to UE inactivity timer expiration on the SgNB side in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.Rel.MeNBTrigger.UeInactive** counter to obtain the total number of SgNB releases due to UE inactivity on the MeNB side in LTE-NR NSA DC scenarios.
- Observe the **N.NsaDc.SgNB.Rel.SgNBTrigger.UeInactive** counter to obtain the total number of SgNB releases due to UE inactivity on the SgNB side in LTE-NR NSA DC scenarios.

5 L4S Congestion Identification

5.1 Principles

In NSA Option 3x networking, delay-sensitive services, such as real-time media, cloud gaming, video conferencing, and Cloud AR/VR, require collaborative cooperation among UEs, base stations, core networks, and application servers to provide congestion status identification for end-to-end transmission links to ensure user experience. The core network and application server are unaware of the air interface congestion status. Therefore, the base station must be able to identify the congestion status of the Low-Latency, Low-Loss, Scalable Throughput (L4S) bearer. To address this issue, Huawei introduces the Adaptive Managed Latency feature, which consists of uplink L4S congestion identification and downlink L4S congestion identification.

5.1.1 Uplink L4S Congestion Identification

To enable uplink L4S congestion identification for L4S bearers with a specific QCI, do as follows:

- On the eNodeB side:
 - Select the **UL_CONGESTION_CTRL_SW** option of the **EnodeBAlgoExtSwitch.L4SAlgoSwitch** parameter to enable uplink congestion identification.
 - Set **QciPara.L4SFlag** to **TRUE** for a specific QCI.
- On the gNodeB side:
 - Select the **UL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter to enable uplink congestion identification.
 - Set **gNBQciBearer.L4sFlag** to **TRUE** for a specific QCI.
 - Set **gNBQciBearer.L4sCongestDetectAlgorithm** to **L4S**.

 NOTE

The MAE allows users to manually trigger an L4S flag parameter consistency check. The MAE checks whether the values of the following parameters are consistent for the same QCI (specified by **QciPara.Qci** and **gNBQciBearer.Qci**):

- **QciPara.L4SFlag** corresponding to **QciPara.Qci** of the eNodeB
- **gNBQciBearer.L4SFlag** corresponding to **gNBQciBearer.Qci** of the gNodeB that has neighbor relationships with the eNodeB

If the values are inconsistent, the MAE displays a message indicating the inconsistency of the L4S flags configured for the QCI.

After uplink L4S congestion identification is enabled, the handling procedure works as follows:

1. The eNodeB RLC layer evaluates the average processing delay of uplink packets carried by the L4S bearer and reports the evaluation result to the gNodeB PDCP layer at a period specified by the **EnodeBAlgoExtSwitch.L4SCongestCheckPeriod** parameter.
2. The gNodeB RLC layer evaluates the average processing delay of uplink packets carried by the L4S bearer and reports the evaluation result to the gNodeB PDCP layer at a period specified by the **gNBQciBearer.L4SCongestMarkPeriod** parameter.
3. The gNodeB PDCP layer calculates the average uplink delay of the L4S bearer on the NSA network based on the average processing delay and number of uplink packets received from the gNodeB RLC layer and/or eNodeB RLC layer as well as the average processing delay of uplink packets at the gNodeB PDCP layer, and compares it with the L4S uplink delay upper threshold (specified by **gNBQciBearer.L4SUIDelayUpperThld**) and L4S uplink delay lower threshold (specified by **gNBQciBearer.L4SUIDelayLowerThld**) configured on the RAN side.
 - If the average uplink delay of the L4S bearer is greater than the L4S uplink delay upper threshold and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of the current and all subsequent packets carried by the L4S bearer will be set to **11**.
 - If the average uplink delay of the L4S bearer is less than or equal to the L4S uplink delay lower threshold, the ECN field in the IP header of the current and all subsequent packets carried on this bearer remains unchanged.
 - If the average uplink delay of the L4S bearer is greater than the L4S uplink delay lower threshold but less than or equal to the L4S uplink delay upper threshold and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of the current and all subsequent packets carried by this bearer will be randomly set to **11** at a proportion calculated by $(\text{Average uplink delay of the L4S bearer} - \text{L4S uplink delay lower threshold}) / (\text{L4S uplink delay upper threshold} - \text{L4S uplink delay lower threshold})$.

 NOTE

- If the NSA uplink data split path is MCG only, only steps 1 and 3 are performed. If the NSA uplink data split path is SCG only, only steps 2 and 3 are performed. If the NSA uplink data split path is MCG and SCG, steps 1 to 3 are performed.
- If the L4S uplink delay upper threshold and L4S uplink delay lower threshold are set to the same value, the average uplink delay of the L4S bearer is greater than the L4S uplink delay upper threshold, and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of the current and all subsequent packets carried by this bearer will be set to **11**.

5.1.2 Downlink L4S Congestion Identification

To enable downlink L4S congestion identification for L4S bearers with a specific QCI, do as follows:

- On the eNodeB side:
 - Select the **DL_CONGESTION_CTRL_SW** option of the **EnodeBAlgoExtSwitch.L4SAlgoSwitch** parameter to enable downlink congestion identification.
 - Set **QciPara.L4SFlag** to **TRUE** for a specific QCI.
- On the gNodeB side:
 - Select the **DL_CONGEST_CTRL_SW** option of the **gNodeBParam.L4sAlgoSwitch** parameter to enable downlink congestion identification.
 - Set **gNBQciBearer.L4sFlag** to **TRUE** for a specific QCI.
 - Set **gNBQciBearer.L4sCongestDetectAlgorithm** to **L4S**.

 NOTE

The MAE allows users to manually trigger an L4S flag parameter consistency check. The MAE checks whether the values of the following parameters are consistent for the same QCI (specified by **QciPara.Qci** and **gNBQciBearer.Qci**):

- **QciPara.L4SFlag** corresponding to **QciPara.Qci** of the eNodeB
- **gNBQciBearer.L4sFlag** corresponding to **gNBQciBearer.Qci** of the gNodeB that has neighbor relationships with the eNodeB

If the values are inconsistent, the MAE displays a message indicating the inconsistency of the L4S flags configured for the QCI.

After downlink L4S congestion identification is enabled, the handling procedure works as follows:

1. The eNodeB RLC layer evaluates the average processing delay of downlink packets carried by the L4S bearer and reports the evaluation result to the gNodeB PDPC layer at a period specified by the **EnodeBAlgoExtSwitch.L4SCongestCheckPeriod** parameter.
2. The gNodeB RLC layer evaluates the average processing delay of downlink packets carried by the L4S bearer and reports the evaluation result to the gNodeB PDPC layer at a period specified by the **gNBQciBearer.L4sCongestMarkPeriod** parameter.
3. The gNodeB PDPC layer calculates the average downlink delay of the L4S bearer on the NSA network based on the average processing delay and number of downlink packets received from the gNodeB RLC layer and/or eNodeB RLC layer as well as the average processing delay of downlink packets

at the gNodeB PDCP layer, and compares it with the L4S downlink delay upper threshold (specified by **gNBQciBearer.L4sDlDelayUpperThld**) and L4S downlink delay lower threshold (specified by **gNBQciBearer.L4sDlDelayLowerThld**) configured on the RAN side.

- If the average downlink delay of the L4S bearer is greater than the L4S downlink delay upper threshold and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of the current and all subsequent packets carried by the L4S bearer will be set to **11**.
- If the average downlink delay of the L4S bearer is less than or equal to the L4S downlink delay lower threshold, the ECN field in the IP header of the current and all subsequent packets carried on this bearer will not be processed.
- If the average downlink delay of the L4S bearer is greater than the L4S downlink delay lower threshold but less than or equal to the L4S downlink delay upper threshold and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of the current and all subsequent packets carried by this bearer will be randomly set to **11** at a proportion calculated by (Average downlink delay of the L4S bearer – L4S downlink delay lower threshold)/(L4S downlink delay upper threshold – L4S downlink delay lower threshold).

NOTE

- If the NSA downlink data split path is MCG only, only steps 1 and 3 are performed. If the NSA downlink data split path is SCG only, only steps 2 and 3 are performed. If the NSA downlink data split path is MCG and SCG, steps 1 to 3 are performed.
- If the L4S downlink delay upper threshold and L4S downlink delay lower threshold are set to the same value, the average downlink delay of the L4S bearer is greater than the L4S downlink delay upper threshold, and the ECN field in the IP header of the packets carried by the L4S bearer is **01**, then the ECN field in the IP header of the current and all subsequent packets carried by this bearer will be set to **11**.

5.2 Network Analysis

5.2.1 Benefits

After L4S congestion identification is enabled, the uplink and downlink latency of UEs may decrease, improving user experience.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

After L4S congestion identification is enabled, the **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** may decrease.

5.3 Requirements

5.3.1 Licenses

RAT	Feature ID	Feature Name	Model	NE	Sales Unit
NR	FOFD-071203	Adaptive Managed Latency	NR0S00AM L000	Macro gNode B LampSite gNode B	Per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with a BBU model other than the BBU5900C.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. The LBBPd board on the LTE side does not support this function. For details, see technical specifications of the related BBUs in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

No requirements

Cells

None

5.3.4 Networking

NSA Option 3x networking is used.

5.3.5 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
L4S Flag	QciPara.L4SFlag	If this parameter is set to TRUE , the L4S function takes effect for the corresponding bearer.

Parameter Name	Parameter ID	Setting Notes
L4S Algorithm Switch	EnodebAlgoExtSwitch.L4SAlgoSwitch	To enable uplink L4S congestion identification, select the UL_CONGESTION_CTRL_SW option. To enable downlink L4S congestion identification, select the DL_CONGESTION_CTRL_SW option.
L4S congestion detection algorithm	gNBQciBearer.L4sCongestDetectAlgorithm	This parameter specifies the congestion detection algorithm for L4S-related functions. Set this parameter to its recommended value.
L4S Congestion Check Period	EnodeBAlgoExtSwitch.L4SCongestCheckPeriod	This parameter specifies the check period for L4S congestion identification. Set this parameter to its recommended value.
L4S Flag	gNBQciBearer.L4sFlag	If this parameter is set to TRUE , the L4S function takes effect for the corresponding bearer.
L4S Algorithm Switch	gNodeBParam.L4sAlgoSwitch	To enable uplink L4S congestion identification, select the UL_CONGEST_CTRL_SW option. To enable downlink L4S congestion identification, select the DL_CONGEST_CTRL_SW option.
L4S Congestion Check Period	gNBQciBearer.L4sCongestMarkPeriod	This parameter specifies the check period for L4S congestion identification. Set this parameter to its recommended value.
L4S Uplink Delay Lower Threshold	gNBQciBearer.L4sUldelayLowerThld	Set this parameter to its recommended value.
L4S Uplink Delay Upper Threshold	gNBQciBearer.L4sUldelayUpperThld	Set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
L4S Downlink Delay Lower Threshold	gNBQciBearer.L4sDlDelayLowerThld	Set this parameter to its recommended value.
L4S Downlink Delay Upper Threshold	gNBQciBearer.L4sDlDelayUpperThld	Set this parameter to its recommended value.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the uplink congestion identification switches if uplink congestion identification is required
MOD ENODEBALGOEXTSWITCH: L4SAlgoSwitch=UL_CONGESTION_CTRL_SW-1;
MOD GNODEBPARAM: L4SAlgoSwitch=UL_CONGEST_CTRL_SW-1;
//Turning on the downlink congestion identification switches if downlink congestion identification is required
MOD ENODEBALGOEXTSWITCH: L4SAlgoSwitch=DL_CONGESTION_CTRL_SW-1;
MOD GNODEBPARAM: L4SAlgoSwitch=DL_CONGEST_CTRL_SW-1;
//Configuring the L4S bearer flag
MOD QCIPARA: Qci=100, L4SFlag=TRUE;
MOD GNBQCIBEARER: Qci=100, L4SFlag=TRUE;
//Configuring the L4S congestion detection algorithm
MOD GNBQCIBEARER:Qci=100,L4sCongestDetectAlgorithm=L4S;
//Configuring the check period for L4S congestion identification
MOD ENODEBALGOEXTSWITCH: L4SCongestCheckPeriod=1;
MOD GNBQCIBEARER:Qci=100,L4sCongestMarkPeriod=10;
//Configuring L4S Uplink Delay Lower Threshold
MOD GNBQCIBEARER:Qci=100,L4sUlDelayLowerThld=20;
//Configuring L4S Uplink Delay Upper Threshold
MOD GNBQCIBEARER:Qci=100,L4sUlDelayUpperThld=60;
//Configuring L4S Downlink Delay Lower Threshold
MOD GNBQCIBEARER: Qci=100, L4sDlDelayLowerThld=20;
//Configuring L4S Downlink Delay Upper Threshold
MOD GNBQCIBEARER: Qci=100, L4sDlDelayUpperThld=60;
```

Deactivation Command Examples

```
//Turning off the downlink congestion identification switches
MOD ENODEBALGOEXTSWITCH: L4SAlgoSwitch=DL_CONGESTION_CTRL_SW-0;
MOD GNODEBPARAM: L4SAlgoSwitch=DL_CONGEST_CTRL_SW-0;
//Turning off the uplink congestion identification switches
MOD ENODEBALGOEXTSWITCH: L4SAlgoSwitch=UL_CONGESTION_CTRL_SW-0;
MOD GNODEBPARAM: L4SAlgoSwitch=UL_CONGEST_CTRL_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Use a packet capture tool to capture uplink and downlink IP packets at the egress of the base station. If the ECN field in the IP packet header corresponding to the L4S bearer is set to **01** before the L4S congestion identification function is enabled and set to **11** after the function is enabled, the base station has set the L4S congestion flag. This means that the L4S congestion identification function has taken effect. [Table 5-2](#) describes the ECN value information.

Table 5-2 ECN value information

Binary Codepoint	Codepoint Name	Meaning
00	Non-ECT	Not ECN-capable Transport
01	ECT(1)	L4S-Capable Transport
10	ECT(0)	ECN-Capable Transport
11	CE	Congestion Experienced

5.4.3 Network Monitoring

None

6 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

- Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.
- End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.101: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 37.340: "E-UTRA and NR; Multi-connectivity; Stage-2"
- 3GPP TS 36.331: "E-UTRA; Radio Resource Control (RRC) ; Protocol specification"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.425: "NG-RAN; NR user plane protocol"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.323: "NR; Packet Data Convergence Protocol (PDCP) specification"
- *Transmission Resource Management* in 5G RAN feature documentation
- *Power Control* in 5G RAN feature documentation
- *Multi-Frequency Smart Aggregation* in 5G RAN feature documentation
- *Uplink Scheduling* in 5G RAN feature documentation
- *Downlink Scheduling* in 5G RAN feature documentation
- *Cell Combination* in 5G RAN feature documentation
- *Hyper Cell* in 5G RAN feature documentation
- *CA Experience Improvement* in 5G RAN feature documentation
- *Extended Cell Range* in 5G RAN feature documentation
- *DRX* in 5G RAN feature documentation
- *URLLC* in 5G RAN feature documentation
- *Multi-Operator Sharing* in 5G RAN feature documentation
- *UL and DL Decoupling* in 5G RAN feature documentation
- *Remote Interference Management (Low-Frequency TDD)* in 5G RAN feature documentation
- *Ultimate Cloud X Experience* in 5G RAN feature documentation
- *Adaptive Managed Latency* in 5G RAN feature documentation
- *LTE AHR (TDD)* in eRAN feature documentation
- *eMTC* in eRAN feature documentation
- *Massive MIMO Enhancements (FDD)* in eRAN feature documentation
- *VoLTE* in eRAN feature documentation

- *Terminal Awareness Differentiation* in eRAN feature documentation
- *High Speed Mobility* in eRAN feature documentation
- *AQM* in eRAN feature documentation
- *Emergency Call* in eRAN feature documentation
- *RAN Sharing* in eRAN feature documentation
- *ROHC* in eRAN feature documentation
- *Energy Conservation and Emission Reduction* in eRAN feature documentation
- *Relay* in eRAN feature documentation
- *Video Experience Optimization* in eRAN feature documentation
- *Super Combined Cell (FDD)* in eRAN feature documentation
- *ANR Management* in eRAN feature documentation
- *Multi-carrier Unified Scheduling* in eRAN feature documentation
- *DRX and Signaling Control* in eRAN feature documentation
- *Carrier Aggregation* in eRAN feature documentation
- *Idle Mode Management* in eRAN feature documentation
- *Uplink Scheduling* in eRAN feature documentation
- *Downlink Scheduling* in eRAN feature documentation
- *QoS Management* in eRAN feature documentation
- *Breathing Pilot* in eRAN feature documentation
- *LTE and NR Zero Bufferzone* in eRAN feature documentation
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *SRAN Networking and Evolution Overview*
- *IP Performance Monitor*
- *X2 and S1 Self-Management in NSA Networking*
- *NSA Mobility Management*
- *EPC-based NSA Performance Enhancement*
- *NSA-SA Selection Based on User Experience*
- *EPC-based NSA Experience Improvement*
- *Common Transmission*
- *Synchronization*
- *BBU Interconnection* in 3900 & 5900 Series Base Station Product Documentation
- *WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management*
- *License Control Item Lists (FDD)* in eRAN feature documentation
- *License Control Item Lists (CIoT)* in eRAN feature documentation
- *License Control Item Lists (TDD)* in eRAN feature documentation
- *License Control Item Lists* in 5G RAN feature documentation
- *LTE FDD and NR Dynamic Spectrum Sharing*
- *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
- *Multi-RAT Coordinated Energy Saving*